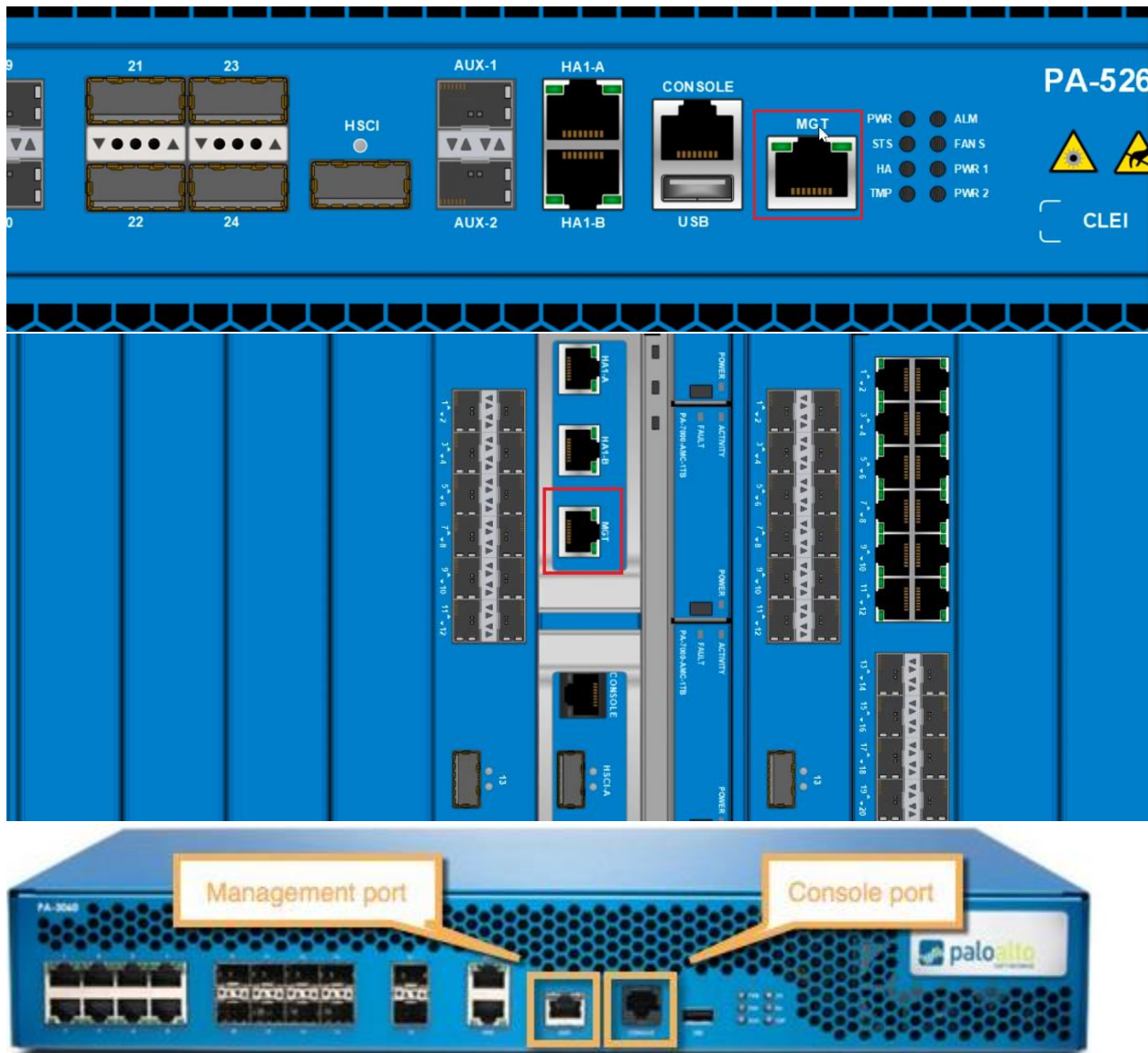
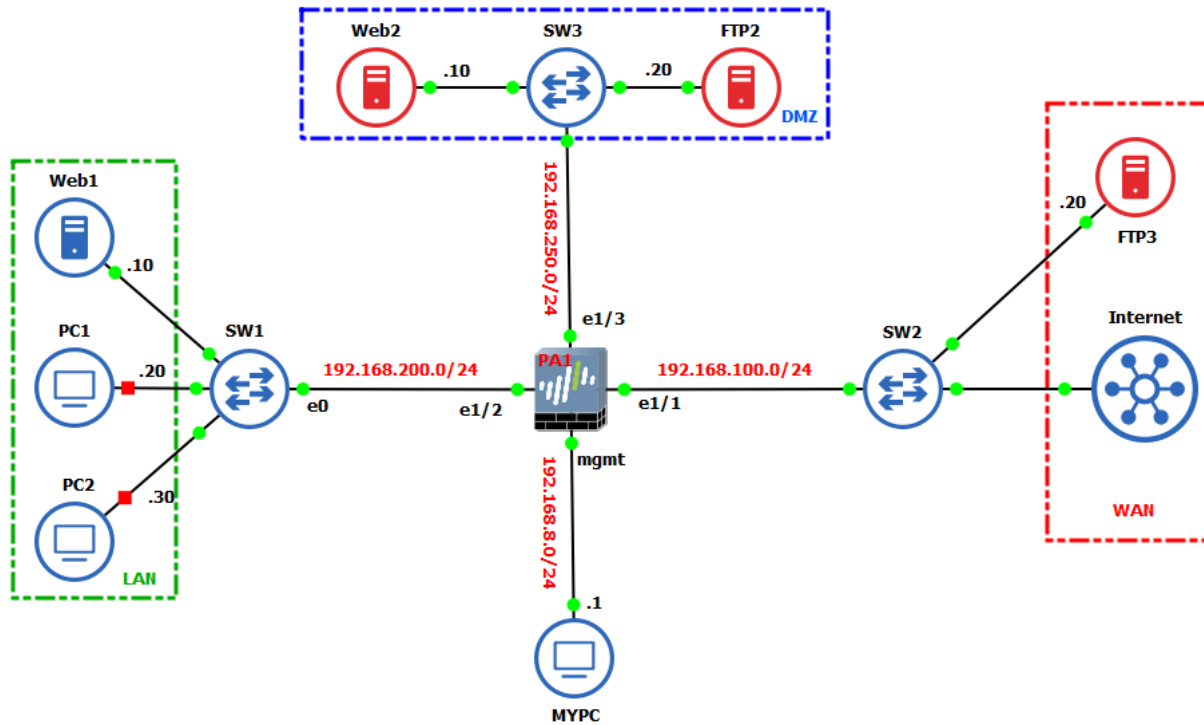


PA Initial Configuration:

- o To configure and access first time Palo Alto Networks Next-Generation Firewalls.
- o PA Firewalls can be accessed by either out-of-band management port labelled as MGT.
- o Or Palo Alto Firewalls can be accessed by a Serial Console port (similar to Cisco devices).
- o MGT port, separate management functions of firewall from data processing functions.
- o All initial configurations be performed either on out-of-band management interface.
- o Or all initial configurations of firewall be performed by using a serial console port.
- o The serial port need standard roll over cable to used to connect to Palo Alto Firewall.
- o To access the Palo Alto Networks Firewall for the first time through the MGT port,
- o You need to connect a laptop to the MGT port using a straight-thru Ethernet cable.
- o By default, the web GUI interface is accessed through 192.168.1.1 /24 IP Address.
- o By default, Web GUI & CLI login credentials Username: admin and Password: admin.





LAN and Management IP Address Configuration	
Web1 (WordPress Docker)	PC1 (Webterm Docker)
<pre># Static config for eth0 auto eth0 iface eth0 inet static address 192.168.200.10 netmask 255.255.255.0 gateway 192.168.200.100 up echo nameserver 8.8.8.8 > /etc/resolv.conf</pre>	<pre># Static config for eth0 auto eth0 iface eth0 inet static address 192.168.200.20 netmask 255.255.255.0 gateway 192.168.200.100 up echo nameserver 8.8.8.8 > /etc/resolv.conf</pre>
PC2 (Webterm Docker)	MGMT (VMWare Network Adapter VMnet8)
<pre># Static config for eth0 auto eth0 iface eth0 inet static address 192.168.200.30 netmask 255.255.255.0 gateway 192.168.200.100 up echo nameserver 8.8.8.8 > /etc/resolv.conf</pre>	<pre>Physical Address 00-50-56-C0-00-08 DHCP Enabled No IPv4 Address 192.168.8.1 IPv4 Subnet Mask 255.255.255.0 IPv4 Default Gateway IPv4 DNS Server</pre>

DMZ IP Address Configuration	
Web2 (WordPress Docker)	FTP2 (Toolbox Docker)
# Static config for eth0 auto eth0 iface eth0 inet static address 192.168.250.10 netmask 255.255.255.0 gateway 192.168.250.100 up echo nameserver 8.8.8.8 > /etc/resolv.conf	# Static config for eth0 auto eth0 iface eth0 inet static address 192.168.250.20 netmask 255.255.255.0 gateway 192.168.250.100 up echo nameserver 8.8.8.8 > /etc/resolv.conf
WAN IP Address Configuration	
FTP3 (Toolbox Docker)	WAN Adopter (Interent Interface)
# Static config for eth0 auto eth0 iface eth0 inet static address 192.168.100.20 netmask 255.255.255.0 gateway 192.168.100.1 up echo nameserver 8.8.8.8 > /etc/resolv.conf	DHCP Enabled Yes IPv4 Address 192.168.100.23 IPv4 Subnet Mask 255.255.255.0 Lease Obtained Friday, December 13, 2019 8:45:03 PM Lease Expires Monday, December 16, 2019 8:44:49 PM IPv4 Default Gateway 192.168.100.1 IPv4 DHCP Server 192.168.100.1 IPv4 DNS Server 192.168.100.1

Console to Palo Alto Firewall:

Login with default username and password **admin**.

PA-VM login: **admin**

Password: **admin**

Last login: Fri Dec 13 22:43:10 from 192.168.8.1

The Management Interface get IP address automatically from DHCP server.

Name: **Management Interface**

Link status:

Runtime link speed/duplex/state: 1000/full/up

Configured link speed/duplex/state: auto/auto/auto

MAC address:

Port MAC address 0c:c8:d3:97:19:00

Ip address: **192.168.8.186**

Netmask: 255.255.255.0

Default gateway: **192.168.8.2**

Ipv6 address: unknown

Ipv6 link local address: fe80::ec8:d3ff:fe97:1900/64

Ipv6 default gateway:

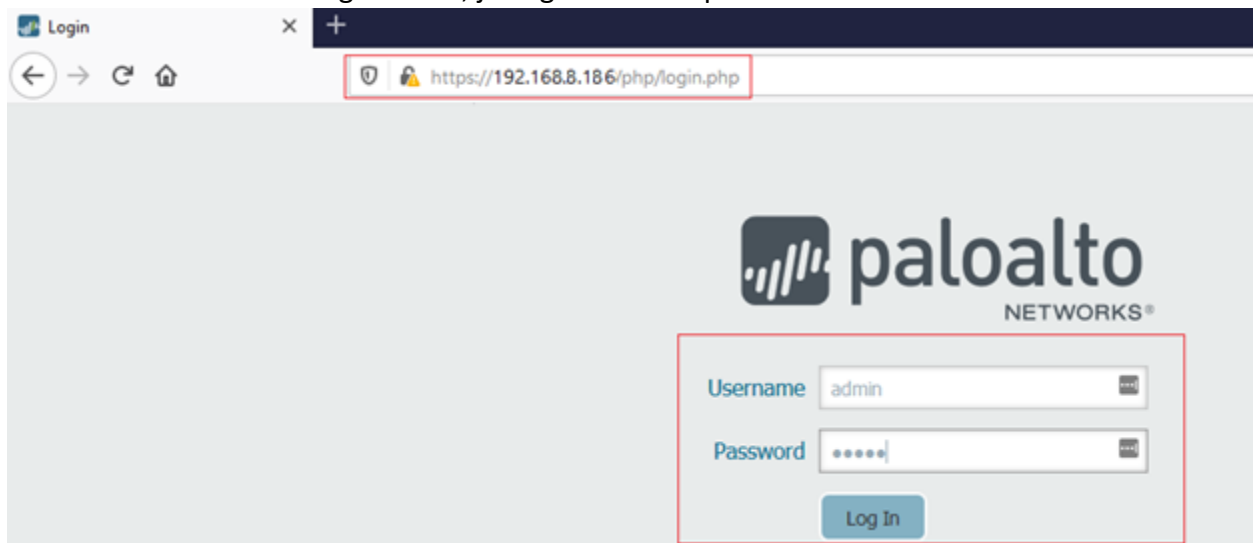
**Get IP address
Configuration from DHCP**

Set Static IP Address through CLI

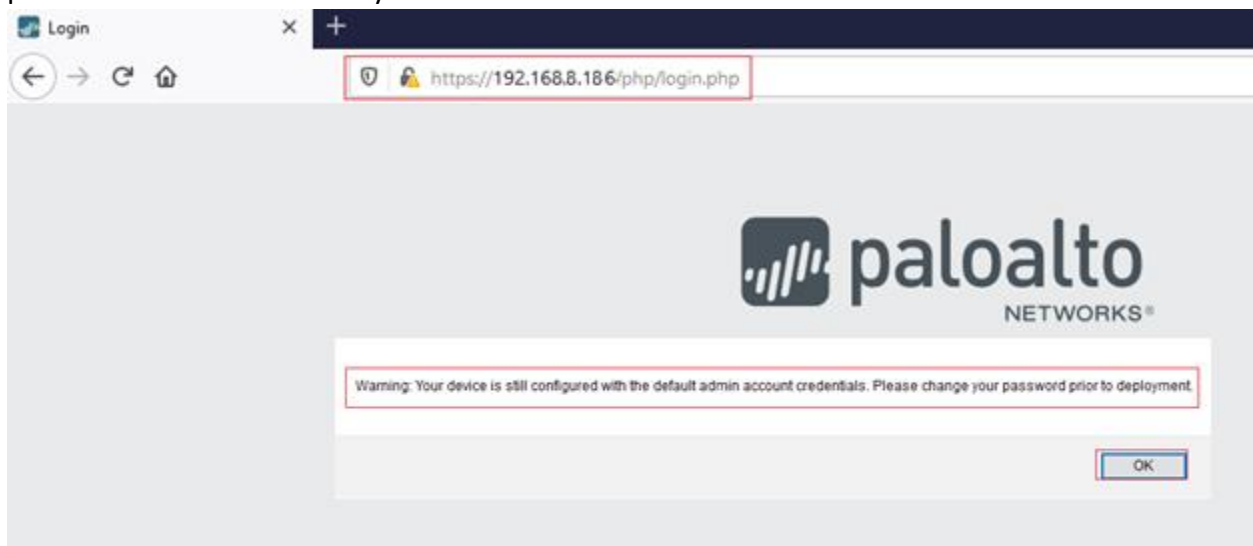
```
admin@PA-VM> configure
admin@PA-VM# delete deviceconfig system type dhcp-client
admin@PA-VM# set deviceconfig system type static
admin@PA-VM# set deviceconfig system ip-address 192.168.8.100 netmask 255.255.255.0
admin@PA-VM# commit
admin@PA-VM# run show interface management
```

Set Static IP Address through GUI:

Login to PA-VM via GUI, Use the computer that connected to management network; then use web browser to navigate <https://192.168.8.186> Login using **admin/admin**
This is the error of local signed cert, just ignore it and proceed with the connection.



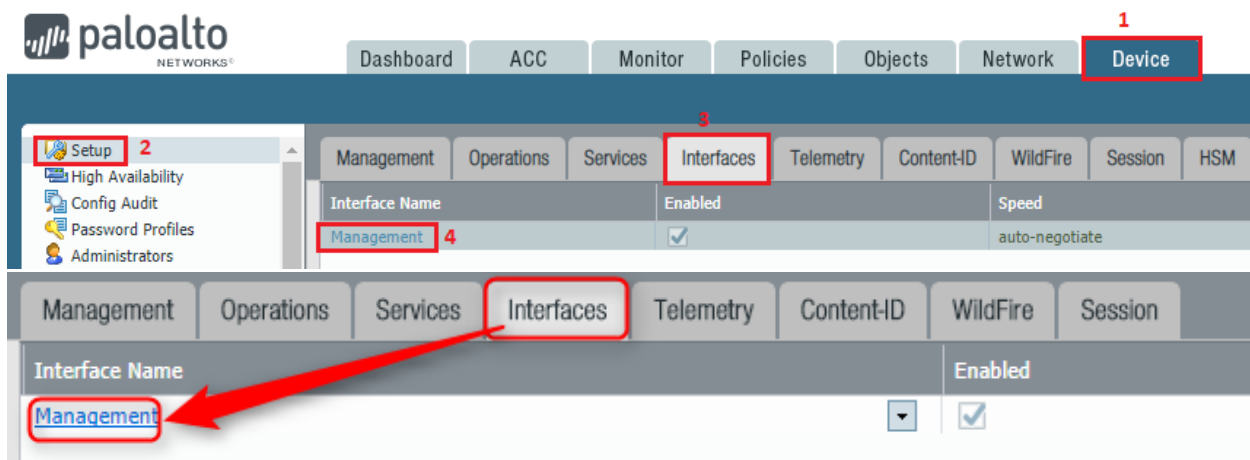
There will be pop-up asking to reset the password to new one as you logged in with default password. Need not to worry and click on **OK**.



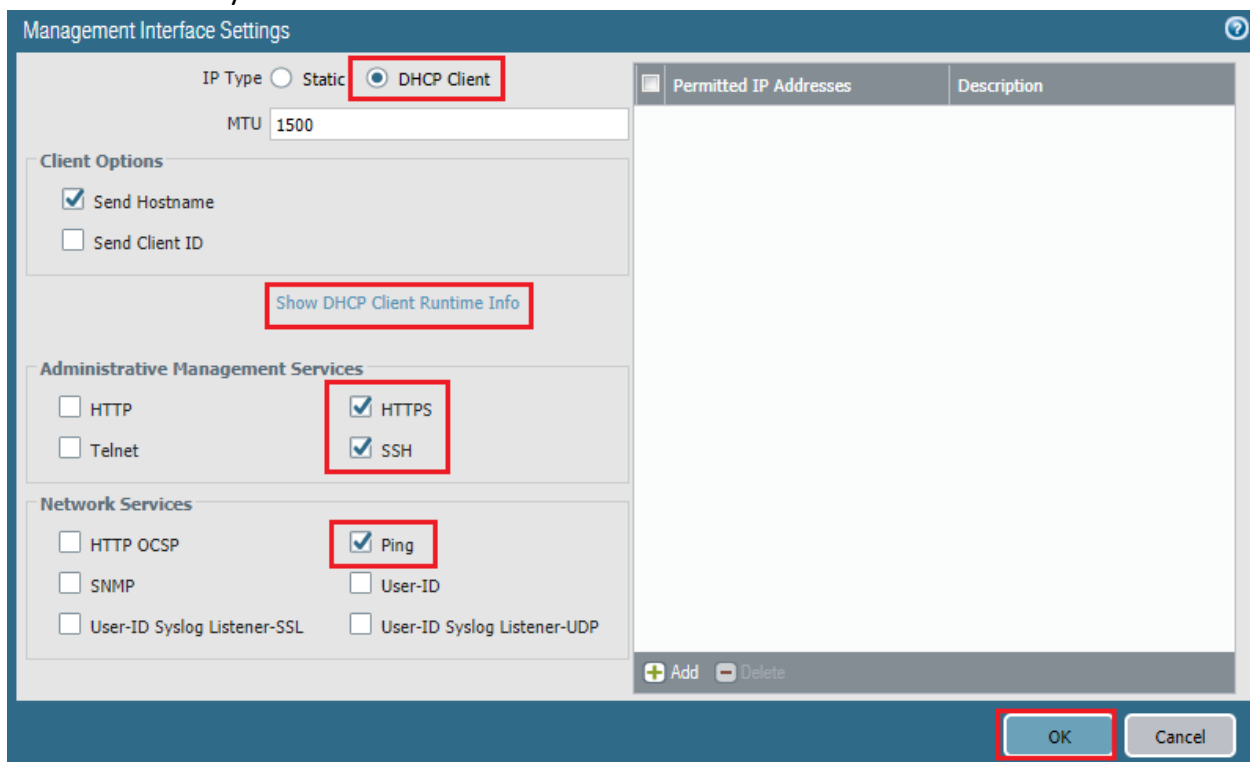
Click the **Device Tab**

To change the IP address of the Management Port. **Device>Setup >Interfaces**

Click the Interfaces TAB – Click **Management** Interface.



We can specify the IP Address to be DHCP or a static IP address. By default, Virtual Machine take IP address by DHCP server.



Change the IP Address accordingly and enable or disable any management services as required. **HTTPS, SSH and Ping (ICMP)** are enabled by default. When ready **click OK**:

Management Interface Settings

IP Type ☒ Static ☐ DHCP Client

IP Address: 192.168.8.100
 Netmask: 255.255.255.0
 Default Gateway: 192.168.8.2

IPv6 Address/Prefix Length:
 Default IPv6 Gateway:
 Speed: auto-negotiate
 MTU: 1500

Administrative Management Services
☐ HTTP ☒ HTTPS
☐ Telnet ☒ SSH

Network Services
☐ HTTP OSCP ☒ Ping
☐ SNMP ☐ User-ID
☐ User-ID Syslog Listener-SSL ☐ User-ID Syslog Listener-UDP

Permitted IP Addresses | Description

+ Add - Delete

OK Cancel

Now **Click** on **Commit** on the top right corner to save the changes to the new configuration.

Commit Status

Commit Config

Operation: Commit
 Status: Active
 Result: Pending

Progress: 98%

Details
 Warnings

Close

Since we change the IP address from 192.168.8.186 network to the 192.168.8.100 Network the progress bar will never reach 100%. The reason for this is the browser won't be able to update progress bar to 100%. If you were using the command console it would complete as expected. In my instance, the device got to 98%. Click **Close**. Switch computer's IP address back to its normal network and be able to access the Palo Alto on the new IP address.

Management Interface Settings

IP Type: ☒ Static ☐ DHCP Client

IP Address: 192.168.8.100
 Netmask: 255.255.255.0
 Default Gateway: 192.168.8.2

IPv6 Address/Prefix Length:
 Default IPv6 Gateway:

Speed: auto-negotiate
 MTU: 1500

Administrative Management Services

☐ HTTP ☒ HTTPS
☐ Telnet ☒ SSH

Network Services

☐ HTTP OCSP ☒ Ping
☐ SNMP ☐ User-ID
☐ User-ID Syslog Listener-SSL ☐ User-ID Syslog Listener-UDP

Permitted IP Addresses	Description
------------------------	-------------

+ Add - Delete

OK Cancel

Static IP Address
 Subnet Mask
 Default Gateway
 IPV6 Address and Subnet Mask
 Default IPV6 Gateway Address
 Configure a data rate
 MTU in bytes for packets sent on interface
 Use this service to access the firewall web interface
 Use this Service to access the firewall CLI interface
 Enter the IP addresses from which administrators can access the firewall through the interface.

Item	Description
MGT interface	Select one: Static- Requires you to enter the IP address (IPv4), Netmask (IPv4),and Default Gateway manually. DHCP Client—Configures the MGT interface as a DHCP client. Client Options: Send Hostname—Causes the MGT interface to send its hostname to the DHCP server as part of DHCP Option 12. Send Client ID—Causes the MGT interface to send its client identifier as part of DHCP Option 61. If you select DHCP Client, optionally click Show DHCP Client Runtime Info to view the dynamic IP interface status: Interface—Indicates MGT interface. IP Address—IP address of the MGT interface. Netmask—Subnet mask for the IP address, which indicates which bits are network or subnetwork and which bits are host. Gateway—Default gateway for traffic leaving the MGT interface. Primary/Secondary NTP—IP address of up to two NTP servers serving the MGT interface. Lease Time—Number of days, hours, minutes, and seconds that the DHCP IP address is assigned. Expiry Time—Year/Month/Day, Hours/Minutes/Seconds, and time zone, indicating when DHCP lease will expire.

	DHCP Server—IP address of the DHCP Server responding to MGT interface DHCP Client. Domain—Name of domain to which the MGT interface belongs. DNS Server—IP address of up to two DNS servers serving the MGT interface. Optionally, you can Renew the DHCP lease for the IP address assigned to the MGT interface.
IP Address (IPv4)	If the network uses IPv4, assign an IPv4 address to the interface.
Netmask (IPv4)	If assigned an IPv4 address to interface, also enter network mask.
Default Gateway	If assigned IPv4 address to interface, also assign default gateway.
IPv6 Address/Prefix Length	If the network uses IPv6, assign an IPv6 address to the interface. To indicate the netmask, enter an IPv6 prefix length.
Default IPv6 Gateway	If assigned an IPv6 address to the interface, also assign an IPv6 address to the default gateway.
Speed	Configure a data rate and duplex option for the interface. Use the default auto-negotiate setting to have the firewall determine the interface speed.
MTU	Enter maximum transmission unit in bytes for packets sent on interface.
Administrative Management Services	HTTP—Use this service to access the firewall web interface. Telnet—Use this service to access the firewall CLI. HTTPS—Use this service for secure access to firewall web interface. SSH—Use this service for secure access to the firewall CLI.
Network Services	Select the services you want to enable on the interface: HTTP OCSP—Use this service to configure the firewall as an Online Certificate Status Protocol (OCSP) responder. Ping—Use this service to test connectivity with external services. SNMP—Use this service to process firewall statistics queries from an SNMP manager. User-ID—Enable Redistribution of user mappings among firewalls. User-ID Syslog Listener-SSL—To enable the PAN-OS integrated User-ID agent to collect syslog messages over SSL. User-ID Syslog Listener-UDP—To enable the PAN-OS integrated User-ID agent to collect syslog messages over UDP.
Permitted IP Addresses	Enter the IP addresses from which administrators can access the firewall through the interface. An empty list (default) specifies that access is available from any IP address.

Interface Mgmt:

- o Interface Management profile protects the firewall from unauthorized access.
- o By defining the services and IP addresses that a firewall interface permits.
- o Can assign Interface Management profile to Layer 3 Ethernet interfaces.
- o Can assign to sub interfaces and to logical interfaces, VLAN, loopback, & tunnel.
- o Do not attach it profile that allows Telnet, SSH, HTTP, or HTTPS Internet Interface.
- o Do not attach Interface Management profile to GlobalProtect portal or gateway.
- o Because these will expose the management interface to the outside Internet.
- o To allow Ping & other management traffic, configure Interface Management Profile.

Network>Network Profiles > Interface Mgmt > Add

Interface Management Profile

Name: Allow-Ping-SSH

Administrative Management Services

- ☐ HTTP
- ☐ HTTPS
- ☐ Telnet
- ☒ SSH

Network Services

- ☒ Ping
- ☐ HTTP OCSP
- ☐ SNMP
- ☐ Response Pages
- ☐ User-ID
- ☐ User-ID Syslog Listener-SSL
- ☐ User-ID Syslog Listener-UDP

Permitted IP Addresses

Restricted the access to specific IP Address if needed

+ Add - Delete

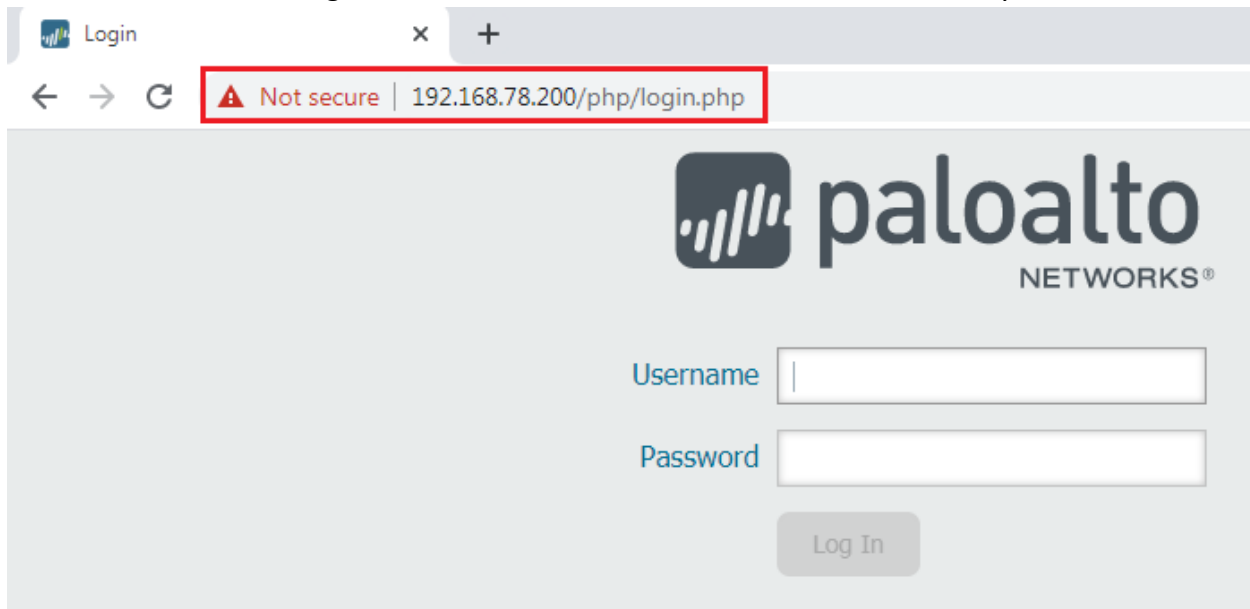
Ex. IPv4 192.168.1.1 or 192.168.1.0/24 or IPv6 2001:db8:123:1::1 or 2001:db8:123:1::/64

OK Cancel

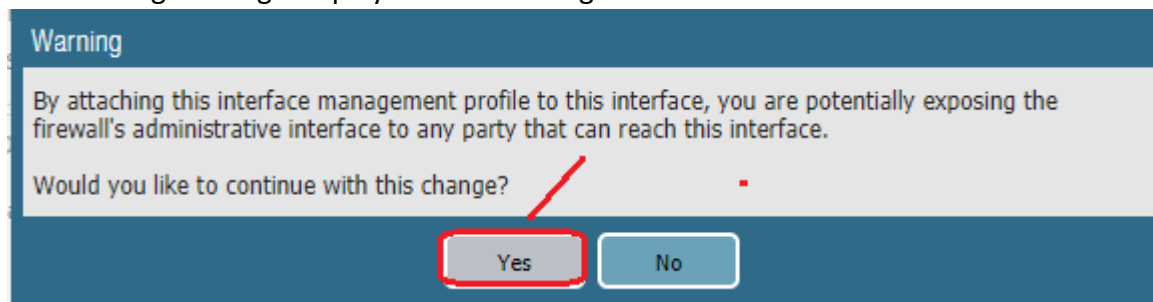
- o Assign Management Profile to any Layer 3 interface where want to manage PA Firewall.
- o Quite handy want to allow management functions on any layer 3 interface.
- o Network>Interfaces>Ethernet> ethernet1/2>Advance>Management Profile.

Field	Description
Name	Enter profile name. This name appears in list of Interface Management profiles when configuring interfaces.
Administrative Management Services	Telnet —Use to access firewall CLI. Telnet uses plaintext, not secure. SSH —Use for secure access to the firewall CLI. HTTP —Use to access the firewall web interface. HTTP uses plaintext, which is not as secure as HTTPS. HTTPS —Use for secure access to the firewall web interface.
Network Services	Ping —Use to test connectivity with external services. HTTP OCSP —Use to configure the firewall as an Online Certificate Status Protocol (OCSP) responder. SNMP —Use to process firewall statistics queries from SNMP manager. Response Pages —Use to enable response pages for: Captive Portal —The ports used to serve Captive Portal response pages are left open on Layer 3 interfaces: User-ID Syslog Listener-SSL —Use to allow the PAN-OS integrated User-ID agent to collect syslog messages over SSL. User-ID Syslog Listener-UDP —Use to allow the PAN-OS integrated User-ID agent to collect syslog messages over UDP.
Permitted IP Addresses	Enter the list of IPV4 or IPV6 addresses from which the interface allows access.

- o After attached Management Profile, to inside interface now can access by inside IP.



- o Warning Message display when attaching to other interfaces.

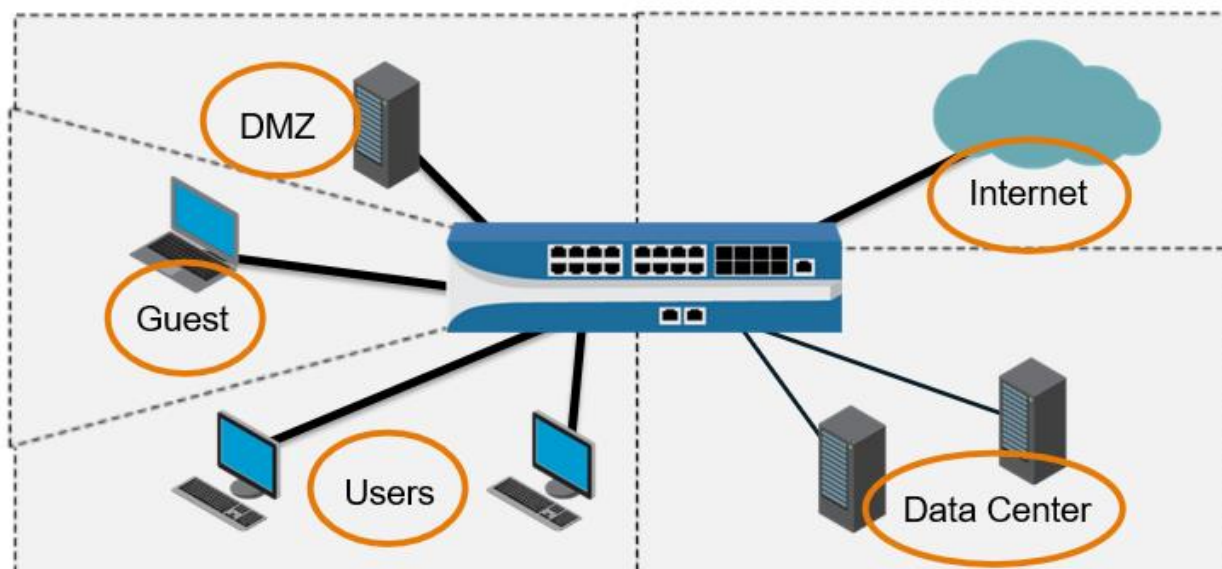


Inside interface where management Profile has been attached.

Interface	Interface Type	Management Profile	Link State	IP Address	Virtual Router	Tag	VLAN / Virtual-Wire	Security Zone
ethernet1/1	Layer3	Ping		192.168.17.200/24	default	Untagged	none	Outside
ethernet1/2	Layer3	HTTPS-SSH		192.168.78.200/24	default	Untagged	none	Inside

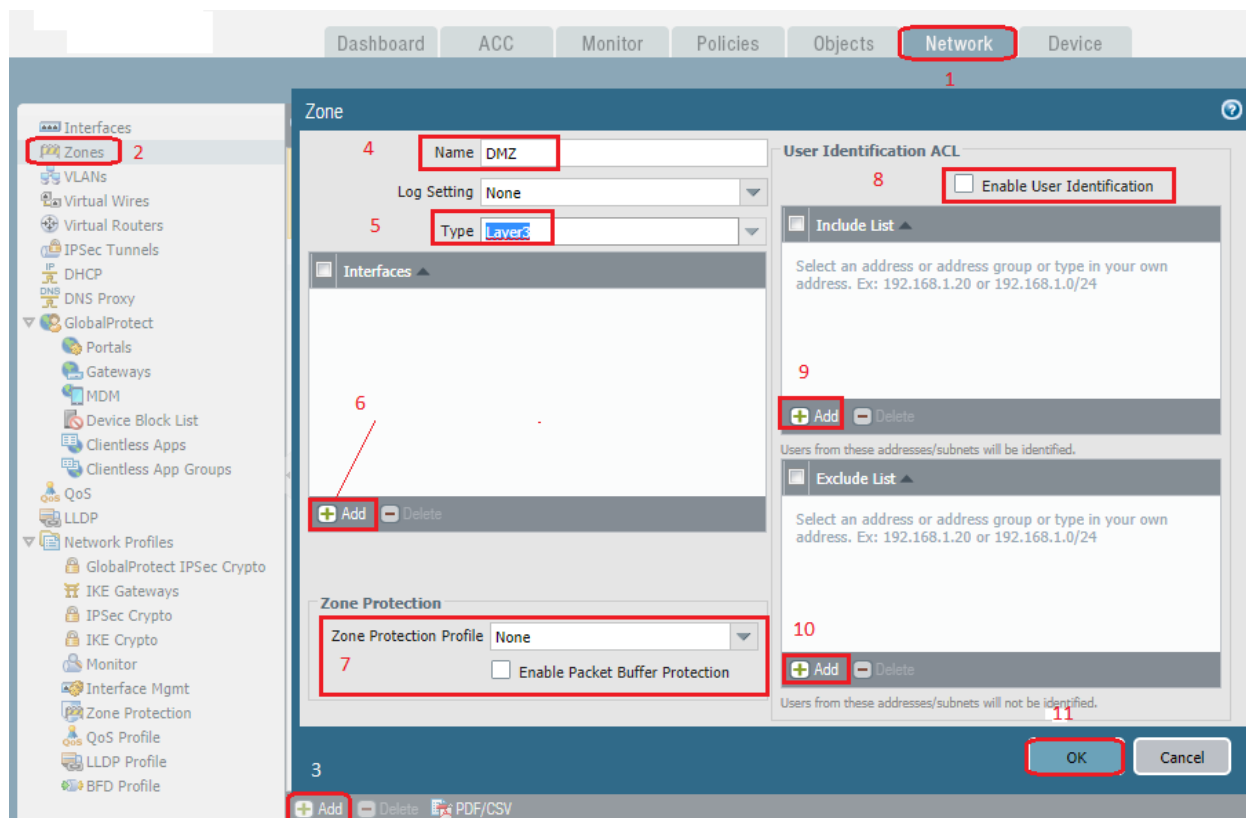
Firewall Zones:

- o Security zones are logical way to group physical and virtual interfaces on the Firewall.
- o Security Zones is used to control and log the traffic that traverses specific interfaces.
- o Interface on Firewall must be assigned to security zone before interface process traffic.
- o Zone can have multiple interfaces of same type, but **interface belong to only one zone**.
- o Palo Alto Firewalls zone names have no predefined meaning or policy associations.
- o Palo Alto Firewalls rely on concept of security zones in order to apply security policies.
- o It means that Security Policies (Firewall Rules) are applied to zones & not to interfaces.
- o This zone feature is similar to Cisco's Zone-Based Firewall supported by IOS Routers.
- o Policy rules on Firewall use zones to identify where traffic comes from & where going.
- o Traffic can flow freely within a zone, but traffic cannot flow between different zones.
- o Traffic between different zones can't flow until define Security policy rule that allows it.
- o Creating a security zone in the Palo Alto Networks NG Firewalls involves three steps.
- o Specify the Zone Name, Select the Zone Type and Assign the Interface to the given Zone.



Name	Type	Interfaces / Virtual Systems	Zone Protection Profile	Packet Buffer Protection	Log Setting	User-ID		
						Enabled	Included Networks	Excluded Networks
☐ Inside	layer3	ethernet1/2	Zone Protection Profile is enable Or not	☒ Packet Buffer Protection enable or not	Log-FW	☒	1.1.1.1	2.2.2.2
☐ Outside	layer3	ethernet1/1		☐ Protection enable or not	Log Forwarding is enable or not	User ID is enable or not	any	none

Go to Palo Alto Networks firewall WebUI and select **Network>Zones** and then click Add to create a new zone, Provide the **name** for the **new Zone** and select the **zone type** and **click OK**. In a similar manner we can repeat to create **Tap**, **Virtual Wire** or **Layer 2 Security Zones**. Click **Add** and create a Zone and name it **DMZ** and type should be **Layer 3**. Assign an interface to the newly created zone by clicking **Add** and then select the interface and click **OK**.



Settings	Description
Name	Enter a zone name, this name appears in the list of zones when defining security policies and configuring interfaces. The name is case-sensitive.
Type	Select a zone type Tap, Virtual Wire, Layer2, Layer3, External, or Tunnel. Add the interfaces that you want to assign to the zone.
Interfaces	Add one or more interfaces to this zone.
Zone Protection Profiles	Select a profile that defines how the firewall responds to attacks on the zone.
Enable Packet Buffer Protection	Configure Packet Buffer globally and apply it to each zone. The firewall applies Packet Buffer Protection to the ingress zone only.
Log Setting	Select a Log Forwarding profile for forwarding zone protection logs.
Enable User Identification	If configured User-ID to perform IP address-to-username mapping Enable User-ID on trusted zones only.
User Identification ACL Include List	By default, if you do not specify subnetworks in this list, the firewall applies the user mapping information it discovers to all the traffic of this zone for use in logs, reports, and policies. To limit the application of user mapping information to specific subnetworks within the zone.
User Identification ACL Exclude List	To exclude user mapping information for a subset of the subnetworks in the Include List, Add, an address (or address group) object or type the IP address range for each subnetwork to exclude.

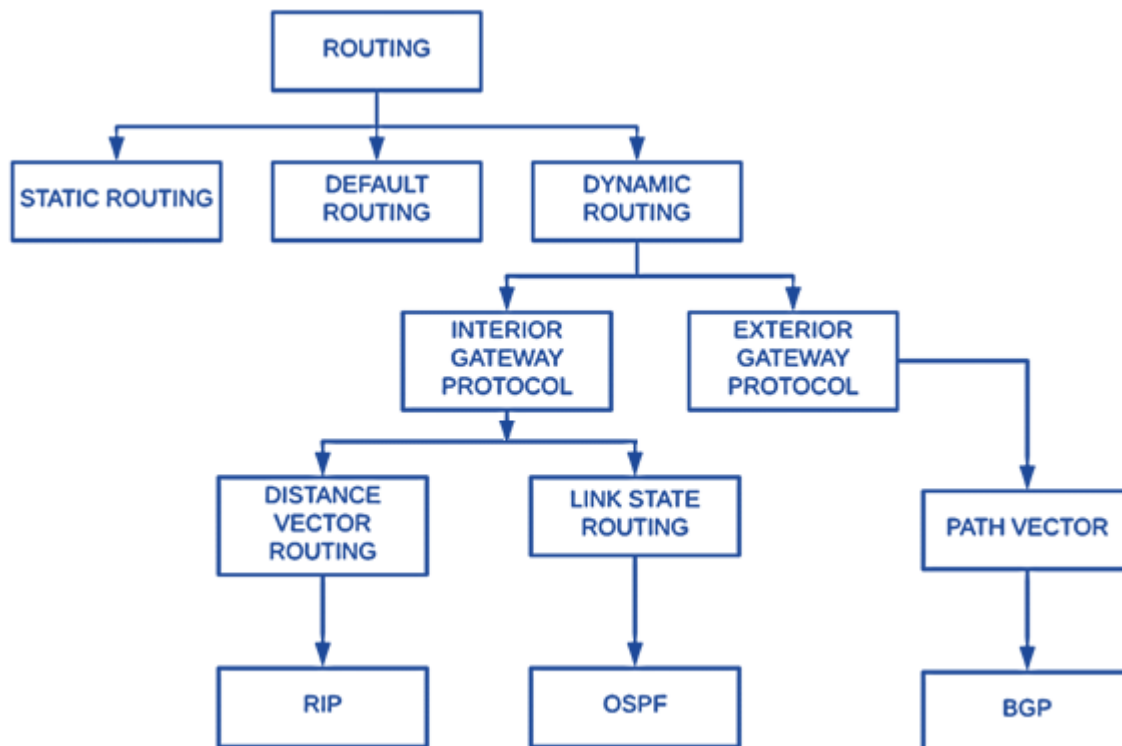
Routed Protocols:

- o Routed protocols are the actual data that is transferred from router to router.
- o Examples of routed protocols are Internet Protocol (IP) such as IPV4 and IPV6.
- o Routed Protocol is used to send user data from one network to another network.
- o Routed Protocol carries user traffic such as e-mails, file transfers, web traffic etc.
- o Used between routers to direct user traffic, it is also called network protocols.
- o That are signs at intersections that point to nearby cities, giving mileage to each.

Routing Protocols:

- o Routing protocols are used by routers to exchange information about known networks.
- o Routers will initially only know the existence of directly connected networks.
- o With the use of routing protocols routers communicate with & learn from other routers.
- o Routing protocols are used to distribute routing information across routers on a network.
- o Routing protocol is Google Maps: It tells you the best way to get where you want to go.

Routed Protocol	Routing Protocols
IP, IPV4, IPV6	RIPV2, OSPF,BGP



Connected:

- o Connected routes are IP Subnet belonging to Firewall interfaces itself.
- o All interfaces configured in the Firewall will belong to this category.
- o Connected are the most commonly routes used for Redistribution.
- o any /32 interface configured on Firewall are not considered Connected.
- o Any /32 interface is consider Host, and therefore cannot be redistributed.

Interface	Interface Type	Management Profile	Link State	IP Address	Virtual Router
 ethernet1/1	Layer3			192.168.17.100/24	default
 ethernet1/2	Layer3	HTTPS-Inside		192.168.78.100/24	default
 ethernet1/3	Layer3			none	none
 ethernet1/4				none	none
 ethernet1/5				none	none

Static Routing:

- o Static Routing administrator manually inputs all routing table information.
- o A static route is a route that is created manually by a network administrator.
- o Router's routing table entries are populated manually by network administrator.
- o Static route tells the network devices about exact location for the routes.
- o Static routes are typically and can only be used in smaller networks.
- o Configure static routes between devices to allow data transfer between devices.
- o Major advantages of static routing are reduced routing protocol router overhead.
- o Another advantage of static routing it reduced routing protocol network traffic.
- o Disadvantages of static routing are network changes require manual reconfiguration.
- o Easy to Configure, High Secure, Use Low Resources, Bandwidth & in small network.
- o In PA Static Routing does not require Advanced Knowledge to configure them.
- o By default, in PA Firewall the static routes have an Administrative Distance of 10.

Default Routing:

- o A Default Route also known as the gateway of last resort is a special type of static route.
- o This method where all routers are configured to send all packets towards single device.
- o Default Route is network route used by device when there is no other known route exists.
- o All the IP datagrams with unknown destination address are sent to the default route.

	Name	Destination	Interface	Next Hop		Admin Distance	Metric	BFD	Route Table
				Type	Value				
☒	Outside-Route	0.0.0.0/0	ethernet1/1	ip-address	192.168.17.2	default	10	None	unicast

Dynamic Routing:

- o Dynamic routing protocols can dynamically respond to changes in the network.
- o Routing protocol is configured on each device & device learn about both each other.
- o Dynamic routing table is created, maintained and updates by routing protocol.
- o Examples of Dynamic routing protocols includes RIPv2, OSPFV3 and OSPF and BGP.
- o Dynamic routing protocols share routing updates with neighbors and find best path.
- o Dynamically choose a different route if a link goes also updates are dynamically.
- o Also, Dynamic Protocols has the ability to load balance between multiple links.
- o Dynamic Routing protocols put additional load on devices CPU and RAM.
- o The choice of the best route is on the hands of the Dynamic Routing Protocol.

Administrative Distance:

- o Administrative distance is the feature that routers use to select best path.
- o Administrative Distance (AD) defines the reliability of a routing protocol.
- o The administrative Distance (AD) number is start from 0 and end on 240.
- o AD has only local significance and is not advertised in routing updates.
- o The smaller Administrative Distance value, the more reliable the protocol.
- o In multiple routing protocols, AD is used to determine trustable protocol.
- o Routing Protocol with lowest Administrative Distance wins put in routing table.
- o Administrative distance of 255 will cause the router to Disbelieve route entirely.
- o AD value of 255 router disbelieve & will be excluded from route selection process.
- o AD are set from Network > Virtual Routers > vr_name > General screen & are applicable.

Administrative Distances		
Static Routes	Range is 10-240	Default is 10
OSPF Internal	Range is 10-240	Default is 30
OSPF External	Range is 10-240	Default is 110
IBGP	Range is 10-240	Default is 200
EBGP	Range is 10-240	Default is 20
RIP	Range is 10-240	Default is 120

Interfaces ▲

ethernet1/1

ethernet1/2

Administrative Distances

Static	10
Static IPv6	10
OSPF Int	30
OSPF Ext	110
OSPFv3 Int	30
OSPFv3 Ext	110
IBGP	200
EBGP	20
RIP	120

Route Metric:

- o The Route Metric is similar to AD in helping to select or predict the best route.
- o However, route metrics are only applicable within a single routing protocol.
- o Route Metric are not used when forwarding traffic through multiple protocols.
- o To forwarding traffic through multiple protocols Administrative Distance is used.
- o Route Metric on Palo Alto Firewalls can be set to determine the route selection.
- o The Route Metric is used when there are multiple paths to the same destination.
- o In Palo Alto Network Firewall Lower metric values are preferred over higher values.

Protocol or Route Type	Valid Range
Static Route Metric	1 - 65535
OSPF Interface Metric	1 - 65535
BGP Local Preference	0 - 4294967295
BGP MED Metric	0 - 4294967295
BGP Weight	0 – 65535
RIPv2 Metric	1 – 16

The screenshot shows the configuration interface for a Palo Alto Firewall. It displays two sections for route metrics. The first section is for 'Static Route Metric', where the 'Metric' is set to 10, and the valid range is 1-65535. The second section is for 'RIP Metric Value', where the 'Metric' is set to 10, and the valid range is 1-15. Both sections have a dropdown menu for 'Route Table' set to 'Unicast' and an 'Auth Profile' set to 'None'. The interface also shows a 'BFD' section with a 'None' selection.

RIB:

- o RIB is a PA Firewall term which is stand for Routing Information Base.
- o All routes learned are first place in the Routing Information Base (RIB).
- o **Show routing route**, command will display the RIB table from the Firewall.
- o All the routes that are “active” are marked with an “A” in the flags column.

Flags: A:active, ?:loose, C:connect, H:host, S:static, ~:internal, R:rip, O:ospf, B:bgp,
Oi:ospf intra-area, Oo:ospf inter-area, O1:ospf ext-type-1, O2:ospf ext-type-2, E:ecmp, M:multicast

```
VIRTUAL ROUTER: default (id 1)
=====
destination      nexthop          metric flags    age    interface
next-AS
0.0.0.0/0         192.168.17.2    10    A S          ethernet1/1
6.6.6.0/24        0.0.0.0         10    oi          13660 vlan.1
6.6.6.0/24        6.6.6.6         0     A C          vlan.1
6.6.6.6/32        0.0.0.0         0     A H
192.168.17.0/24   192.168.17.100  0     A C          ethernet1/1
192.168.17.100/32 0.0.0.0         0     A H
192.168.78.0/24   0.0.0.0         10    oi          13660 ethernet1/2
192.168.78.0/24   192.168.78.100  0     A C          ethernet1/2
192.168.78.100/32 0.0.0.0         0     A H
```

FIB:

- o FIB is a PA Firewall term which is stand for Forwarding Information Base.
- o Best route obtained through protocols are used to populate the firewall's FIB.
- o **Show routing fib**, command will display the FIB table from the PA Firewall.
- o Firewall chooses the best route toward destination network to place in the FIB.

```
virtual-router name: default
interfaces:
  ethernet1/1 ethernet1/2 vlan.1

route table:
flags: u - up, h - host, g - gateway, e - ecmp, * - preferred path

maximum of fib entries for device:          2500
maximum of IPv4 fib entries for device:     2500
maximum of IPv6 fib entries for device:     2500
number of fib entries for device:           7
maximum of fib entries for this fib:        2500
number of fib entries for this fib:         7
number of fib entries shown:                7
```

id	destination	nexthop	flags	interface	mtu
15	0.0.0.0/0	192.168.17.2	ug	ethernet1/1	1500
18	6.6.6.0/24	0.0.0.0	u	vlan.1	1500
5	6.6.6.6/32	0.0.0.0	uh	vlan.1	1500
14	192.168.17.0/24	0.0.0.0	u	ethernet1/1	1500
13	192.168.17.100/32	0.0.0.0	uh	ethernet1/1	1500
19	192.168.78.0/24	0.0.0.0	u	ethernet1/2	1500
16	192.168.78.100/32	0.0.0.0	uh	ethernet1/2	1500

Redistribute:

- o Redistribute Rule allows to “Redistribute” route from one protocol into another.
- o Redistribution allows devices to share routes info among routing protocols.
- o Redistribution profiles are only way to exchange routing info between protocols.
- o Palo Alto Firewalls provide flexible, profile-based approach for route redistribution.
- o PA Firewall also provide filtering between protocols, static routes, connected & hosts.
- o In PA Firewall Without redistribution profiles, each protocol functions on its own.
- o Does not exchange any route info with other protocols running on same virtual router.
- o Redistribute static route into dynamic routing protocol like OSPF is common use case.

Redistribution Profile IPv4

Name: Redistribute: ☒ No Redist ☐ Redist

Priority:

General Filter | OSPF Filter | BGP Filter

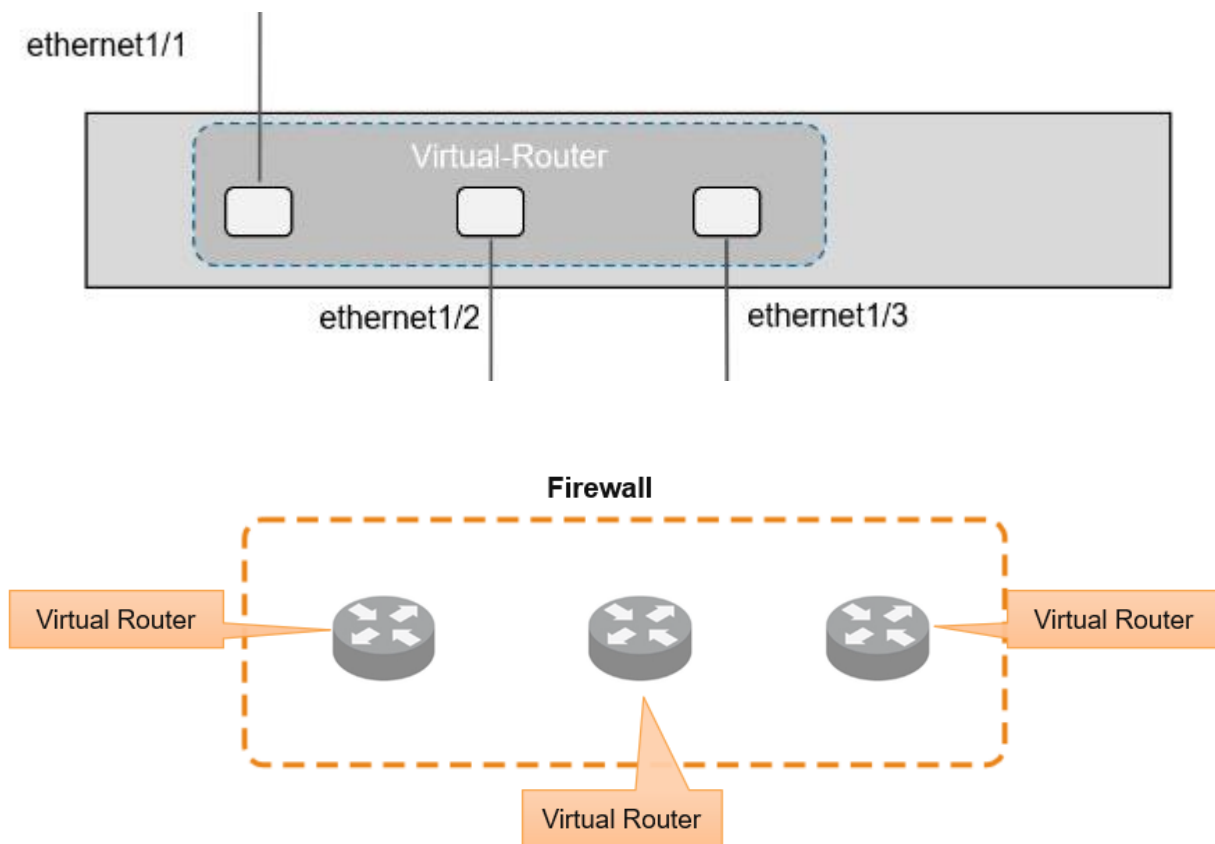
Source Type

- ☐ bgp
- ☐ connect
- ☐ ospf
- ☐ rip
- ☐ static

Interface	Destination	Next Hop
	Ex. 10.1.7.1 or 10.1.7.0/24	Ex. 10.1.7.1 or 10.1.7.0/24

Virtual Routers:

- o When it comes to routing traffic between different networks one needs a Router.
- o Palo Alto Networks Firewalls are capable of routing the traffic between networks.
- o Uses concept of “Virtual Routers” to route traffic be it static or dynamic routing.
- o Virtual Router uses virtualized or partitioned routing tables to do the routing job.
- o Uses virtual routers to obtain routes & uses best route to populates its routing table.
- o PA Firewall capable of supporting Dynamic routing protocols like RIPv2, OSPF & BGPv4.
- o The Palo Alto NG Network Firewalls comes with a Virtual router named the default.
- o Can be used for routing provided layer3 interfaces or VLANs are part of that default VR.
- o Can also create new Virtual Router & name & use it for both static & dynamic routing.
- o In Palo Alto Firewall Layer 3 deployment, the Firewall routes traffic between ports.
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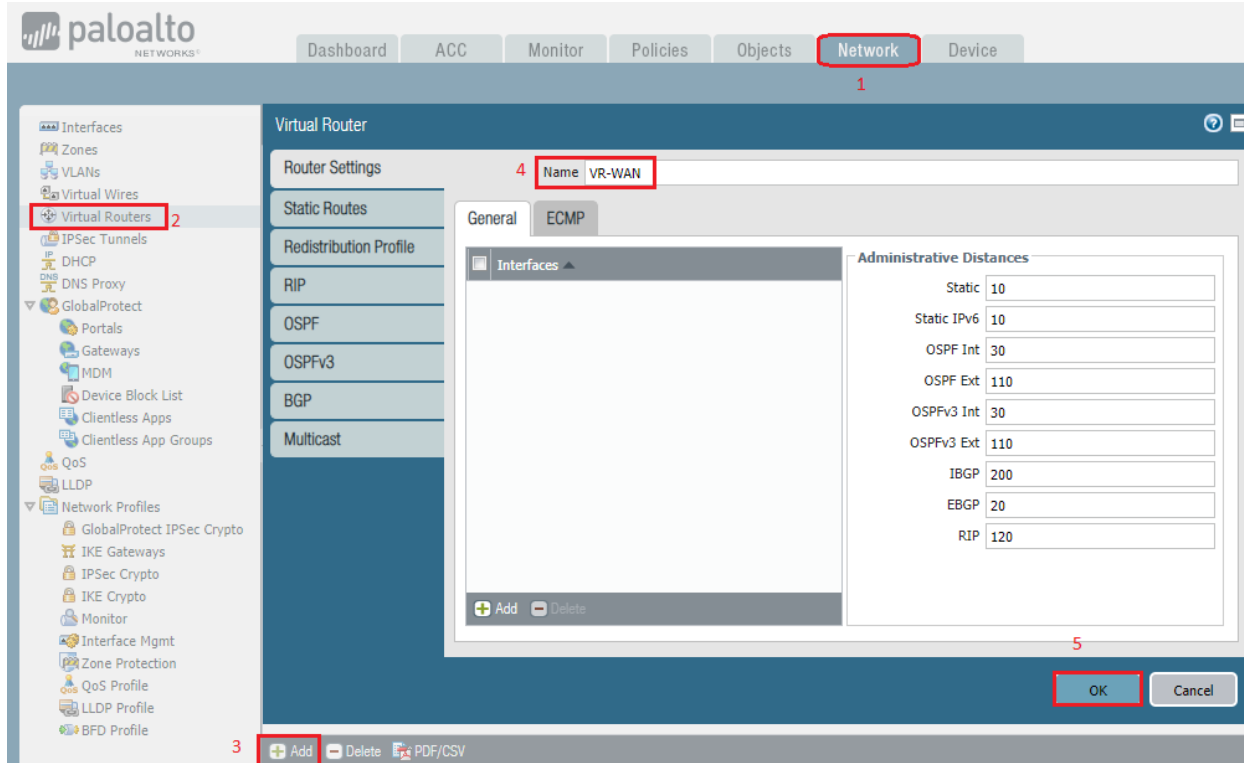


Network > Virtual Routers > Router Settings > General

Settings	Description
Name	Specify a name to describe the Virtual Router.
Interfaces	Select the interfaces that want to include in the Virtual Router.
Administrative Distances	Specify the Administrative Distances.

Administrative Distances		
Static Routes	Range is 10-240	Default is 10
OSPF Internal	Range is 10-240	Default is 30
OSPF External	Range is 10-240	Default is 110
IBGP	Range is 10-240	Default is 200
EBGP	Range is 10-240	Default is 20
RIP	Range is 10-240	Default is 120

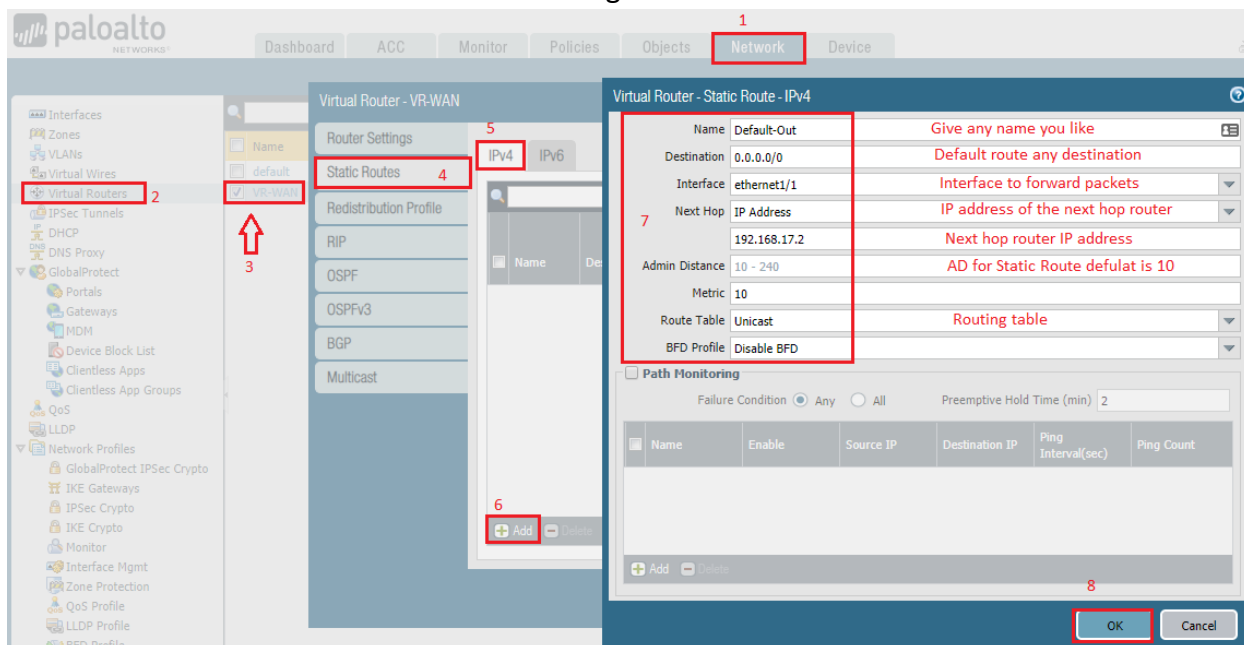
Network -> Virtual Routers -> Add Type name "VR-WAN" -> click **OK**



Let's configure Static Route; I will route all traffic to ISP (Default Route); assuming that the ISP router IP address is 192.168.17.2

Static Routes → Add

Type name, select the Interface (ethernet1/1) and Next Hop keep the default Metric 10
You will need to click "Commit" then the changes take effect.



Settings	Description
Name	Enter a name to identify the static route .
Destination	Enter an IP address and network mask in Classless Inter-domain Routing (CIDR) notation for example, 192.168.2.0/24
Interface	Select the interface to forward packets to the destination, or configure the next hop settings, or both.
Next Hop	Select one of the following: IP Address—Select to enter the IP address of the next hop router. Next VR—Select to select a virtual router in the firewall as the next hop. To route internally between virtual routers within a single firewall. Discard—Select if want to drop traffic that is addressed to destination. None—Select if there is no next hop for the route.
Admin Distance	Specify administrative distance for static route (10-240; default is 10).
Metric	Specify a valid metric for the static route (1 - 65535).
Route Table	Select the route table into which the firewall installs the static route: Unicast—Installs the route into the unicast route table. Multicast—Installs the route into the multicast route table. Both—Installs the route into the unicast and multicast route tables. No Install—Does not install the route in the route table (RIB).
BFD Profile	To enable Bidirectional Forwarding Detection (BFD) for static route. default (default BFD settings) A BFD profile that you have created on the firewall New BFD Profile to create a new BFD profile Select None (Disable BFD) to disable BFD for the static route. To use BFD on a static route: Both firewall & peer at opposite end of static route must support BFD sessions. Static route Next Hop type must be IP Address & must enter valid IP address. The Interface setting cannot be None; you must select an interface.
Path Monitoring	Select to enable path monitoring for the static route.
Failure Condition	Select the condition under which the firewall considers the monitored path down and thus the static route down: Any—If any one of the monitored destinations for the static route is unreachable by ICMP. All—If all of the monitored destinations for the static route are unreachable by ICMP. Select All to avoid the possibility of a single monitored destination signaling a static route failure when that monitored destination is simply offline for maintenance, for example.
Preemptive Hold Time (min)	Enter the number of minutes a downed path monitor must remain in Up state— Range is 0-1,440; default is 2.
Name	Enter a name for the monitored destination.
Enable	Select to enable path monitoring of this specific destination for the static route; the firewall sends ICMP pings to this destination.

Source IP	Select the IP address that the firewall will use as the source in the ICMP ping to the monitored destination:
Destination IP	Enter IP address for which the firewall will monitor the path.
Ping Interval (sec)	Specify ICMP ping interval in seconds to determine how frequently firewall monitors path range is 1-60; default is 3.
Ping Count	Specify number of consecutive ICMP ping packets that do not return from the monitored destination before firewall considers link down.

Virtual Router - VR-WAN

Router Settings

Static Routes

Redistribution Profile

RIP

OSPF

OSPFv3

IPv4 IPv6

1 item

Name	Destination	Interface	Next Hop		Admin Distance	Metric	BFD	Route Table
			Type	Value				
Default-Out	0.0.0.0/0	ethernet...	ip-address	192.168....	default	10	None	unicast

More Runtime Stats for a Virtual Router:

After configuring static routes or routing protocols for a virtual router, select **Network > Virtual Routers**, and select **More Runtime Stats** in the last column to see detailed information about the virtual router, such as the route table, forwarding table, and the routing protocols and static routes you configured.

Routing

RIP

OSPF

OSPFv3

BGP

Multicast

BFD Summary Information

Route Table

Forwarding Table

Static Route Monitoring

Route Table Unicast Multicast

Display Address Family IPv4 and IPv6

9 items

Destination	Next Hop	Metric	Weight	Flags	Age	Interface
0.0.0.0/0	192.168.17.2	10		A S		ethernet1/1
6.6.6.0/24	0.0.0.0	10		Oi	15402	vlan.1
6.6.6.0/24	6.6.6.6	0		A C		vlan.1
6.6.6.6/32	0.0.0.0	0		A H		
192.168.17.0/24	192.168.17.100	0		A C		ethernet1/1
192.168.17.100/32	0.0.0.0	0		A H		
192.168.78.0/24	0.0.0.0	10		Oi	15402	ethernet1/2
192.168.78.0/24	192.168.78.100	0		A C		ethernet1/2
192.168.78.100/32	0.0.0.0	0		A H		

Refresh Active, ?loose, Cconnect, H:host, S:static, ~:internal, R:rip, O:OSPF, B:bpg, Oi:OSPF intra-area, Oo:OSPF inter-area, O1:OSPF ext-type-1, O2:OSPF ext-type-2

Route Table Flags and Description	
Flags	Description
A?B	Active and learned via BGP
A C	Active and a result of an internal interface (connected) - Destination = network
AH	Active and a result of an internal interface (connected) - Destination = Host only
AR	Active and learned via RIP
AS	Active and Static
S	Inactive (Because this route has a higher metric) and static
01	OSPF external type-1
02	OSPF external type-2
Oi	OSPF intra-area
Oo	OSPF inter-area

Routing

RIP

OSPF

OSPFv3

BGP

Multicast

BFD Summary Information

Route Table

Forwarding Table

Static Route Monitoring

Display Address Family

IPv4 and IPv6

7 items

Destination	Next Hop	Flags	Interface	MTU
0.0.0.0/0	192.168.17.2	ug	ethernet1/1	1500
6.6.6.0/24	0.0.0.0	u	vlan.1	1500
6.6.6.6/32	0.0.0.0	uh	vlan.1	1500
192.168.17.0/24	0.0.0.0	u	ethernet1/1	1500
192.168.17.100/32	0.0.0.0	uh	ethernet1/1	1500
192.168.78.0/24	0.0.0.0	u	ethernet1/2	1500
192.168.78.100/32	0.0.0.0	uh	ethernet1/2	1500

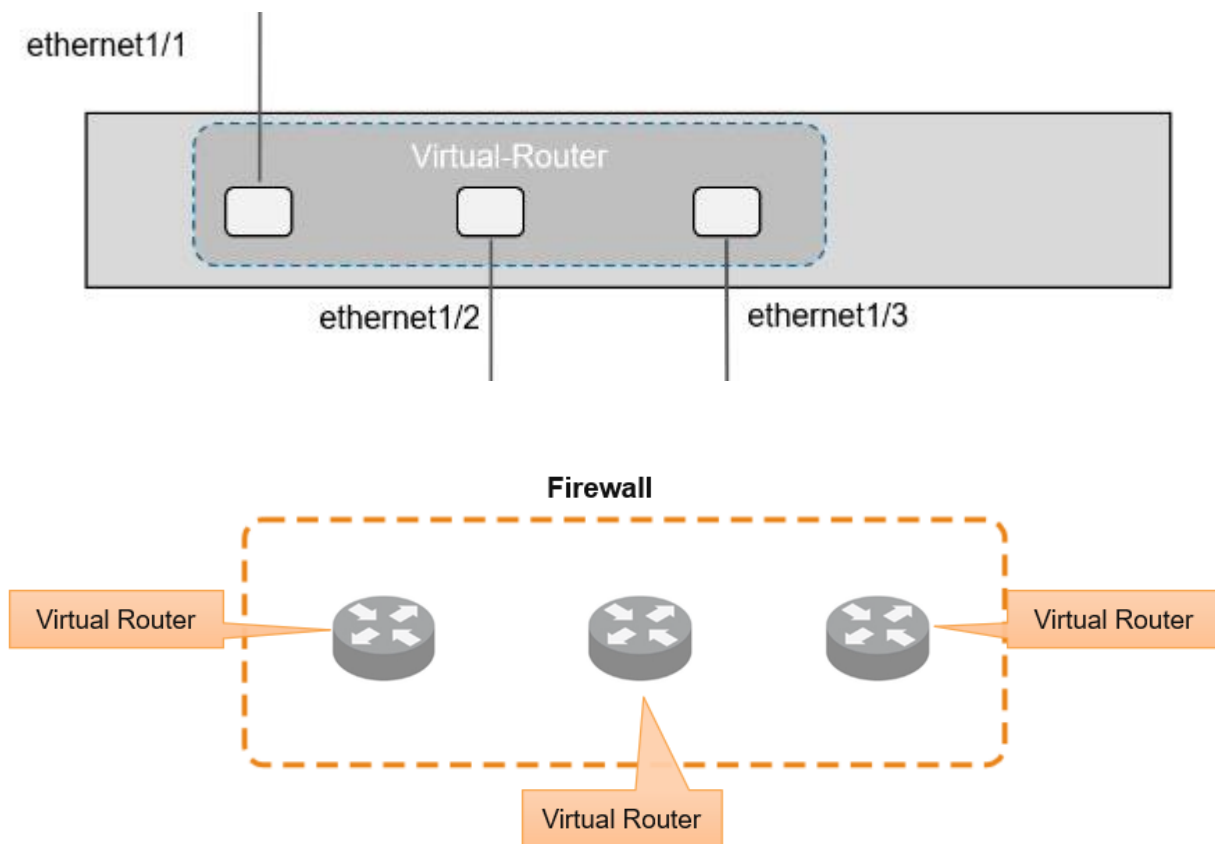
Refresh

u - up, h - host, g - gateway, e - ECMP, * - preferred path

Forwarding Table Flags and Description	
Flags	Description
u	Route is up.
h	Route is to a host.
g	Route is to a gateway.
e	Firewall selected this route using Equal Cost Multipath.
*	Route is the preferred path to a destination network.
ug	Route is up and a gateway.
uh	Route is up and a host.

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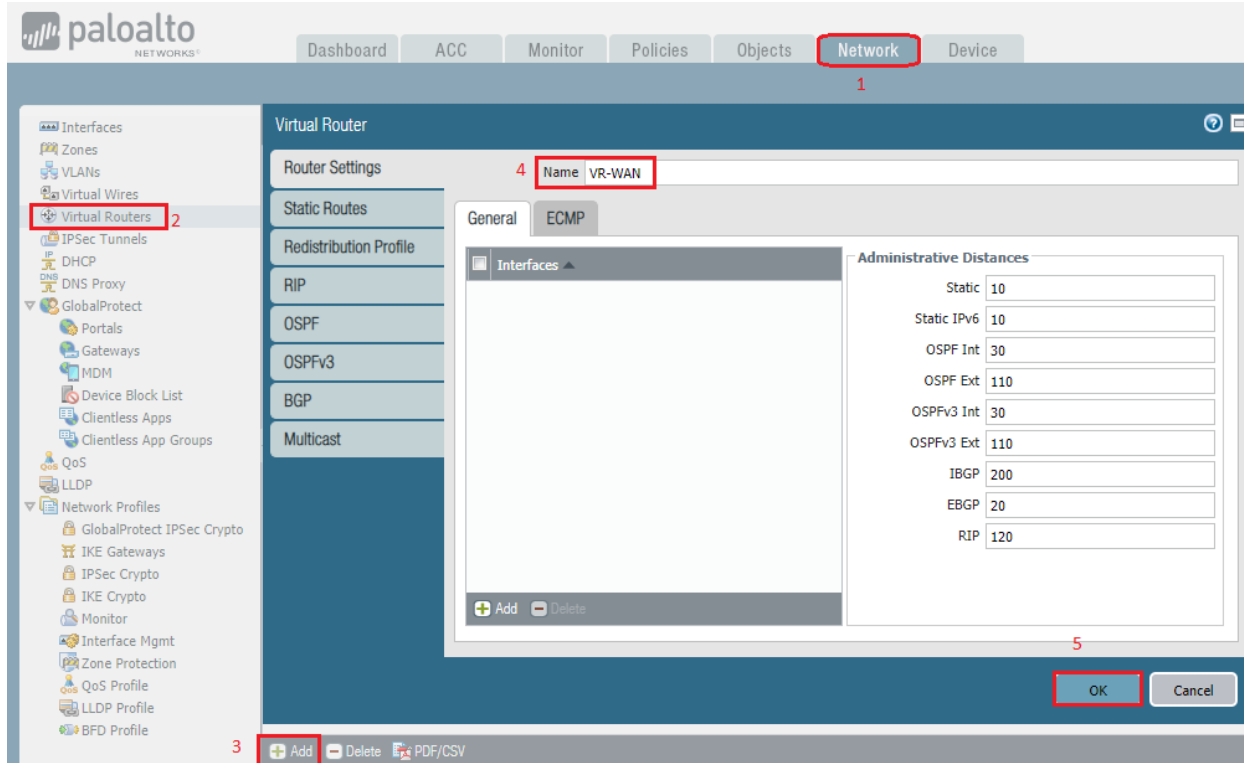


Network > Virtual Routers > Router Settings > General

Settings	Description
Name	Specify a name to describe the Virtual Router.
Interfaces	Select the interfaces that want to include in the Virtual Router.
Administrative Distances	Specify the Administrative Distances.

Administrative Distances		
Static Routes	Range is 10-240	Default is 10
OSPF Internal	Range is 10-240	Default is 30
OSPF External	Range is 10-240	Default is 110
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EBGP	Range is 10-240	Default is 20
RIP	Range is 10-240	Default is 120

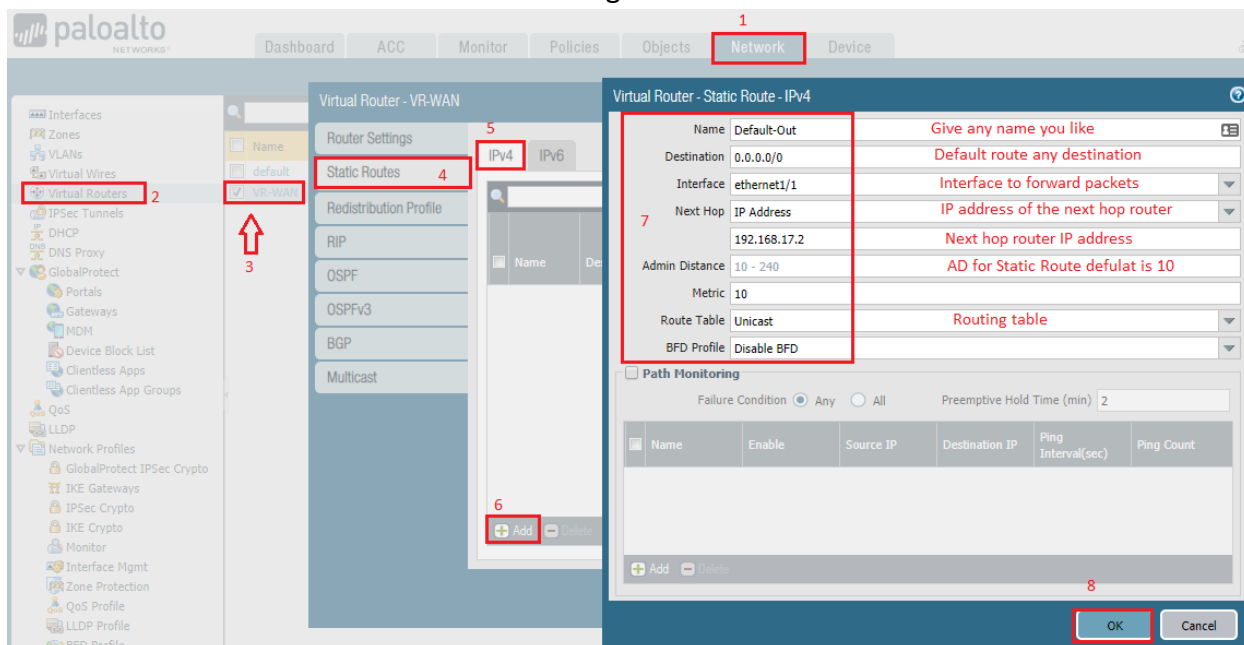
Network -> Virtual Routers -> Add Type name "VR-WAN" -> click **OK**



Let's configure Static Route; I will route all traffic to ISP (Default Route); assuming that the ISP router IP address is 192.168.17.2

Static Routes → Add

Type name, select the Interface (ethernet1/1) and Next Hop keep the default Metric 10
You will need to click “Commit” then the changes take effect.



Settings	Description
Name	Enter a name to identify the static route .
Destination	Enter an IP address and network mask in Classless Inter-domain Routing (CIDR) notation for example, 192.168.2.0/24
Interface	Select the interface to forward packets to the destination, or configure the next hop settings, or both.
Next Hop	Select one of the following: IP Address—Select to enter the IP address of the next hop router. Next VR—Select to select a virtual router in the firewall as the next hop. To route internally between virtual routers within a single firewall. Discard—Select if want to drop traffic that is addressed to destination. None—Select if there is no next hop for the route.
Admin Distance	Specify administrative distance for static route (10-240; default is 10).
Metric	Specify a valid metric for the static route (1 - 65535).
Route Table	Select the route table into which the firewall installs the static route: Unicast—Installs the route into the unicast route table. Multicast—Installs the route into the multicast route table. Both—Installs the route into the unicast and multicast route tables. No Install—Does not install the route in the route table (RIB).
BFD Profile	To enable Bidirectional Forwarding Detection (BFD) for static route. default (default BFD settings) A BFD profile that you have created on the firewall New BFD Profile to create a new BFD profile Select None (Disable BFD) to disable BFD for the static route. To use BFD on a static route: Both firewall & peer at opposite end of static route must support BFD sessions. Static route Next Hop type must be IP Address & must enter valid IP address. The Interface setting cannot be None; you must select an interface.
Path Monitoring	Select to enable path monitoring for the static route.
Failure Condition	Select the condition under which the firewall considers the monitored path down and thus the static route down: Any—If any one of the monitored destinations for the static route is unreachable by ICMP. All—If all of the monitored destinations for the static route are unreachable by ICMP. Select All to avoid the possibility of a single monitored destination signaling a static route failure when that monitored destination is simply offline for maintenance, for example.
Preemptive Hold Time (min)	Enter the number of minutes a downed path monitor must remain in Up state— Range is 0-1,440; default is 2.
Name	Enter a name for the monitored destination.
Enable	Select to enable path monitoring of this specific destination for the static route; the firewall sends ICMP pings to this destination.

Source IP	Select the IP address that the firewall will use as the source in the ICMP ping to the monitored destination:
Destination IP	Enter IP address for which the firewall will monitor the path.
Ping Interval (sec)	Specify ICMP ping interval in seconds to determine how frequently firewall monitors path range is 1-60; default is 3.
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Virtual Router - VR-WAN

Router Settings

Static Routes

Redistribution Profile

RIP

OSPF

OSPFv3

IPv4 IPv6

1 item

Name	Destination	Interface	Next Hop		Admin Distance	Metric	BFD	Route Table
			Type	Value				
Default-Out	0.0.0.0/0	ethernet...	ip-address	192.168....	default	10	None	unicast

More Runtime Stats for a Virtual Router:

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Routing

RIP

OSPF

OSPFv3

BGP

Multicast

BFD Summary Information

Route Table

Forwarding Table

Static Route Monitoring

Route Table

Unicast

Multicast

Display Address Family

IPv4 and IPv6

9 items

Destination	Next Hop	Metric	Weight	Flags	Age	Interface
0.0.0.0/0	192.168.17.2	10		A S		ethernet1/1
6.6.6.0/24	0.0.0.0	10		Oi	15402	vlan.1
6.6.6.0/24	6.6.6.6	0		A C		vlan.1
6.6.6.6/32	0.0.0.0	0		A H		
192.168.17.0/24	192.168.17.100	0		A C		ethernet1/1
192.168.17.100/32	0.0.0.0	0		A H		
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192.168.78.100/32	0.0.0.0	0		A H		

Refresh

A:active, ? :loose, C:connect, H:host, S:static, ~:internal, R:rip, O:OSPF, B:bgp, Oi:OSPF intra-area, Oo:OSPF inter-area, O1:OSPF ext-type-1, O2:OSPF ext-type-2

Route Table Flags and Description	
Flags	Description
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01	OSPF external type-1
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Routing

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Route Table

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Static Route Monitoring

Display Address Family

IPv4 and IPv6

7 items

Destination	Next Hop	Flags	Interface	MTU
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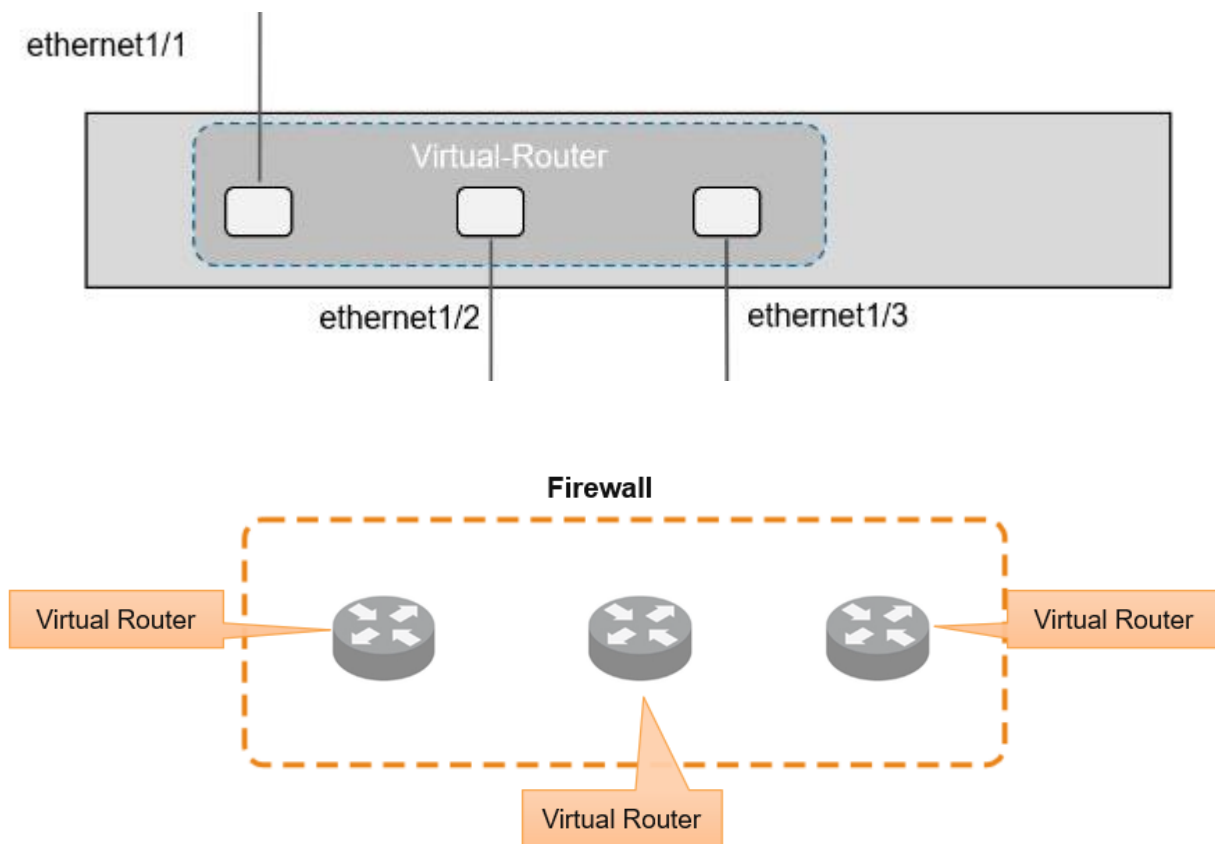
Refresh

u - up, h - host, g - gateway, e - ECMP, * - preferred path

Forwarding Table Flags and Description	
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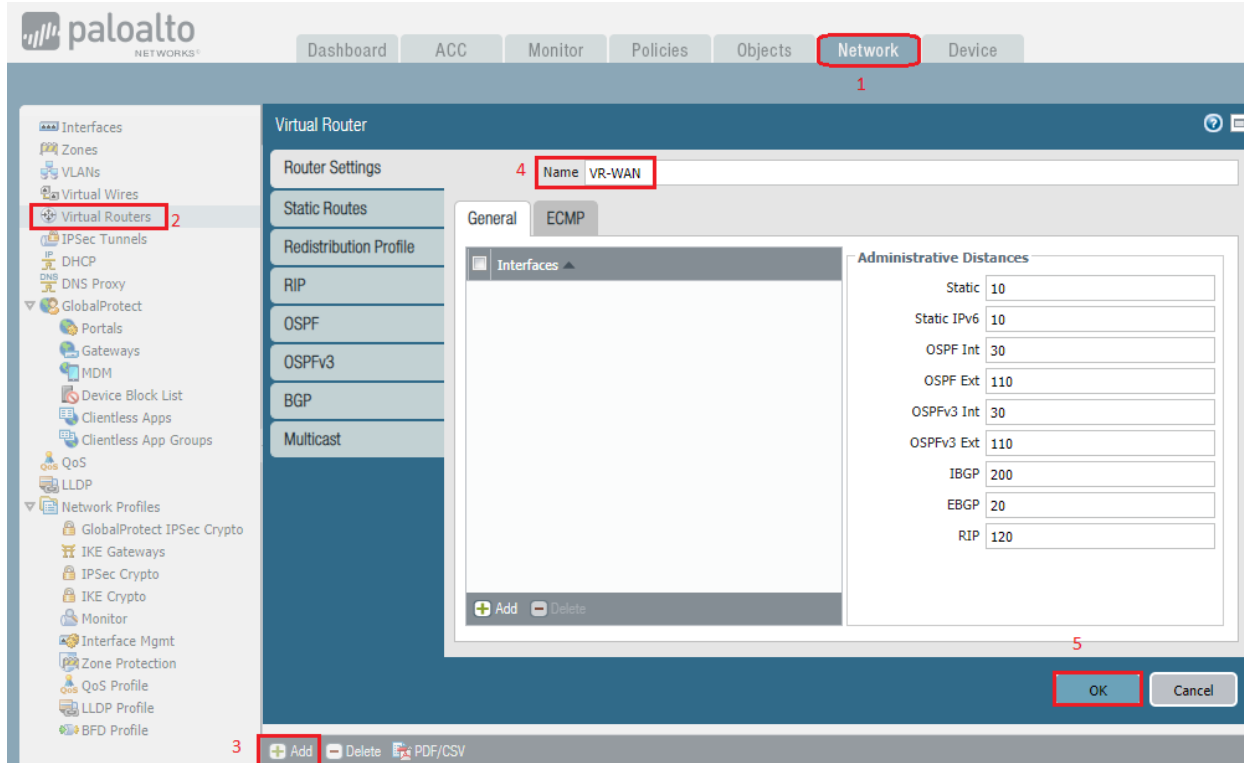


Network > Virtual Routers > Router Settings > General

Settings	Description
Name	Specify a name to describe the Virtual Router.
Interfaces	Select the interfaces that want to include in the Virtual Router.
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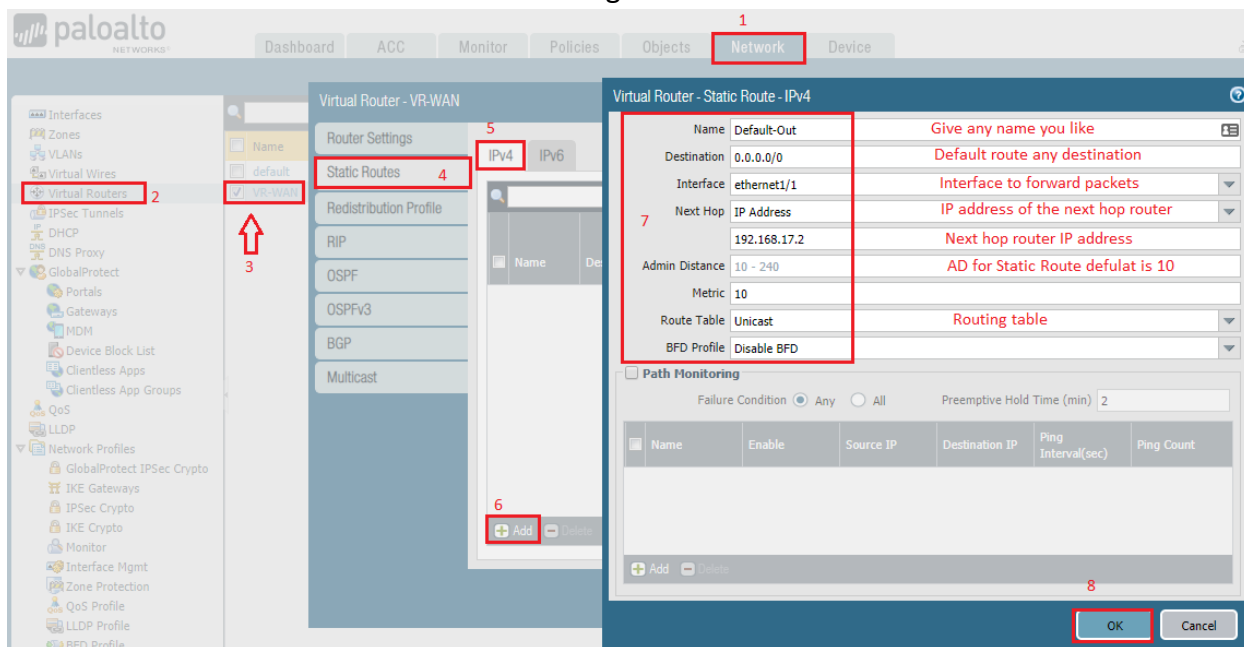
Network -> Virtual Routers -> Add Type name "VR-WAN" -> click **OK**



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Static Routes → Add

Type name, select the Interface (ethernet1/1) and Next Hop keep the default Metric 10
You will need to click "Commit" then the changes take effect.



Settings	Description
Name	Enter a name to identify the static route .
Destination	Enter an IP address and network mask in Classless Inter-domain Routing (CIDR) notation for example, 192.168.2.0/24
Interface	Select the interface to forward packets to the destination, or configure the next hop settings, or both.
Next Hop	Select one of the following: IP Address—Select to enter the IP address of the next hop router. Next VR—Select to select a virtual router in the firewall as the next hop. To route internally between virtual routers within a single firewall. Discard—Select if want to drop traffic that is addressed to destination. None—Select if there is no next hop for the route.
Admin Distance	Specify administrative distance for static route (10-240; default is 10).
Metric	Specify a valid metric for the static route (1 - 65535).
Route Table	Select the route table into which the firewall installs the static route: Unicast—Installs the route into the unicast route table. Multicast—Installs the route into the multicast route table. Both—Installs the route into the unicast and multicast route tables. No Install—Does not install the route in the route table (RIB).
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Virtual Router - VR-WAN

Router Settings

Static Routes

Redistribution Profile

RIP

OSPF

OSPFv3

IPv4 IPv6

1 item

Name	Destination	Interface	Next Hop		Admin Distance	Metric	BFD	Route Table
			Type	Value				
Default-Out	0.0.0.0/0	ethernet...	ip-address	192.168....	default	10	None	unicast

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Routing

RIP

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OSPFv3

BGP

Multicast

BFD Summary Information

Route Table

Forwarding Table

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Route Table Unicast Multicast

Display Address Family IPv4 and IPv6

9 items

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192.168.17.0/24	192.168.17.100	0		A C		ethernet1/1
192.168.17.100/32	0.0.0.0	0		A H		
192.168.78.0/24	0.0.0.0	10		Oi	15402	ethernet1/2
192.168.78.0/24	192.168.78.100	0		A C		ethernet1/2
192.168.78.100/32	0.0.0.0	0		A H		

Refresh Active, ?loose, Cconnect, Hhost, Sstatic, ~internal, Rrip, O:OSPF, B:ospf, O:OSPF intra-area, O:OSPF inter-area, O1:OSPF ext-type-1, O2:OSPF ext-type-2

Route Table Flags and Description	
Flags	Description
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S	Inactive (Because this route has a higher metric) and static
01	OSPF external type-1
02	OSPF external type-2
Oi	OSPF intra-area
Oo	OSPF inter-area

Routing

RIP

OSPF

OSPFv3

BGP

Multicast

BFD Summary Information

Route Table

Forwarding Table

Static Route Monitoring

Display Address Family

IPv4 and IPv6

7 items

Destination	Next Hop	Flags	Interface	MTU
0.0.0.0/0	192.168.17.2	ug	ethernet1/1	1500
6.6.6.0/24	0.0.0.0	u	vlan.1	1500
6.6.6.6/32	0.0.0.0	uh	vlan.1	1500
192.168.17.0/24	0.0.0.0	u	ethernet1/1	1500
192.168.17.100/32	0.0.0.0	uh	ethernet1/1	1500
192.168.78.0/24	0.0.0.0	u	ethernet1/2	1500
192.168.78.100/32	0.0.0.0	uh	ethernet1/2	1500

Refresh

u - up, h - host, g - gateway, e - ECMP, * - preferred path

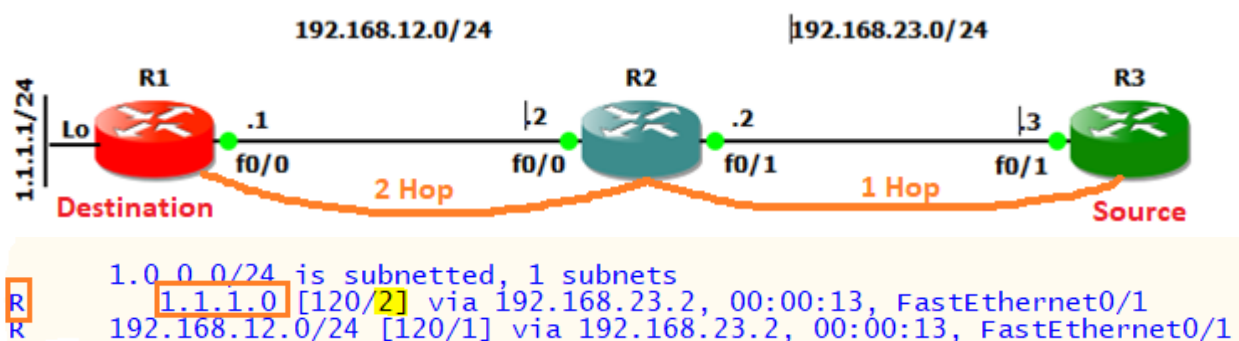
Forwarding Table Flags and Description	
Flags	Description
u	Route is up.
h	Route is to a host.
g	Route is to a gateway.
e	Firewall selected this route using Equal Cost Multipath.
*	Route is the preferred path to a destination network.
ug	Route is up and a gateway.
uh	Route is up and a host.

RIP:

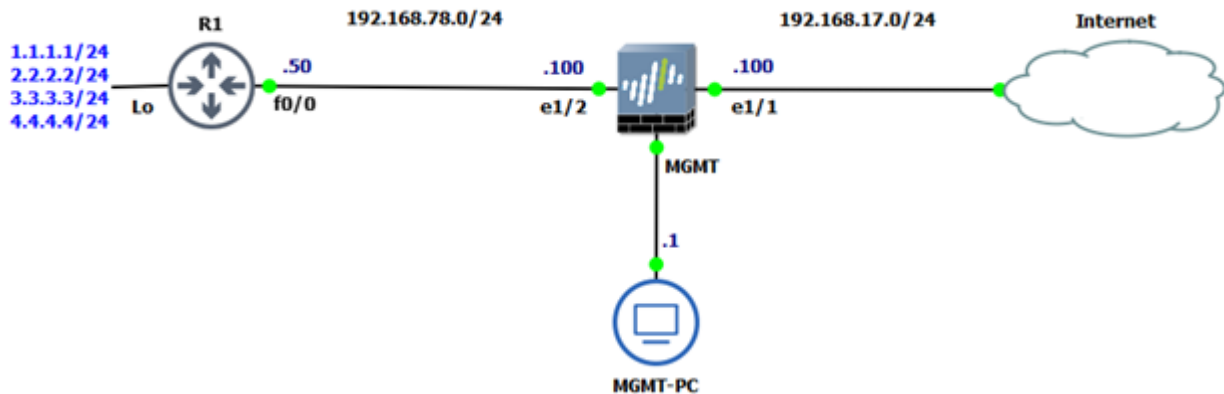
- o RIP is a term which stands for Routing Information Protocol.
- o Routing Information Protocol is Distance Vector Routing Protocol.
- o Default Administrative Distance of Routing Information Protocol is 120.
- o Routing Information Protocol Supports VLSM and subnetting as well.
- o Routing Information Protocol Uses split horizon and poison reverse.
- o Routing Information Protocol Sends full updates every 30 seconds.
- o Routing Information Protocol The invalid timer default is 180.
- o Routing Information Protocol, the hold down timer default is 180.
- o Routing Information Protocol The flush timer default is 240.
- o Routing Information Protocol Uses triggered updates only in v2.
- o Routing Information Protocol (RIP) Uses hop count as the metric.
- o RIPv1 and V2 support maximum hop count of 15 & 16 is Unreachable.
- o Routing Information Protocol (RIP) Supports route tags as well.
- o RIPv1 uses broadcast & RIPv2 uses multicast 224.0.0.9 to send updates.
- o RIPv1 doesn't support while RIPv2 supports MD5 authentication & plain text.
- o Routing Information Protocol V2 Supports both classful and classless routing.
- o Routing Information Protocol (RIP) Uses UDP as the layer 4 protocol.
- o Routing Information Protocol supports load balancing over same-cost paths.
- o The default is four equal-cost paths for Routing Information Protocol (RIP).

RIP Metric:

- o Each routing protocol has its own way to calculate the metric RIP uses hop counts.
- o Hop count is number of routers occurring in between source & destination network.
- o The path with lowest hop count is considered as the best route to reach a network.
- o The path with lowest hop count is therefore, placed in the routing table by RIP.
- o If there are multiple paths with different bandwidth to the destination network.
- o Then Routing Information Protocol best path calculation may become wrong.
- o 1.1.1.1 Network is two (2) hops away from R3 source. As shown in below routing table.



Routing RIP Lab:



Router R1 Configuration

```
R1(config)#interface fastEthernet0/0
R1(config-if)#ip add 192.168.78.50 255.255.255.0
R1(config-if)#no shutdown

R1(config)#interface loopback 1
R1(config-if)#ip add 1.1.1.1 255.255.255.0

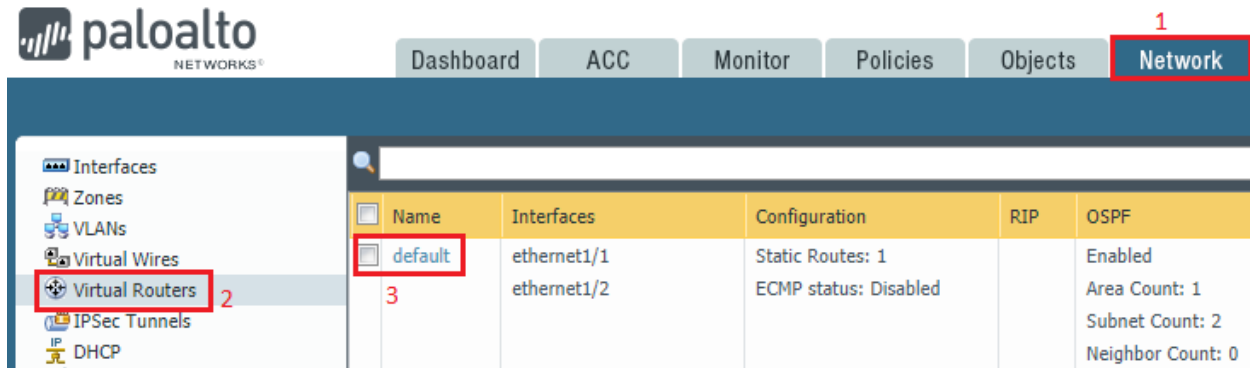
R1(config)#interface loopback 2
R1(config-if)#ip add 2.2.2.2 255.255.255.0

R1(config)#interface loopback 3
R1(config-if)#ip add 3.3.3.3 255.255.255.0

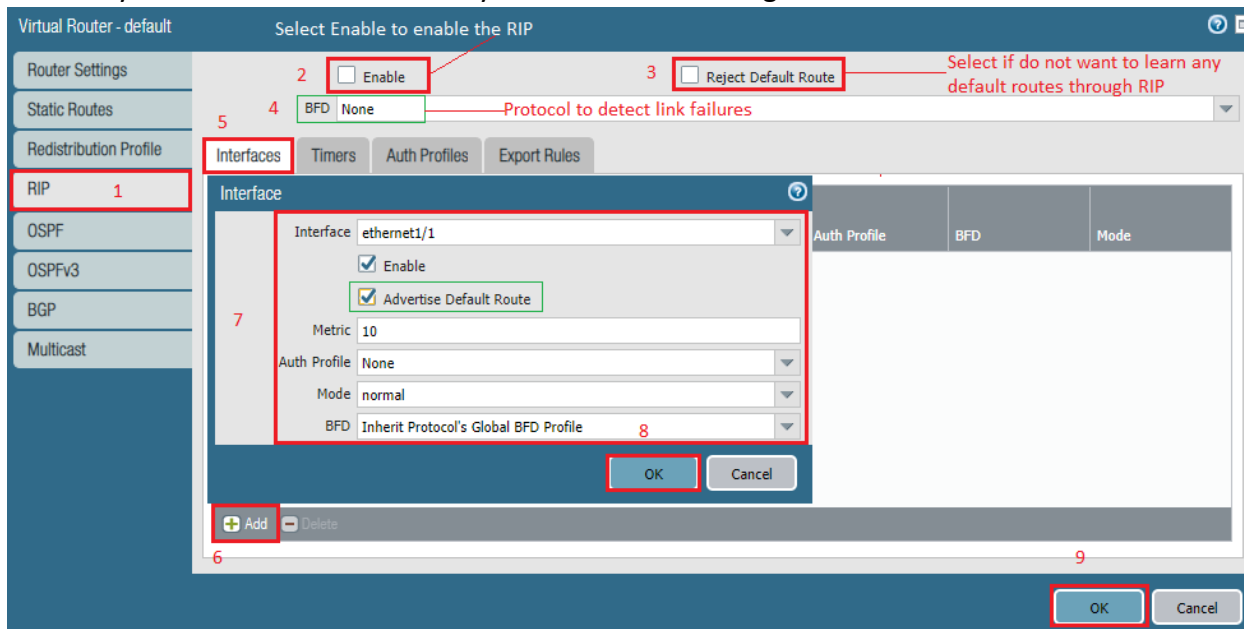
R1(config)#interface loopback 4
R1(config-if)#ip add 4.4.4.4 255.255.255.0

R1(config)#router rip
R1(config-router)#network 0.0.0.0
R1(config-router)#version 2
R1(config-router)#no auto-summary
```

Go to **Network > Virtual Router** and check **default**.



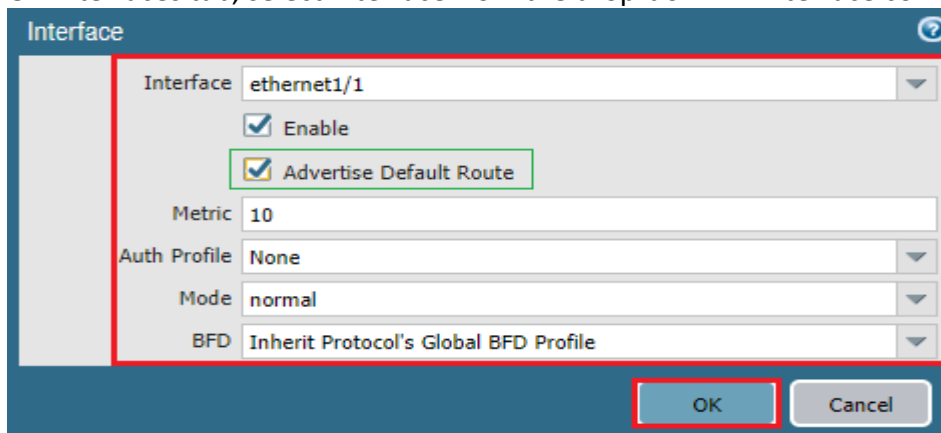
Select the RIP tab on left side, Select **Enable** to enable the RIP protocol. Select **Reject Default Route** if you do not want to learn any default routes through RIP.



RIP Settings	Description
Enable	Select Enable to enable the RIP protocol.
Reject Default Route	Select if you do not want to learn any default routes through RIP. De-select Reject Default Route if you want to permit redistribution of default routes through RIP.
BFD	To enable Bidirectional Forwarding Detection (BFD) for RIP globally. Default (profile with the default BFD settings) A BFD profile that you have created on the firewall. New BFD Profile to create a new BFD profile. Select None (Disable BFD) to disable BFD for all RIP interfaces.

Network > Virtual Routers > RIP > Interfaces

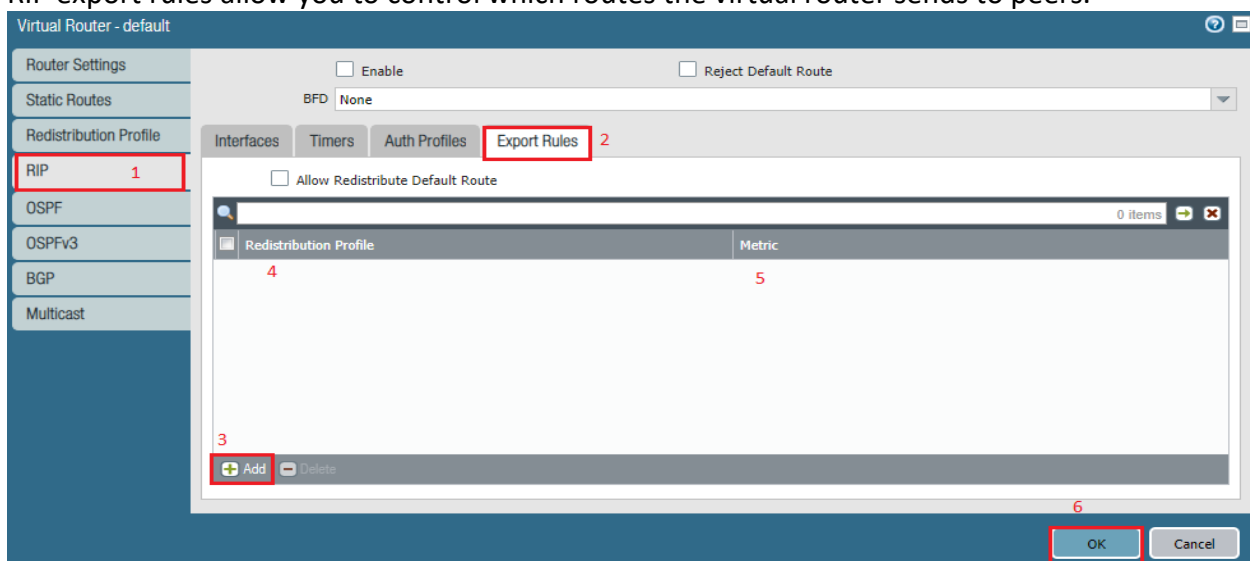
On Interfaces tab, select interface from the drop-down in Interface configuration section.



Interface Settings	Description
Interface	Select the interface that runs the RIP protocol.
Enable	Select to enable these settings.
Advertise	Select to enable advertisement of default route to RIP peers.
Metric	Specify a metric value for the router advertisement. This field is visible only if you enable Advertise.
Auth Profile	Select a profile from the Auth Profile drop-down.
Mode	Select normal, passive or send-only from the Mode drop-down.
BFD	To enable BFD for a RIP interface. override the BFD setting for RIP.

Network > Virtual Router > RIP > Export Rules

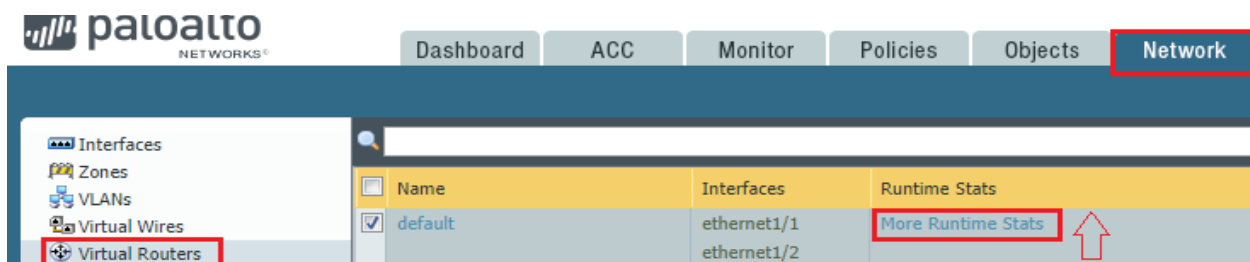
RIP export rules allow you to control which routes the virtual router sends to peers.



Export Rules Settings	Description
Allow Redistribute Default Route	Select to permit the firewall to redistribute its default route to peers.
Redistribution Profile	Click Add and select or create a redistribution profile that allows you to modify route redistribution, filter, priority, and action based on the desired network behavior.

Verification:

Go to **Network>Virtual Routers** click on **More Runtime Stats**



Go to **Routing** Tab> The following table describes the virtual router's runtime stats for the **Route Table**, Forwarding Table, and the Static Route Monitoring table.

Routing RIP OSPF OSPFv3 BGP Multicast BFD Summary Information				
Route Table Forwarding Table Static Route Monitoring				
Route Table ☒ Unicast ☐ Multicast Display Address Fan				
Destination	Next Hop	Metric	Flags	Interface
0.0.0.0/0	192.168.17.2	10	A S	ethernet1/1
1.1.1.0/24	192.168.78.50	2	A R	ethernet1/2
2.2.2.0/24	192.168.78.50	2	A R	ethernet1/2
3.3.3.0/24	192.168.78.50	2	A R	ethernet1/2
4.4.4.0/24	192.168.78.50	2	A R	ethernet1/2
6.6.6.0/24	0.0.0.0	1	R	vlan.1
6.6.6.0/24	6.6.6.6	0	A C	vlan.1
6.6.6.6/32	0.0.0.0	0	A H	
7.7.7.0/24	0.0.0.0	1	R	vlan.2
7.7.7.0/24	7.7.7.7	0	A C	vlan.2
7.7.7.7/32	0.0.0.0	0	A H	
192.168.17.0/24	192.168.17.100	0	A C	ethernet1/1
192.168.17.100/32	0.0.0.0	0	A H	
192.168.78.0/24	0.0.0.0	1	R	ethernet1/2
192.168.78.0/24	192.168.78.100	0	A C	ethernet1/2
192.168.78.100/32	0.0.0.0	0	A H	

Refresh A:active, ?loose, C:connect, H:host, S:static, ~:internal, R:rip, O:OSPF, B:bgp, O:OSPF intra-area, O:OSPF inter-area, O1:OSPF ext-type-1, O2:OSPF ext-type-2, E:ECMP

Routing RIP OSPF OSPFv3 BGP Multicast BFD Summary Information					
Route Table Forwarding Table Static Route Monitoring					
Destination	Next Hop	Flags	Interface	MTU	
0.0.0.0/0	192.168.17.2	ug	ethernet1/1	1500	
1.1.1.0/24	192.168.78.50	ug	ethernet1/2	1500	
2.2.2.0/24	192.168.78.50	ug	ethernet1/2	1500	
3.3.3.0/24	192.168.78.50	ug	ethernet1/2	1500	
4.4.4.0/24	192.168.78.50	ug	ethernet1/2	1500	
6.6.6.0/24	0.0.0.0	u	vlan.1	1500	
6.6.6.6/32	0.0.0.0	uh	vlan.1	1500	
7.7.7.0/24	0.0.0.0	u	vlan.2	1500	
7.7.7.7/32	0.0.0.0	uh	vlan.2	1500	
192.168.17.0/24	0.0.0.0	u	ethernet1/1	1500	
192.168.17.100/32	0.0.0.0	uh	ethernet1/1	1500	
192.168.78.0/24	0.0.0.0	u	ethernet1/2	1500	
192.168.78.100/32	0.0.0.0	uh	ethernet1/2	1500	

Refresh u - up, h - host, g - gateway, e - ECMP, * - preferred path

Virtual Router - default

Routing **RIP** OSPF OSPFv3 BGP Multicast BFD Summary Information

Summary **Interface** Peer

Address	Auth Type	Send Allowed	Receive Allowed	Advertise Default Route	Default Route Metric
6.6.6.6	none	☒	☒		0
7.7.7.7	none	☒	☒		0
192.168.78.100	none	☒	☒	☒	10

Summary Interface **Peer**

Interval Seconds 1
Update Intervals 30
Expire Intervals 180
Delete Intervals 120

Peer Address	Last Update	RIP Version	Invalid Packets	Invalid Routes
192.168.78.50	Fri Jan 10 12:06:52 2020	2	0	0

Stats	Description
Summary Tab	
Interval Seconds	Number of seconds in an interval.
Update Intervals	Number of intervals between RIP route advertisement updates that the virtual router sends to peers.
Expire Intervals	Number of intervals since the last update the virtual router received from a peer, after which the virtual router marks the routes from the peer as unusable.
Delete Intervals	Number of intervals after route has been marked as unusable that, if no update is received, the firewall deletes the route from the routing table.
Interface Tab	
Address	IP address of an interface on the virtual router where RIP is enabled.
Auth Type	Type of authentication: simple password, MD5, or none.
Send Allowed	Check mark indicates this interface is allowed to send RIP packets.
Receive Allowed	Check mark indicates this interface is allowed to receive RIP packets.
Advertise Default Route	Check mark indicates that RIP will advertise its default route to its peers.
Default Route Metric	Metric (hop count) assigned to the default route.
Key Id	Authentication key used with peers.
Preferred	Preferred key for authentication.
Peer Tab	
Peer Address	IP address of a peer to the virtual router's RIP interface.
Last Update	Date and time that the last update was received from this peer.
RIP Version	RIP version the peer is running.
Invalid Packets	Count of invalid packets received from this peer.
Invalid Routes	Count of invalid routes received from this peer.


```
admin@PA-VM> show routing protocol rip peer
```

```
=====
```

```
virtual router: default
peer address: 192.168.78.50
last update: Fri Jan 10 11:57:27 2020
```

```
RIP version: 2
invalid packets: 0
```

flags: A:active, ?:loose, C:connect, H:host, S:static, ~:internal, R:rip, O:ospf, B:bgp,
Oi:ospf intra-area, Oo:ospf inter-area, O1:ospf ext-type-1, O2:ospf ext-type-2, E:ecmp, M:multicast

```
VIRTUAL ROUTER: default (id 1)
```

destination	next-hop	metric	flags	age	interface
0.0.0.0/0	192.168.17.2	10	A S		ethernet1/1
1.1.1.0/24	192.168.78.50	2	A R		ethernet1/2
2.2.2.0/24	192.168.78.50	2	A R		ethernet1/2
3.3.3.0/24	192.168.78.50	2	A R		ethernet1/2
4.4.4.0/24	192.168.78.50	2	A R		ethernet1/2
6.6.6.0/24	0.0.0.0	1	R		vlan.1
6.6.6.0/24	6.6.6.6	0	A C		vlan.1
6.6.6.6/32	0.0.0.0	0	A H		
7.7.7.0/24	0.0.0.0	1	R		vlan.2
7.7.7.0/24	7.7.7.7	0	A C		vlan.2
7.7.7.7/32	0.0.0.0	0	A H		

```
admin@PA-VM> show routing protocol rip summary
```

```
=====
```

```
virtual router: default
reject default route: yes
redist default route: allow
interval seconds: 1
update intervals: 30
expire intervals: 180
delete intervals: 120
  interface: 6.6.6.6
  interface: 7.7.7.7
  interface: 192.168.78.100
```

```
admin@PA-VM> ping source 192.168.78.100 host 1.1.1.1
```

```
PING 1.1.1.1 (1.1.1.1) from 192.168.78.100 : 56(84) bytes of data.
64 bytes from 1.1.1.1: icmp_seq=1 ttl=255 time=9.80 ms
64 bytes from 1.1.1.1: icmp_seq=2 ttl=255 time=8.93 ms
```

```
admin@PA-VM> ping source 192.168.78.100 host 2.2.2.2
PING 2.2.2.2 (2.2.2.2) from 192.168.78.100 : 56(84) bytes of data.
64 bytes from 2.2.2.2: icmp_seq=1 ttl=255 time=7.37 ms
64 bytes from 2.2.2.2: icmp_seq=2 ttl=255 time=6.43 ms
```

```
admin@PA-VM> show routing protocol rip database
```

```
VIRTUAL ROUTER: default
```

```
=====
```

```
Flags: S:stale path, L:linked, P:programmed
destination      metric tag  flag path-detail
1.1.1.0/24       2         0    LP RIP route, next hop 192.168.78.50, from peer 192.168.78.50, age 2
2.2.2.0/24       2         0    LP RIP route, next hop 192.168.78.50, from peer 192.168.78.50, age 2
3.3.3.0/24       2         0    LP RIP route, next hop 192.168.78.50, from peer 192.168.78.50, age 2
4.4.4.0/24       2         0    LP RIP route, next hop 192.168.78.50, from peer 192.168.78.50, age 2
6.6.6.0/24       1         0    LP RIP connected route on vlan.1
6.6.6.0/24       5         0    L  RDIST connected route on vlan.1
7.7.7.0/24       1         0    LP RIP connected route on vlan.2
7.7.7.0/24       5         0    L  RDIST connected route on vlan.2
192.168.17.0/24  5         0    L  RDIST connected route on ethernet1/1
192.168.78.0/24  1         0    LP RIP connected route on ethernet1/2
192.168.78.0/24  5         0    L  RDIST connected route on ethernet1/2
192.168.78.0/24  10        0    L  RIP local route
```

Verification on Router:

```
R1#show ip route rip
```

```
Codes: L - local, C - connected, S - static, R - RIP, M - mobile, B - BGP
        D - EIGRP, EX - EIGRP external, O - OSPF, IA - OSPF inter area
        N1 - OSPF NSSA external type 1, N2 - OSPF NSSA external type 2
        E1 - OSPF external type 1, E2 - OSPF external type 2
        i - IS-IS, su - IS-IS summary, L1 - IS-IS level-1, L2 - IS-IS level-2
        ia - IS-IS inter area, * - candidate default, U - per-user static route
        o - ODR, P - periodic downloaded static route, H - NHRP, l - LISP
        + - replicated route, % - next hop override
```

Gateway of last resort is not set

```
R      6.0.0.0/24 is subnetted, 1 subnets
R      6.6.6.0 [120/1] via 192.168.78.100, 00:00:10, FastEthernet0/0
R      7.0.0.0/24 is subnetted, 1 subnets
R      7.7.7.0 [120/1] via 192.168.78.100, 00:00:10, FastEthernet0/0
```

```
R1#ping 6.6.6.6
```

Type escape sequence to abort:

Sending 5, 100-byte ICMP Echos to 6.6.6.6, timeout is 2 seconds:

!!!!

Success rate is 100 percent (5/5), round-trip min/avg/max = 8/12/24 ms

```
R1#ping 7.7.7.7
```

Type escape sequence to abort.

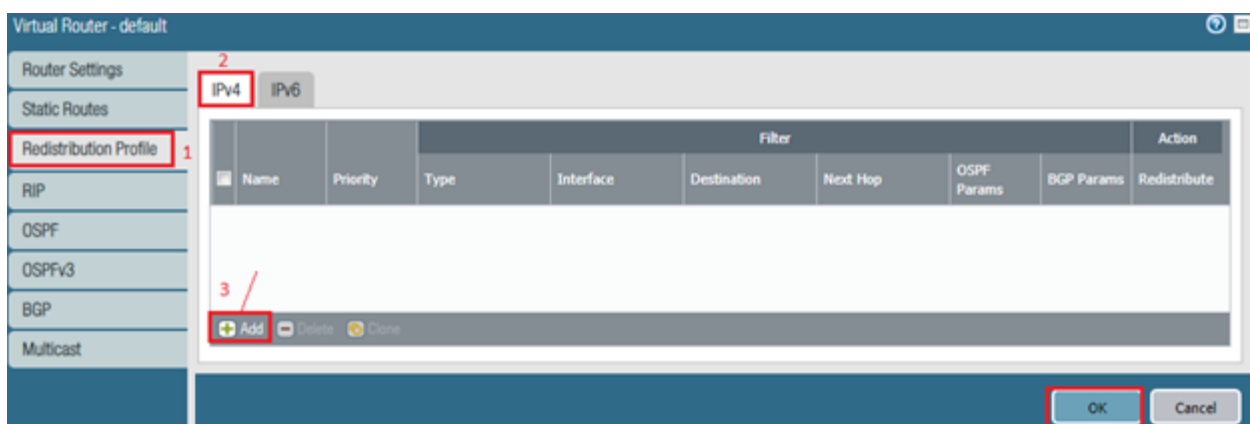
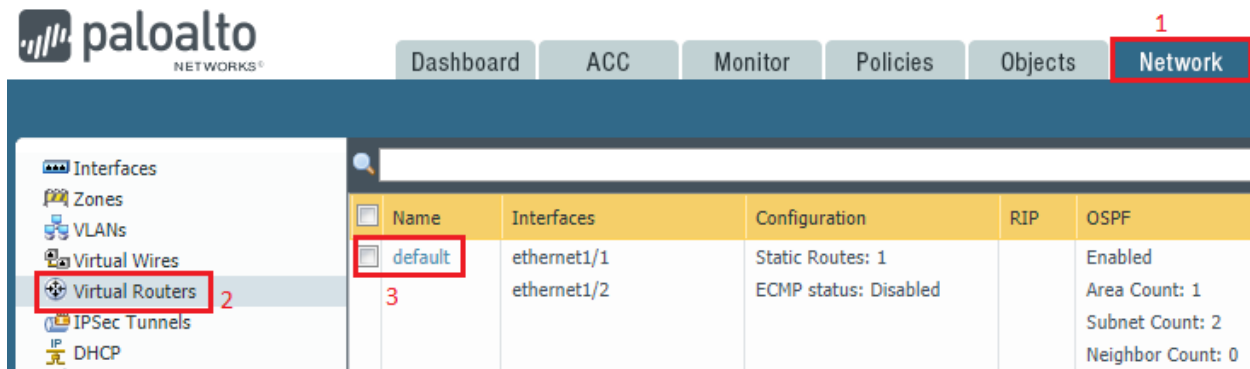
Sending 5, 100-byte ICMP Echos to 7.7.7.7, timeout is 2 seconds:

!!!!

Success rate is 100 percent (5/5), round-trip min/avg/max = 12/18/24 ms

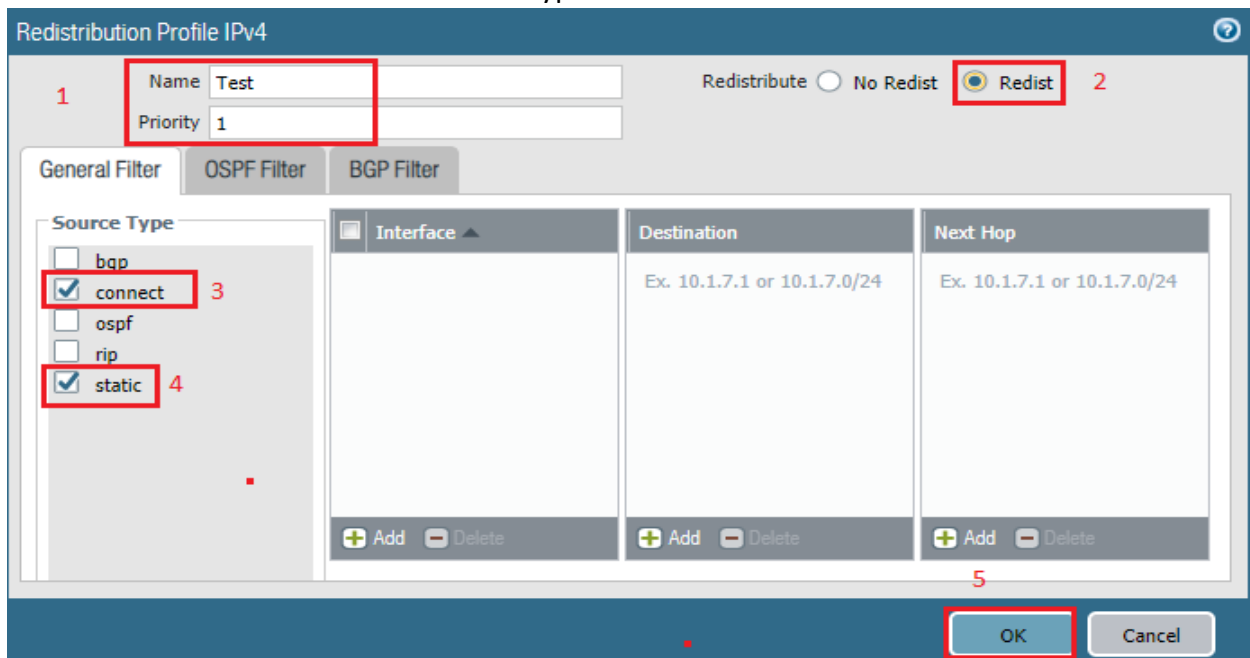
Redistribute Static Route into RIP:

Go to **Network > Virtual router** and click "**default**". Then create a redistribution profile.

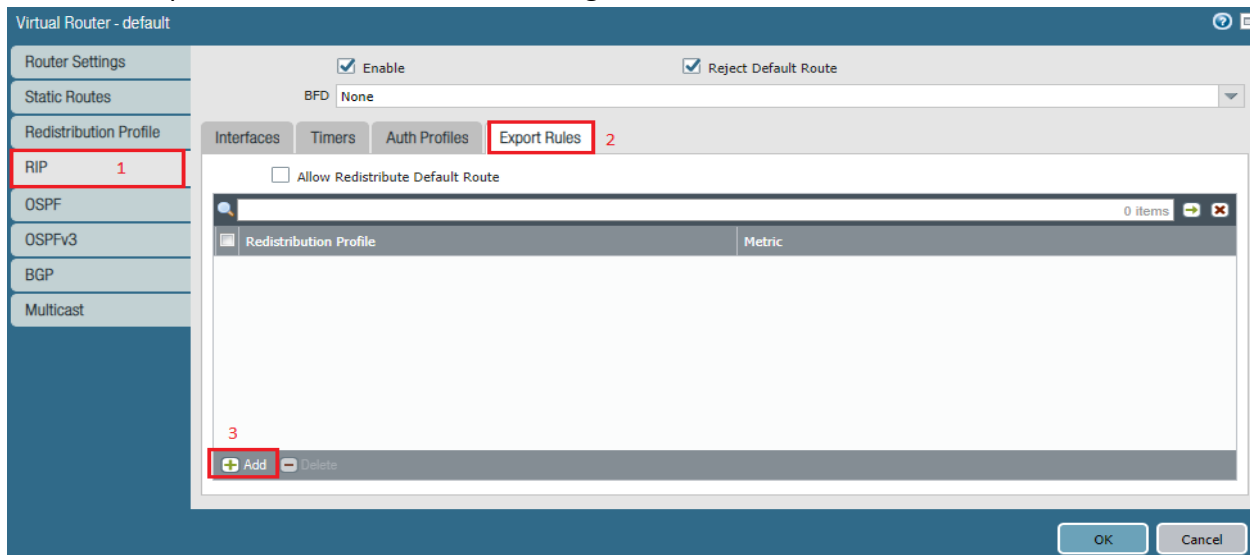


Configure the profile name and enter Priority. Check the **Redist** option for Redistribute.

Go to General Filter and select the filter type as **static** and **connected**. Click "OK"

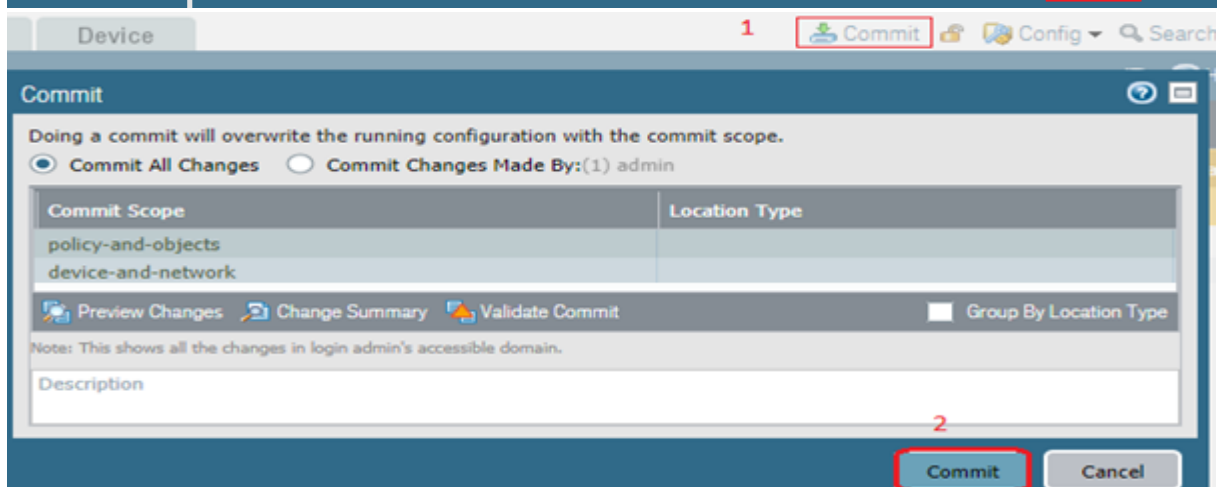
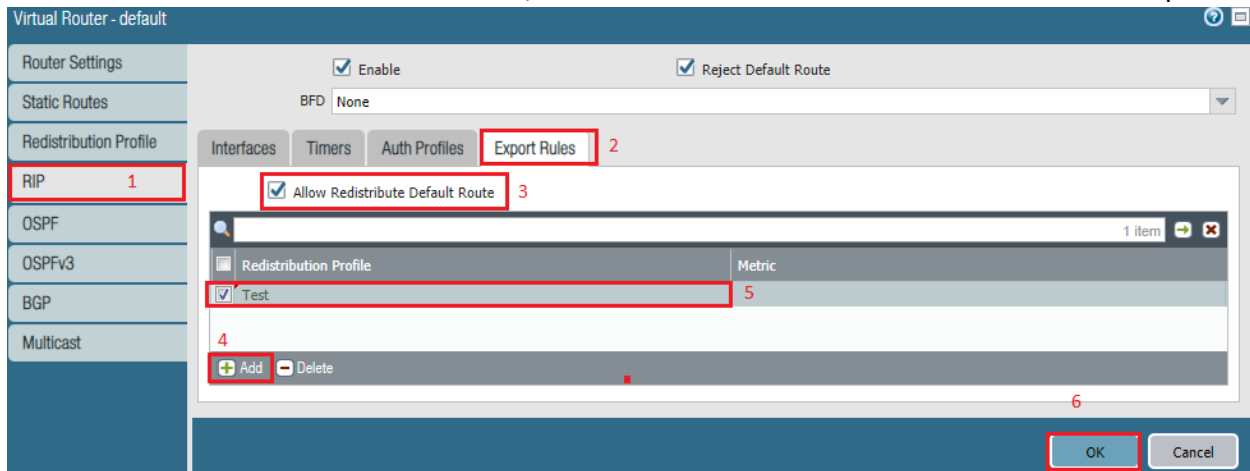


Create an Export Rule under RIP router configuration.



Add the Redistribution Profile and click "OK". Then commit.

If there is a default route to redistribute, enable the "Allow Redistribute Default Route" option.



Verification :

R1#show ip route rip

Codes: L - local, C - connected, S - static, R - RIP, M - mobile, B - BGP
D - EIGRP, EX - EIGRP external, O - OSPF, IA - OSPF inter area
N1 - OSPF NSSA external type 1, N2 - OSPF NSSA external type 2
E1 - OSPF external type 1, E2 - OSPF external type 2
i - IS-IS, su - IS-IS summary, L1 - IS-IS level-1, L2 - IS-IS level-2
ia - IS-IS inter area, * - candidate default, U - per-user static route
o - ODR, P - periodic downloaded static route, H - NHRP, l - LISP
+ - replicated route, % - next hop override

Gateway of last resort is 192.168.78.100 to network 0.0.0.0

```
R* 0.0.0.0/0 [120/1] via 192.168.78.100, 00:00:25, FastEthernet0/0
    6.0.0.0/24 is subnetted, 1 subnets
R    6.6.6.0 [120/1] via 192.168.78.100, 00:00:25, FastEthernet0/0
    7.0.0.0/24 is subnetted, 1 subnets
R    7.7.7.0 [120/1] via 192.168.78.100, 00:00:25, FastEthernet0/0
R    192.168.17.0/24 [120/1] via 192.168.78.100, 00:00:25, FastEthernet0/0
```

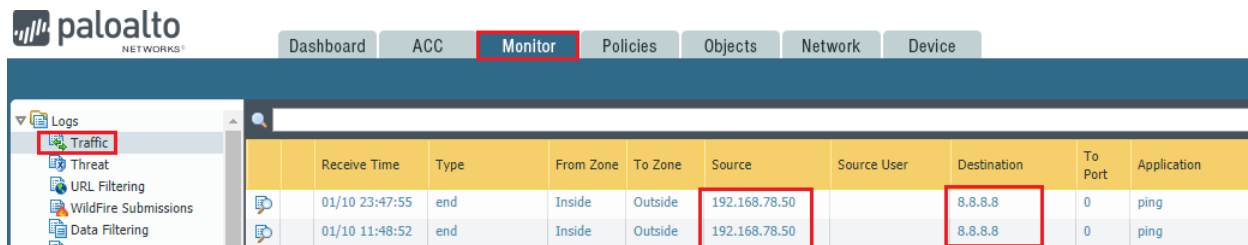
R1#ping 8.8.8.8

Type escape sequence to abort.

Sending 5, 100-byte ICMP Echos to 8.8.8.8, timeout is 2 seconds:

!!!!

Success rate is 100 percent (5/5), round-trip min/avg/max = 80/83/88 ms



	Receive Time	Type	From Zone	To Zone	Source	Source User	Destination	To Port	Application
	01/10 23:47:55	end	Inside	Outside	192.168.78.50		8.8.8.8	0	ping
	01/10 11:48:52	end	Inside	Outside	192.168.78.50		8.8.8.8	0	ping

Securing Routing Protocols:

- o By default, all, the routing information is visible to all interested parties.
- o By default, routing information is not encrypted therefore open to an attack.
- o Routing protocols can be configured to prevent receiving false routing updates.
- o Routing Protocol Authentication is a security mechanism prevent such attacks.
- o To prevent router from accepting unauthorized or malicious routing updates.
- o Routing protocols RIP, EIGRP and OSPF can be secure by putting an authentication
- o Two options for authentication are Plain text authentication & MD5 authentication.

Authentication on OSPF:

- o OSPF Dynamic Routing protocols can authenticate every OSPF message to protect.
- o This is basically to prevent a rogue Router from injecting false routing information.
- o This is also to prevent the Denial-of-Service (DoS) attack on Cisco Devices as well.
- o Two types of authentication can be used one of them is Plain text authentication.
- o Second is MD5 authentication, MD5 type of Authentication is stronger and secure.
- o There are all together three types of OSPF authentication Type 1, Type 2 and Type 3.
- o Type 1 – No Authentication, Type 2 – Clear Text & Type 3 – Cryptographic (MD5 or SHA).
- o OSPF Authentication can be configured on a per area or per interface on OSPF Protocols.
- o But any OSPF authentication passwords has to be configured on the interface only.
- o OSPF authentication type 1 means no authentication and is the default on any link.
- o OSPF Authentication type 2 First enable authentication & create password on interface.

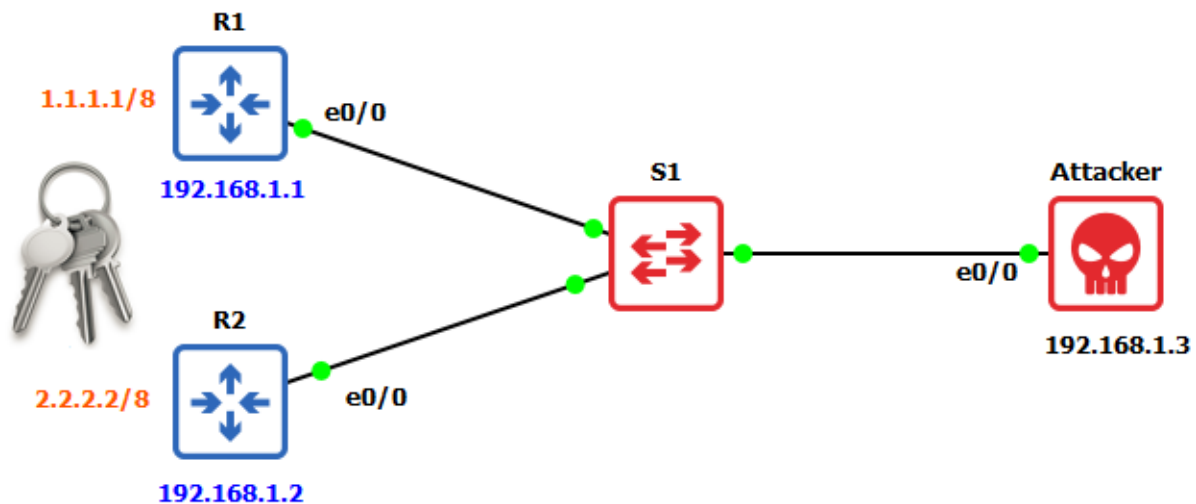


Key ID 1
String: cisco1

Keychain: cisco

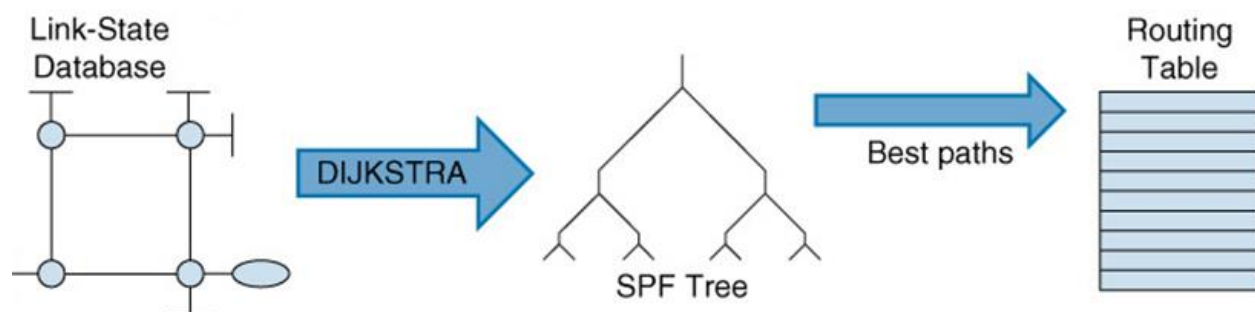
Authentication on RIP:

- o Routing Information Protocol Version 1 (RIPv1) doesn't support authentication.
- o Routing Information Protocol Version 2 (RIPv2) does support authentication.
- o RIP authentication is one of the protocol enhancements that appeared in Version 2.
- o RIPv2 allows packets to be authenticated plain text password or secure MD5 hash.
- o Authentication ensure that unauthorized equipment cannot affect how traffic is routed.
- o Authentication in RIP comes in two variations, the plain text mode and the MD5 mode.

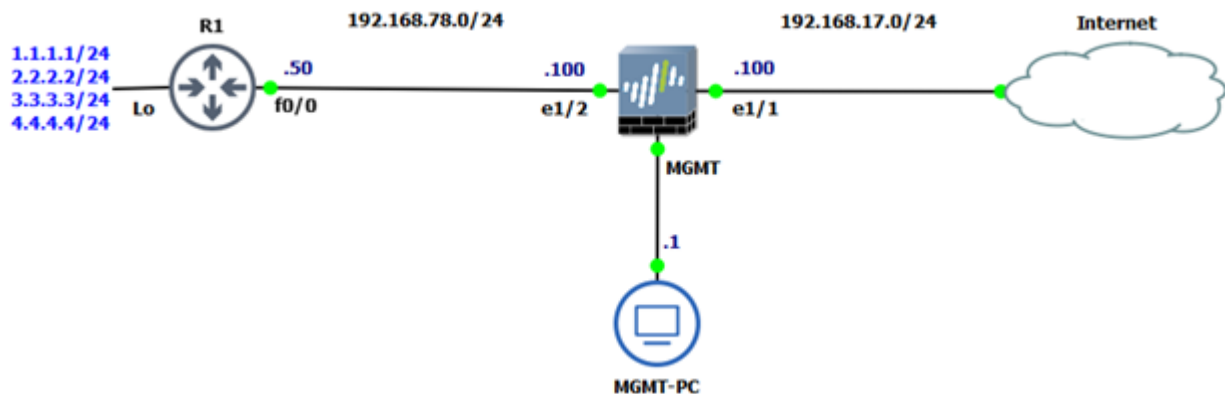


OSPF (Open Shortest Path First) protocol:

- o OSPF is term which stands for Open Shortest Path First.
- o Link State protocol sends update based in state of the link.
- o When a link comes up and goes down it sends the updates.
- o OSPF protocols is a dynamic open standard Routing Protocol.
- o OSPF protocols is the Link-State dynamic routing protocol.
- o OSPF protocols uses the Shortest Path First (SPF) algorithm.
- o OSPF protocols uses IP protocol type **89** (not TCP or UDP).
- o OSPF External and Internal Administrative Distance is **110**.
- o OSPF is a classless Routing protocol and OSPF metric is Cost.
- o OSPF use multicast address **224.0.0.5** to send the hello packet.
- o OSPF use multicast address **224.0.0.6** for all designated routers.
- o OSPF default Hello time is **10 seconds** and dead time is **40 seconds**.
- o OSPF protocols supports both the VLSM and route summarization.
- o OSPF protocols supports both MD5 and clear text authentication.
- o OSPF protocols supports the summarization at ABRs router only.
- o OSPF are requires more memory and CPU processing to run.
- o OSPF work on area, Area 0 is the backbone of OSPF technology.
- o OSPF have Neighbor table, Topology table and Routing table.
- o OSPF packets are only sends to the neighbor of own Area.
- o OSPF protocols supports both IPv4 and IPv6 routed protocols.
- o OSPF load balancing with equal cost routes for same destination.
- o OSPF dynamic routing protocols supports unlimited hop counts.
- o OSPF protocols supports trigger updates for fast convergence.
- o OSPF sends update with a sequence number of **0x80000001**.
- o The sequence number ends with **0x7FFFFFFF** and start again.
- o The sequence number will increment by one in every update.
- o OSPF protocol is more complex to setup and hard to troubleshoot.



OSPF Lab:



Router R1 Configuration

```
R1(config)#interface fastEthernet0/0
R1(config-if)#ip add 192.168.78.50 255.255.255.0
R1(config-if)#no shutdown

R1(config)#interface loopback 1
R1(config-if)#ip add 1.1.1.1 255.255.255.0

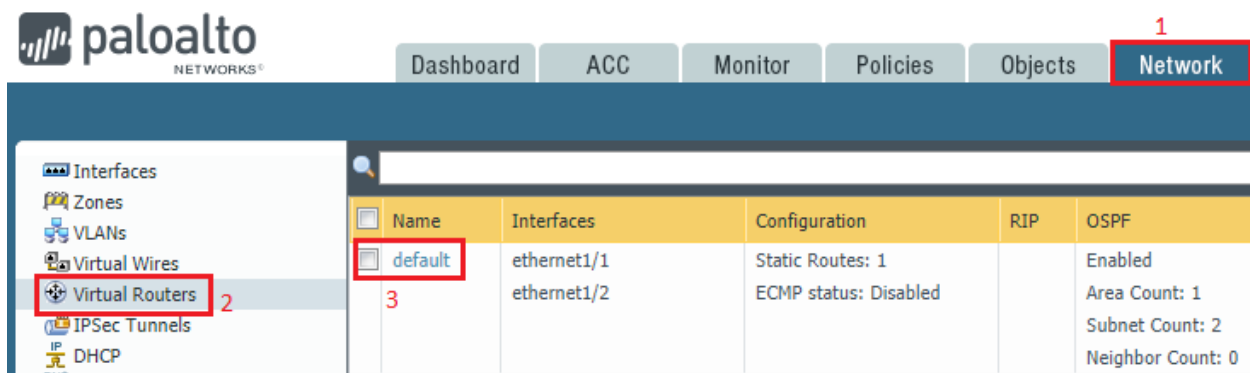
R1(config)#interface loopback 2
R1(config-if)#ip add 2.2.2.2 255.255.255.0

R1(config)#interface loopback 3
R1(config-if)#ip add 3.3.3.3 255.255.255.0

R1(config)#interface loopback 4
R1(config-if)#ip add 4.4.4.4 255.255.255.0

R1(config)#router ospf 1
R1(config-router)#network 1.1.1.0 0.0.0.255 area 0
R1(config-router)#network 2.2.2.0 0.0.0.255 area 0
R1(config-router)#network 3.3.3.0 0.0.0.255 area 0
R1(config-router)#network 4.4.4.0 0.0.0.255 area 0
R1(config-router)#network 192.168.78.0 0.0.0.255 area 0
```

Go to **Network > Virtual Router** and check **default**.



Select OSPF from the menu on the left. Select **Enable** to enable the OSPF protocol. Select **Reject Default Route** if you do not want to learn any default routes through OSPF. Enter the **Router ID**.

Enter an Area ID for the area. This is the identifier that each neighbor must accept to be part of the same area. Area ID 0.0.0.0 is the same as Cisco's Area 0. On the Type tab, select Normal.

Select Interface tab and click add. Interface—Select an interface from the drop-down. Click **OK**

Commit change by click **commit** button on right top Conner.

Final Configuration after adding all three interfaces **ethernet1/2** inside, **Vlan.1** & **Vlan.2**.

Virtual Router - OSPF - Area

Area ID: 0.0.0.0

Interface

Type	Range	Interface	Enable	Passive	Link Type	Metric	Priority	Timing	Auth Profile	BFD
☐		ethernet1/2	☒	☐	broadcast	10	1	Hello Interval: 10 Dead Counts: 4 Retransmit Intervals: 5 Transit Delay: 1 Graceful Restart Hello Delay: 10		☐
☒		vlan.1	☒	☐	broadcast	10	1	Hello Interval: 10 Dead Counts: 4 Retransmit Intervals: 5 Transit Delay: 1 Graceful Restart Hello Delay: 10		☐

+ Add - Delete

Virtual Router - default

Router Settings

Static Routes

Redistribution Profile

RIP

OSPF

OSPFv3

BGP

Multicast

Router ID: 192.168.78.100

BFD: None

Enable ☒ Reject Default Route ☒

Areas

Area ID	Type	Range	Interface
0.0.0.0	normal		ethernet1/2 vlan.1 vlan.2

+ Add - Delete

OK Cancel

Verification:

Go to **Network>Virtual Routers** click on **More Runtime Stats**

paloalto NETWORKS®

Dashboard ACC Monitor Policies Objects **Network**

Interfaces Zones VLANs Virtual Wires **Virtual Routers**

Name	Interfaces	Runtime Stats
default	ethernet1/1 ethernet1/2	More Runtime Stats

Go to **Routing** Tab> The following table describes the virtual router's runtime stats for the **Route Table**, Forwarding Table, and the Static Route Monitoring table.

Routing

RIP

OSPF

OSPFv3

BGP

Multicast

BFD Summary Information

Route Table

Forwarding Table

Static Route Monitoring

Route Table

☒ Unicast

☐ Multicast

Display Address Family

IPv4 and IPv6

16 items

Destination	Next Hop	Metric	Flags	Interface
0.0.0.0/0	192.168.17.2	10	A S	ethernet1/1
1.1.1.1/32	192.168.78.50	11	A Oi	ethernet1/2
2.2.2.2/32	192.168.78.50	11	A Oi	ethernet1/2
3.3.3.3/32	192.168.78.50	11	A Oi	ethernet1/2
4.4.4.4/32	192.168.78.50	11	A Oi	ethernet1/2
6.6.6.0/24	0.0.0.0	10	Oi	vlan.1
6.6.6.0/24	6.6.6.6	0	A C	vlan.1
6.6.6.6/32	0.0.0.0	0	A H	
7.7.7.0/24	0.0.0.0	10	Oi	vlan.2
7.7.7.0/24	7.7.7.7	0	A C	vlan.2
7.7.7.7/32	0.0.0.0	0	A H	

Refresh

A:active, ?loose, C:connect, H:host, S:static, ~:internal, R:rip, O:OSPF, B:bgp, Oi:OSPF intra-area, Oo:OSPF inter-area, O1:OSPF ext-type-1, O2:OSPF ext-type-2

Routing

RIP

OSPF

OSPFv3

BGP

Multicast

BFD Summary Information

Route Table

Forwarding Table

Static Route Monitoring

Destination	Next Hop	Flags	Interface	MTU
0.0.0.0/0	192.168.17.2	ug	ethernet1/1	1500
1.1.1.1/32	192.168.78.50	ug	ethernet1/2	1500
2.2.2.2/32	192.168.78.50	ug	ethernet1/2	1500
3.3.3.3/32	192.168.78.50	ug	ethernet1/2	1500
4.4.4.4/32	192.168.78.50	ug	ethernet1/2	1500
6.6.6.0/24	0.0.0.0	u	vlan.1	1500
6.6.6.6/32	0.0.0.0	uh	vlan.1	1500
7.7.7.0/24	0.0.0.0	u	vlan.2	1500
7.7.7.7/32	0.0.0.0	uh	vlan.2	1500
192.168.17.0/24	0.0.0.0	u	ethernet1/1	1500
192.168.17.100/32	0.0.0.0	uh	ethernet1/1	1500
192.168.78.0/24	0.0.0.0	u	ethernet1/2	1500
192.168.78.100/32	0.0.0.0	uh	ethernet1/2	1500

Refresh

u - up, h - host, g - gateway, e - ECMP, * - preferred path

Routing

RIP

OSPF

OSPFv3

BGP

Multicast

BFD Summary Information

Summary

Area

Interface

Neighbor

Virtual Link

Virtual Neighbor

1 item

Area Id	Range	Area Type	NSSA Area	Stub Area	Normal Area
0.0.0.0		normal			

Summary

Area

Interface

Neighbor

Virtual Link

Virtual Neighbor

3 items

Name	Address	Type	Passive	Area Id	Router Priority	Status	Transit
vlan.1	6.6.6.6	broadcast		0.0.0.0	1	DR	1
vlan.2	7.7.7.7	broadcast		0.0.0.0	1	DR	1
ethernet1/2	192.168.78.100	broadcast		0.0.0.0	1	DR	1

Routing

RIP

OSPF

OSPFv3

BGP

Multicast

BFD Summary Information

Summary

Area

Interface

Neighbor

Virtual Link

Virtual Neighbor

1 item

Neighbor Address	Neighbor Router Id	Local Address Binding	Area Id	Neighbor Priority	Remaining Lifetime	Status	Message
192.168.78.50	4.4.4.4	0.0.0.0	0.0.0.0	1	30	full	0

```

admin@PA-VM> show routing route
flags: A:active, ?:loose, C:connect, H:host, S:static, ~:internal, R:rip, O:ospf, B:bgp,
       O1:ospf intra-area, Oo:ospf inter-area, O1:ospf ext-type-1, O2:ospf ext-type-2, E:ecmp, M:multicast

VIRTUAL ROUTER: default (id 1)
=====
destination      nexthop          metric flags    age  interface
next-AS
0.0.0.0/0        192.168.17.2    10   A S           ethernet1/1
1.1.1.1/32       192.168.78.50   11   A O i         4372 ethernet1/2
2.2.2.2/32       192.168.78.50   11   A O i         4372 ethernet1/2
3.3.3.3/32       192.168.78.50   11   A O i         4372 ethernet1/2
4.4.4.4/32       192.168.78.50   11   A O i         4372 ethernet1/2
6.6.6.0/24       0.0.0.0         10   O i           2419 vlan.1
6.6.6.0/24       6.6.6.6         0    A C           vlan.1
6.6.6.6/32       0.0.0.0         0    A H           vlan.1
7.7.7.0/24       0.0.0.0         10   O i           2267 vlan.2
7.7.7.0/24       7.7.7.7         0    A C           vlan.2

```

```
admin@PA-VM> show routing protocol ospf summary
```

```
=====
router id:                        192.168.78.100
virtual router:                   default
reject default route:            yes
redist default route:            block
spf calculation delay (sec):      5.00
LSA interval timer (sec):        5.00
RFC1583 behavior:                no
area border router:              no
AS border router:                yes
LS type 5 count:                 0
LS type 11 count:                0
LS sent count:                   24
LS rcv count:                    4
  area id:                       0.0.0.0
    interface:                   6.6.6.6
    interface:                   7.7.7.7
    interface:                   192.168.78.100
  dynamic neighbors:
                                IP 192.168.78.50 ID 4.4.4.4
```

```
R1#show ip route ospf
```

```
Codes: L - local, C - connected, S - static, R - RIP, M - mobile, B - BGP
       D - EIGRP, EX - EIGRP external, O - OSPF, IA - OSPF inter area
       N1 - OSPF NSSA external type 1, N2 - OSPF NSSA external type 2
       E1 - OSPF external type 1, E2 - OSPF external type 2
       i - IS-IS, su - IS-IS summary, L1 - IS-IS level-1, L2 - IS-IS level-2
       ia - IS-IS inter area, * - candidate default, U - per-user static route
       o - ODR, P - periodic downloaded static route, H - NHRP, l - LISP
       + - replicated route, % - next hop override
```

```
Gateway of last resort is not set
```

```

O    6.0.0.0/24 is subnetted, 1 subnets
O      6.6.6.0 [110/11] via 192.168.78.100, 00:37:11, FastEthernet0/0
O    7.0.0.0/24 is subnetted, 1 subnets
O      7.7.7.0 [110/11] via 192.168.78.100, 00:34:40, FastEthernet0/0
```

```
admin@PA-VM> ping source 192.168.78.100 host 1.1.1.1
```

```
PING 1.1.1.1 (1.1.1.1) from 192.168.78.100 : 56(84) bytes of data.
64 bytes from 1.1.1.1: icmp_seq=1 ttl=255 time=9.80 ms
64 bytes from 1.1.1.1: icmp_seq=2 ttl=255 time=8.93 ms
```

```
admin@PA-VM> ping source 192.168.78.100 host 2.2.2.2
```

```
PING 2.2.2.2 (2.2.2.2) from 192.168.78.100 : 56(84) bytes of data.
64 bytes from 2.2.2.2: icmp_seq=1 ttl=255 time=7.37 ms
64 bytes from 2.2.2.2: icmp_seq=2 ttl=255 time=6.43 ms
```

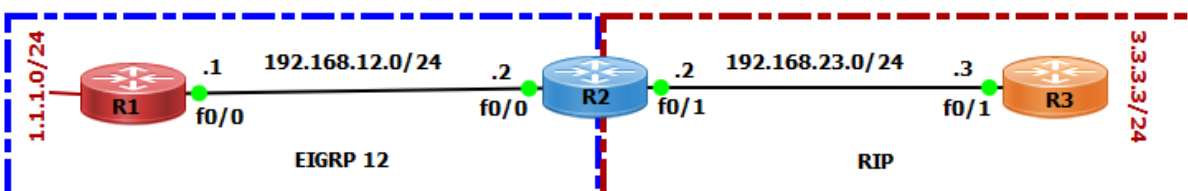
Routing Protocols Redistribution:

- o Redistribution allows a Cisco Router to run more than one Routing Protocol.
- o Exchange routing information between different protocols is called redistribution.
- o Redistribution allows Cisco Routers to share Routes among the routing protocols.
- o Redistribution is process of distributing routes learned from one source to another.
- o For Redistribution, at least one working physical link in each routing domain need.
- o The Redistribution is very useful when the networks are expanding or are merging.
- o Routes can only be redistributed at Routers that run both the Routing Protocols.
- o The Static Routes and Connected Routes and interfaces can also be redistributed.
- o Only Routes present in Cisco Router, Routing Tables can be redistributed to other.
- o Whenever want to redistributing routes, make ensure metric compatibility issue.
- o Route redistribution runs on the Cisco Router that connects two different networks.
- o Configure default hop count for everything redistribute into Dynamic Routing RIP.
- o Configure K-Values for everything redistribute into Dynamic Routing Protocol EIGRP.
- o By default, redistribute into Dynamic Protocol Open Shortest Path First cost is 20.
- o For each Static and Dynamic Routing protocol, the redistribution is slightly different.
- o Keep in mind Routing Protocols Redistribution always happens outbound direction.

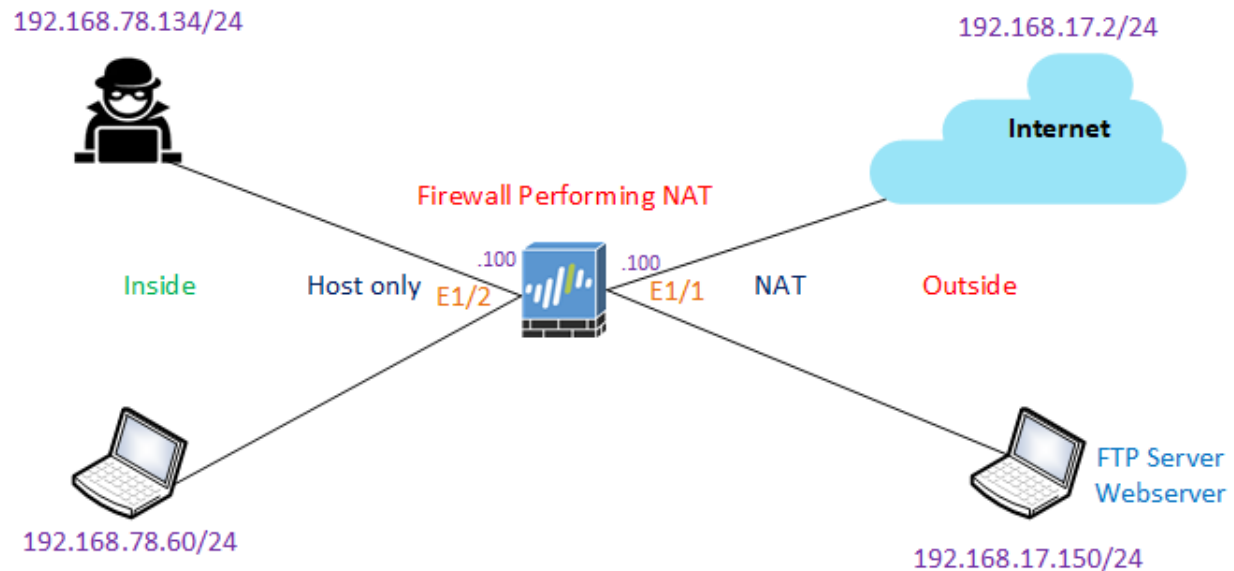
Seed Metric:

- o The metric with which a protocol receives the Routes learned by another metric.
- o When redistribute from one routing protocol into another, we use a seed metric.
- o Each Static or Dynamic Routing protocol uses their own default seed metric value.
- o The default seed metric can also be modified using the “Default Metric” command.

Protocols	Default Seed Metric
RIP	Infinity
EIGRP	Infinity
OSPF	20 except BGP 1
BGP	BGP metric is set to IGP metric



Lab Time:



Zone Protection Testing and Verification:

Go to **Network > Network Profiles > Zone Protection >> Flood Protection**

Zone Protection Profile

Name: Zone-Protection-Profile

Description:

Flood Protection | Reconnaissance Protection | Packet Based Attack Protection | Protocol Protection

☒ SYN

Action: Random Early Drop

Alarm Rate (connections/sec): 10

Activate (connections/sec): 12

Maximum (connections/sec): 15

☒ UDP

Alarm Rate (connections/sec): 10

Activate (connections/sec): 12

Maximum (connections/sec): 15

☒ ICMP

Alarm Rate (connections/sec): 10

Activate (connections/sec): 12

Maximum (connections/sec): 15

☒ ICMPv6

Alarm Rate (connections/sec): 10000

Activate (connections/sec): 10000

Maximum (connections/sec): 40000

☒ Other IP

Alarm Rate (connections/sec): 10000

Activate (connections/sec): 10000

Maximum (connections/sec): 40000

Unchecked everything, even in Packet Based Attack Protection Tab To verify TCP Flood

OK Cancel

Let's launch attack from Kali Linux Inside PC to on Outside Windows XP PC, Open terminal type:

```
root@kali:~# hping3 -c 15000 -d 120 -S -w 64 -p 80 --flood --rand-source 192.168.17.150
```

```
File Actions Edit View Help
root@kali: ~
root@kali:~# hping3 -c 15000 -d 120 -S -w 64 -p 80 --flood --rand-source 192.168.17.150
```


Now Let's check TCP Flood logs go to **Monitor > Logs>Threat**

Receive Time	Type	Name	From Zone	To Zone	Source address	Destination address	To Port	Application	Action	Severity
01/28 08:31:54	flood	TCP Flood	Inside	Inside	0.0.0.0	0.0.0.0	0	not-applicable	drop	critical
01/28 08:31:43	flood	TCP Flood	Inside	Inside	0.0.0.0	0.0.0.0	0	not-applicable	drop	critical
01/28 08:25:04	flood	TCP Flood	Inside	Inside	0.0.0.0	0.0.0.0	0	not-applicable	drop	critical
01/28 08:24:52	flood	TCP Flood	Inside	Inside	0.0.0.0	0.0.0.0	0	not-applicable	drop	critical
01/28 08:24:42	flood	TCP Flood	Inside	Inside	0.0.0.0	0.0.0.0	0	not-applicable	drop	critical
01/28 08:24:36	flood	TCP Flood	Inside	Inside	0.0.0.0	0.0.0.0	0	not-applicable	drop	critical
01/28 08:24:36	flood	TCP Flood	Inside	Inside	0.0.0.0	0.0.0.0	0	not-applicable	drop	critical

Receive Time	Type	Name	From Zone	To Zone	Source address	Destination address	To Port	Application	Action	Severity
01/28 08:31:54	flood	TCP Flood	Inside	Inside	0.0.0.0	0.0.0.0	0	not-applicable	drop	critical
01/28 08:31:43	flood	TCP Flood	Inside	Inside	0.0.0.0	0.0.0.0	0	not-applicable	drop	critical
01/28 08:25:04	flood	TCP Flood	Inside	Inside	0.0.0.0	0.0.0.0	0	not-applicable	drop	critical
01/28 08:24:52	flood	TCP Flood	Inside	Inside	0.0.0.0	0.0.0.0	0	not-applicable	drop	critical
01/28 08:24:42	flood	TCP Flood	Inside	Inside	0.0.0.0	0.0.0.0	0	not-applicable	drop	critical
01/28 08:24:36	flood	TCP Flood	Inside	Inside	0.0.0.0	0.0.0.0	0	not-applicable	drop	critical
01/28 08:24:36	flood	TCP Flood	Inside	Inside	0.0.0.0	0.0.0.0	0	not-applicable	drop	critical

Go to **Network > Network Profiles > Zone Protection>>Food Protection** tab.

Zone Protection Profile

Name: Zone-Protection-Profile

Description:

Flood Protection | Reconnaissance Protection | Packet Based Attack Protection | Protocol Protection

☐ SYN

 Action: Random Early Drop

 Alarm Rate (connections/sec): 10

 Activate (connections/sec): 12

 Maximum (connections/sec): 15

☒ ICMP

 Alarm Rate (connections/sec): 10

 Activate (connections/sec): 12

 Maximum (connections/sec): 15

☐ Other IP

 Alarm Rate (connections/sec): 10000

 Activate (connections/sec): 10000

 Maximum (connections/sec): 40000

☐ UDP

 Alarm Rate (connections/sec): 10

 Activate (connections/sec): 12

 Maximum (connections/sec): 15

☐ ICMPv6

 Alarm Rate (connections/sec): 10000

 Activate (connections/sec): 10000

 Maximum (connections/sec): 40000

Unchecked everything lets verify ICMP Flood attack

OK Cancel

Let's launch attack from Kali Linux Inside PC to on Outside Windows XP PC, Open terminal type:
 root@kali:~# **hping3 -1 --flood -a 192.168.17.150 192.168.17.255**

```
File Actions Edit View Help
root@kali: ~
root@kali:~# hping3 -1 --flood -a 192.168.17.150 192.168.17.255
```

Now Let's check ICMP Flood logs, go to **Monitor > Logs>Threat**

The screenshot shows the Palo Alto Networks Monitor > Logs > Threat interface. The left sidebar has 'Threat' selected. The main table displays logs for ICMP Flood attacks.

	Receive Time	Type	Name	From Zone	To Zone	Source address	Destination address	To Port	Application	Action	Severity
	01/28 10:45:32	flood	ICMP Flood	Inside	Inside	0.0.0.0	0.0.0.0	0	not-applicable	drop	critical
	01/28 10:45:22	flood	ICMP Flood	Inside	Inside	0.0.0.0	0.0.0.0	0	not-applicable	drop	critical
	01/28 10:45:11	flood	ICMP Flood	Inside	Inside	0.0.0.0	0.0.0.0	0	not-applicable	drop	critical
	01/28 10:45:00	flood	ICMP Flood	Inside	Inside	0.0.0.0	0.0.0.0	0	not-applicable	drop	critical
	01/28 10:44:51	flood	ICMP Flood	Inside	Inside	0.0.0.0	0.0.0.0	0	not-applicable	drop	critical
	01/28 10:44:41	flood	ICMP Flood	Inside	Inside	0.0.0.0	0.0.0.0	0	not-applicable	drop	critical

Go back to **Network > Network Profiles > Zone Protection>>Reconnaissance Protection** tab.

The screenshot shows the 'Zone Protection Profile' configuration window. The 'Reconnaissance Protection' tab is selected. The 'Scan' table shows that TCP Port Scan, Host Sweep, and UDP Port Scan are all enabled.

Scan	Enable	Action	Interval (sec)	Threshold (events)
TCP Port Scan	☒	alert	2	100
Host Sweep	☒	alert	10	100
UDP Port Scan	☒	alert	2	100

Unchecked everything, Even in Flood & Packet Based Attack Protection Tab

OK Cancel

Let's launch attack from Kali Linux Inside PC to on Outside Windows XP PC, Open terminal type:
 root@kali:~# nmap -sU -p0-65535 192.168.17.150

```

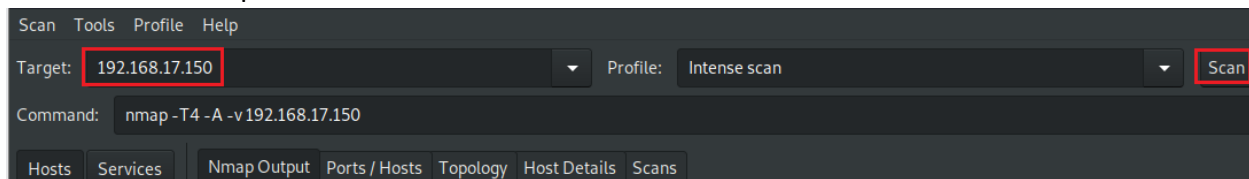
File Actions Edit View Help
root@kali: ~
root@kali:~#
root@kali:~# nmap -sU -p0-65535 192.168.17.150
  
```

Now Let's check UDP Port Scan logs, go to **Monitor > Logs>Threat**

The screenshot shows the Palo Alto Networks Monitor > Logs > Threat interface. The main table displays logs for UDP Port Scans.

	Receive Time	Type	Name	From Zone	To Zone	Source address	Destination address	To Port	Application	Action	Severity
	01/28 10:55:51	scan	SCAN: UDP Port Scan	Inside	Outside	192.168.78....	192.168.17.150	24921	not-applicable	alert	medium
	01/28 10:55:39	scan	SCAN: UDP Port Scan	Inside	Outside	192.168.78....	192.168.17.150	26987	not-applicable	alert	medium
	01/28 10:55:24	scan	SCAN: UDP Port Scan	Inside	Outside	192.168.78....	192.168.17.150	19197	not-applicable	alert	medium
	01/28 10:54:54	scan	SCAN: UDP Port Scan	Inside	Outside	192.168.78....	192.168.17.150	27473	not-applicable	alert	medium

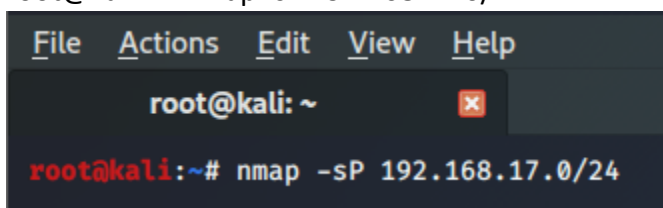
Let's launch attack from Kali Linux Inside PC to on Outside Windows XP PC, Open terminal type:
 root@kali:~# nmap -O 192.168.17.150



Now Let's check TCP Port Scan logs, go to **Monitor > Logs>Threat**

	Receive Time	Type	Name	From Zone	To Zone	Source address	Destination address	To Port	Application	Action	Severity
	01/28 11:02:37	scan	SCAN: TCP Port Scan	Inside	Outside	192.168.78....	192.168.17.150	1187	not-applicable	alert	medium
	01/28 11:00:38	scan	SCAN: TCP Port Scan	Inside	Outside	192.168.78....	192.168.17.150	2718	not-applicable	alert	medium
	01/28 10:59:36	scan	SCAN: TCP Port Scan	Inside	Outside	192.168.78....	192.168.17.150	6580	not-applicable	alert	medium

Let's launch attack from Kali Linux Inside PC to on Outside Windows XP PC, Open terminal type:
 root@kali:~# nmap -sP 192.168.17.0/24



Now Let's check Host Sweep logs, go to **Monitor > Logs>Threat**

	Receive Time	Type	Name	From Zone	To Zone	Source address	Destination address	To Port	Application	Action	Severity
	01/28 11:17:12	flood	ICMP Flood	Inside	Inside	0.0.0.0	0.0.0.0	0	not-applicable	drop	critical
	01/28 11:17:07	scan	SCAN: Host Sweep	Inside	Outside	192.168.78....	192.168.17.106	0	not-applicable	alert	medium
	01/28 11:17:05	scan	SCAN: Host Sweep	Inside	Outside	192.168.78....	192.168.17.99	443	not-applicable	alert	medium

DOS Protection Testing and Verification:

DoS Protection working the like Zone Protection few test below.

root@kali:~# hping3 -c 15000 -d 120 -S -w 64 -p 80 --flood --rand-source 192.168.17.150



Now Let's check TCP Flood logs, go to **Monitor > Logs>Threat**

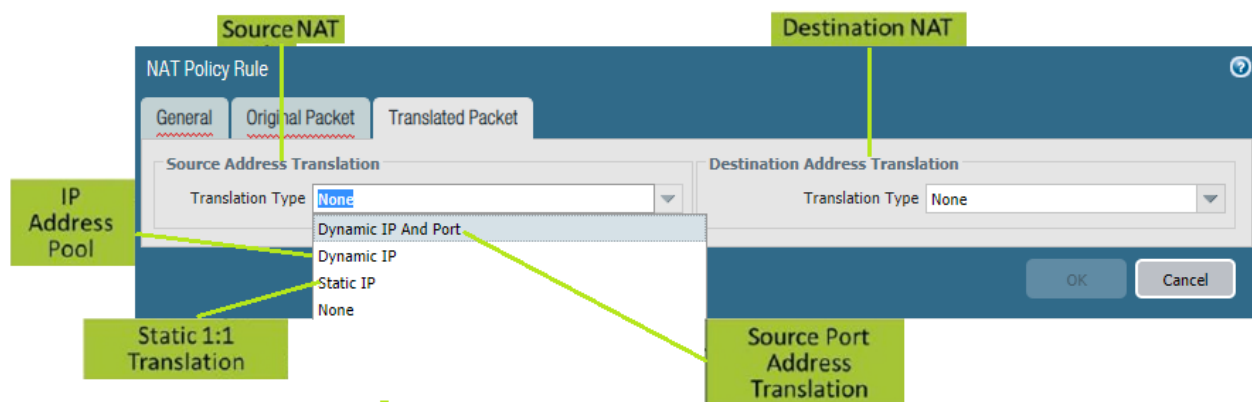
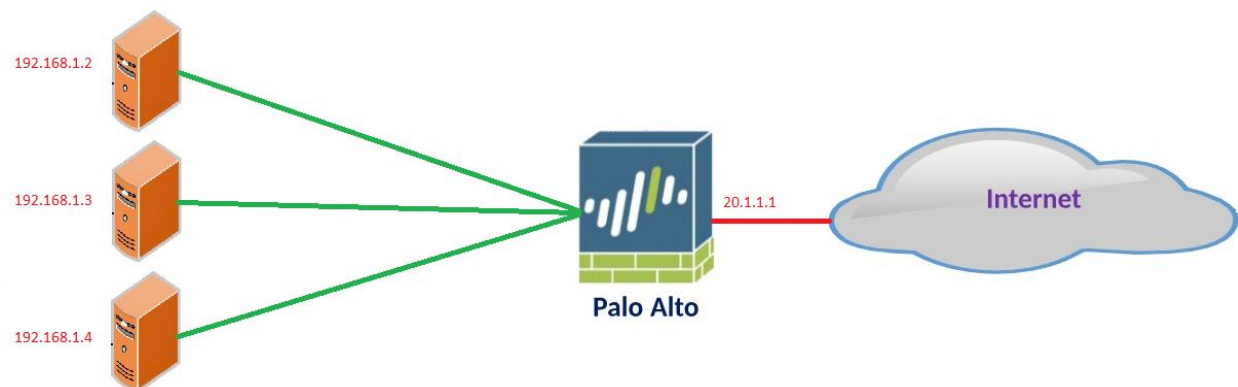
	Receive Time	Type	Name	From Zone	To Zone	Source address	Destination address	To Port	Application	Action	Severity
	01/28 11:50:30	flood	TCP Flood			0.0.0.0	0.0.0.0	0	not-applicable	drop	critical
	01/28 11:50:19	flood	TCP Flood			0.0.0.0	0.0.0.0	0	not-applicable	drop	critical
	01/28 11:50:10	flood	TCP Flood			0.0.0.0	0.0.0.0	0	not-applicable	drop	critical
	01/28 11:49:59	flood	TCP Flood			0.0.0.0	0.0.0.0	0	not-applicable	drop	critical
	01/28 11:49:49	flood	TCP Flood			0.0.0.0	0.0.0.0	0	not-applicable	drop	critical
	01/28 11:49:40	flood	TCP Flood			0.0.0.0	0.0.0.0	0	not-applicable	drop	critical
	01/28 11:49:29	flood	TCP Flood			0.0.0.0	0.0.0.0	0	not-applicable	drop	critical
	01/28 11:49:20	flood	TCP Flood			0.0.0.0	0.0.0.0	0	not-applicable	drop	critical

Go to **Policies > DoS Protection** to see the rule Hit Counters.

</

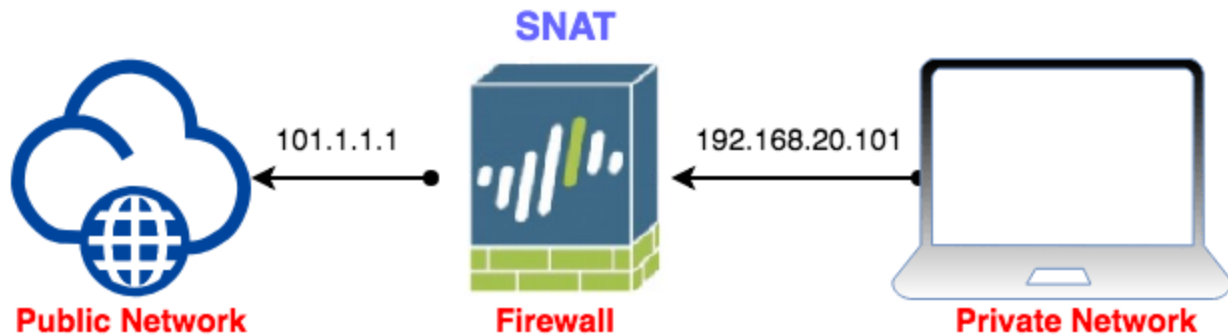
NAT on Palo Alto Firewall:

- o NAT stands for Network Address Translation.
- o NAT is often used to translate private IP addresses to public IP addresses.
- o NAT is a process that involves translating Private IP addresses into Public IP addresses.
- o The process of translating one IP address to another is known as NAT.
- o Router and Firewall is a device, which is used for network Address Translation.
- o NAT can provide Internet connectivity to many LAN users over single public IP address.
- o Network Address Translation (NAT) technique helps a lot to save IPv4 address space.
- o There are many forms of Network Address Translation (NAT) & Port Address Translation.
- o Network Address Translation used to reduce the requirement of the Public IP address.
- o Network Address Translation increase security of Internal Computer Networks.
- o NAT Translate Private IP into Public IP address & Public IP address into Private IP address.
- o NAT used to connect a device with Private IP address to the Wide Area Network Internet.
- o Network Address Translation hide an organization internal network from external network.
- o Network Address Translation modifies only the Layer 3 header of OSI reference model.
- o PAT, translation of an IP address and Port to another Internet Protocol address and Port.
- o Port Address Translation (PAT) modifies both the Layer 3 and Layer 4 header of OSI model.
- o Palo Alto Firewalls supports both source & destination address and/or port translation.



Source Network Address Translation (SNAT):

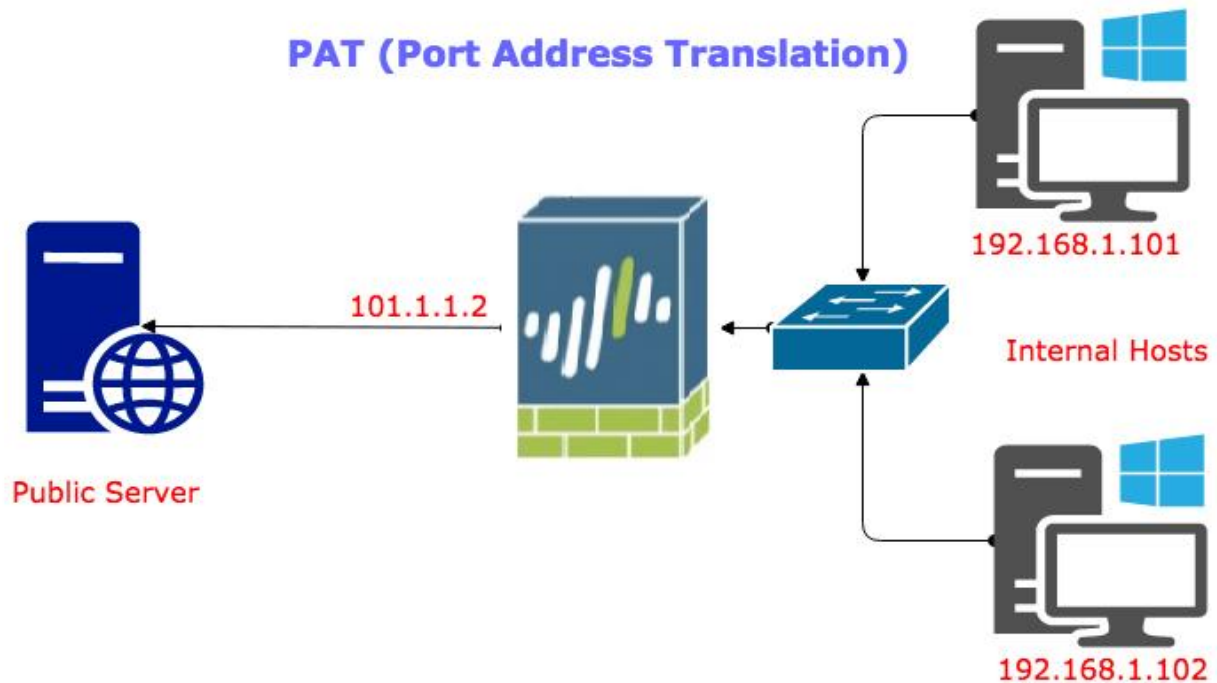
- o SNAT is abbreviation for Source Network Address Translation.
- o Source NAT changes the source address in Internet Protocol header of a packet.
- o Source NAT may also change the source port in the TCP/UDP headers.
- o It is used when an internal host needs to initiate a connection to an external host.
- o It is used when a private host needs to initiate a connection to a public host.
- o Device performing NAT changes private IP address of source host to public IP address.
- o There are three types of Source Network Address Translation (SNAT).



1-Dynamic IP and Port (DIPP):

- o Multiple hosts source IP addresses translated to same public IP address with different port.
- o Palo Alto Networks firewall translates the source IP address or range to a single IP address.
- o Multiple clients use the same public IP addresses with different source port numbers.
- o DIPP is sometimes referred to interface-based NAT or Network Address Port Translation.
- o DIPP has default NAT oversubscription rate, same translated IP address & port can be used.
- o Mapping is based on source port; multiple source IPs can share a single translated address.
- o When only having a single public IP address to be shared among many private IP addresses.
- o It is common to choose the IP address assigned to the interface connecting to your ISP.
- o It is also called Hide NAT, it hides all internal subnets behind a single external public IP.

The screenshot shows the 'NAT Policy Rule' configuration window. The 'General' tab is active. Under 'Source Address Translation', the following settings are visible: Translation Type is 'Dynamic IP And Port', Address Type is 'Interface Address', Interface is 'ethernet1/1', and IP Address is '192.168.100.100/24'. To the right, under 'Destination Address Translation', the checkbox is unchecked. At the bottom right, the 'OK' button is highlighted with a red box.



2-Dynamic IP (DIP):

- o Dynamic Internet Protocol (IP), allows One-to-One (1 to 1) Network translations only.
- o Mapping is not port based; this is One-to-One mapping as long as the session lasts.
- o Each concurrent session uses address from pool, making it unavailable to other source IPs.
- o The size of the Dynamic-IP pool defines the number of the hosts that can be translated.
- o Private source addresses translate to the next available address in specified address range.
- o Dynamic translation of source IP address only no port number to next available address.
- o Size of NAT pool should be equal to number of internal hosts require address translations.
- o If source address pool is larger than NAT address pool & all of NAT addresses are allocated.
- o The new source address connections that need address translation are dropped.
- o To override this default behavior, use Advanced (Dynamic IP/Port Fallback) method.
- o If sessions terminate & addresses in the pool become available, they can be allocated.

NAT Policy Rule

General Original Packet Translated Packet

Source Address Translation

Translation Type: **Dynamic IP**

Translated Address:

- ☐ 192.168.100.150
- ☒ 192.168.100.151

+ Add - Delete

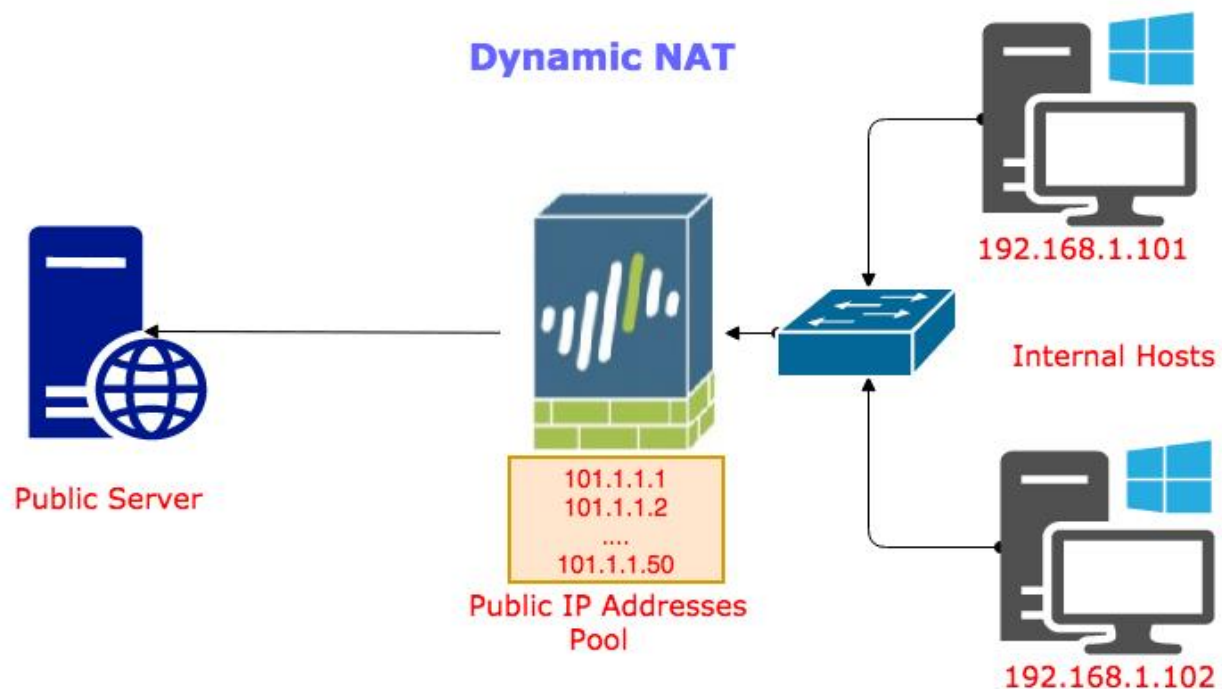
☐ Advanced (Dynamic IP/Port Fallback)

Destination Address Translation

Translated Address:

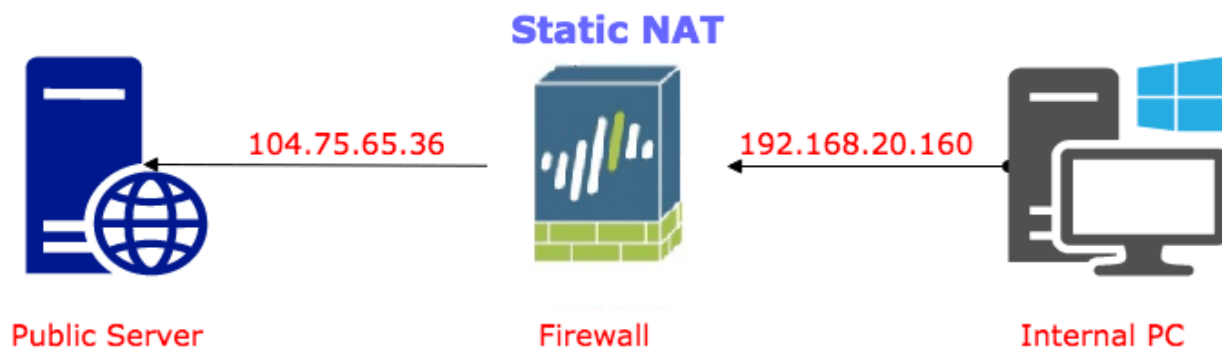
Translated Port:

OK Cancel



3-Static IP:

- o Static IP Allows the One-to-One (1-to-1), Static translation of a source IP address.
- o Static NAT implies a translation of single IP address to another single IP address.
- o Static IP Network Address Translation (NAT) leaves the source port unchanged.
- o Use this translation type to translate single source address to a specific public address.
- o Use static IP to change the source IP address while leaving the source port unchanged.
- o Size of SNAT pool must be the same as the size of the source addresses to be translated.
- o Common scenario for a static IP translation is internal server that available to Internet.
- o When network device inside a private network needs to be accessible from Internet.
- o Common configuration, when an internal server needs to be available on the internet.
- o Bi-directional translation, destination IP can also be translated for inbound connections.



NAT Policy Rule

General Original Packet Translated Packet

Source Address Translation

Translation Type Static IP

Translated Address 192.168.100.150

☐ Bi-directional

Destination Address Translation

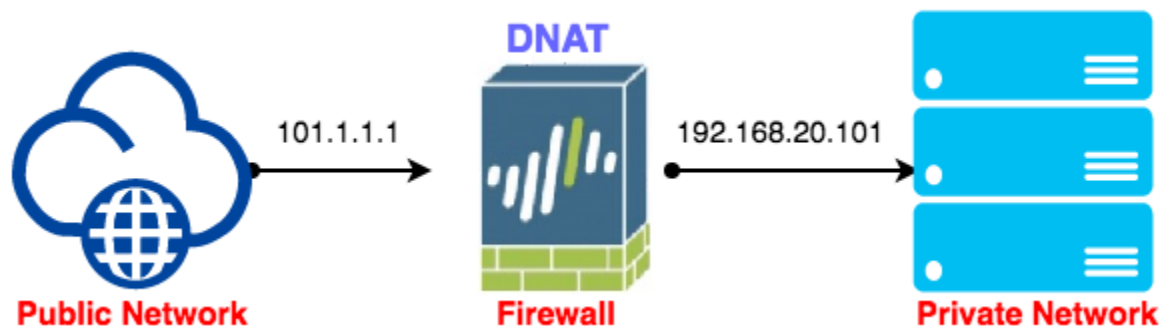
Translated Address

Translated Port [1 - 65535]

OK Cancel

Destination Network Address Translation (DNAT):

- o DNAT stand for Destination Network Address Translation.
- o Destination NAT changes the destination address in IP header of a packet.
- o Destination NAT may also change the destination port in the TCP/UDP headers.
- o Redirect incoming packets with destination of public address to private IP address.
- o Destination Network Address Translation (DNAT) is performed on incoming packets.
- o Where the PA firewall translates a public destination address to a private address.
- o DNAT is a 1-to-1, static translation with option to perform port forwarding or translation.
- o Users over Internet Accessing a Web Server hosted in a Data Center is a typical example.
- o Destination NAT also offers the option to perform port forwarding or port translation.



1-Static IP:

- o Use static IP to change destination IP address while leaving destination port unchanged.
- o Statically translates original destination address to same translated destination address.
- o Original packet have single destination IP address, range of IP addresses, or list.

The screenshot shows the 'NAT Policy Rule' configuration window. The 'Translated Packet' tab is active. In the 'Destination Address Translation' section, the 'Translated Address' is set to '192.168.250.10' and the 'Translated Port' is set to '[1 - 65535]'. The 'OK' button is highlighted with a red box.

2-Port Forwarding:

- o Can translate public destination address and port number to private destination address.
- o Technique used to manage traffic through NAT policies based on destination port numbers.
- o It is used to map a single public IP address to multiple private servers and services.
- o The destination ports can stay the same or be directed to different destination ports.

The screenshot shows the 'Original Packet' tab of a NAT Policy Rule configuration window. The 'Destination Zone' is set to 'Outside'. The 'Destination Interface' is 'ethernet1/1'. The 'Service' is 'service-http'. The 'Destination Address Translation' section is checked and highlighted with a red box, showing 'Translated Address' as '192.168.250.10' and 'Translated Port' as '80'. The 'OK' button is also highlighted with a red box.

3-Port Translation:

- o Translate public destination address & port no to private destination address different port.
- o Port Translation keeping the real port number private and change the destination port.
- o It is configured by entering Translated Port on Translated Packet tab in NAT policy rule.
- o Example is suppose the web server is configured to listen for HTTP traffic on port 8080.
- o The clients access the web server using the public Internet Protocol (IP) and TCP Port 80.
- o The destination NAT rule is configured to translate both IP address and TCP port to 8080.

The screenshot shows the 'Translated Packet' tab of a NAT Policy Rule configuration window. The 'Source Address Translation' section shows 'Translation Type' as 'None'. The 'Destination Address Translation' section is checked and highlighted with a red box, showing 'Translated Address' as '192.168.250.10' and 'Translated Port' as '8080'. The 'OK' button is also highlighted with a red box.

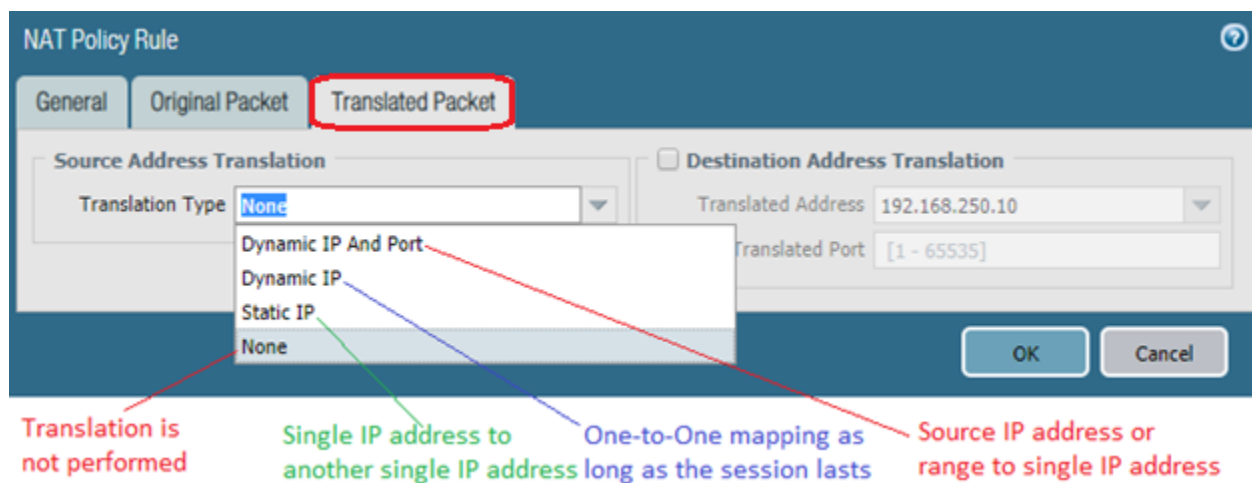
NAT Policy Rule:

The screenshot shows the 'NAT Policy Rule' configuration window with the 'General' tab selected. The 'Name' field is 'Outside-To-DMZ', the 'Description' is 'NAT Policy from outside to DMZ', the 'Tags' field is empty, and the 'NAT Type' is 'ipv4'. The 'OK' button is highlighted with a red box.

Settings	Description
Name	Enter a name to identify the rule. The name is case-sensitive and can have up to 31 characters, which can be letters, numbers, spaces, hyphens etc.
Description	Enter a description for the rule (Up to 255 characters).
Tag	If you want to tag the policy, Add and specify the tag.
NAT Type	Specify the type of translation: Ipv4-Translation between IPv4 addresses. nat64-Translation between IPv6 and IPv4 addresses nptv6-Translation between IPv6 prefixes.

The screenshot shows the 'NAT Policy Rule' configuration window with the 'Original Packet' tab selected. The 'Source Zone' is 'Outside', the 'Destination Zone' is 'Outside', the 'Destination Interface' is 'ethernet1/1', and the 'Service' is 'service-http'. The 'Add' button for the Source Zone list is highlighted with a red box. The 'OK' button is also highlighted with a red box.

Settings	Description
Source Zone / Destination Zone	Select one or more source and destination zones for the original (non-NAT) packet (default is Any).
Destination Interface	Specify the destination interface of packets the firewall translates.
Service	Specify the service for which the firewall translates the source or destination address.
Source Address / Destination Address	Specify a combination of source and destination addresses for the firewall to translate.

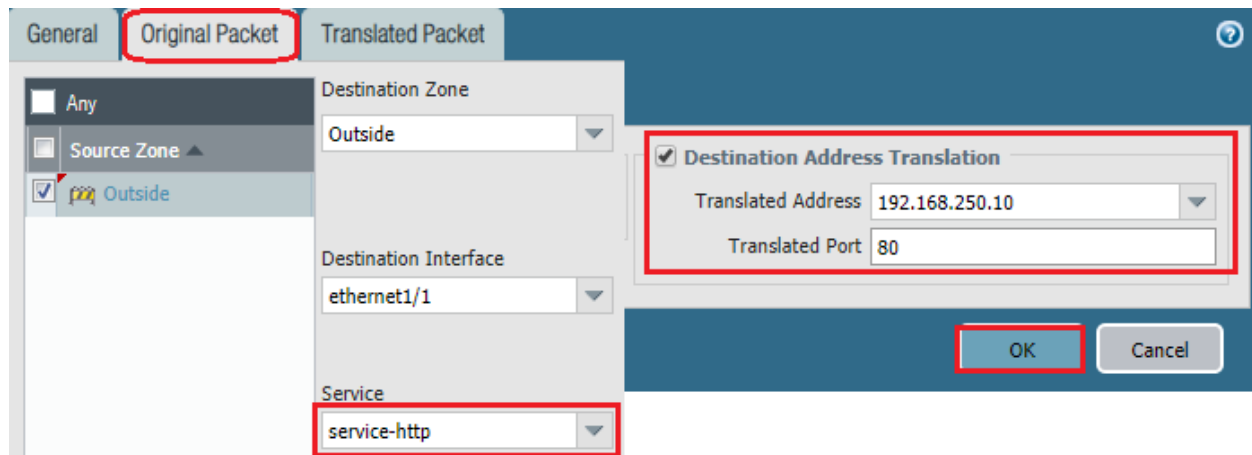


Destination Network Address Translation (DNAT):

- o DNAT stand for Destination Network Address Translation.
- o Destination NAT changes the destination address in IP header of a packet.
- o Destination NAT may also change the destination port in the TCP/UDP headers.
- o Redirect incoming packets with destination of public address to private IP address.
- o Destination Network Address Translation (DNAT) is performed on incoming packets.
- o Where the PA firewall translates a public destination address to a private address.
- o DNAT is a 1-to-1, static translation with option to perform port forwarding or translation.
- o Users over Internet Accessing a Web Server hosted in a Data Center is a typical example.
- o Destination NAT also offers the option to perform port forwarding or port translation.

2-Port Forwarding:

- o Can translate public destination address and port number to private destination address.
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3-Port Translation:

- o Translate public destination address & port no to private destination address different port.
- o Port Translation keeping the real port number private and change the destination port.
- o It is configured by entering Translated Port on Translated Packet tab in NAT policy rule.
- o Example is supposing the web server is configured to listen for HTTP traffic on port 8080.
- o The clients access the web server using the public Internet Protocol (IP) and TCP Port 80.
- o The destination NAT rule is configured to translate both IP address and TCP port to 8080.

NAT Policy Rule

General Original Packet Translated Packet

Source Address Translation

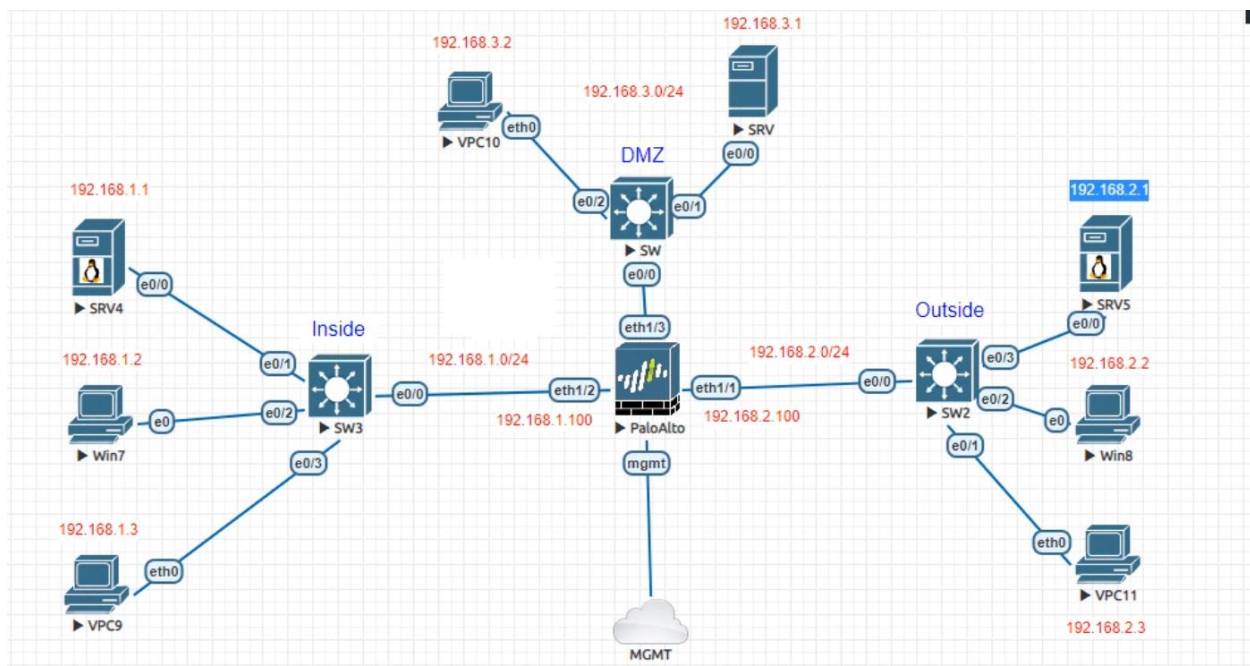
Translation Type None

☒ Destination Address Translation

Translated Address 192.168.250.10

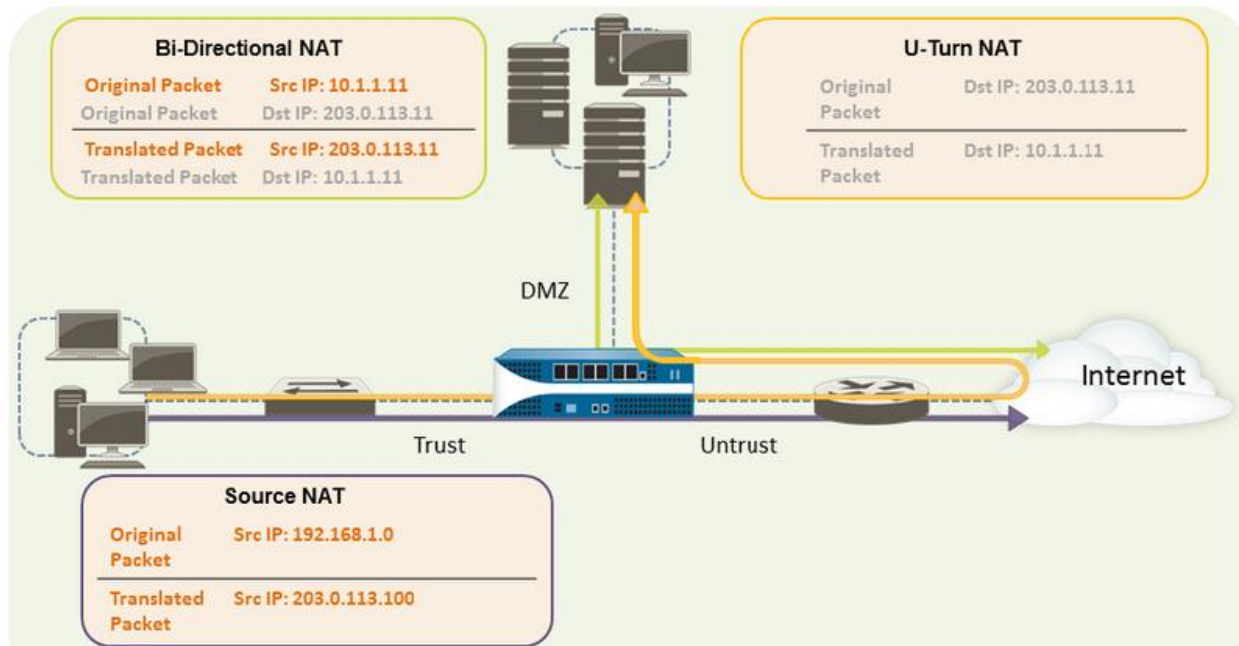
Translated Port 8080

OK Cancel

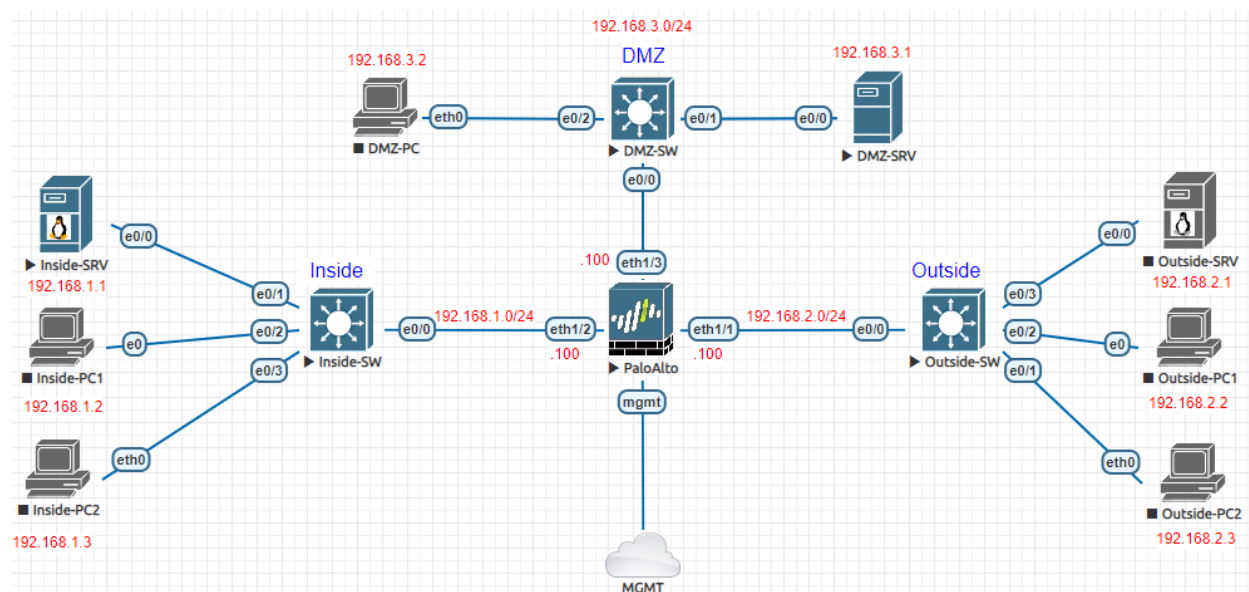


U-Turn NAT:

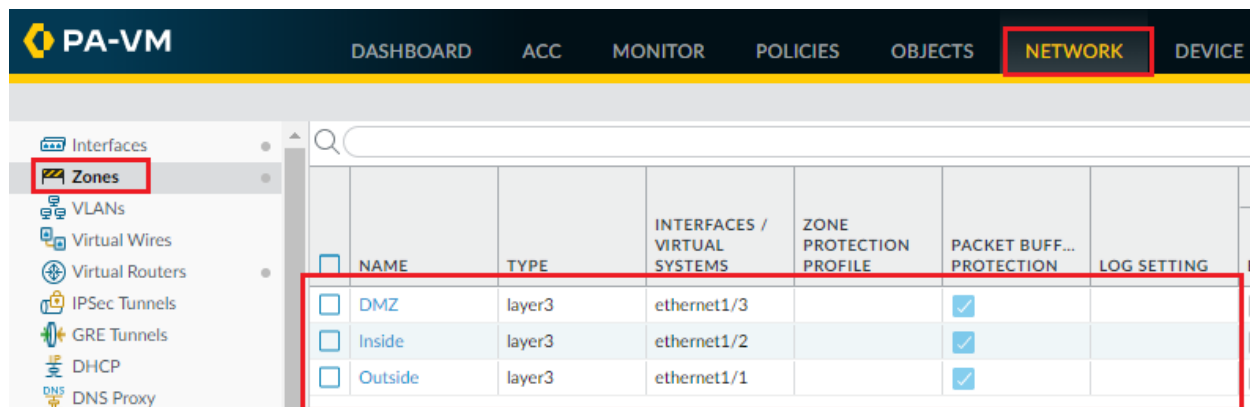
U-Turn NAT refers to the logical path that traffic appears to travel when accessing an internal resource when they resolve their external address. U-turn NAT is often used in a network where internal users need to access an internal DMZ server using the server's external public IP address. UNAT NAT is a special type of NAT which is configured when internet users want to access internal servers using their public IP address.



Lab Time:

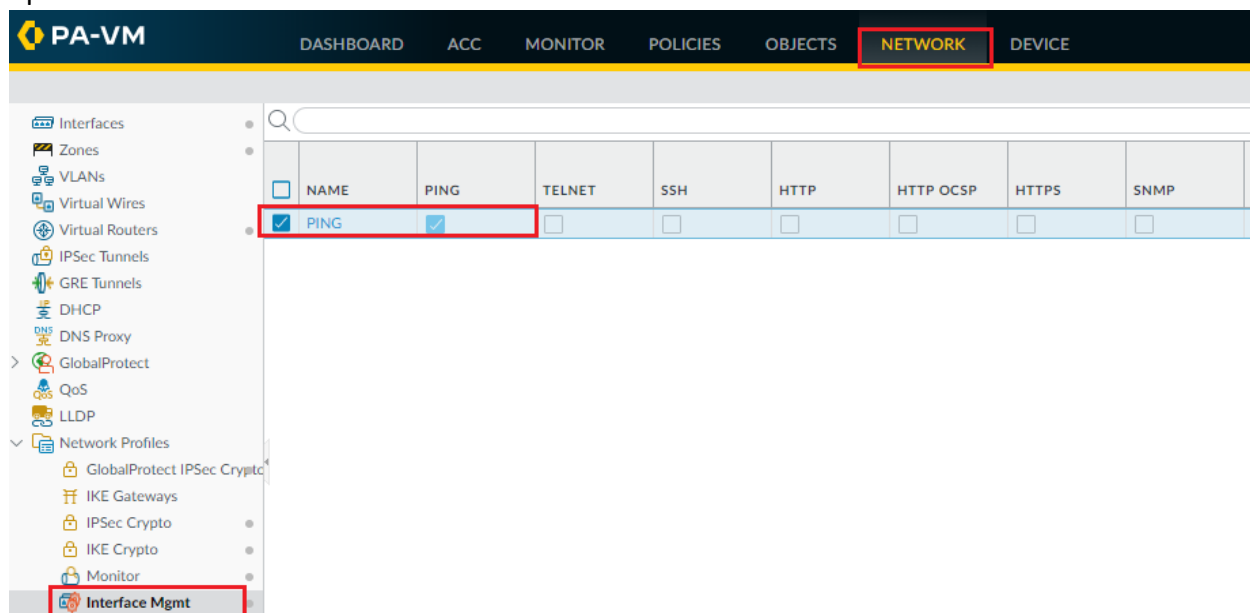


Let's configure three zones names **Inside**, **Outside** and **DMZ**. Go to **Network> Zone>Add**, Give the name **Inside**, select Type to be **Layer3** and click **OK**. Create the same way other two Zones **Outside** and **DMZ**.



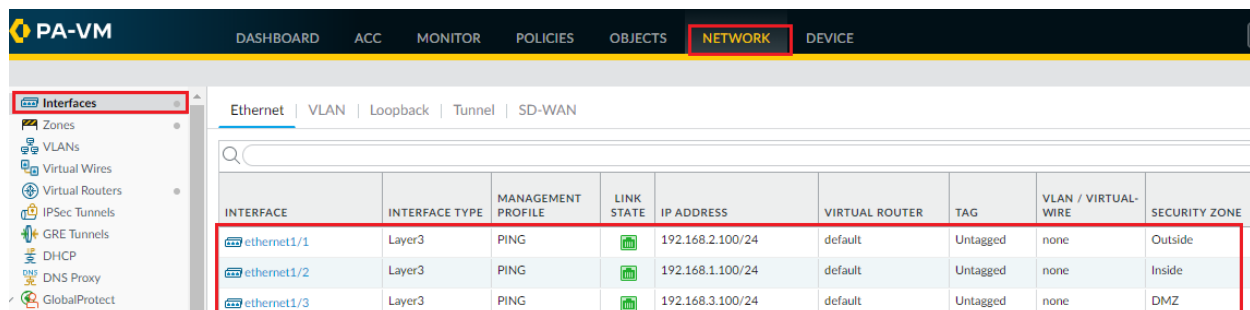
	NAME	TYPE	INTERFACES / VIRTUAL SYSTEMS	ZONE PROTECTION PROFILE	PACKET BUFF... PROTECTION	LOG SETTING
☐	DMZ	layer3	ethernet1/3		☒	
☐	Inside	layer3	ethernet1/2		☒	
☐	Outside	layer3	ethernet1/1		☒	

In order to make it able to be ping, create Interface Management Profile to assing all three interfaces. Go to **Network>Interface Mgmt** And **add** give any name: **Ping**, then check ping option and click **OK**.



	NAME	PING	TELNET	SSH	HTTP	HTTP OCSP	HTTPS	SNMP
☒	PING	☒	☐	☐	☐	☐	☐	☐

The final interfaces should look like this



INTERFACE	INTERFACE TYPE	MANAGEMENT PROFILE	LINK STATE	IP ADDRESS	VIRTUAL ROUTER	TAG	VLAN / VIRTUAL-WIRE	SECURITY ZONE
ethernet1/1	Layer3	PING	☒	192.168.2.100/24	default	Untagged	none	Outside
ethernet1/2	Layer3	PING	☒	192.168.1.100/24	default	Untagged	none	Inside
ethernet1/3	Layer3	PING	☒	192.168.3.100/24	default	Untagged	none	DMZ

Let's Create few policies to allow traffic between three Different Zones Inside, Outside & DMZ.

	NAME	TYPE	ZONE	ADDRESS	USER	DEVICE	Destination
1	Allow_Inside-To-Outside	universal	Inside	192.168.1.0/24	any	any	Outside 192.168.2.0/24
2	Allow-Inside-To-DMZ	universal	Inside	192.168.1.0/24	any	any	DMZ 192.168.3.0/24
3	Allow-DMZ-to-Outside	universal	DMZ	192.168.3.0/24	any	any	Outside 192.168.2.0/24
4	Allow-DMZ-to-Inside	universal	DMZ	any	any	any	Inside any
5	Allow-Outside-to-DMZ	universal	Outside	any	any	any	DMZ any
6	Allow-Any	universal	any	any	any	any	any

Let's configure U-Turn NAT, to enable clients on the internal network to access the public web server in the DMZ zone, we will need to configure a NAT rule that redirects the packet from the external network, where the original routing table lookup will determine it should go based on the destination address of 192.168.2.100 within the packet, to the actual address of the web server on the DMZ network of 192.168.3.1. Create a NAT rule from the Inside zone (where the source address in the packet is) to the Outside zone (where the original destination address is) to translate the destination address to an address in the DMZ zone.

NAT Policy Rule

General | Original Packet | Translated Packet

Name: U-Turn-NAT

Description:

Tags:

Group Rules By Tag: None

NAT Type: ipv4

Audit Comment:

Audit Comment Archive

OK Cancel

Original Packet details packet will go from Inside destination zone is Outside.

NAT Policy Rule ?

General | **Original Packet** | Translated Packet

☐ Any ☒ SOURCE ZONE ^ ☒ Inside	Destination Zone Outside	☐ Any ☒ SOURCE ADDRESS ^ ☒ 192.168.1.0/24	☐ Any ☒ DESTINATION ADDRESS ^ ☒ 192.168.2.100/32
Destination Interface any			
Service any			
+ Add - Delete		+ Add - Delete	

Translated Packet is DMZ Zone interface while original destination is DMZ Server IP 192.168.3.1.

NAT Policy Rule ?

General | Original Packet | **Translated Packet**

Source Address Translation Translation Type: Dynamic IP And Port Address Type: Interface Address Interface: ethernet1/3 IP Address: 192.168.3.100/24	Destination Address Translation Translation Type: Static IP Translated Address: 192.168.3.1 Translated Port: [1 - 65535] ☐ Enable DNS Rewrite Direction: reverse
---	---

The show command display NB means Both NAT source and Destination NAT is used.

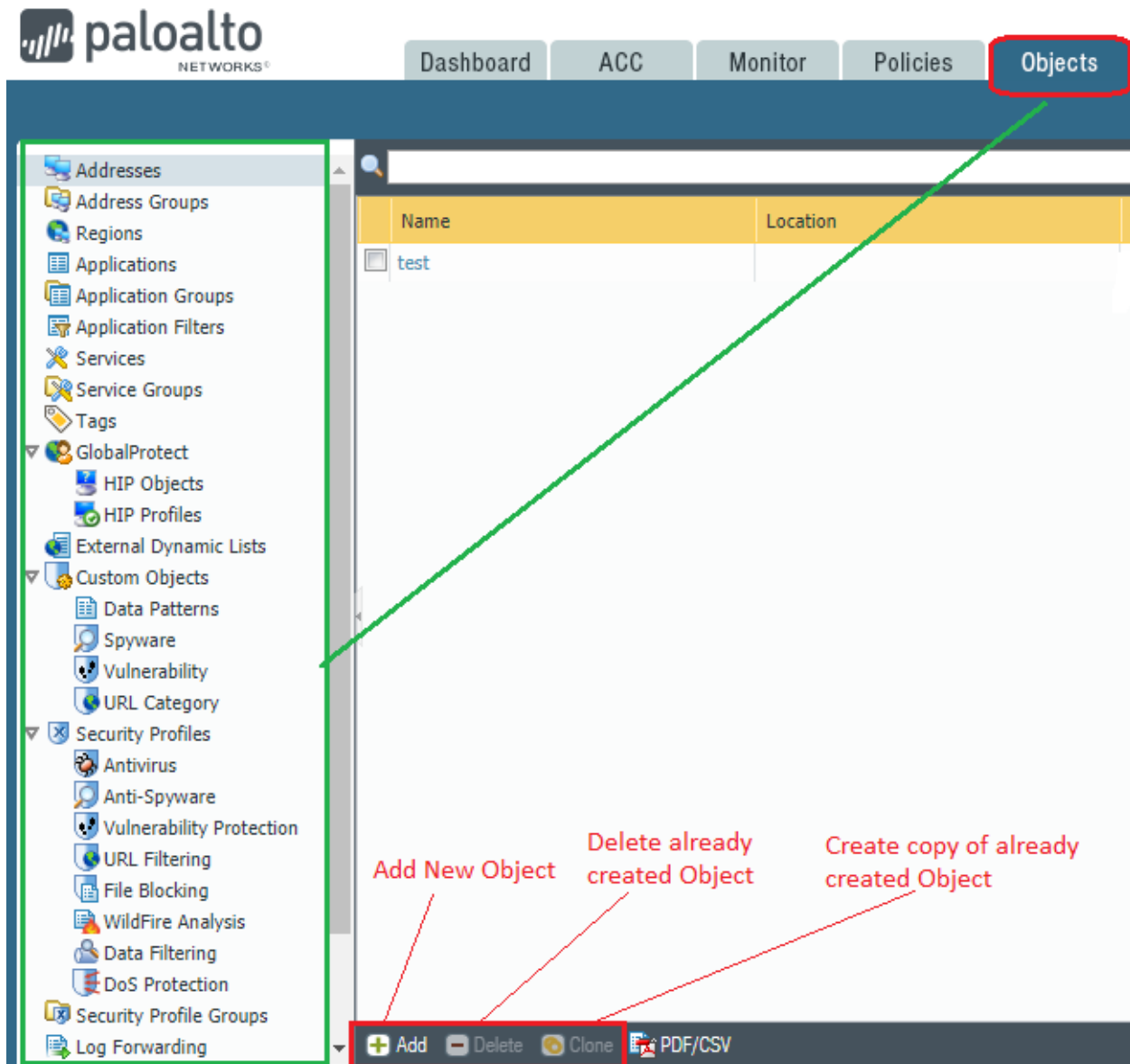
```
admin@PA-VM> show session all
```

ID	Application	State	Type	Flag	Src[Sport]/Zone/Proto (translated IP[Port])	Dst[Dport]/Zone (translated IP[Port])
25	traceroute	ACTIVE	FLOW	NB	192.168.1.1[49168]/Inside/17	(192.168.3.100[31516])
vsys1					192.168.2.100[33434]/DMZ	(192.168.3.1[33434])
30	undecided	ACTIVE	FLOW	NB	192.168.1.1[46063]/Inside/6	(192.168.3.100[16647])
vsys1					192.168.2.100[80]/DMZ	(192.168.3.1[80])
29	undecided	ACTIVE	FLOW	NB	192.168.1.1[49172]/Inside/17	(192.168.3.100[61271])
vsys1					192.168.2.100[33438]/DMZ	(192.168.3.1[33438])
27	traceroute	ACTIVE	FLOW	NB	192.168.1.1[49170]/Inside/17	(192.168.3.100[24131])
vsys1					192.168.2.100[33436]/DMZ	(192.168.3.1[33436])
26	traceroute	ACTIVE	FLOW	NB	192.168.1.1[49169]/Inside/17	(192.168.3.100[50623])
vsys1					192.168.2.100[33435]/DMZ	(192.168.3.1[33435])
28	undecided	ACTIVE	FLOW	NB	192.168.1.1[49171]/Inside/17	(192.168.3.100[44399])
vsys1					192.168.2.100[33437]/DMZ	(192.168.3.1[33437])

Both NAT

Objects Tab:

- o Palo Alto Firewall, Object is a container that groups specific policy filter values.
- o Example of Object are such as IP addresses, URLs, applications, or services etc.
- o Palo Alto Network Firewall, the Object are used for simplified rule definition.
- o Address object might contain specific IP address definitions for web in DMZ zone.
- o Palo Alto Firewall is Object Based device where Objects are configuration elements.



Objects > Addresses:

- o Just like all other firewalls , Palo Alto Networks Firewall supports Address objects.
- o Address object include an IPv4 or IPv6 address (single IP, range, & Subnet) or FQDN.
- o Allows to reuse same object as a source or destination address across all the policy.
- o Allows to reuse the same object without having to add it manually each & every time.

The screenshot shows the 'Address' configuration window in the Palo Alto Networks Firewall GUI. The 'Name' field is 'Server-Inside', 'Description' is 'Inside Server Address', 'Type' is 'IP Netmask', and the 'Address' field contains '192.168.78.200'. A 'Resolve' button is located to the right of the address field. Below the address field is a 'Tags' field. At the bottom right are 'OK' and 'Cancel' buttons. A red box highlights the main configuration area, and another red box highlights the 'OK' button.

Settings	Description
Name	Enter name describes the addresses you will include as part of this object.
Description	Enter a description for the object.
Type	Specify an IPv4 or IPv6 address or address range, or an FQDN. IP Netmask: Examples: 192.168.80.150/32—Indicates one address. 192.168.80.0/24—all addresses from 192.168.80.0 through 192.168.80.255. IP Range: Enter a range of addresses using the following format: ip_address-ip_address 192.168.80.100-192.168.80.200 where both ends of the range are IPv4 addresses or both are IPv6 addresses. FQDN: To specify an address using the FQDN, select FQDN and enter the domain name. The FQDN is resolved by the system DNS server or DNS Proxy object.
Resolve	After selecting the address type and entering an IP address or FQDN, click Resolve to see the associated FQDN or IP addresses.
Tags	Select or enter the tags that you wish to apply to this address object.

Name	Description	Type	Address
☐ DMZ-Server	test	IP Netmask	3.3.3.3
☐ LAN	Inside Subnet	IP Netmask	192.168.78.0/24
☐ Outside-Range	Outside Zone Range	IP Range	192.168.1.1-192.168.1.100
☐ Server-Inside	Inside Server Address	IP Netmask	192.168.78.200

Objects > Address Groups:

Address Group simplify the creation of security policies, addresses that require the same security settings can be combined into address groups.

Static Address Groups:

A static address group can include address objects that are static, dynamic address groups, or it can be a combination of both address objects and dynamic address groups.

Dynamic Address Groups:

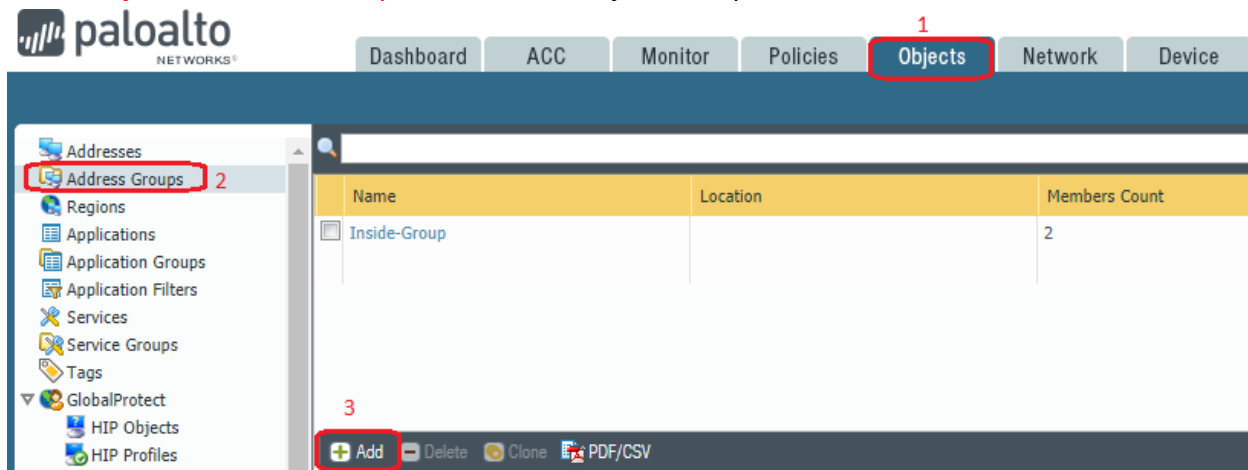
A dynamic address group populates its members dynamically using looks ups for tags and tag-based filters.

To create an address group, click Add and fill in the following fields.

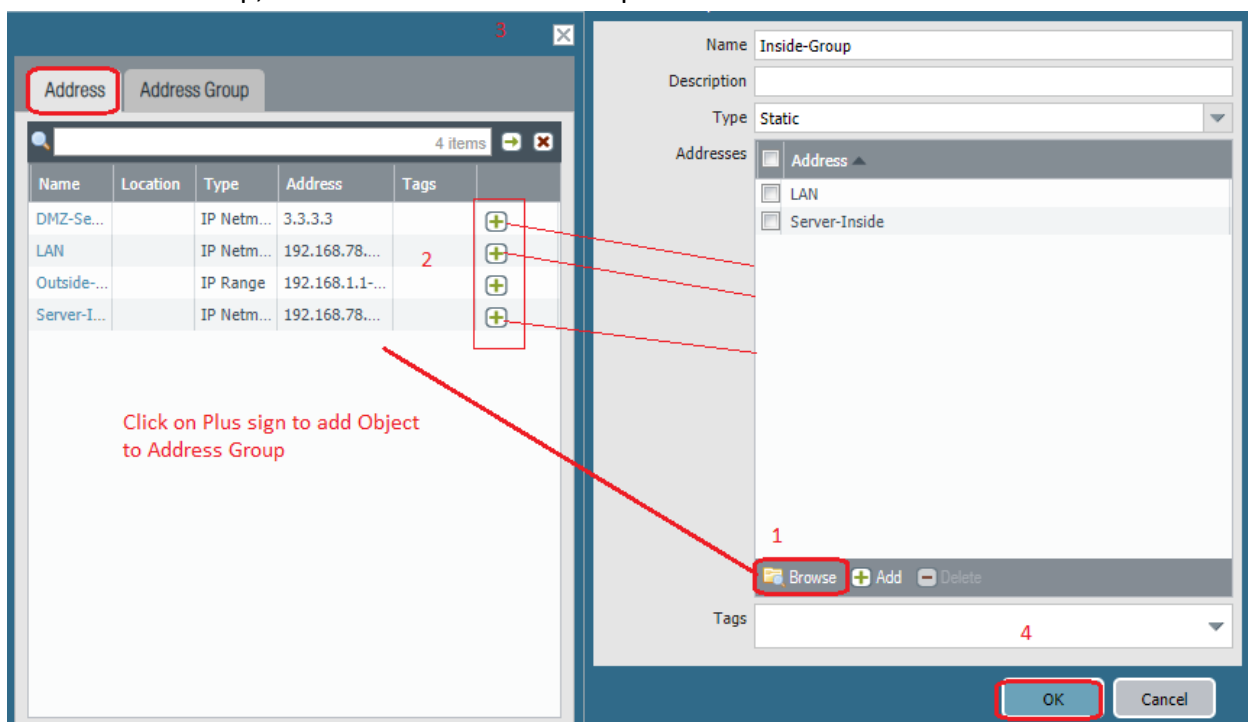
The screenshot shows the 'Address Group' configuration window. The 'Name' field is set to 'Inside-Group'. The 'Type' is set to 'Static'. The 'Addresses' list contains 'Address', 'LAN', and 'Server-Inside' (checked). The 'Add' button is highlighted. The 'OK' button is also highlighted.

Settings	Description
Name	Enter a name that describes the address group.
Description	Enter a description for the object.
Type	Select Static or Dynamic.
Addresses	For static address group, click Add & select one or more Addresses. Click Add to add an object or an address group to the address group. The group can contain address objects.
Tags	Select or enter the tags that you wish to apply to this address group.

Go to **Object > Address Groups>Add** to add Object Groups.



Click the **Add** button to add Address objects or address Group, alternatively, click on **Browse** button to open a new window from there click on plus sign to add address objects. If want to add Address Group, then click on Address Group tab to add.



Objects > Regions:

- o Firewall supports creation of policy rules that apply to specified countries or other regions.
- o Region is available as an option when specifying source & destination for security policies.
- o Region is also available as an option in the decryption policies, and DoS policies.

Objects > Services:

- o Define security policies for specific applications, can select one or more services.
- o This Service Object is used to limit the port numbers the applications can use.
- o The default service is any, which allows all TCP and UDP ports from 1 to 65535.
- o The HTTP & HTTPS services are predefined but can add additional service definitions.
- o On firewall web interface navigate to **Objects > Services**. Click on **Add** to bring up dialog.

- o For TCP or UDP service, configure the timeout values to "Inherit from application".
- o For TCP or UDP service also can be set the timeout values by using "Override".
- o Click **OK** to add the service and click **commit** to save & apply the configuration.
- o Now can reference the newly added service when configuring the security policy.

Service

Name: RDP-3389

Description:

Protocol: ☒ TCP ☐ UDP ☐ SCTP

Destination Port: 3389

Source Port: [>= 0]

Port can be a single port #, range (1-65535), or comma separated (80, 8080, 443)

Session Timeout: ☐ Inherit from application ☒ Override

TCP Timeout (sec): 3600 TCP Half Closed (sec): 120 TCP Time Wait (sec): 15

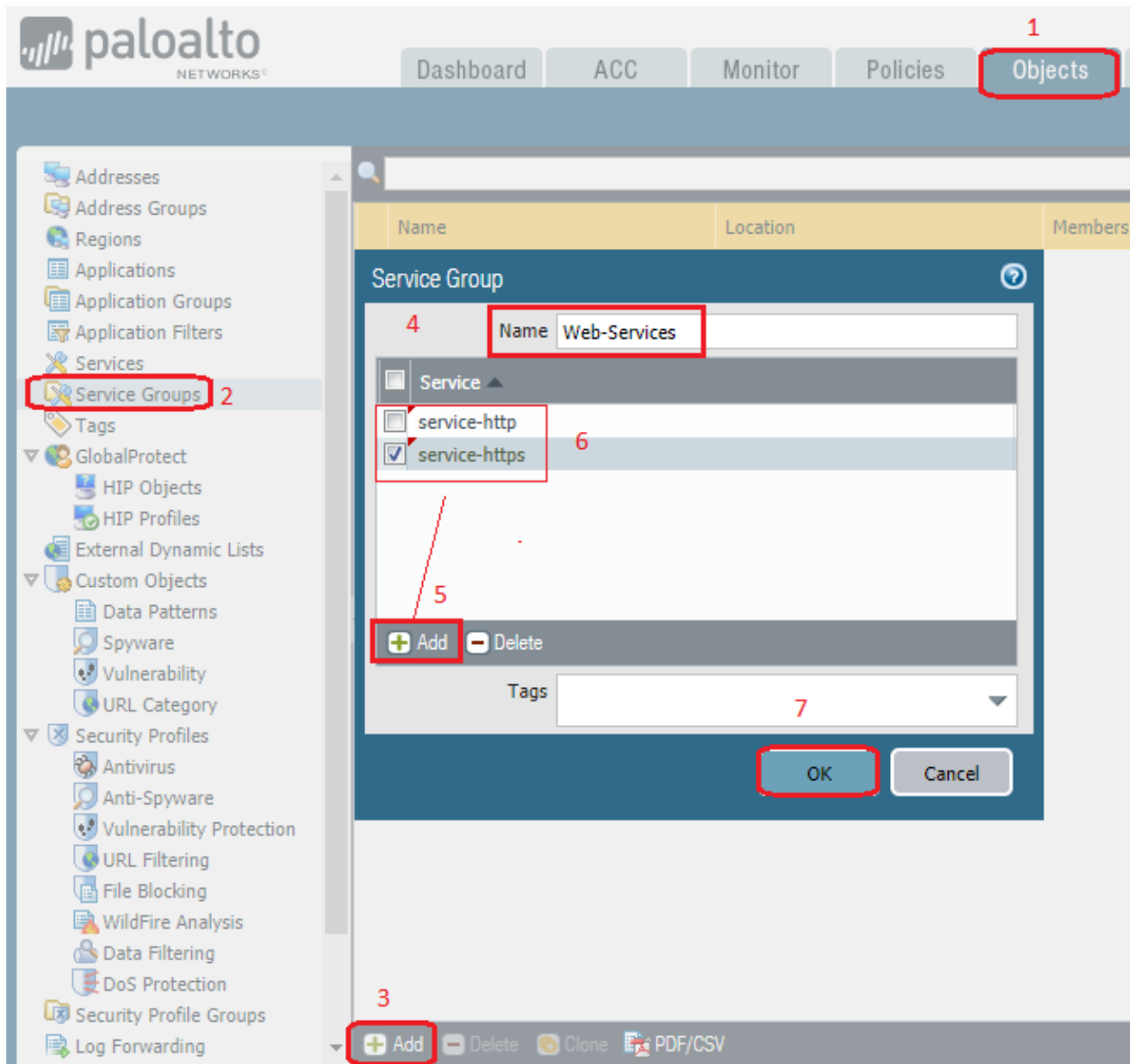
Tags:

OK Cancel

Settings	Description
Name	Enter the service name. This name appears in the services list when defining security policies.
Description	Enter a description for the service.
Protocol	Select the protocol used by the service (TCP, UDP, or SCTP). Can specify SCTP (Voice) if have enabled SCTP (Device > Setup > Management).
Destination Port	Enter destination port number (0 to 65535) or range of port numbers. Multiple ports or ranges must be separated by commas & destination port is required.
Source Port	Enter the source port number (0 to 65535) or range of port numbers. Multiple ports or ranges must be separated by commas & source port is optional.
Session Timeout	Define the session timeout for the service: Inherit from application (default)—No service-based timeouts are applied. Override—Define a custom session timeout for the service. Continue to populate the TCP Timeout, TCP Half Closed, and TCP Wait Time fields.
TCP Timeout	Set the maximum length of time in seconds that a TCP session can remain open after data transmission has started. When this time expires, the session closes. Range is 1 - 604800. Default value is 3600 seconds.
TCP Half Closed	Set the maximum length of time in seconds that a session remains open when only one side of the connection has attempted to close the connection. Range is 1 - 604800. Default value is 120 seconds.
TCP Wait Time	Set the maximum length of time in seconds that a session remains open after receiving the second of the two FIN packets required to terminate a session, or after receiving an RST packet to reset a connection. When the timer expires, the session closes. Range is 1 - 600. Default value is 15 seconds.

Objects > Service Groups:

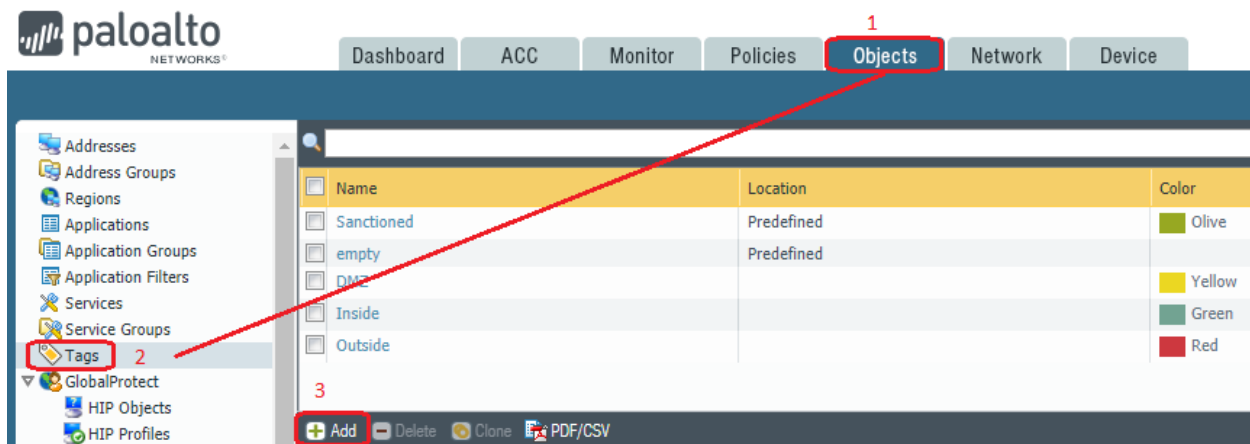
To simplify the creation of security policies, you can combine services that have the same security settings into service groups. Go to **Objects > Services Groups**, and then click the **Add** button at the bottom of the screen. create a Service Group named **Web-Services**. Click the **Add** button to add the Web-services to the group. Select and add all services.



Settings	Description
Name	Enter the service group name.
Service	Click Add to add services to the group. Select from the drop-down or click Service at the bottom of the drop-down and specify the settings.

Objects > Tags:

- o Object Tags to create a tag, assign a color, delete, rename, and clone tags.
- o Tags can be used to sort, or filter objects, & visually distinguish object have color.
- o When color is applied to a tag, Policies tab displays the object with background color.
- o Tags applied to address objects, groups, zones, services, service groups & policy rules.
- o Go to **Objects > Tags**, and then click the **Add** button at the bottom of the screen.
- o Color value of the tag object can be selected from color palette of predefined colors.
- o Predefined colors only, cannot create custom colors, multiple tags can use same color.
- o If an item has multiple tags with different colors, then first tag color will be displayed.
- o Configuration logs are generated for add/edit/delete of tag objects and setting of tags.
- o Palo Alto network Firewall, tag objects has three fields: Name, Color and Comments.

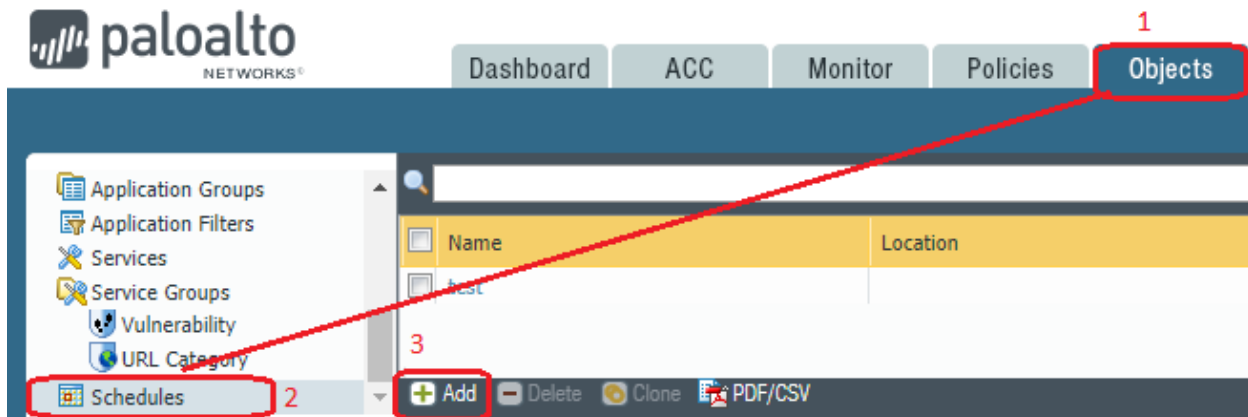


- o Go to **Objects > Tags**, and then click the **Add** button at the bottom of the screen.

Tag Settings	Description
Name	Enter a unique tag name, the name is not case-sensitive.
Color	Select color from color palette in drop-down. The default value is None.
Comments	Add a label or description to remind you what the tag is used for.

Objects > Schedules:

- o In PA Firewall by default, Security policy rules are always in effect all dates and times.
- o To limit a Security policy in firewall rule to specific times, you can define schedules.
- o After, define schedules, can then apply them to the appropriate policies in action tab.
- o you can specify a fixed date and time range or a recurring daily or weekly schedule.
- o Go to **Objects > Schedules**, and then click the **Add** button at the bottom of the screen.



The 'Schedule' configuration window is shown. The 'Name' field is set to 'Daily-Schedule' (labeled 1). The 'Recurrence' dropdown is set to 'Daily' (labeled 2). The 'Start Time' is set to '00:00' (labeled 3) and the 'End Time' is set to '23:30' (labeled 4). At the bottom, the 'Add' button is circled in red (labeled 5). The 'OK' and 'Cancel' buttons are at the bottom right.

Settings	Description
Name	Enter a schedule name. This name appears in the schedule list when defining security policies.
Recurrence	Select the type of schedule (Daily, Weekly, or Non-Recurring).
Daily	Click Add and specify a Start Time and End Time in 24-hour format (HH:MM).
Weekly	Click Add, select a Day of Week, and specify the Start Time and End Time in 24-hour format (HH:MM).
Non-recurring	Click Add and specify a Start Date, Start Time, End Date, and End Time.

On the **WebGUI** go to **Policies > Security > Security Policy Rule > Schedule > Actions**

The screenshot shows the 'Security Policy Rule' configuration page with the 'Actions' tab selected. The 'Action' is set to 'Allow'. The 'Log Setting' section has 'Log at Session End' checked. The 'Profile Setting' section has various security features configured. The 'Other Settings' section has 'Schedule' set to 'None'. A red box highlights the 'Schedule' dropdown, and a red arrow points to the 'Daily-Schedule' option in the dropdown menu. The 'OK' button is highlighted with a red box.

Security Policy Rule

General Source User Destination Application Service/URL Category **Actions**

Action Setting

Action: Allow

☐ Send ICMP Unreachable

Profile Setting

Profile Type: Profiles

Antivirus: default

Vulnerability Protection: strict

Anti-Spyware: strict

URL Filtering: default

File Blocking: strict file blocking

Data Filtering: None

WildFire Analysis: default

Log Setting

☐ Log at Session Start

☒ Log at Session End

Log Forwarding: None

Other Settings

Schedule: None

QoS Marking: None

Daily-Schedule

New Schedule

OK Cancel

Let's create a Schedule Object for Twitter named: **Twitter-Schedule**, Recurrence: **Daily**, Start Time: **2PM** and End Time: **11:30PM**

The screenshot shows the 'Schedule' configuration page. The 'Name' field is 'Twitter-Schedule'. The 'Recurrence' is set to 'Daily'. The 'Start Time' is '14:00' and the 'End Time' is '23:30'. The 'Add' button is highlighted with a red box and a red arrow. The 'OK' button is highlighted with a red box.

Schedule

Name: Twitter-Schedule

Recurrence: Daily

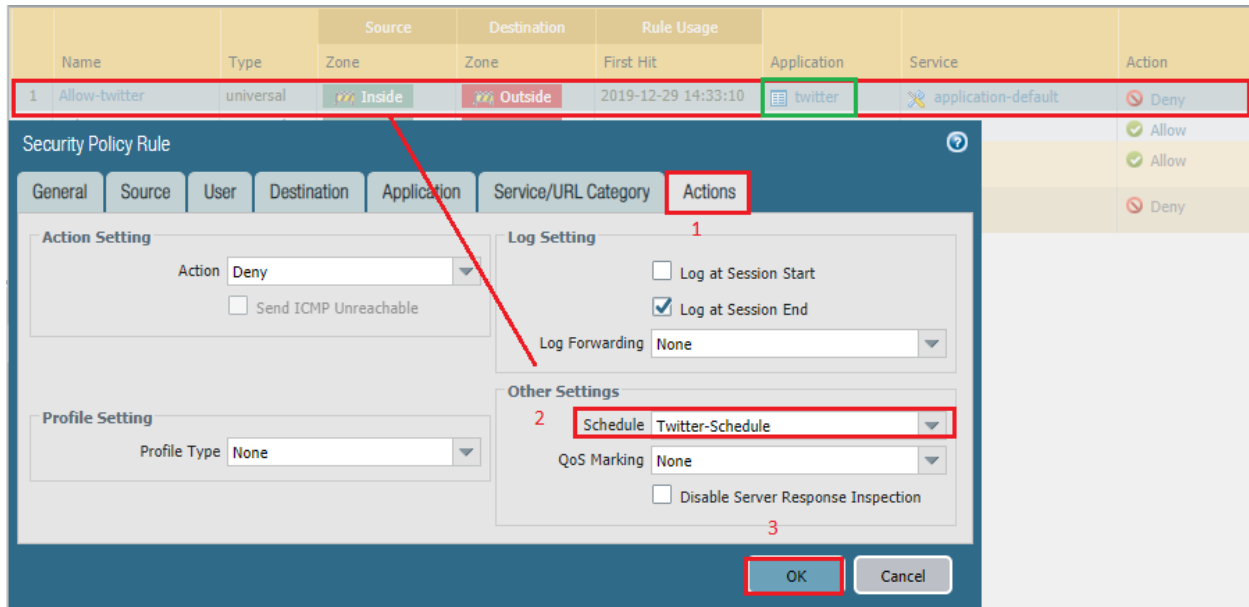
Start Time: 14:00

End Time: 23:30

+ Add - Delete

OK Cancel

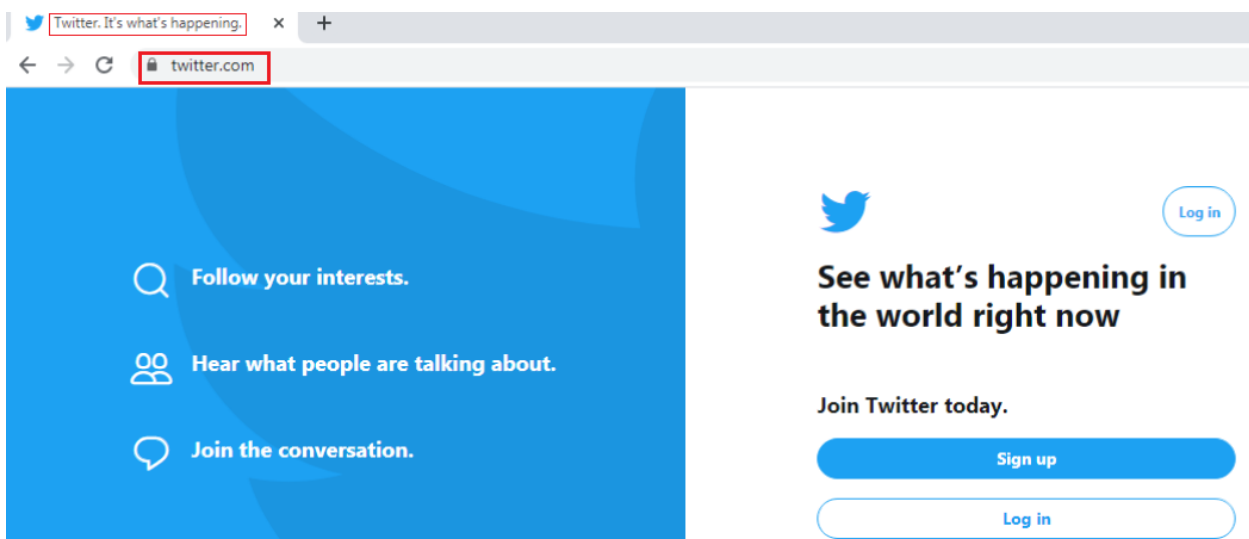
go to **Policies > Security > Security Policy Rule > Schedule > Actions** to apply



After 2PM till 11:30PM night Twitter application or URL will be blocked.

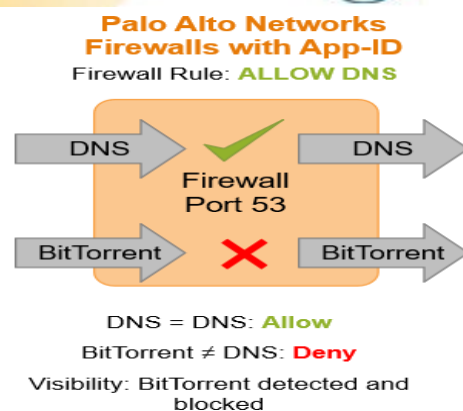
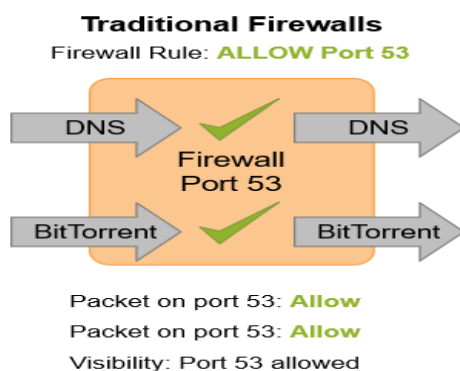


Before 2PM it will work.



APP-ID:

- o Application Identification or App-ID is a main component of Palo Alto devices.
- o Traditional Firewalls classify traffic by port and protocol, PA Firewall use App-ID.
- o Application is specific program who's communication can be labeled & monitored.
- o Application is specific feature whose communication can be labeled and controlled.
- o Port-based rules use Service, App-ID or Application-based rules use the application.
- o Application rules allow only the application traffic that is allowed & no other traffic.
- o Unknown traffic trying to pass application policy is blocked, because it doesn't match.
- o Application Identity for UDP can generally identify the application on the first packet.
- o App-ID for TCP take several packets to identify, as 3-way handshake needs to be done.
- o Then the application data will need to be examined, depending on the application data.
- o Application Database is updated weekly with new and updated application identifiers.
- o App-ID identification of applications ensures the success of proper Layer 7 inspections.
- o Application Signatures today over 3,000, Application Protocol Decoders, and heuristics.
- o These elements are responsible for visibility of L7 traffic traversing P Networks firewalls.
- o Apps are categorized and classified by App-ID engine, allowing proper identification.
- o During classification process, Palo Alto Networks defines main applications (Parent App).
- o Some directly dependent (or Child App), which are part of these main applications.
- o Link to verify Application Identity or inside firewall applipedia.paloaltonetworks.com.



Identifying the Applications:

- o The below App-ID (Application Identity) technologies are used during policy inspection.
- o Uses multiple identification mechanism to determine the exact identity of applications.
- o Accurate traffic classification, then it determine if it is going to allow or block the traffic.

Application Signature:

- o Database of application signature updated weekly as part of firewall content updates.
- o Application Signature can also be manually downloaded and installed in PA Firewall.

Unknown Protocol Decoder:

- o An App-ID heuristic engine used to look up the pattern of the communication.
- o It attempts to identity the application based on its network behavior.
- o Example this type of detection required for proprietary end-to-end encryption like Skype.

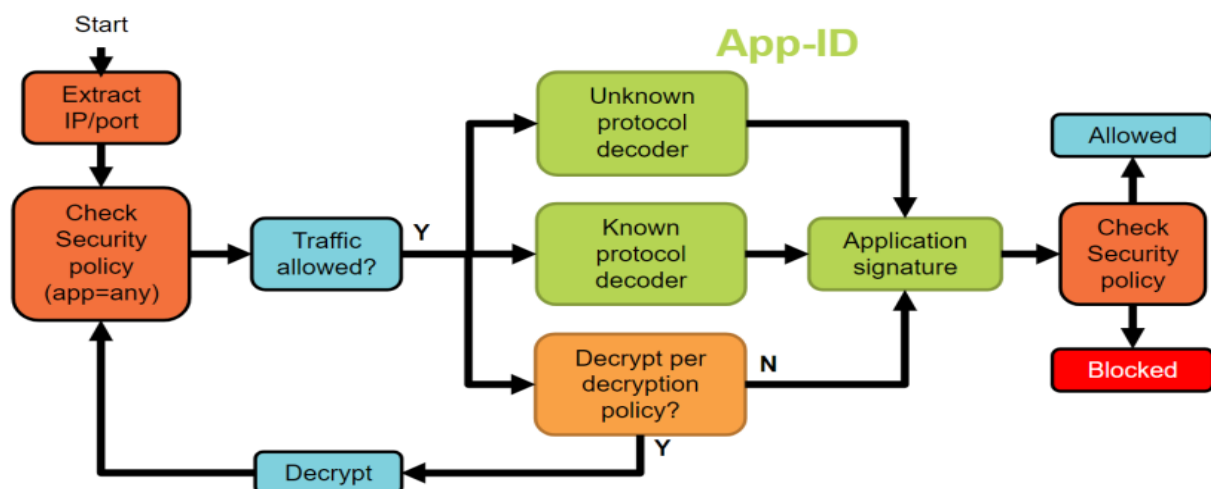
Known Protocol Decoder:

- o Set of applications that coders understand syntax & commands of common applications.

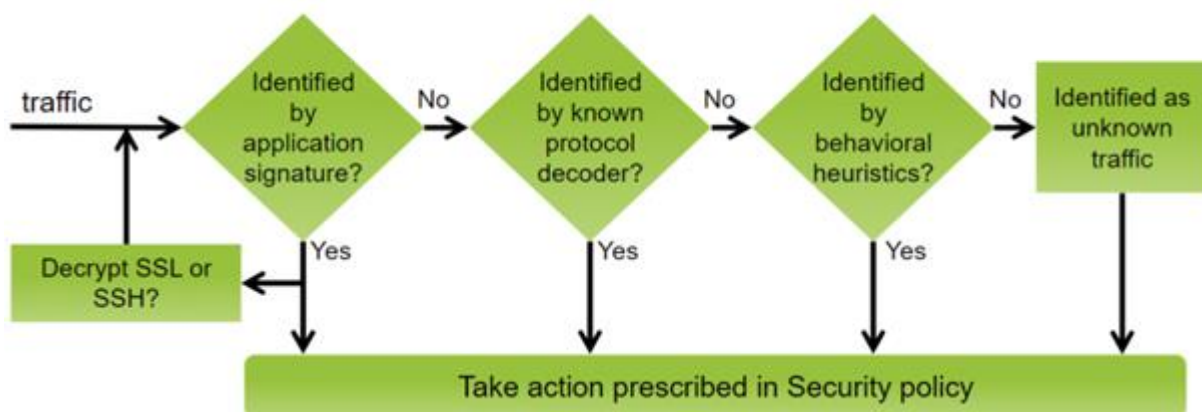
Protocol Decryption:

- o Uses SSH (Secure Shell) and SSL (Secure Socket Layer) for encryption capabilities.

App-ID Operation



- App-ID™ identifies applications in traffic observed by the firewall.



Application Window Details:

Application

Name: ssh

Standard Ports: tcp/22

Depends on:

Implicitly Uses:

Additional Information: [Wikipedia](#) [Google](#) [Yahoo!](#)

Description:

Secure Shell is a set of standards and an associated network protocol that allows establishing a secure channel between a local and a remote computer.

Characteristics

Evasive: no

Excessive Bandwidth Use: no

Used by Malware: yes

Capable of File Transfer: yes

Has Known Vulnerabilities: yes

Tunnels Other Applications: yes

Prone to Misuse: no

Widely Used: yes

Options

TCP Timeout (seconds): 3600 [Customize...](#)

TCP Half Closed (seconds): 120 [Customize...](#)

TCP Time Wait (seconds): 15 [Customize...](#)

App-ID Enabled: yes

Classification

Category: networking

Subcategory: encrypted-tunnel

Technology: client-server

Risk: 4 [Customize...](#)

Application Details	Description
Name	Name of the application.
Description	Description of the application.
Additional Information	Links to web sources contain additional info about app.
Standard Ports	Ports that application uses to communicate with network.
Depends on	List of other apps that are required for this application to run.
Implicitly Uses	These applications are supported implicitly.
Previously Identified As	This indicates what application was previously identified as.
Deny Action	Default deny action can specify either silent drop or TCP reset.
Characteristics	
Evasive	Uses a port or protocol for something other than its originally.
Excessive Bandwidth	Consumes atleast 1 Mbps on regular basis through normal use.
Prone to Misuse	Easily set up to expose more than the user intended.
SaaS	On the firewall, Software as a Service (SaaS) is characterized.
Capable of File Transfer	Has capability to transfer a file from one system to another.
Tunnels Other Applications	Is able to transport other applications inside its protocol.
Used by Malware	Use app for propagation, attack or data theft, or distributed.
Has Known Vulnerabilities	Has publicly reported vulnerabilities.
Widely used	Likely has more than 1,000,000 users.
Continue Scanning for Other Applications	Instructs the firewall to continue to try and match against other application signatures.

Applipedia Windows:

Inside the Palo Alto WebGUI, go to the **Objects tab > Applications** section.

The screenshot shows the Palo Alto Networks WebGUI interface. The 'Objects' tab is selected, and the 'Applications' section is active. The main table lists various applications with their attributes. A red box highlights the 'Applications Section' in the table header. Another red box highlights the 'Clear Filters' button. A note indicates that to reset any filters, the user should click the 'Clear Filters' button. The total number of applications is 3190.

From the website link: <https://applipedia.paloaltonetworks.com/>

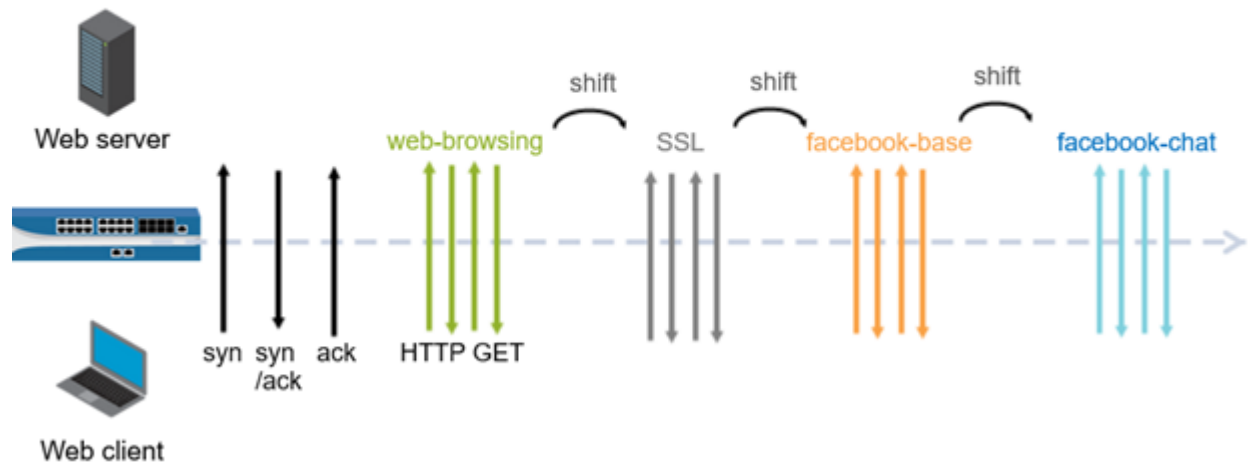
Search:		Search any App		Total number of apps		3189 Applications		Clear filters	
CATEGORY	SUBCATEGORY	TECHNOLOGY	RISK	CHARACTERISTIC					
1238 business-systems	54 audio-streaming	1174 browser-based	133 1	633 Evasive					
632 collaboration	23 auth-service	1385 client-server	827 2	634 Excessive Bandwidth					
501 general-internet	38 database	481 network-protocol	531 3	654 Excessive Bandwidth					
320 media	85 email	147 peer-to-peer	359 4	36 FEDRAMP					
496 networking	66 encrypted-tunnel		142 5	1 FINRA					
	45 erp-crm			98 HIPAA					
	344 file-sharing			99 IP Based Restrictions					
	68 gaming			23 New App-ID					
	192 general-business			519 No Certifications					
NAME	CATEGORY	SUBCATEGORY	RISK	TECHNOLOGY					
100bao	general-internet	file-sharing	5	peer-to-peer					
1c-enterprise	business-systems	erp-crm	1	client-server					
1und1-mail	collaboration	email	3	browser-based					
24sevenoffice	business-systems	erp-crm	2	browser-based					
2ch									
2ch-base	collaboration	social-networking	2	browser-based					
2ch-posting	collaboration	web-posting	2	browser-based					
360-safeguard-update	business-systems	software-update	2	client-server					
3pc	networking	ip-protocol	1	network-protocol					
4shared	general-internet	file-sharing	4	browser-based					
4sync	general-internet	file-sharing	3	client-server					
51.com									

Search and Category Browser section - WebGUI

Search:		Clear Filters		3190 matching applications	
Category	Subcategory	Technology	Risk	Characteristic	
1238 business-systems	54 audio-streaming	1175 browser-based	1331 1	41 Data Breaches	
633 collaboration	23 auth-service	1385 client-server	827 2	634 Evasive	
501 general-internet	38 database	481 network-protocol	531 3	654 Excessive Bandwidth	
320 media	85 email	147 peer-to-peer	359 4	36 FEDRAMP	
496 networking	66 encrypted-tunnel		142 5	1 FINRA	
	45 erp-crm			98 HIPAA	
	344 file-sharing			99 IP Based Restrictions	
	68 gaming			23 New App-ID	
	192 general-business			519 No Certifications	

Application Shifts:

- o Applications can change during lifetime of a session called an application shift.
 - o example, user types www.facebook.com into web browser to access Facebook-chat.
 - o Initial request goes out as HTTP request, & application is recognized as web-browsing.
 - o After the HTTP request is completed, the application is changed to Facebook-base.
 - o After Facebook-base application is processed, application changes to Facebook-chat.
- Network traffic can shift from one application to another during a session.



Dependent Applications:

- o Some applications within PAN-OS software are dependent on other applications.
- o Which means that if Application no 1 is dependent on Application no 2, then both.
- o Application no 1 and Application no 2 need to be allowed in security policies to work.
- o For Example, www.youtube.com is dependent on the [google-base](#) application to work.
- o Therefore, both applications need to be allowed in same security policy or other rules.

	Name	Type	Source			Destination		Rule Usage		Application	Service	Action
			Zone	Address	User	Zone	Hit Count					
1	Youtube-Allow	universal	Inside	any	any	Outside	0		youtube-base	application-default	Allow	
2	HR-Rule	Commit Status Operation Commit Status Completed Result Successful Details Configuration committed successfully Warnings vsys1 vsys1: Rule 'Youtube-Allow' application dependency warning: Application 'youtube-base' requires 'google-base' be allowed (Module: device) Youtube requires that this google-base app be allowed in this same policy or other security rule to work										
3	Outside-to-Inside-Server								dns	application-default	Allow	
4	Inside-to-Outside								google-base		Allow	
5	Inside-To-Outside								ssl		Allow	
									web-browsing		Allow	
									youtube-base		Allow	
									any	application-default	Allow	
									any	any	Allow	
									any	application-default	Allow	

Determining Dependent Applications:

To determine applications and their dependencies, navigate to **Objects > Applications**

Or go this link type the application & search <https://applipedia.paloaltonetworks.com/>

Go to **Objects > Applications > application detail window** is a list of other applications that are required for this application to properly work.

Objects 1

Search: **youtube** 3

Applications 2

Name: youtube-base

Standard Ports: tcp/80,443

Depends on: google-base 5

Implicitly Uses:

Additional Information: Wikipedia Google Yahoo!

Name	Tagged	Category	Subcategory
☐ google-play		general-internet	internet-utility
☐ khan-academy		media	photo-video
☐ newrow		media	photo-video
☐ youtube			
☐ youtube-base 4		media	photo-video
☐ youtube-posting		collaboration	web-posting
☐ youtube-safety-mode		media	photo-video
☐ youtube-streaming		media	photo-video

Click

https://applipedia.paloaltonetworks.com 1

youtube-base

Description

YouTube is a popular free video sharing website which lets users upload, view, and share video clips. Videos can be rated, and the average rating and the number of times a video has been watched are both published. Based on recent research we have noticed that for the photo-upload function to work properly on youtube, customers will have to allow google-docs-base App-ID as well.

Reference

Wikipedia Google Yahoo!

Depends on Applications: 4

google-base

Characteristics

Category	media	Evasive	yes
Subcategory	photo-video	Excessive Bandwidth	yes
Risk	4	Prone to Misuse	no
Standard Ports	tcp/80,443	Capable of File Transfer	no
Technology	browser-based	Tunnels Other Applications	no
		Used by Malware	yes
		Has Known Vulnerabilities	yes
		Widely Used	yes
		SaaS	no

Search: **youtube** 2

CATEGORY

1	business-systems
8	collaboration
16	media

NAME

teachertube
torch-browser
torch-browser-music
tudou
veohTV
youtube
youtube-posting
youtube-tv
youtube-tv-streaming
youtube-base 3
youtube-safety-mode

Implicitly Use Applications:

- o The Implicitly Use Applications is a subset of the Depends on Applications but allowed.
- o If allow Facebook-base, it will also allow SSL & Web-Browsing applications implicitly.
- o Sometimes don't have to explicitly allow access to dependent applications for traffic.
- o Because the PA firewall can determine the dependencies and allow them implicitly.
- o Example is to be able to use **google-base**, don't have to add ssl or web-browsing to rule.
- o When we look at the application detail window for '**google-base**', can see 2 things listed.
- o Depends on is blank, means it doesn't need any other applications to be allowed to work.
- o Implicitly Use Applications has SSL and web-browsing listed for google-base application.
- o Means if allow **google-base**, it will also allow SSL & Web-Browsing applications implicitly.

To determine applications that specifically are used, navigate to **Objects > Applications**.

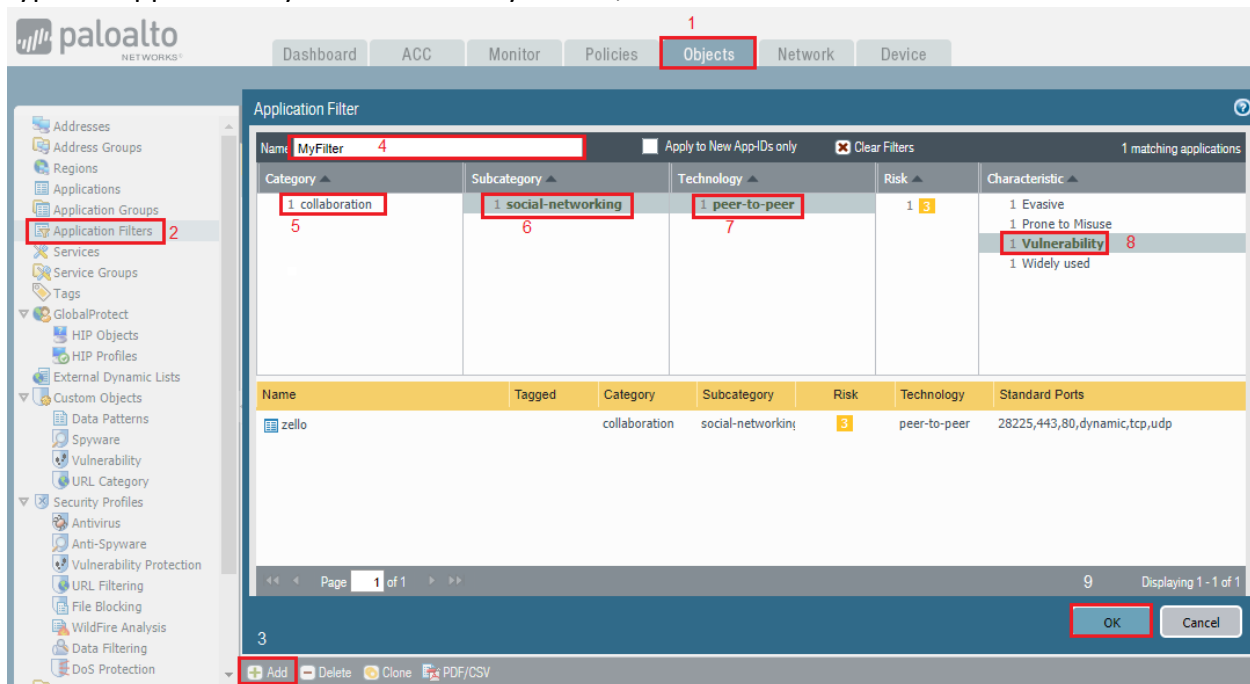
The screenshot shows the 'Application' detail window for 'google-base'. The 'Name' is 'google-base'. The 'Standard Ports' are 'tcp/80,443,5222-5224,5228,5229'. The 'Depends on' field is empty. The 'Implicitly Uses' field is highlighted with a red box and contains 'ssl, web-browsing'. The 'Deny Action' is 'drop-reset'. The 'Additional Information' field contains links to 'Wikipedia Google Yahoo!'. The 'Description' states: 'This App-ID covers common service and infrastructure traffic generated by all Google services and applications. To safely enable Google's services and applications, this App-ID is required to be permitted by policy.'

The screenshot shows the Palo Alto Networks application management interface. At the top, the URL 'https://applipedia.paloaltonetworks.com' is highlighted with a red box and labeled '1'. On the left, a search bar contains 'google' and is labeled '2'. Below the search bar is a 'CATEGORY' list with items like 'business-systems', 'collaboration', 'general-internet', 'media', and 'networking'. A table lists applications with columns 'NAME' and 'CATEGORY'. The 'google-base' application is highlighted with a red box and labeled '3'. On the right, the 'google-base' application detail window is open. The 'Implicit use Applications' field is highlighted with a red box and labeled '4', showing 'ssl, web-browsing'. The 'Description' and 'Reference' (Wikipedia Google Yahoo!) are also visible. The 'Characteristics' section lists various attributes: Category (general-internet), Subcategory (internet-utility), Risk (4), Standard Ports (tcp/80,443,5222-5224,5228,5229), Technology (browser-based), Evasive (no), Excessive Bandwidth (no), Prone to Misuse (no), Capable of File Transfer (yes), Tunnels Other Applications (yes), Used by Malware (yes), Has Known Vulnerabilities (yes), Widely Used (no), and SaaS (no).

Application Filters:

- o Administrator can dynamically categorize multiple applications into an application filter.
- o Based on specific attributes Category, Subcategory, Technology, Risk, and Characteristic.
- o In PA the application filter is a dynamic item that is created by selecting filter options.
- o Application filter, such as Category, Subcategory, Technology etc in application browser.
- o Any new applications coming to PAN-OS in a content update that match the same filters.
- o The application set will automatically be added to the Application Filter created earlier.

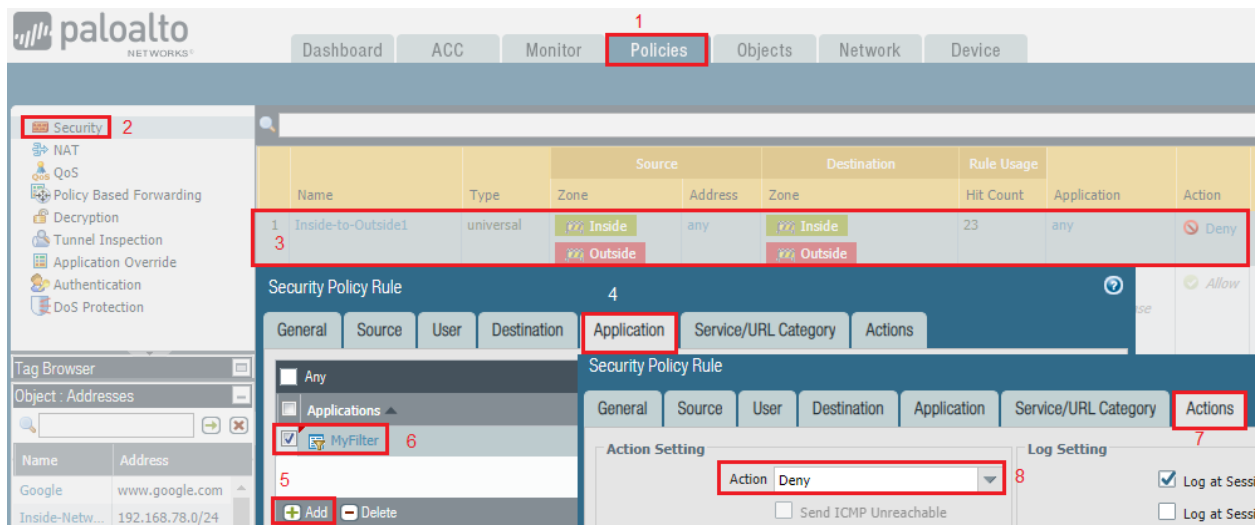
Select **Objects > Application Filters**, add a filter and give it descriptive Name in my case **MyFilter**. Define the filter by selecting attribute values from the Category, Subcategory, Technology, Risk, and Characteristic sections. As you select values, notice that the list of matching applications at the bottom of the dialog narrows. When you have adjusted the filter attributes to match the types of applications you want to safely enable, click **OK**.



Finally, it will look like below.

Name	Subcategory	Technology	Risk	Characteristic
MyFilter	social-networking	peer-to-peer		Vulnerability

Go to **Policies > Security** modified **Inside-to-Outside** policy rule in the **application** tab add recently created Filter named: MyFilter to the policy rules. Now go to Action tab **deny** the policy rules.



Go to Test Inside-PC and try to browse any Social networking website it show blocking page.

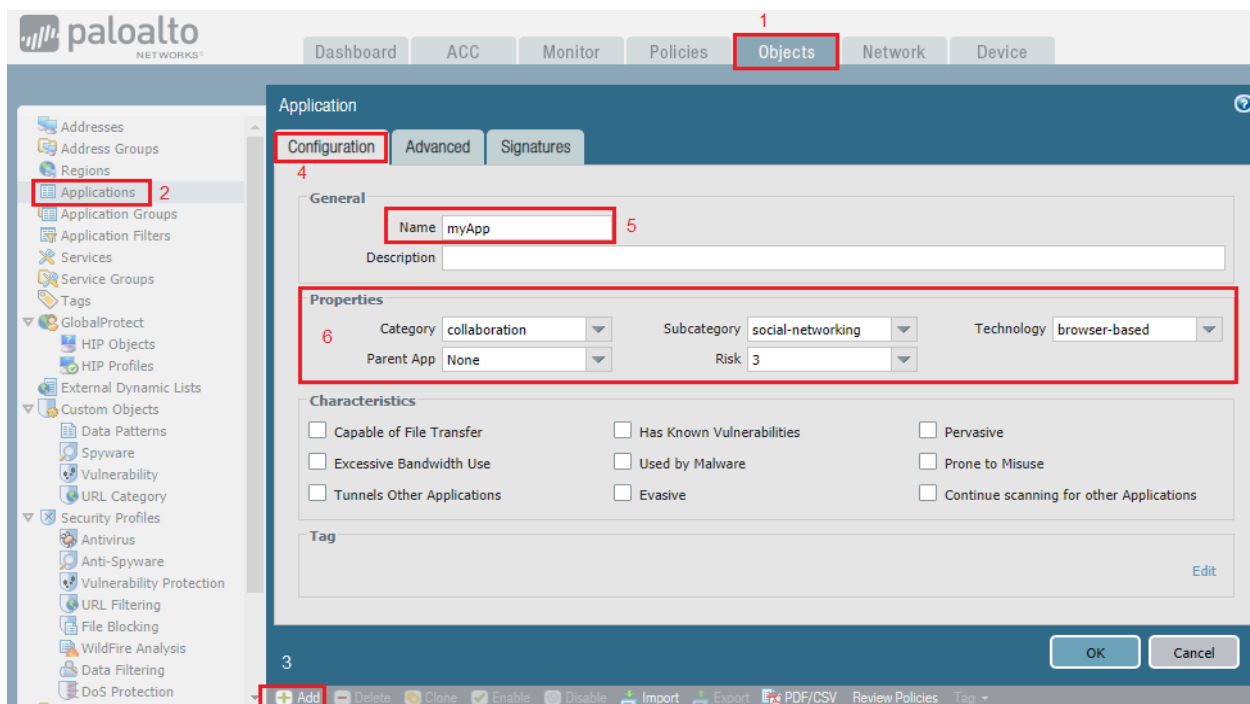
Application Blocked

Access to the application you were trying to use has been blocked in accordance with company policy. Please contact your system administrator if you believe this is in error.

User: 192.168.78.50

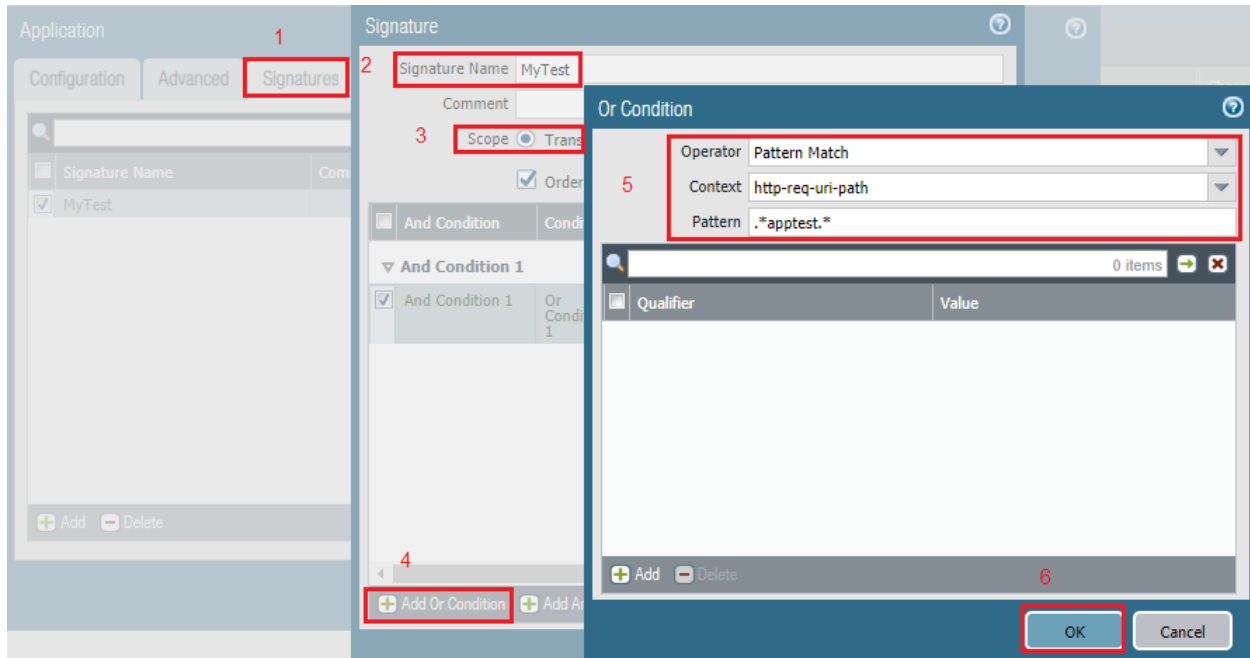
Application: myApp

Select **Objects> Applications** and click **Add**. On the **Configuration** tab, enter a **Name** and a **Description** for the custom application. Define the application Properties and Characteristics.

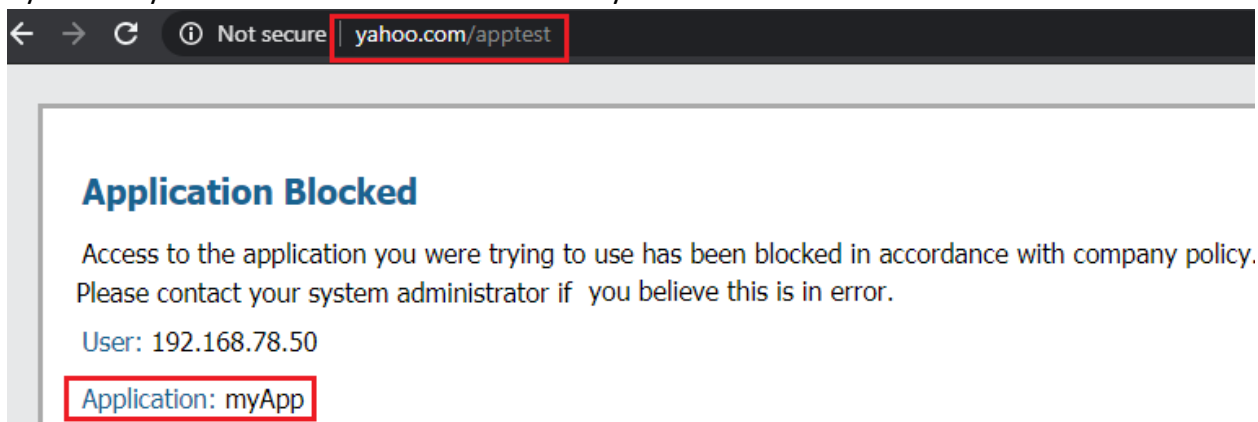


Custom Application:

Let's create custom Signature, on the same our create Filter category. On the **Signatures** tab, click **Add** and define a Signature Name. Specify the Scope of the signature: whether it matches to a full Session or a single **Transaction**. Specify conditions to define signatures by clicking **Add And Condition** or **Add Or Condition**. Select an Operator to define the type of match conditions.



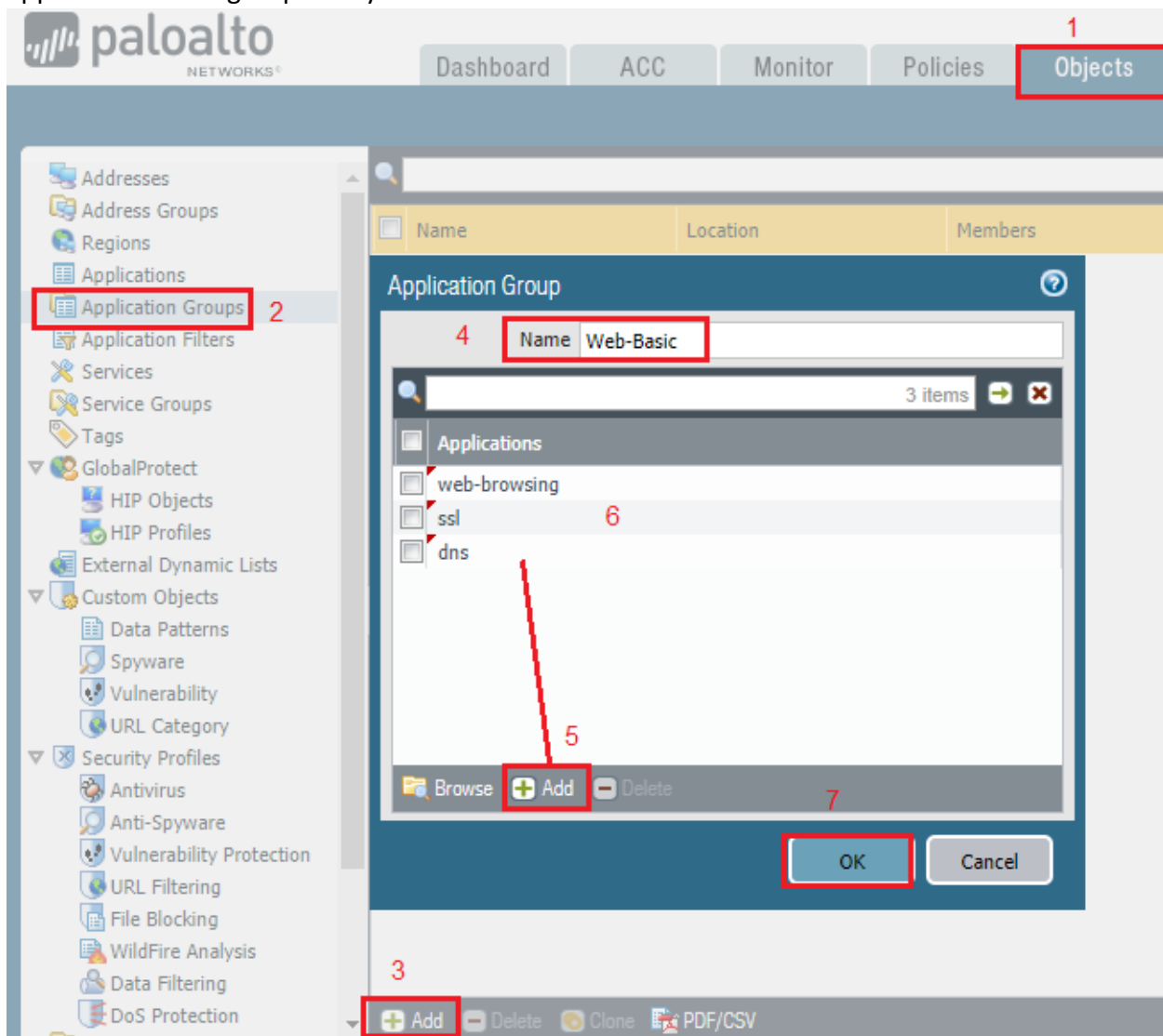
Lets tested go to Inside-PC and type **yahoo.com/apptest** it will show block page because our custom created app is under social networking category, which is block in our filter, so it dynamically added to that Filter and blocked by Palo Alto Network Firewall.



Application Groups:

- o Administrator can manually categorize multiple applications into an application group.
- o This manually created application group categorize multiple apps based on App-ID.
- o This application group then is added to one or more security policy rules as required.
- o Instead of firewall admin individually adding different applications into a security policy.
- o The Firewall Admin can only added the application group needs to the security policy.
- o Application groups are used to simplify security, QoS, & PBF policy rule implementation.
- o Multiple applications and application filters can be combined into an application group.
- o One or more application groups then also can be combined into one application group.
- o The final application group which is created then can be added to a security policy rule.

Go to **Objects > Application Groups**, click **Add** enter group name then click add or browse to add application to the group finally click **OK** to finished.



Name	Members	Applications	Filters	Groups
Web-Basic	3	web-browsing ssl dns		
Socail_Network	4	linkedin-base facebook-base twitter-base instagram-base	application filters is added to Group	application groups is added to group
Combine-Group	3		MyFilter	Web-Basic Socail_Network

Go to **Policies > Security** modified **Inside-to-Outside** policy rule in the **application** tab add recently created Application Groups named: Web-Basic, Socail_Network and Combine-Group to the policy rules. Now go to **Action** tab **deny** the policy rules.

The screenshot shows the Palo Alto Networks GUI with the 'Policies' tab selected. The 'Security Policy Rule' configuration is shown for the 'Inside-to-Outside' rule. The 'Application' tab is active, and the 'Action' is set to 'Deny'. The 'Applications' list includes 'Web-Basic', 'Socail_Network', and 'Combine-Group'. The 'Log Setting' is checked.

Go to Test Inside-PC and try to browse any Social networking website it show blocking page.

Application Blocked

Access to the application you were trying to use has been blocked in accordance with company policy. Please contact your system administrator if you believe this is in error.

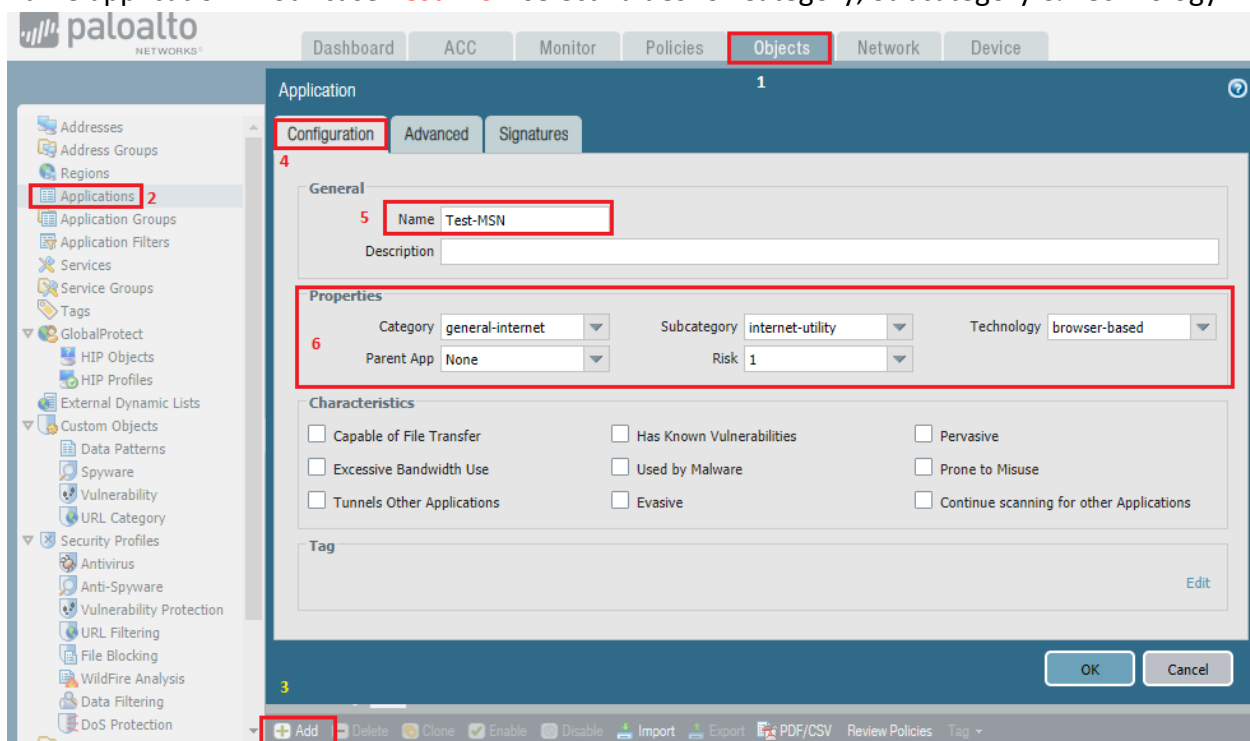
User: 192.168.78.50

Application: myApp

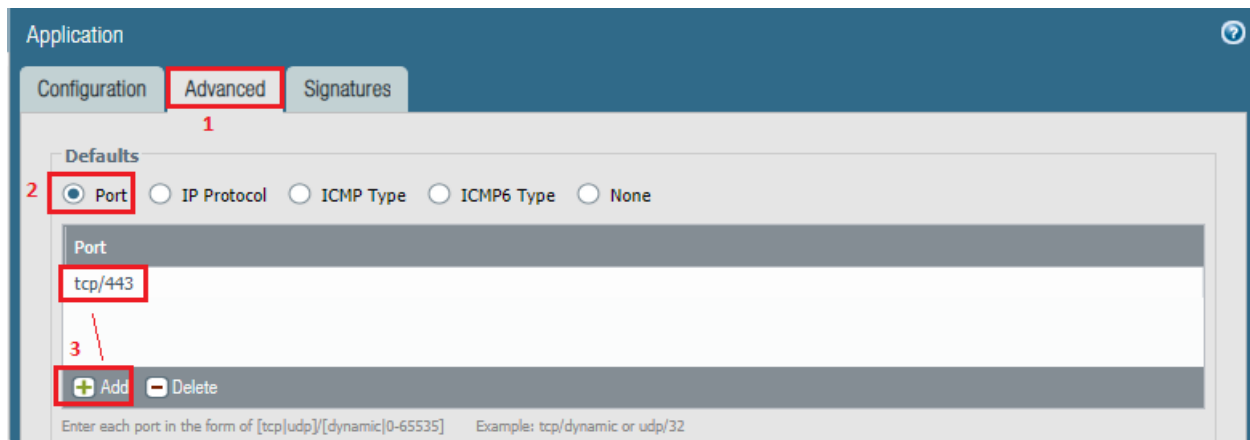
Application Override:

- o Firewall is configured to override normal App-ID of specific traffic passing through firewall.
- o Application Override policy takes effect, all further App-ID inspection of traffic is stopped.
- o Change the firewall classifies network traffic into apps, specify application override policies.
- o Application override with custom application will prevent session from being processed.
- o Application override prevent session to process by App-ID engine, which is L7 inspection.
- o In simple words, with Application Override Policy, Firewall stops doing Layer 7 processing.
- o Instead it forces firewall to handle the session as regular stateful inspection firewall at L4.
- o So, in other words application override policy thereby saves application processing time.
- o Customers build own custom applications to address specific needs unique to the company.
- o May not have signatures to properly identify expected behavior and identify traffic & app.
- o Creating application override to allow easier identification & reporting & prevent confusion.
- o Network applications that are classified as unknown can create new application definitions.

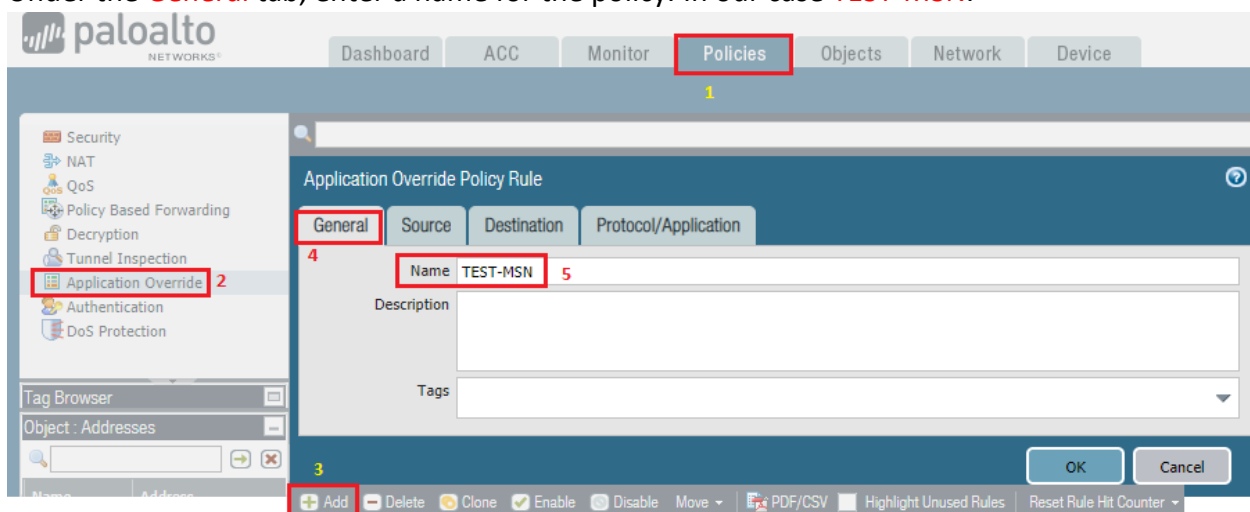
Let's create custom application go to **Objects>Applications** then click **Add** in lower left corner name application in our case **Test-MSN** select values for Category, Subcategory & Technology.



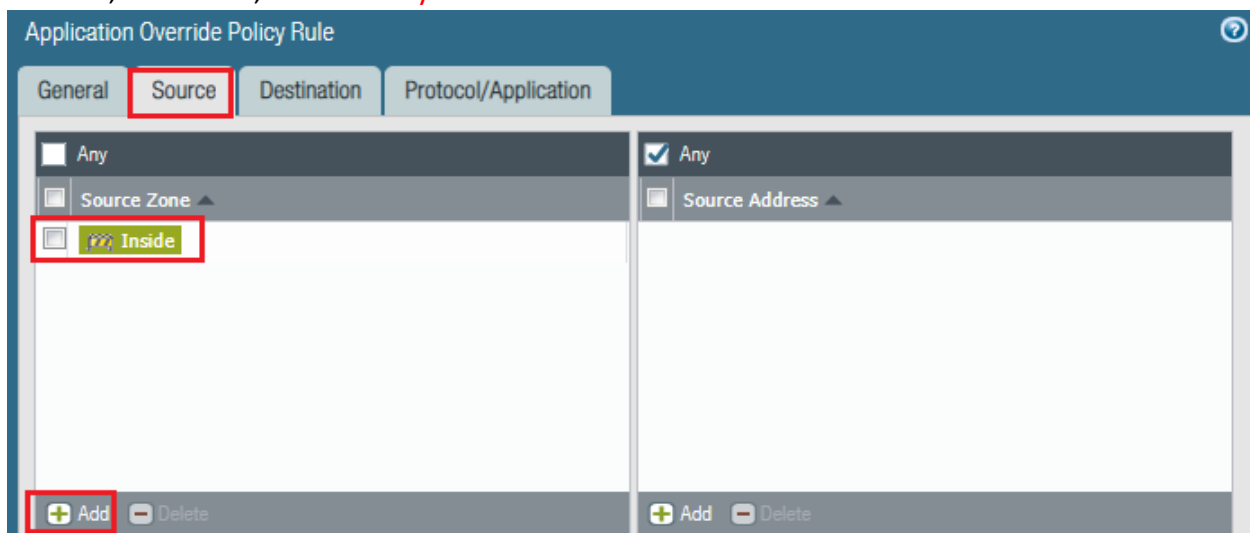
Go to **Advanced > Defaults** and select Port to list the ports in the application. click **Add** type the port **tcp/443** and press **OK**. We don't need a signature, so go ahead and click **OK** to complete this custom application.



To create an Application Override policy, go to **Policies > Application Override**, then click **Add**: Under the **General** tab, enter a name for the policy. In our case **TEST-MSN**.



Go to **Source** and add the **Source Zone**. Specify a Source Address if the source is a static address; otherwise, leave as **Any**.



Go to **Destination** and add the Destination Zone. Specify a Destination Address if the destination is a static address; otherwise, leave as **Any**.

The screenshot shows the 'Application Override Policy Rule' configuration window with the 'Destination' tab selected. The 'Destination Zone' list contains 'Outside'. The 'Destination Address' is set to 'Any'. The 'Add' button is highlighted.

Go to **Protocol/Application** and select the **Protocol**, enter the Port number in our case 443, and select the custom application created in our case **Test-MSN**.

The screenshot shows the 'Application Override Policy Rule' configuration window with the 'Protocol/Application' tab selected. The 'Protocol' is set to 'TCP', the 'Port' is '443', and the 'Application' is 'Test-MSN'. The 'OK' button is highlighted.

Create Security Policy to allow this new application through firewall or modify an existing rule.

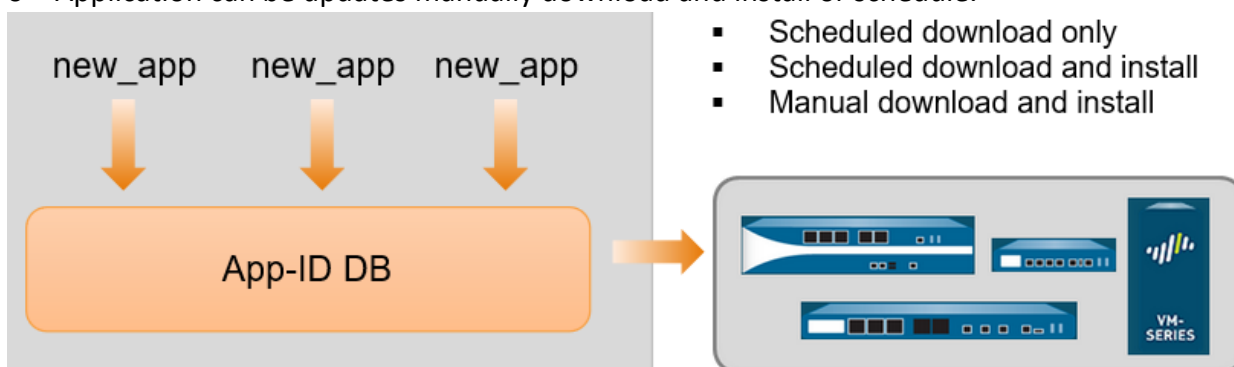
Now commit all the changes and let's test the new created Application Override policy.

	Receive Time	Type	Decr...	From Zone	To Zone	Source	Destination	To Port	Application	Action	Rule	Session End Reason	Bytes
	02/13 23:53:26	end	no	Inside	Outside	Inside-PC	a-0003.a-msedge.net	443	Test-MSN	allow	Inside-to-Outside1	tcp-rst-from-server	15.0k
	02/13 23:48:33	end	no	Inside	Outside	Inside-PC	69.172.200.235	443	Test-MSN	allow	Inside-to-Outside1	tcp-rst-from-client	7.3k
	02/13 23:48:22	end	no	Inside	Outside	Inside-PC	a-0003.a-msedge.net	443	Test-MSN	allow	Inside-to-Outside1	tcp-rst-from-server	15.0k
	02/13 23:48:17	end	no	Inside	Outside	Inside-PC	ham02s12-in-f42.1e100.net	443	Test-MSN	allow	Inside-to-Outside1	tcp-rst-from-client	7.7k

admin@PA-VM> show session all filter application Test-MSN

Application Updates:

- o Palo Alto Networks regularly posts updates for application detection.
- o Application can be updates manually download and install or schedule.



- Scheduled download only
- Scheduled download and install
- Manual download and install

From the WebGUI, go to **Device > Dynamic Updates**

Dynamic Updates

Version	File Name	Features	Type	Size	Release Date	Downloaded	Currently Installed	Action	Documentation
Applications and Threats 3 Last checked: 2020/02/13 21:21:56 PST Schedule: Every Wednesday at 01:02 (Download only)									
8226-5859	panupv2-all-apps-8226-5859	Apps	Full	37 MB	2020/01/14 10:25:48 PST			Download	Release Notes
8227-5876	panupv2-all-apps-8227-5876	Apps	Full	42 MB	2020/01/17 17:03:29 PST			Download	Release Notes
8228-5880	panupv2-all-apps-8228-5880	Apps	Full	42 MB	2020/01/20 10:00:49 PST		✓	Review Policies Review Apps	Release Notes
8229-5889	panupv2-all-apps-8229-5889	Apps	Full	42 MB	2020/01/22 20:08:48 PST			Download	Release Notes
8230-5894	panupv2-all-apps-8230-5894	Apps	Full	42 MB	2020/01/24 16:25:16 PST			Download Review Policies Review Apps	Release Notes
8231-5919	panupv2-all-apps-8231-5919	Apps	Full	42 MB	2020/01/29 20:25:38 PST			Download	Release Notes
8232-5926	panupv2-all-apps-8232-5926	Apps	Full	43 MB	2020/01/31 21:18:52 PST			Download	Release Notes
8233-5931	panupv2-all-apps-8233-5931	Apps	Full	43 MB	2020/02/04 20:24:09 PST			Download	Release Notes
8234-5935	panupv2-all-apps-8234-5935	Apps	Full	43 MB	2020/02/06 18:36:31 PST	✓		Install Review Policies Review Apps	Release Notes
8235-5938	panupv2-all-apps-8235-5938	Apps	Full	43 MB	2020/02/11 10:30:09 PST			Download	Release Notes
8236-5939	panupv2-all-apps-8236-5939	Apps	Full	43 MB	2020/02/12 13:25:22 PST			Download	Release Notes
8237-5941	panupv2-all-apps-8237-5941	Apps	Full	43 MB	2020/02/13 17:05:24 PST			Download	Release Notes
GlobalProtect Clientless VPN Last checked: 2020/02/08 11:22:44 PST Schedule: None									
82-175	panup-all-gp-82-175	GlobalProtectCle...	Full	73 KB	2019/12/19 13:52:49 PST	✓	✓		Release Notes
84-177	panup-all-gp-84-177	GlobalProtectCle...	Full	73 KB	2020/01/31 14:26:13 PST			Download	Release Notes

Applications and Threats Update Schedule

Recurrence: Weekly
Day: wednesday
Time: 01:02
Action: download-only
Threshold (hours): [1 - 336]

Allow Extra Time to Review New App-IDs

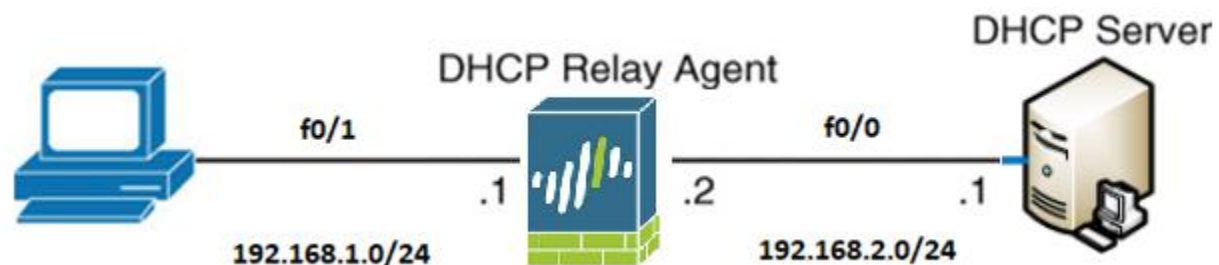
Set the amount of time the firewall waits before installing content updates that contain new App-IDs. You can use this wait period to assess and adjust your security policy based on the new App-IDs.

New App-ID Threshold (hours): [1 - 336]

OK Cancel

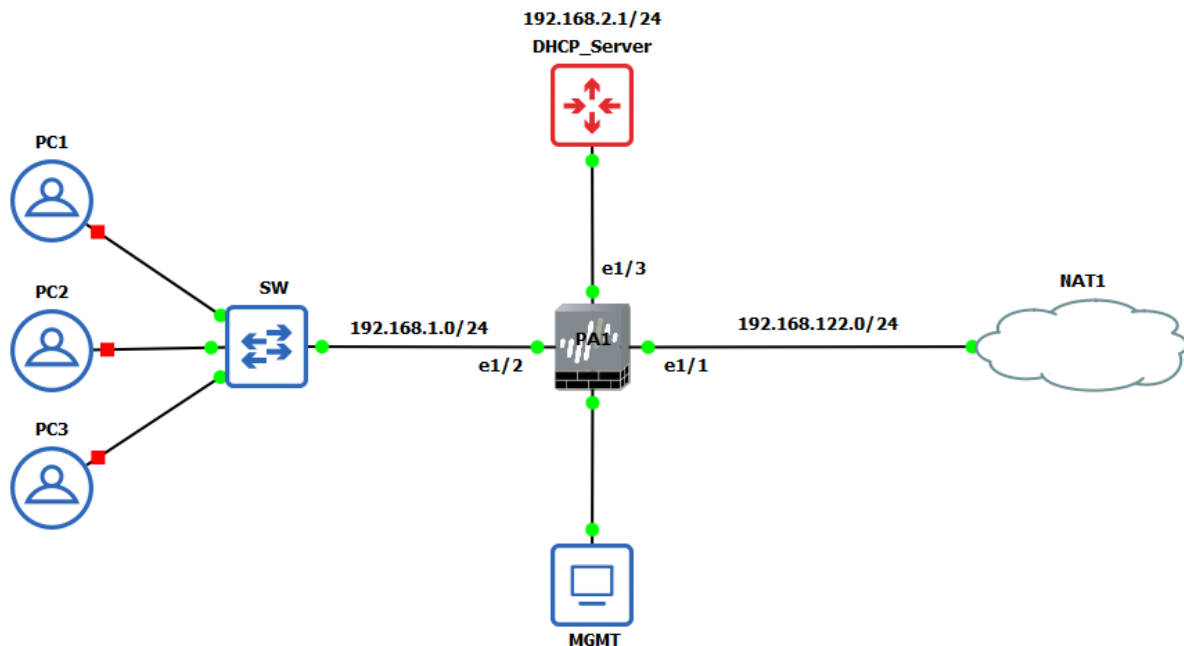
DHCP Relay:

- o DHCP relay agent is any host that forwards DHCP packets between clients & servers.
- o DHCP Relay allows clients to obtain DHCP info from a server on a different subnet.
- o Relay agents are used to forward requests and replies between clients and servers.
- o Relay agents are used to forward request when they are not on same physical subnet.
- o The devices that do the forwarding are referred or called it as the DHCP relay agents.
- o The DHCP Relay agents forward packets differently than the normal IP forwarding.
- o Relay agents receive DHCP message & generate new message out another interface.
- o The DHCP Relay agent adds a GIADDR (Gateway Address of Packet) field to packet.
- o The DHCP Relay also add the Relay agent information DHCP option 82 if enabled.



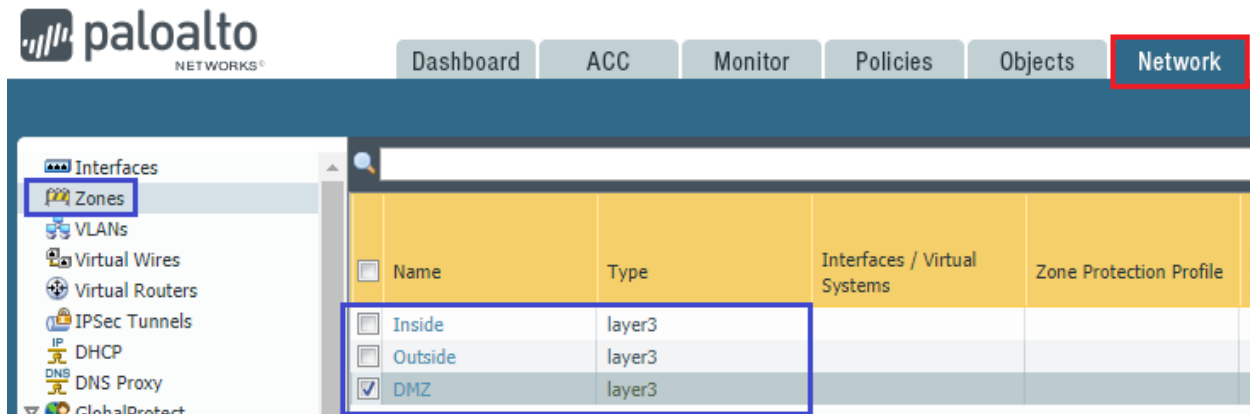
Server Lab Time:

Same topology we will use for DHCP Server and DHCP Relay, we going to enable DHCP Server on e1/2 to provide DHCP services.



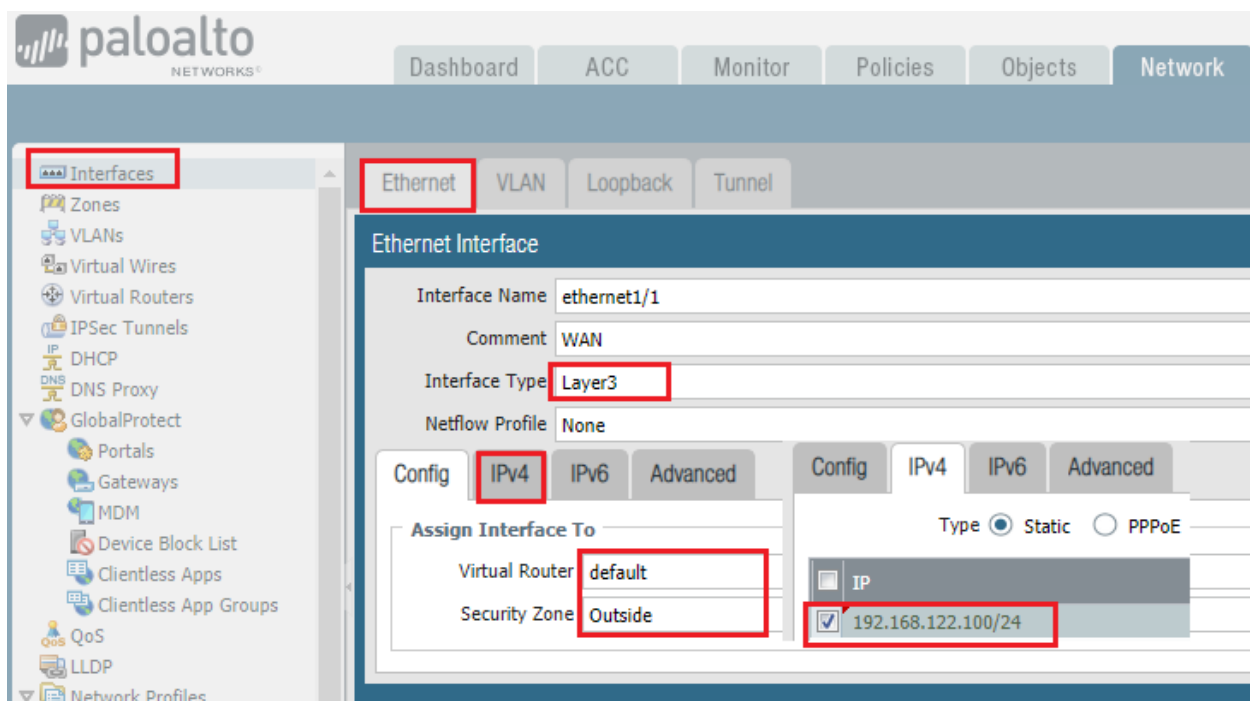
Create Zones:

Let's configure three zones names **Inside** , **Outside** and **DMZ**. Go to **Network> Zone>Add**, Give the name **Inside**, select Type to be **Layer3** and click **OK**. Create the same way other two Zones



Configure Interfaces:

Go to **Network>Interfaces** Click on **ethernet1/1** interface change Interface Type: **Layer3**, set Virtual Router: **default**, set Security Zone: **Outside** , Click on IPv4 tab Assign IP Address: **192.168.122.100/24** and Click **OK**.



Go to **Network>Interfaces** Click on **ethernet1/2** interface change Interface Type: **Layer3**, set Virtual Router: **default**, set Security Zone: **Inside** , Click on IPv4 tab Assign IP Address: **192.168.1.100/24** and Click **OK**.

The screenshot shows the 'Ethernet Interface' configuration window for 'ethernet1/2'. The 'Interface Name' is 'ethernet1/2' and the 'Comment' is 'LAN'. The 'Interface Type' is set to 'Layer3'. The 'Netflow Profile' is 'None'. The 'Assign Interface To' section shows 'Virtual Router' as 'default' and 'Security Zone' as 'Inside'. The 'IPv4' tab is selected, showing 'Type' as 'Static' and 'IP' as '192.168.1.100/24'. The 'OK' button is highlighted.

Go to **Network>Interfaces** Click on **ethernet1/3** interface change Interface Type: **Layer3**, set Virtual Router: **default**, set Security Zone: **DMZ** , Click on IPv4 tab Assign IP Address: **192.168.2.100/24** and Click **OK**.

The screenshot shows the 'Ethernet Interface' configuration window for 'ethernet1/3'. The 'Interface Name' is 'ethernet1/3' and the 'Comment' is 'DMZ'. The 'Interface Type' is set to 'Layer3'. The 'Netflow Profile' is 'None'. The 'Assign Interface To' section shows 'Virtual Router' as 'default' and 'Security Zone' as 'DMZ'. The 'IPv4' tab is selected, showing 'Type' as 'Static' and 'IP' as '192.168.2.100/24'. The 'OK' button is highlighted.

Configure Routing:

Each interface must be given virtual router. **Network>Virtual Router>default** we will add static routing. **Static Routes>IPv4>Add** we will go by choosing interface> **ethernet1/1**(as Outside), put **192.168.122.2** as the next hop due to our topology.

Virtual Router - d Virtual Router - Static Route - IPv4

Router Settings

Static Routes

Redistribution Pr

RIP

OSPF

OSPFv3

BGP

Multicast

Name Default

Destination 0.0.0.0/0

Interface ethernet1/1

Next Hop IP Address 192.168.122.2

Admin Distance

Metric 10

Route Table Unicast

BFD Profile Disable BFD

☐ Path Monitoring

Failure Condition ☒ Any ☐ All Preemptive Hold Time (min) 2

Name	Enable	Source IP	Destination IP	Ping Interval(sec)	Ping Count
------	--------	-----------	----------------	--------------------	------------

Configure NAT/PAT:

Let's configure NAT using **Dynamic IP and Port** means translate all local LAN to only one IP address. I will NAT my Inside LAN 192.168.1.0/24 to 192.168.122.100 IP address of WAN. **Policies > NAT > Add** Let's name it **Inside-To-Outside**.

patato NETWORKS

Dashboard ACC Monitor Policies

Security

NAT

QoS

Policy Based Forwarding

Decryption

Tunnel Inspection

Application Override

Authentication

DoS Protection

Tag Browser

0 items

NAT Policy Rule

General Original Packet Translated Packet

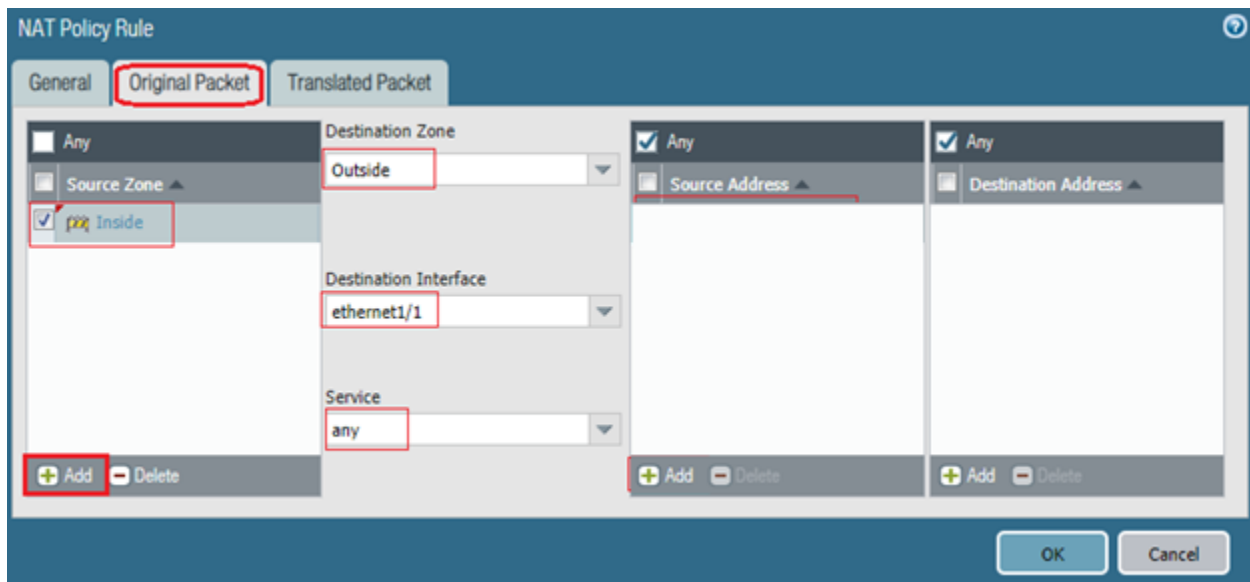
Name Inside-to-Outside

Description

Tags

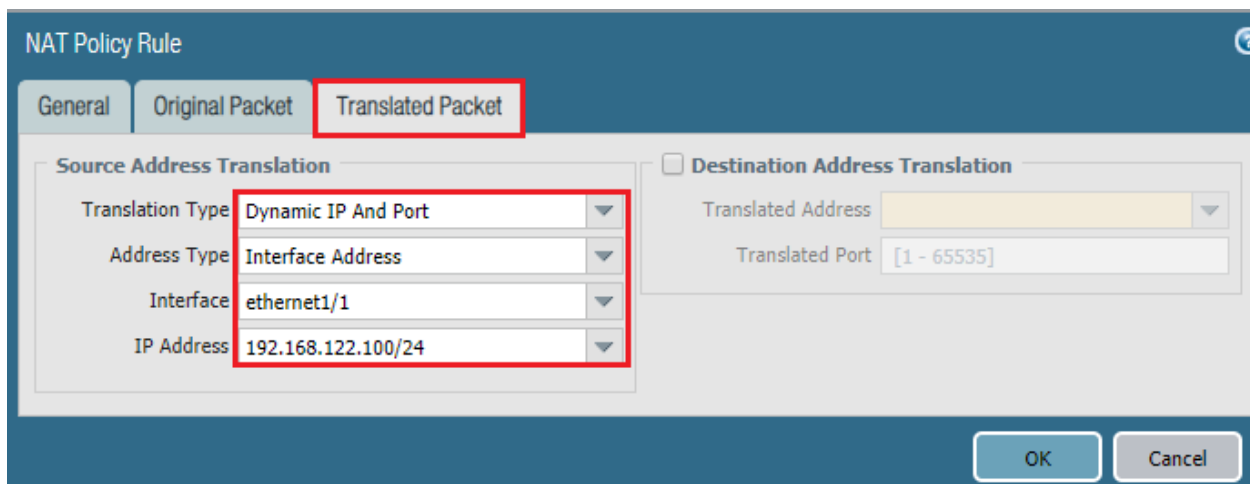
NAT Type ipv4

Go to **Original Packet** and fill since, traffic coming from Inside(192.168.1.0/24 is in Inside) then destination zone Outside(since 192.168.122.100 is going to Internet), destination interface is ethernet1/1 outgoing Interface. Set Service to any.



The screenshot shows the 'NAT Policy Rule' configuration window with the 'Original Packet' tab selected. The 'Source Zone' is set to 'Inside' and the 'Destination Zone' is set to 'Outside'. The 'Destination Interface' is set to 'ethernet1/1' and the 'Service' is set to 'any'. The 'Add' button is highlighted. The 'Translated Packet' tab is also visible, showing 'Source Address' and 'Destination Address' sections.

Then let's go to **Translated Packet**, Translation Type: **Dynamic IP And Port**, Address Type: **Interface Address**, Interface: our WAN interface **ethernet 1/1** and IP Address: WAN IP. **OK**



The screenshot shows the 'NAT Policy Rule' configuration window with the 'Translated Packet' tab selected. The 'Source Address Translation' section is expanded, showing 'Translation Type' set to 'Dynamic IP And Port', 'Address Type' set to 'Interface Address', 'Interface' set to 'ethernet1/1', and 'IP Address' set to '192.168.122.100/24'. The 'Destination Address Translation' section is collapsed. The 'Add' button is highlighted. The 'General' and 'Original Packet' tabs are also visible.

Configure Security Policy:

Now, create a Security Policy to allow access from Inside zone to Outside zone.

Policies>Security>Add, Give the name to your Security Policy (**Inside to Outside**), Add Source Zone (**Inside**), Add Destination Zone (**Outside**), Allow access, in our case allowing all traffic.

The screenshot displays the Palo Alto Networks management console. The 'Policies' tab is selected at the top. On the left, the 'Security' menu item is highlighted. A table lists existing rules:

Name	Tags	Type	Zone	Address	User	HIP Profile
1 intrazone-default	none	intrazone	any	any	any	any
2 interzone-default	none	interzone	any	any	any	any

The 'Security Policy Rule' dialog is open, showing the 'Source' and 'Destination' tabs. The 'Source' tab is active, showing 'Source Zone' with 'Inside' selected. The 'Destination' tab is also visible, showing 'Destination Zone' with 'Outside' selected. The 'Name' field is set to 'Inside to Outside'. The 'OK' button is highlighted.

Admin Roles:

- o Define Admin Role profiles, which are custom roles that determine access privileges.
- o Define Admin Role profiles for custom roles responsibilities of administrative users.
- o The firewall has **three** predefined roles you can use for common criteria purposes.
- o Create custom roles limit administrator access to only what each type of admin needs.
- o Each type of admin, enable, disable, or set read-only access for Web UI, XML API, & CLI.

Auditadmin:

The Audit Administrator is responsible for the regular review of the Firewall's audit data.

Cryptoadmin:

The Cryptographic Administrator is responsible for the configuration and maintenance of cryptographic elements related to the establishment of secure connections to the firewall.

Securityadmin:

The Security Administrator is responsible for all other administrative tasks such as creating Security policy not addressed by the other two administrative roles.

Device > Admin Roles To add an Admin Role profile, click **Add** and specify the settings.

☐	Name	Description	Role
☐	auditadmin	Audit Administrator for Common Criteria	device
☐	cryptoadmin	Crypto Administrator for Common Criteria	device
☐	securityadmin	Security Admin for Common Criteria	device

Admin Role Profile

Profile Name: NetworkAdmin
Description: Network Administration

Web UI | XML API | Command Line

- ✓ Dashboard
- ✓ ACC
- ✓ Monitor
- ✗ Logs
- ✓ Automated Correlation Engine
 - ✓ Correlation Objects
 - ✓ Correlated Events
- ✓ Packet Capture
- ✓ App Scope
- ✓ Session Browser
- ✓ Botnet
- ✓ PDF Reports
 - ✓ Manage PDF Summary
 - ✓ PDF Summary Reports
 - ✓ User Activity Report

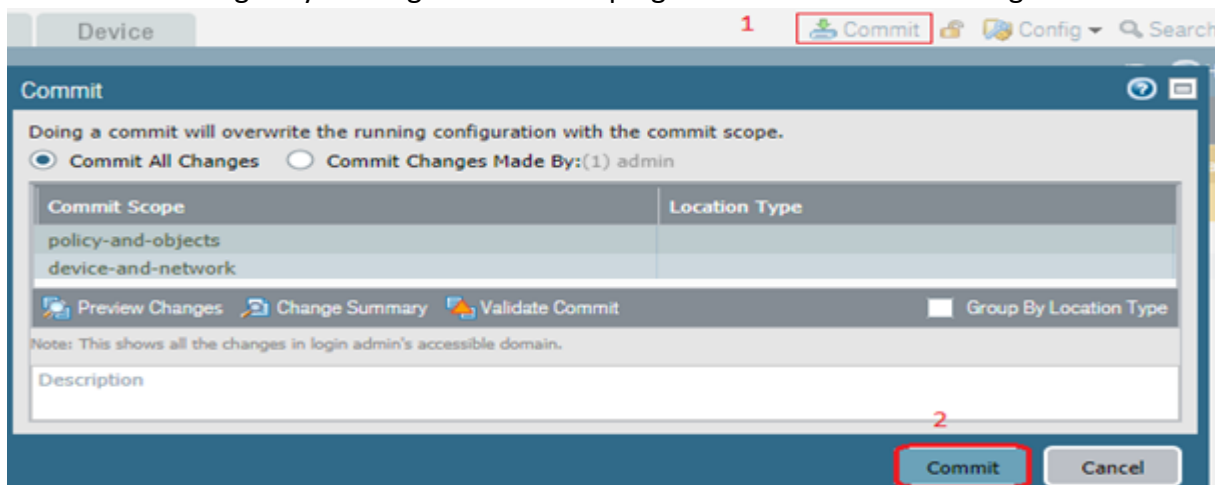
Legend: ✓ Enable 🔒 Read Only ✗ Disable

OK Cancel

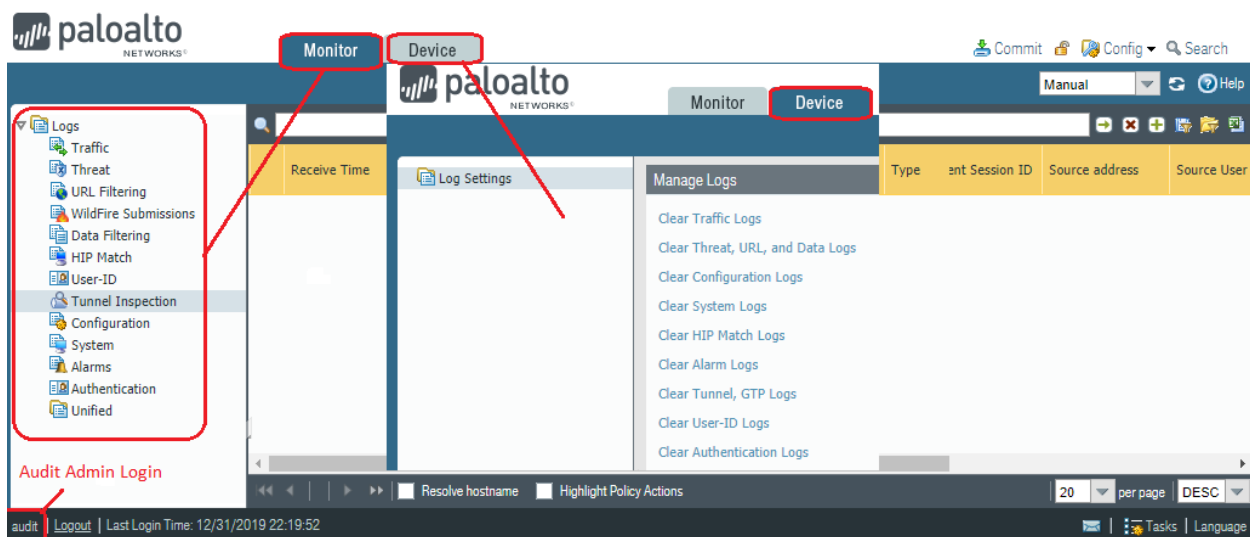
Administrator Name	Password	Admin Role
Audit	123	auditadmin
Crypto	123	cryptoadmin
security	123	securityadmin

☐	Name	Role	Profile	Locked User
☐	admin	Superuser		
☐	audit	Custom role-based administrator	auditadmin	
☐	crypto	Custom role-based administrator	cryptoadmin	
☒	security	Custom role-based administrator	securityadmin	

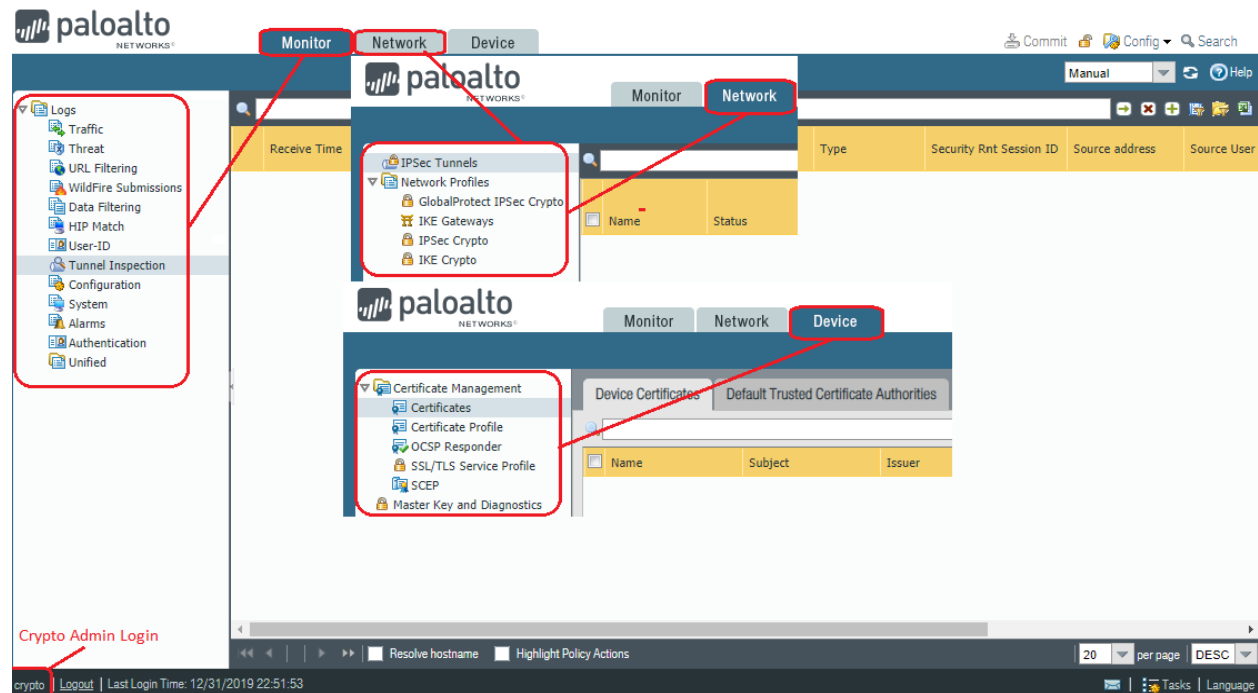
Commit the Changes by Clicking **Commit** on top right corner to save the configuration.



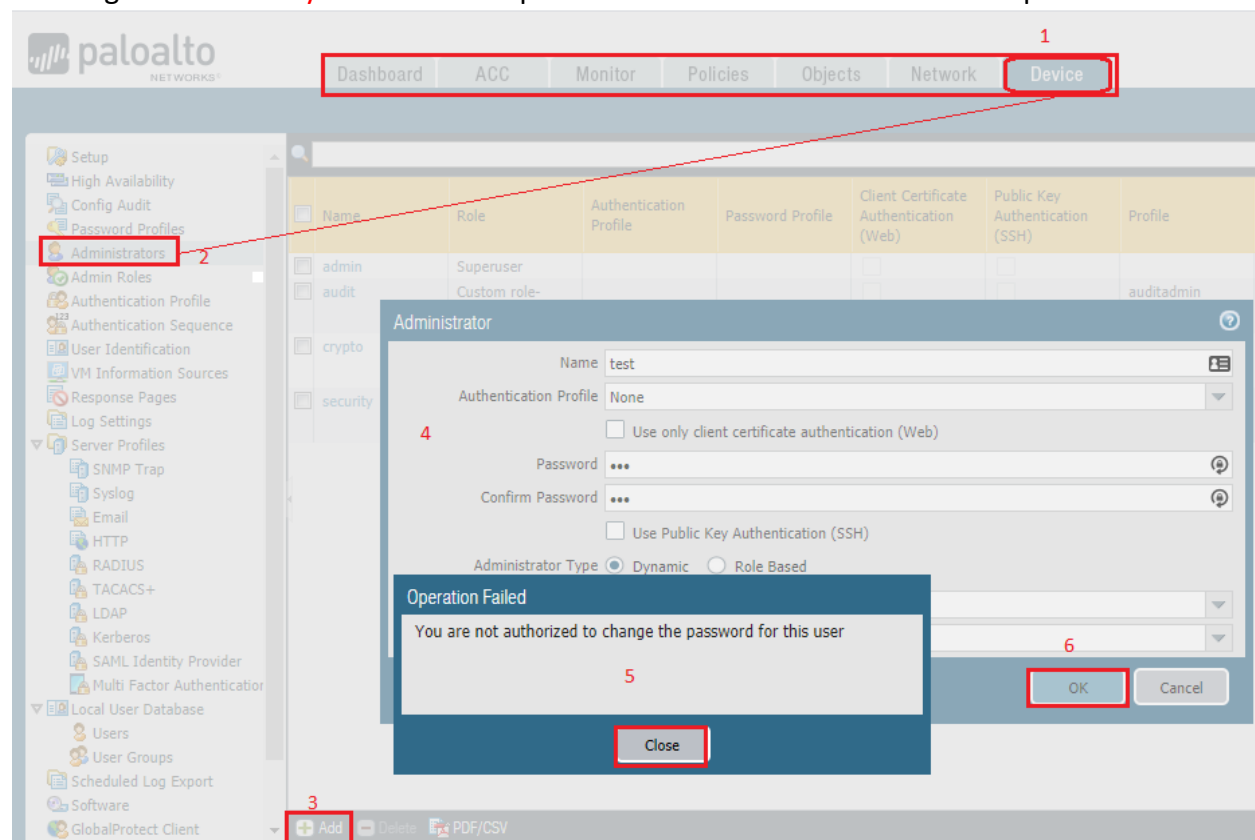
Let's login with **audit** admin can only see two tabs(Monitor & Device) with limited options.



Let's login with **crypto** admin can this time see three tabs(Monitor, Network & Device) with limited options related to logs and security.



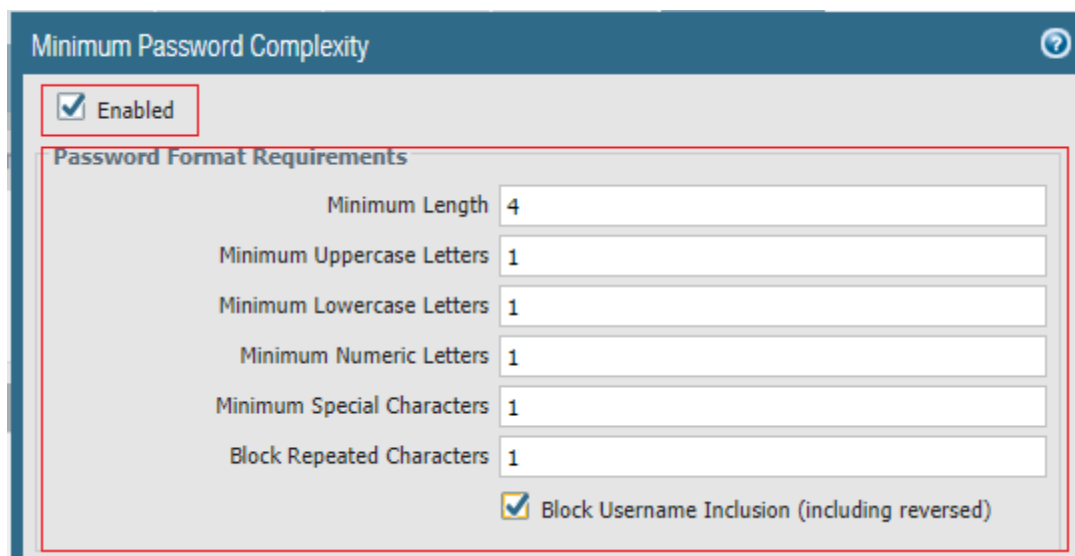
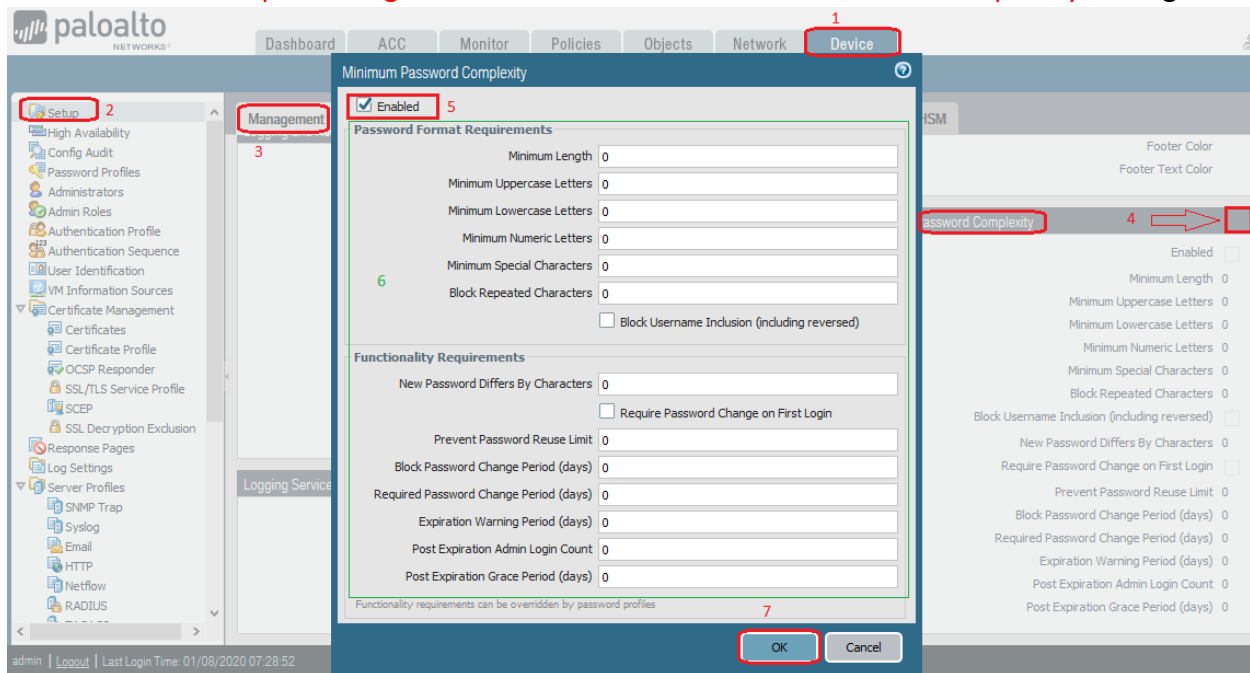
Let's login with **security** admin all the options are available but can set user & password.



Password Complexity:

- o To secure local Admin account, define password complexity requirements.
- o That are enforced when administrators change or create new passwords.
- o Unlike password profiles, which can be applied to individual accounts only.
- o The password complexity rules are firewall-wide and apply to all passwords.
- o This PA Feature Enable minimum password requirements for local accounts.
- o Ensure that local admin accounts follow to defined set of password requirements.

Go to **Device > Setup > Management** click and edit **Minimum Password Complexity** setting.



Options	Description
Enabled	Enable minimum password requirements for local accounts.
Minimum Length	Require minimum password length (range is 1 to 15 characters).
Minimum Uppercase Letters	Require a minimum number of uppercase letters (ranges is 0 to 15 characters).
Minimum Lowercase Letters	Require a minimum number of lowercase letters (range is 0 to 15 characters).
Minimum Numeric Letters	Require a minimum number of numeric letters (range is 0 to 15 numbers).
Minimum Special Characters	Require a minimum number of special (non-alphanumeric) characters (range is 0 to 15 characters).
Block Repeated Characters	Specify the number of sequential duplicate characters permitted in a password (range is 2 to 15).
Block Username Inclusion (including reversed)	Select this option to prevent the account username (or reversed version of the name) from being used in the password.

The screenshot shows the 'Administrator' console interface. In the background, the 'Name' field is set to 'testuser' and the 'Authentication Profile' is set to 'None'. The 'Password' and 'Confirm Password' fields are highlighted with a red box. Below them, the 'Password Requirements' section is also highlighted, showing a requirement for a 'Minimum Password Length (Count) 8'. In the foreground, an 'Operation Failed' dialog box is displayed with the message: 'Invalid Password. Password requirements are: minimum length=8'. The dialog box has a 'Close' button at the bottom.

Functionality Requirements

New Password Differs By Characters

0

☐ Require Password Change on First Login

Prevent Password Reuse Limit

0

Block Password Change Period (days)

0

Required Password Change Period (days)

0

Expiration Warning Period (days)

0

Post Expiration Admin Login Count

0

Post Expiration Grace Period (days)

0

Functionality requirements can be overridden by password profiles

OK

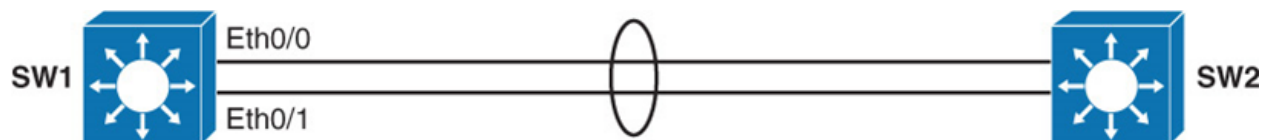
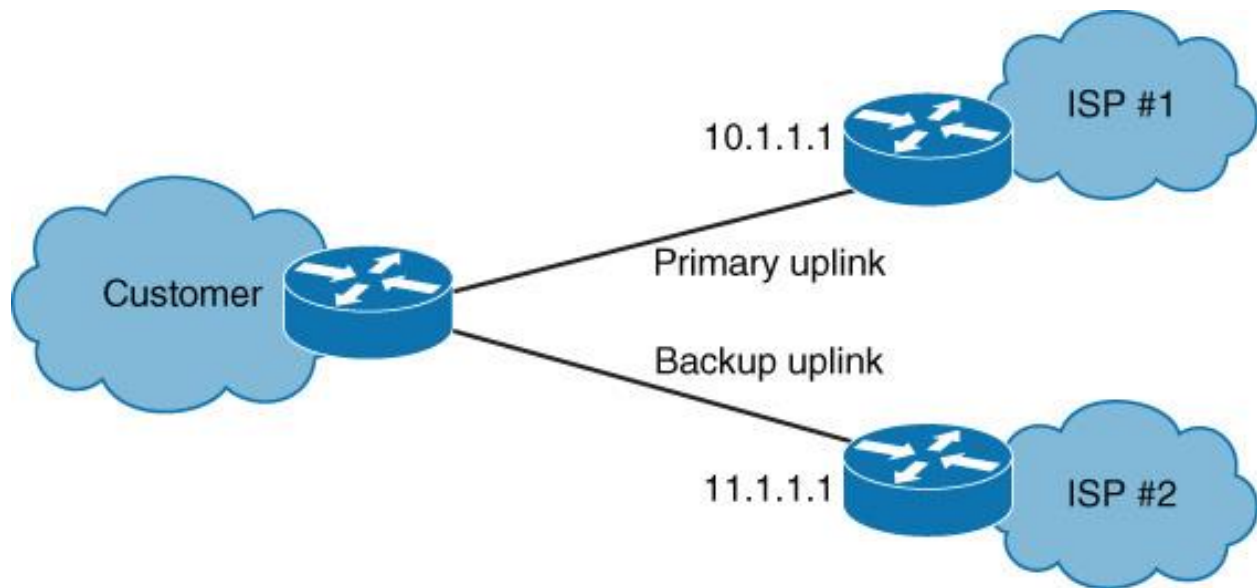
Cancel

Options	Description
New Password Differs By Characters	When administrators change their passwords, the characters must differ by the specified value.
Require Password Change on First Login	Select this option to prompt administrators to change their passwords the first time they log in to the firewall.
Prevent Password Reuse Limit	Require that a previous password is not reused based on the specified count.
Block Password Change Period (days)	User cannot change their passwords until the specified number of days is reached (range is 0 to 365 days).
Required Password Change Period (days)	Require that administrators change their password on a regular basis (in days) (range is 0 to 365).
Expiration Warning Period (days)	If a Required Password Change Period is set, you can use this Expiration Warning Period to prompt users at each log in to change their password when there are less than a specified number of days remaining before required change date (range is 0 to 30).
Post Expiration Admin Login Count (count)	Allow the administrator to log in a specified number of times after the required change date (range is 0 to 3).
Post Expiration Grace Period (days)	Allow the administrator to log in for a specified number of days after the account has expired (range is 0 to 30).

Redundancy:

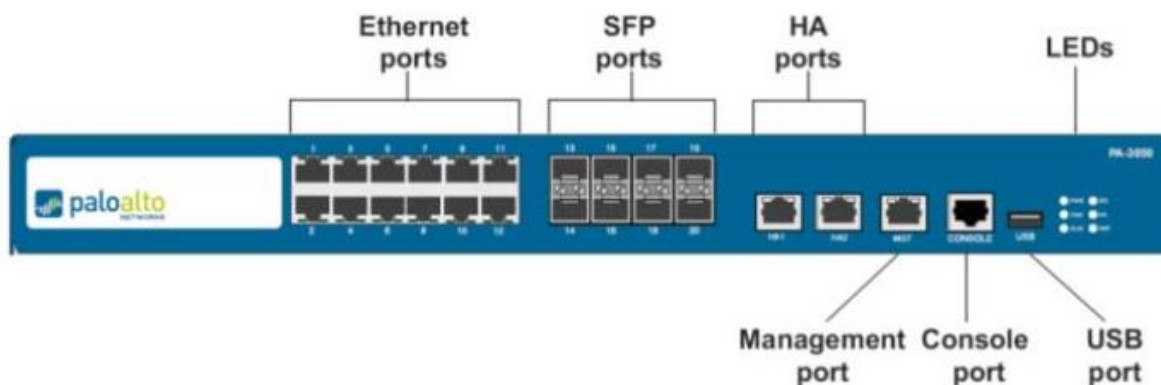
- o Redundancy, Failover, High Availability, Clustering, RAID and Fault-tolerance.
- o A good network design provides the redundancy in devices and network links.
- o Redundancy is basically extra hardware or software that can be used as backup.
- o If the main hardware or software fails or link fail or unavailable in case of emergency.
- o It is method for ensuring network availability in case of network device or path failure.
- o It is method for ensuring network availability in case of network device unavailability.
- o Network redundancy is process through which additional or alternate instances of network devices, equipment & communication mediums are installed within network infrastructure.
- o Redundancy can be achieved via failover, load balancing & high availability in automatically.
- o High availability is a feature which provides redundancy and fault tolerance automatically.
- o High Availability is a number of connected devices processing and providing a services.
- o The goal is to ensure this service is always available even in the event of a failure or down.
- o Clustering is similar to redundant servers & provides fault tolerance in case of emergency.
- o A group of servers are logically combined into a cluster and seen as one device to work.
- o If a device fails within cluster services continue because other devices continue services.
- o One link processing traffic & second link would only become active if primary link fails.
- o Set up to allow company to connect their device to more than one Internet connection.
- o If one connection goes down, all traffic would failover to the other Internet connection.
- o This would eliminate single point of failure and would re-assure availability and reliability.
- o RAID is a fault tolerance solution for hard drives usually implemented in the servers.
- o Redundant Array of Independent Disks providing redundancy and fault tolerance.
- o Automatic failover is process of moving active services from primary device to backup.
- o Usually backup device continues these services until primary device has come back up.
- o When a device fails another device takes over this process which is referred to as a failover.
- o Services failover to backup device which will continue from where primary device left off.
- o Failover feature allows for hardware firewalls to have some redundancy and backup.
- o Have two or more hardware device configured if primary fails, the backup take over.
- o It is implemented on the high-end hardware devices for networks require redundancy.
- o HSRP is a Cisco proprietary protocol for establishing a fault-tolerant default gateway.
- o Redundancy, Fault-tolerance, & High-availability, all refer to some sort of failover or backup.





High Availability Overview:

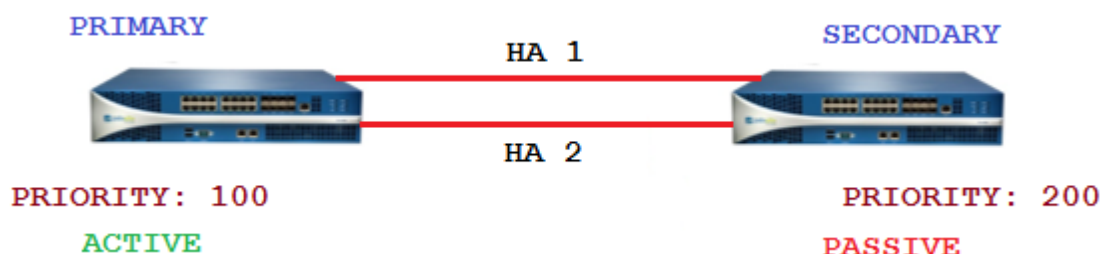
- o The High availability (HA) is a deployment in which two firewalls are placed in a group.
- o Their configuration is synchronized to prevent a single point of failure on your network.
- o Heartbeat connection between firewall peers ensures failover in event peer goes down.
- o Setting up two firewalls in an HA pair provides redundancy & ensure business continuity.
- o Firewalls in an HA pair use HA links to synchronize data and maintain state information.
- o Some models of Firewall have dedicated HA ports—Control link (HA1) & Data link (HA2).
- o While others Palo Alto Network Firewall require you to use the in-band ports as HA links.
- o Firewalls with dedicated HA ports such as PA-3000, PA-4000, PA-5000 & PA-7000 Series.
- o Use dedicated HA ports to manage communication & synchronization between firewalls.
- o For firewalls without dedicated HA ports such as the PA-200, PA-500, and PA-2000 Series.
- o Best practice use the management port for the HA1 link to allow for a direct connection.
- o And use Palo Alto Network Firewall an in-band port or links for the Data Link (HA2) link.



Palo Alto Firewall HA Modes:

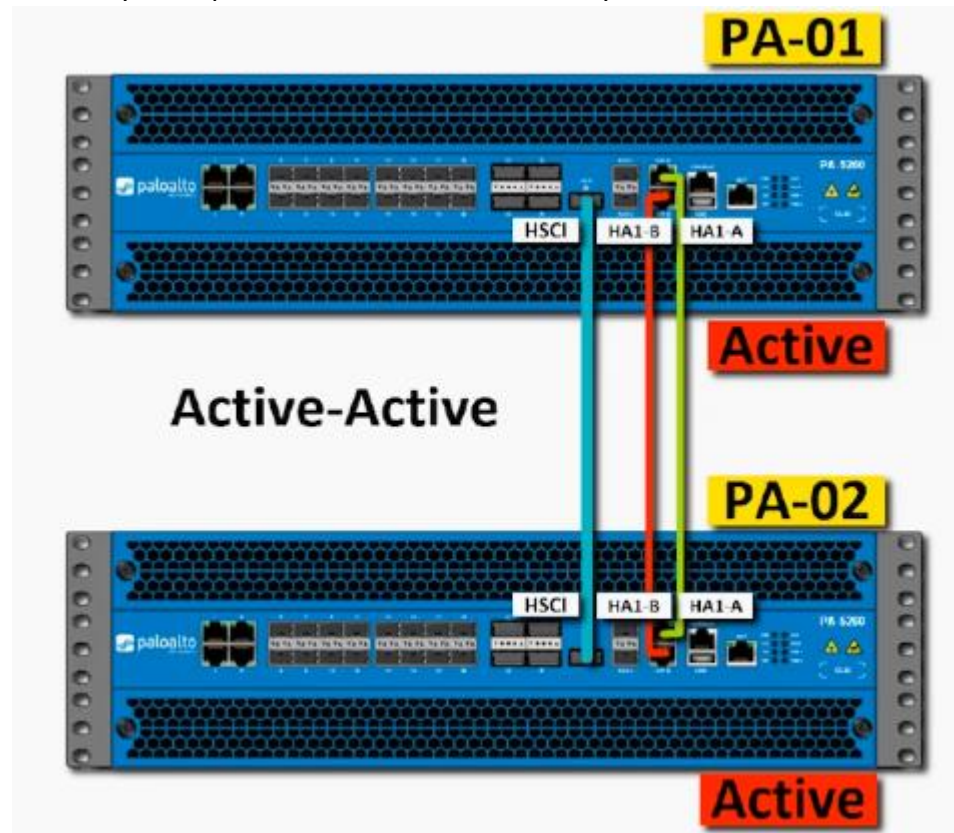
Active-Passive:

- o In Active-Passive one firewall actively manages traffic while other is synchronized.
- o In Active-Passive passive is ready to transition to active state, should a failure occur.
- o One actively manages traffic until a path, link, system, or network failure occurs.
- o When active firewall fails, passive firewall transitions to active state and takes over.
- o Active-passive HA is supported in the virtual wire, Layer 2, and Layer 3 deployments.
- o Active-Passive does not increase session capacity or network throughput in firewall.
- o Active-Passive has simple design concept, so it is easier to troubleshooting routing.



Active-Active:

- o Active-Active deployment, both firewalls in the pair are active and processing traffic.
- o Both firewalls individually maintain session & routing table & synchronize each other.
- o It is primarily designed to support the environments that require asymmetric routing.
- o Active-active PA Firewall HA is supported in virtual wire and Layer 3 deployments only.
- o An active-active Palo Alto Firewall mode configuration does not load-balance traffic.
- o Active-active mode requires advanced design concepts can result more complex.
- o Active-active mode is recommended if each firewall needs its own routing instances.
- o And you require full, real-time redundancy out of both PA firewalls all the time.



HA Pre-Requisite:

- o To set up High Availability HA on firewalls, need a pair of firewalls that meet following.
- o **The same model**—The PA firewalls in the pair must be of the same hardware model.
- o **The same PAN-OS version**—The firewalls must be running the same PAN-OS version.
- o And must each be up-to-date on the application, URL, and threat databases the same.
- o To setup HA in Active-Active & Active-Passive mode the **same type of interfaces** require.
- o Dedicated HA links, or combination of the management port and in-band ports type HA.
- o HA interfaces must be configured with **static IP addresses** only not IP addresses from DHCP.
- o **Licenses** are unique to each firewall & cannot be shared between firewalls same set require.

High Availability Links:

HA1 (Control Link):

- o The HA1 link is used to exchange hellos, heartbeats, and the HA state information.
- o The HA1 link is used to exchange management plane sync for routing & User-ID info.
- o HA1 acts to monitor HA status such configuration synchronization for active-passive.
- o HA1 acts keepalive between HA agents, it senses power cycle, reboot & power down.
- o The PA firewalls also use this link to synchronize configuration changes with its peer.
- o HA1 link is a Layer 3 link and the only HA link that requires an IP address information.
- o Internet Control Message Protocol is used to exchange heartbeats between HA peers.
- o Ports used for HA1 link—TCP port 28769 and TCP 28260 for clear text communication.
- o Port used for HA1 link- Port 28 for encrypted communication Secure Shell over TCP.
- o Default monitor hold time is 3000 ms & HA1 link also called Control or management.

HA2 (Data Link):

- o HA2 link is used to synchronize sessions, forwarding tables, IPsec security associations.
- o The HA2 link is also used to synchronize ARP tables between PA firewalls in the HA pair.
- o HA2 is used to synchronize HA states, routing info, IPsec security association, ARP table.
- o Data flow on the HA2 or data link is always unidirectional except for the HA2 keep-alive.
- o It flows from active or active-primary firewall to the passive or active-secondary firewall.
- o The Data link are unidirectional and flows from the active firewall to the passive firewall.
- o HA2 is layer 2 link, no IP address is required although you can specify layer3 information.
- o A layer 3 link or IP address is required only if the data link are not on the same subnet.

HA Backup Links:

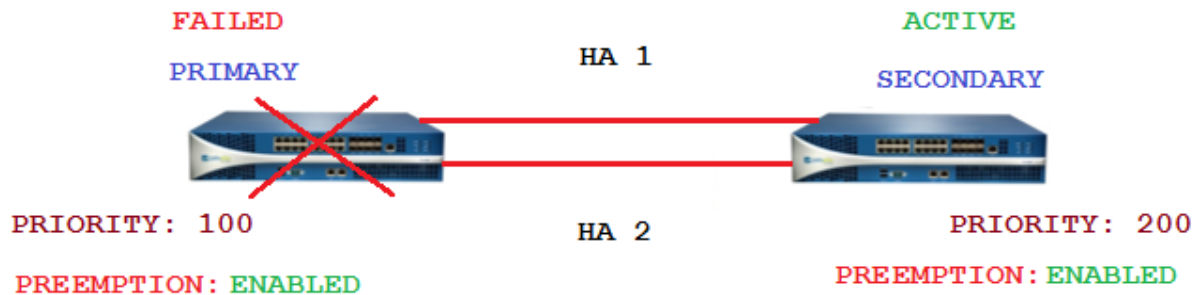
- o In High Availability HA backup links provide redundancy for the control and data links.
- o Backup and primary links need to be on a different subnet from the primary HA Links.
- o The HA1 backup ports and HA2 ports must be configured on separate physical ports.
- o The ip addresses of the primary and the backup HA links must not overlap each other.
- o The purpose of configuration a backup control link is to avoid the split-brain scenario.
- o The Split-Brain occurs when a non-redundant control link or HA1 link goes down.
- o Passive firewall concludes that active firewall is down and attempts to start services.



HA Terminologies:

Failover:

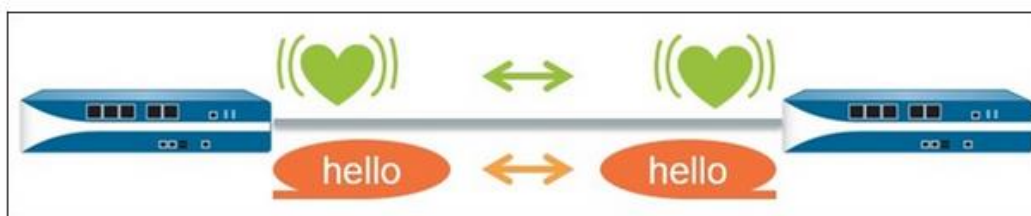
- o When a failure occurs on one firewall and the peer takes over the task of securing traffic.
- o Procedure by which firewall automatically transfers control to peer when it detects a fault.
- o The failover operation is the process of switching production to a backup facility or firewall.
- o A failover is triggered, for example, when a monitored metric on the firewall in HA pair fails.



Heartbeat Polling and Hello messages:

- o Hello Messages, are sent from one peer to the other to verify the state of the firewall.
- o The Heartbeat is an ICMP ping to the HA peer over the Control Link or Management Link.
- o Firewalls use hello message and heartbeats to verify that the peer firewall is responsive.
- o Firewalls use hello message and heartbeats to verify that the peer firewall is operational.
- o Hello messages are sent from one peer to other at the configured Hello Interval to verify.
- o The heartbeat is an Internet Control Message Protocol ping to HA peer over control link.
- o The peer responds to the ping to establish that the firewalls are connected and responsive.
- o By default, In Palo Alto Network Firewall the interval for the heartbeat is 1000 milliseconds.
- o Ping is sent every 1000ms & if there are three consecutive heartbeat losses failovers occurs.

The **heartbeat** is an ICMP ping to the HA peer over the Control link, and the peer responds to establish that the firewalls are connected and responsive.



Hello messages are sent from one peer to the other to verify the state of the firewall.

Link Monitoring:

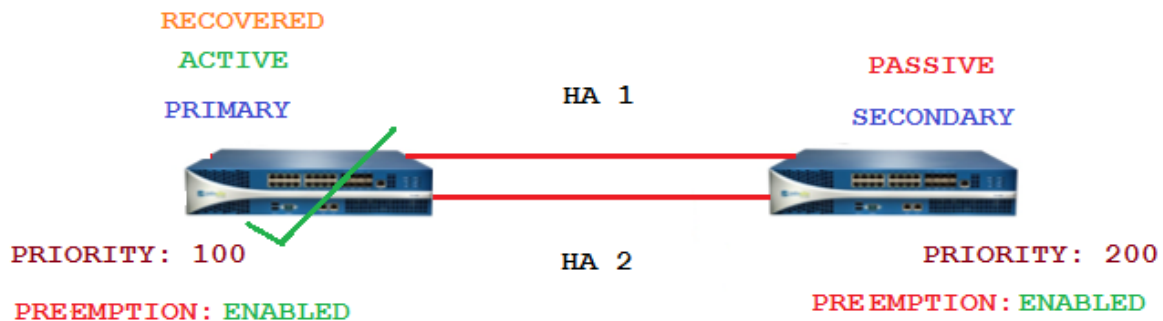
- o Physical interfaces to be monitored are grouped into a link group and their state is monitored.
- o Palo Alto Network Firewall, link group can contain one or more physical interfaces or links.
- o A PA Firewall failure is triggered when any or all of the interfaces or link in the group fail.
- o Default behavior is failure of any one link in the link group will cause the firewall to change.
- o The High Availability (HA) state to non-functional to indicate a failure of a monitored object.

Path Monitoring:

- o Path Monitoring monitors full path through the network to mission-critical IP addresses.
- o Internet Control Message Protocol pings are used to verify reachability of the IP address.
- o In Palo Alto Next Generation Network Firewall, the default interval for the pings is 200ms.
- o The IP address is considered unreachable when 10 consecutive pings the default value fail.
- o Firewall failure is triggered when any or all of IP addresses monitored become unreachable.
- o Default behavior is any one of the IP addresses becoming unreachable will cause firewall.
- o To change High Availability state to non-functional to indicate failure of monitored object.

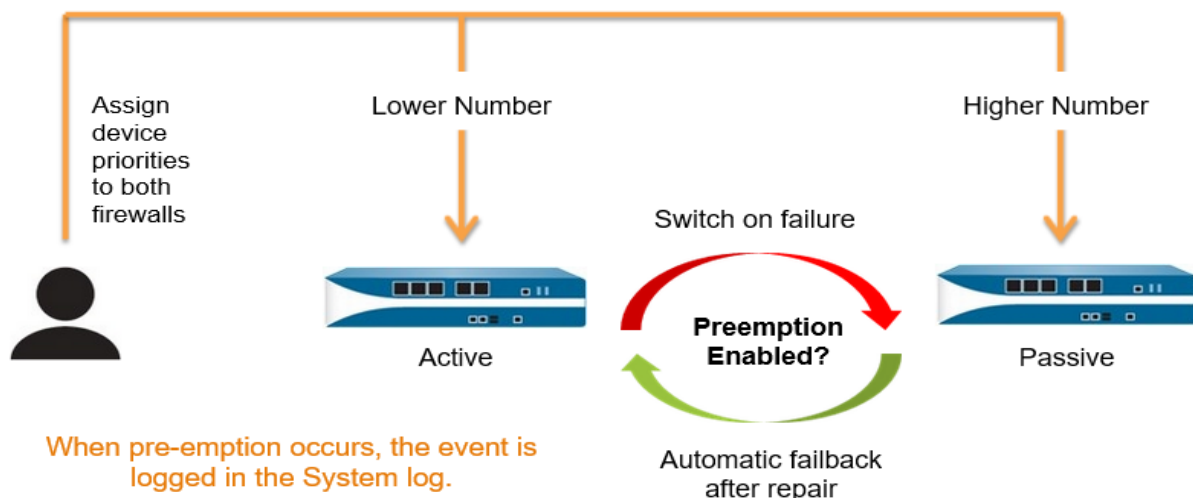
Priority:

- o When two Palo Alto Networks firewalls are deployed in the active-passive cluster.
- o It is mandatory to configure device priority higher priority for passive low for active.
- o Firewall with lower numerical value & therefore higher priority, is designated as active.
- o The device priority decides which Palo Alto firewall will preferably take the active role.
- o Which Palo Alto firewall will take over the passive role when both the firewalls boot up.

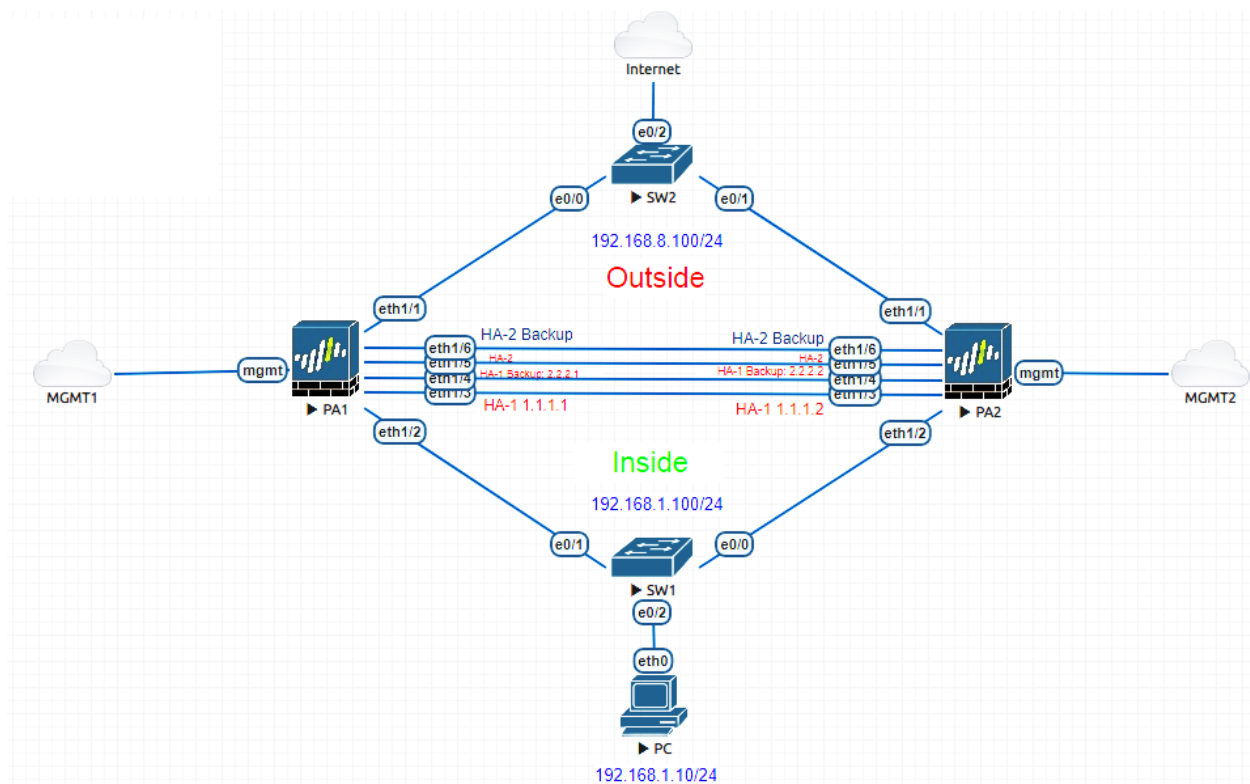


Preemption:

- o The Preemptive behavior allows firewall with lower numerical value to resume as active.
- o By default, preemption is disabled on the firewalls and must be enabled on both firewalls.
- o Preemption which influences this behavior on the event of it being enabled or disabled.
- o When preemption occurs, event is logged in the in Palo Alto Network Firewall system logs.



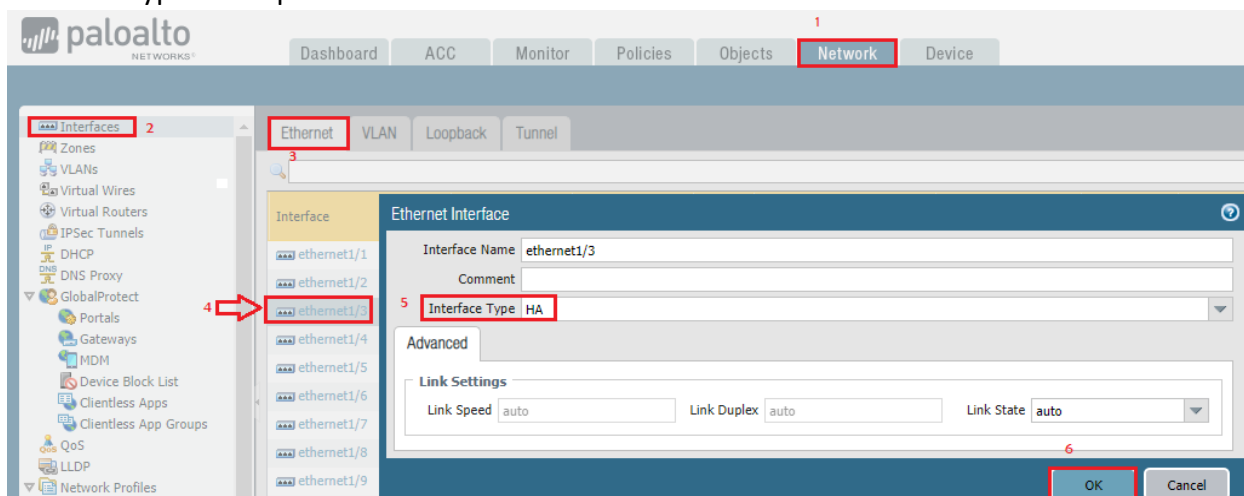
Active-Passive Lab:



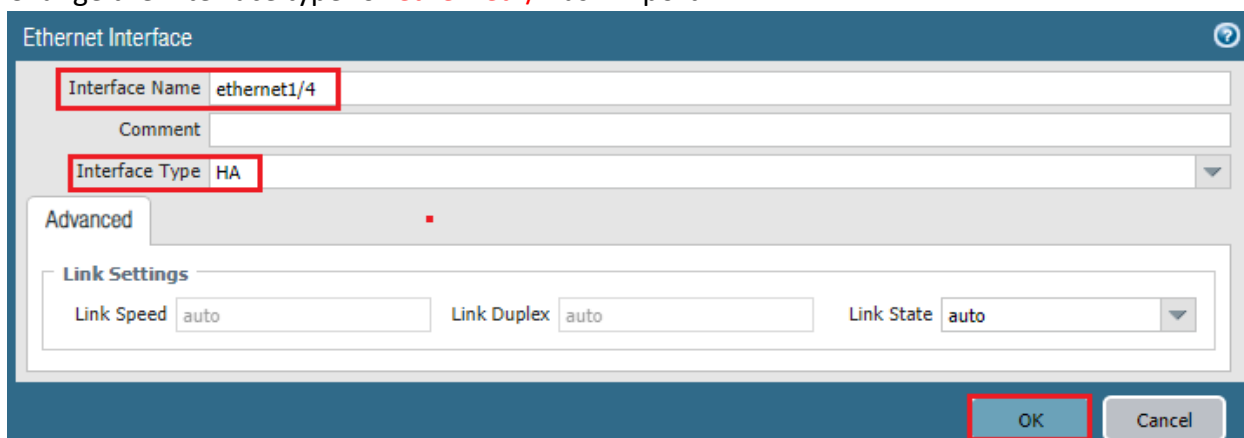
PA1 (Active) IP Schema	
Outside Layer 3 Interface	Ethernet1/1 – 192.168.8.100/24
Inside Layer 3 Interface	Ethernet1/2 – 192.168.1.100/24
HA1 or Control Link	Ethernet1/3 – 1.1.1.1/24
HA1 or Control Link Backup	Ethernet1/4 – 2.2.2.1/24
HA2 or Data Link	Ethernet1/5 – Layer 2
HA1 or Data Link Backup	Ethernet1/6 – Layer 2
PA2 (Passive) IP Schema	
Outside Layer 3 Interface	Ethernet1/1 – 192.168.8.100/24
Inside Layer 3 Interface	Ethernet1/2 – 192.168.1.100/24
HA1 or Control Link	Ethernet1/3 – 1.1.1.2/24
HA1 or Control Link Backup	Ethernet1/4 – 2.2.2.2/24
HA2 or Data Link	Ethernet1/5 – Layer 2
HA1 or Data Link Backup	Ethernet1/6 – Layer 2
LAN PC Details	
LAN PC IP	192.168.1.10/24
LAN PC Default Gateway	192.168.1.100/24
LAN PC DNS	8.8.8.8
Firewall Management IP subnet	192.168.8.0/24

PA1-HA Ports:

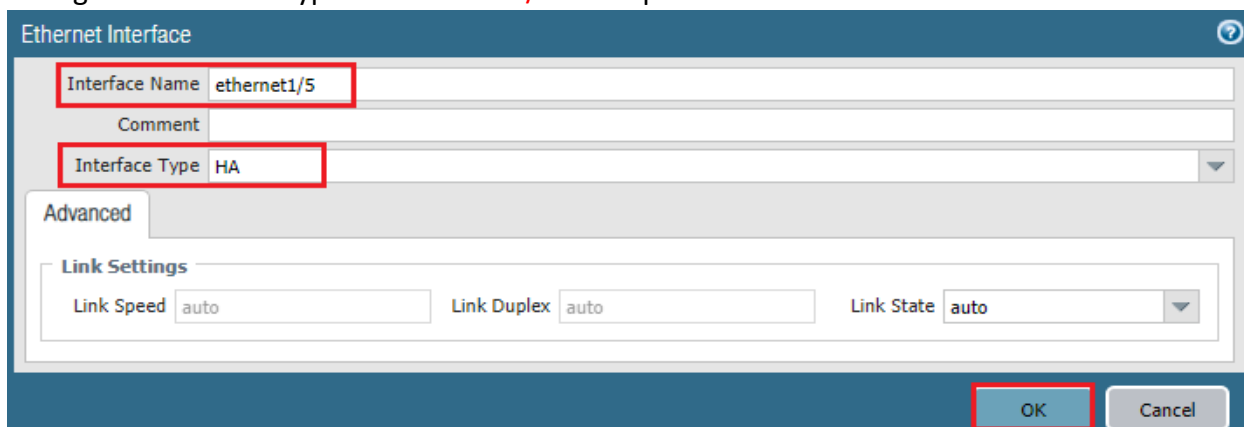
We do not have any dedicated HA1 and HA2 primary and backup ports. So, we are going to make ethernet1/3 as HA1, ethernet1/4 as HA1 backup, ethernet1/5 as HA2 and ethernet1/6 as HA2 backup. To do this, go – **Network >> Interface >> Ethernet**. And, then need to change the interface type to HA port.



Change the interface type for **ethernet1/4** as HA port.



Change the interface type for **ethernet1/5** as HA port.



Change the interface type for **ethernet1/6** as HA port.

Ethernet Interface

Interface Name: **ethernet1/6**

Comment:

Interface Type: **HA**

Advanced

Link Settings

Link Speed: **auto** Link Duplex: **auto** Link State: **auto**

OK **Cancel**

Finally, all four-interface type has been changed to HA mode.

Interface	Interface Type	Management Profile	Link State	IP Address	Virtual Router	Tag	VLAN / Virtual-Wire	Security Zone
ethernet1/1			none	none	none	Untagged	none	none
ethernet1/2			none	none	none	Untagged	none	none
ethernet1/3	HA		none	none	none	Untagged	none	none
ethernet1/4	HA		none	none	none	Untagged	none	none
ethernet1/5	HA		none	none	none	Untagged	none	none
ethernet1/6	HA		none	none	none	Untagged	none	none

PA1-Group Configuration:

Enable HA, add Group ID and put Peer HA1 IP Address. Below are the configuration of PA1 firewall. Select **Device>>High Availability>>General** and edit the **Setup** section.

paloalto NETWORKS

Dashboard ACC Monitor Policies Objects Network **Device**

Setup

High Availability

General Link and Path Monitoring

Setup

☒ **Enable HA**

Group ID: **10**

Description: **HA-Config-Active**

Mode: ☒ **Active Passive** ☐ Active Active

☒ **Enable Config Sync**

Peer HA1 IP Address: **1.1.1.2**

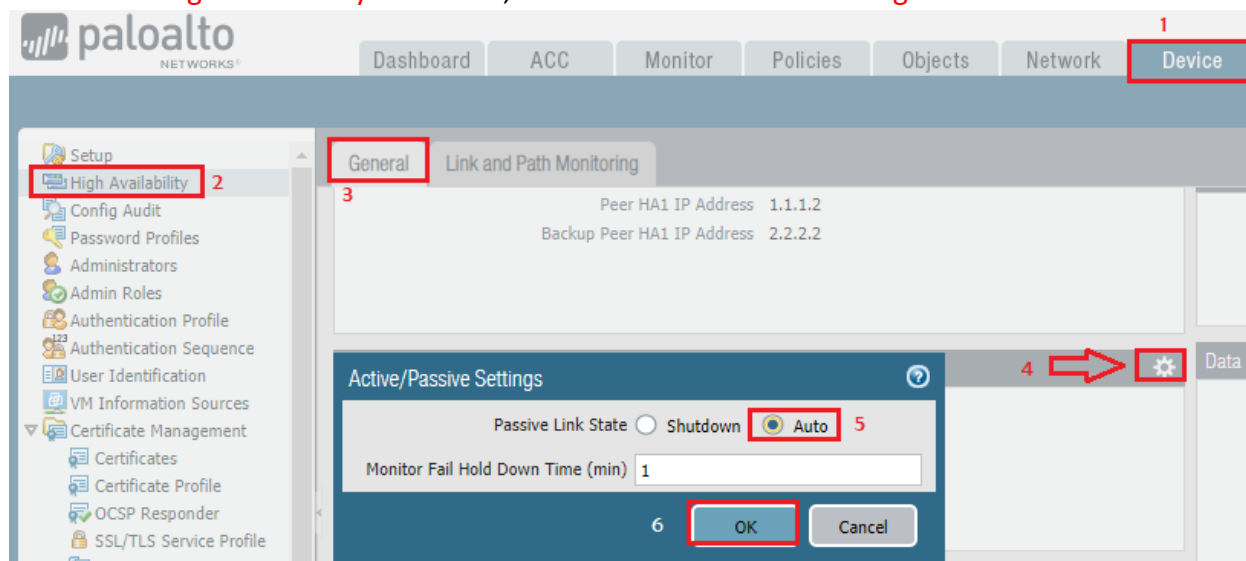
Backup Peer HA1 IP Address: **2.2.2.2**

OK **Cancel**

Settings	Description
Enable HA	Activate HA functionality.
Group ID	Enter a number to identify the HA pair (1 to 63).
Description	Enter a description of the HA pair.
Mode	Set the type of HA deployment: Active-Passive or Active-Active.
Device ID	In active/active configuration, set the Device ID to determine which peer will be active-primary (set Device ID to 0) and which will be active-secondary (set the Device ID to 1).
Enable Config Sync	To enable synchronization of configuration settings.
Peer HA1 IP Address	Enter IP address of the HA1 interface of the peer firewall.
Backup Peer HA1 IP Address	Enter IP address for the peer's backup control link.

PA1-Active-Passive Settings:

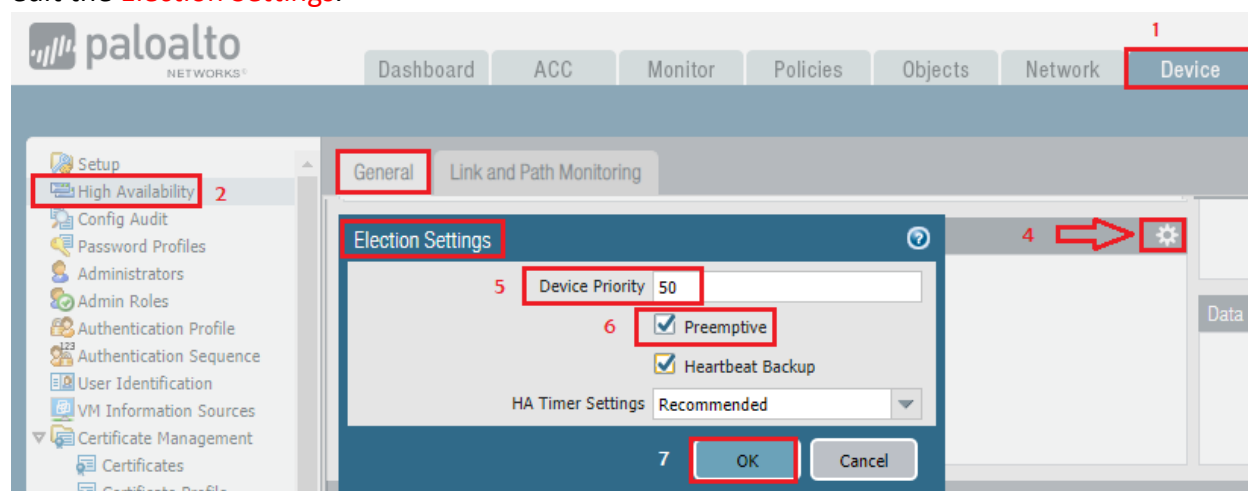
In **Device>>High Availability>>General**, edit the **Active Passive Settings**.



Settings	Description
Passive Link State	Select one of the following options to specify whether the data links on the passive firewall should remain up.
auto	The links that have physical connectivity remain physically up but in a disabled state.
shutdown	Forces the interface link to the down state. This is the default option, which ensures that loops are not created in the network.
Monitor Fail Hold Down Time (min)	This value between 1-60 minutes determines the interval in which a firewall will be in a non-functional state before becoming passive.

PA1-Priority and Preemption:

Add device priority to prefer PA1 as Active unit. And also, preemption will be enabled to make sure whenever PA1 firewall is up and running, it handles the traffic. The firewall with the lower value will be Active and other firewall is Passive firewall. In **Device>>High Availability>>General**, edit the **Election Settings**.



Settings	Description
Device Priority	Enter a priority value to identify the active firewall. The firewall with the lower value (higher priority) becomes the active firewall (range is 0–255).
Preemptive	Enables the higher priority firewall to resume active (active/passive) or active-primary (active/active>) operation after recovering from a failure.
Heartbeat Backup	Uses the management ports on the HA firewalls to provide a backup path for heartbeat and hello messages.
HA Timer Settings	Recommended: Use for typical failover timer settings. Aggressive: Use for faster failover timer settings. Advanced: Allows to customize values to suit network requirement.

PA1-Control Plane Configuration:

Go **Device >> High Availability**. Now, by clicking on top right gear icon in **Control Link (HA1)** section, we will declare ethernet1/3 as our control plane link (HA1) as we decided earlier. We will be using 1.1.1.0/24 for HA1 link. Below are the configuration of Active and Passive nodes. It's Point-to-Point, so we do not need any gateway here.

Settings	Description
Port	Select the HA port for the primary interfaces.
IPv4/IPv6 Address	Enter the IPv4 or IPv6 address of the HA1 interface for the primary.
Netmask	Enter the network mask for the IP address for the primary
Gateway	Enter the IP address of the default gateway for the primary.

PA1-Control Plane Backup Configuration:

Go **Device >> High Availability**. Now, clicking on top right gear icon in **Control Link (HA1 Backup)** section, we will declare ethernet1/4 as our control plane link (HA1 Backup) as we decided earlier. We will be using 2.2.2.0/24 for HA1 backup link.

Settings	Description
Port	Select the HA port for the backup interfaces.
IPv4/IPv6 Address	Enter the IPv4 or IPv6 address of the HA1 interface for backup.
Netmask	Enter the network mask for the IP address for the Backup.
Gateway	Enter the IP address of the default gateway for the backup.

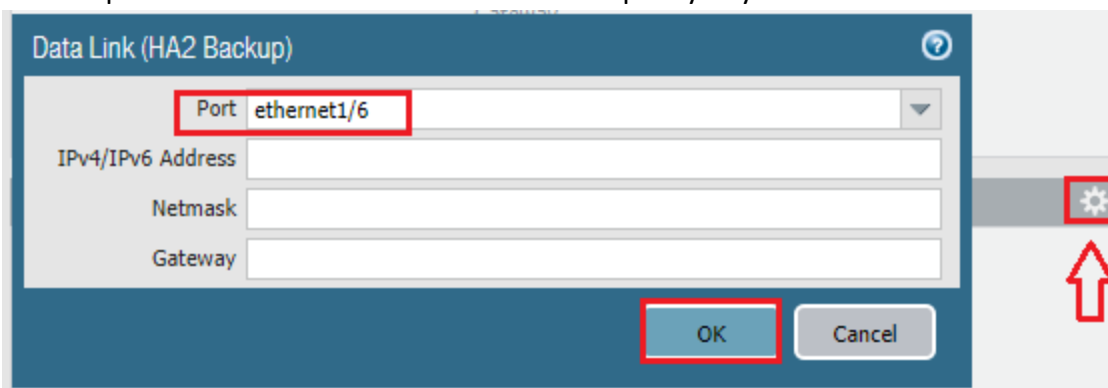
PA1-Data Link Configuration:

On the same page **Device >> High Availability**, need to click on top right gear icon in **Data Link (HA2)** section. In our case, ethernet1/5 is our HA2 link. It's directly connected, so transport mode is **ethernet**. Do not need to specify any IP address.

Settings	Description
Enable Session Synchronization	Enable synchronization of the session information with the passive firewall and choose a transport option.
Port	Select HA port Configure this setting for the primary HA2 interfaces.
IPv4/IPv6 Address	Specify the IPv4 or IPv6 address of the HA interface for the primary.
Netmask	Enter the network mask for the IP address for the Primary.
Gateway	Enter the IP address of the default gateway for the primary.
Transport	Ethernet: Use when the firewalls are connected back-to-back or through a switch.
HA2 keep-alive	If enabled, the peers will use keep-alive messages to monitor the HA2 connection to detect a failure based on the Threshold you set (default is 10,000 ms).
Log Only	Logs the failure of the HA2 interface in the system log as a critical event.
Split Datapath	this option in active/active HA deployments to instruct each peer to take ownership of their local state and session tables when it detects an HA2 interface failure.
Threshold (ms)	The duration in which keep-alive messages have failed before one of the above actions will be triggered

PA1-Data Link Backup Configuration:

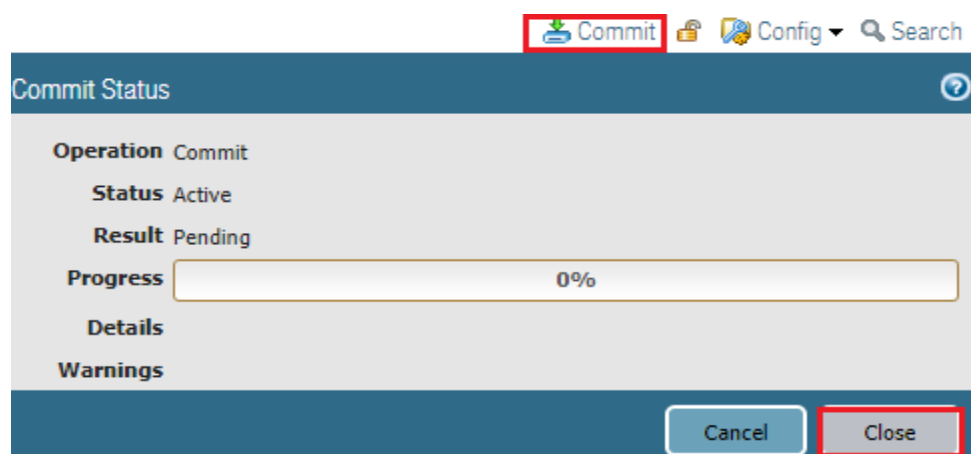
On the same page **Device >> High Availability**, need to click on top right gear icon in **Data Link (HA2 Backup)** section. In our case, ethernet1/6 is our HA2 backup link. It's directly connected, so transport mode is **ethernet**. Do not need to specify any IP address.

A screenshot of the 'Data Link (HA2 Backup)' configuration window. The window has a title bar with a question mark icon. Inside, there are four input fields: 'Port' (with 'ethernet1/6' selected and highlighted by a red box), 'IPv4/IPv6 Address', 'Netmask', and 'Gateway'. At the bottom, there are 'OK' and 'Cancel' buttons, with the 'OK' button highlighted by a red box. To the right of the window, there is a gear icon and a red arrow pointing upwards, both highlighted by red boxes.

Settings	Description
Port	Select the HA port for the backup interfaces.
IPv4/IPv6 Address	Enter the IPv4 or IPv6 address of the HA1 interface for backup.
Netmask	Enter the network mask for the IP address for the Backup.
Gateway	Enter the IP address of the default gateway for the backup.

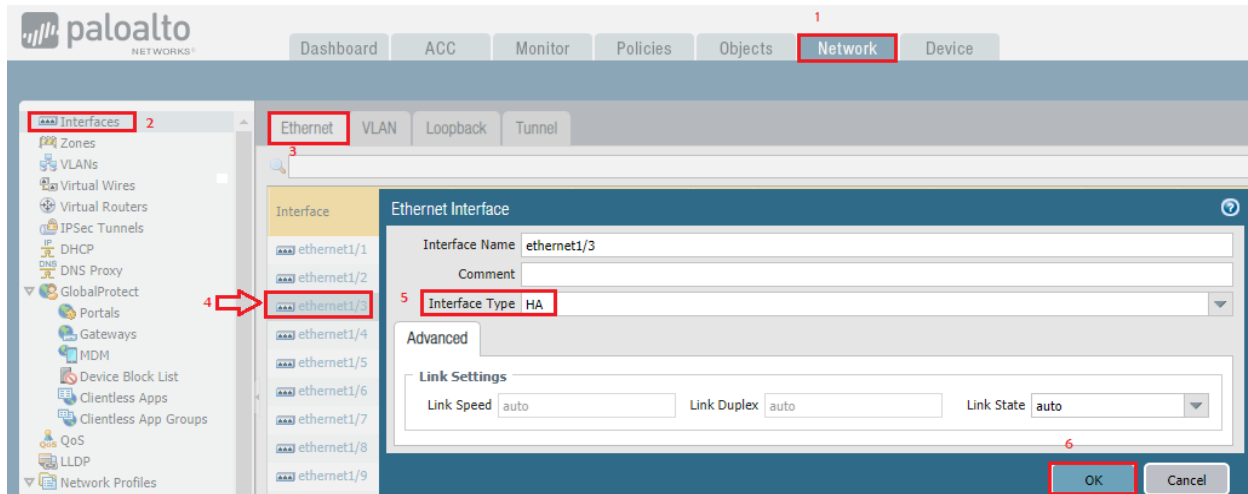
PA1-Save Changes:

Click on top right corner Commit link to save the changes.

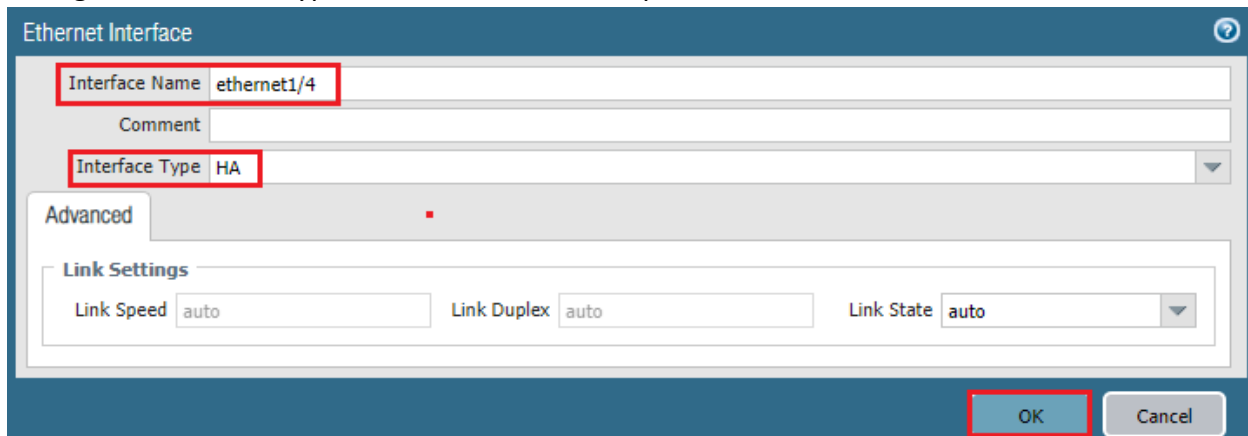
A screenshot of the 'Commit Status' dialog box. The dialog box has a title bar with a question mark icon. Inside, there are several sections: 'Operation' (Commit), 'Status' (Active), 'Result' (Pending), 'Progress' (0%), 'Details', and 'Warnings'. At the bottom, there are 'Cancel' and 'Close' buttons, with the 'Close' button highlighted by a red box. Above the dialog box, there is a 'Commit' button with a green checkmark icon, highlighted by a red box.

PA2-HA Ports:

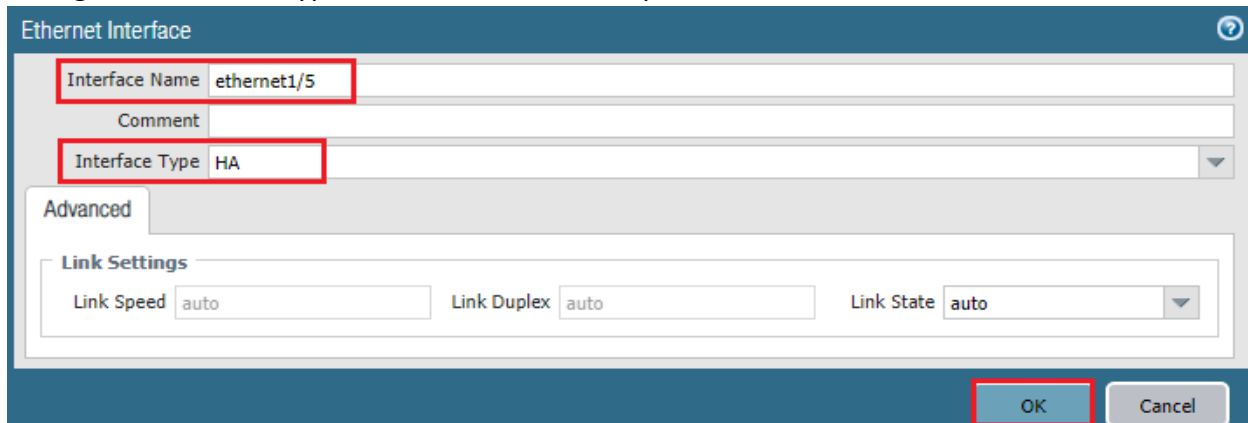
Going to make ethernet1/3 as HA1, ethernet1/4 as HA1 backup, ethernet1/5 as HA2 and ethernet1/6 as HA2 backup. To do this, go – **Network >> Interface >> Ethernet**. And, then need to change the interface type to HA port.



Change the interface type for **ethernet1/4** as HA port.



Change the interface type for **ethernet1/5** as HA port.



Change the interface type for **ethernet1/6** as HA port.

Ethernet Interface

Interface Name: ethernet1/6

Comment:

Interface Type: HA

Advanced

Link Settings

Link Speed: auto Link Duplex: auto Link State: auto

OK Cancel

Finally, all four-interface type has been changed to HA mode.

Interface	Interface Type	Management Profile	Link State	IP Address	Virtual Router	Tag	VLAN / Virtual-Wire	Security Zone
ethernet1/1			none	none	none	Untagged	none	none
ethernet1/2			none	none	none	Untagged	none	none
ethernet1/3	HA		none	none	none	Untagged	none	none
ethernet1/4	HA		none	none	none	Untagged	none	none
ethernet1/5	HA		none	none	none	Untagged	none	none
ethernet1/6	HA		none	none	none	Untagged	none	none

PA2-Group Configuration:

Enable HA, add Group ID and put Peer HA1 IP Address. Below are the configuration of PA1 firewall. Select **Device>>High Availability>>General** and edit the **Setup** section.

General Link and Path Monitoring

Setup

☒ Enable HA

Group ID: 10

Description: HA-Cofnig-Passive

Mode: ☒ Active Passive ☐ Active Active

☒ Enable Config Sync

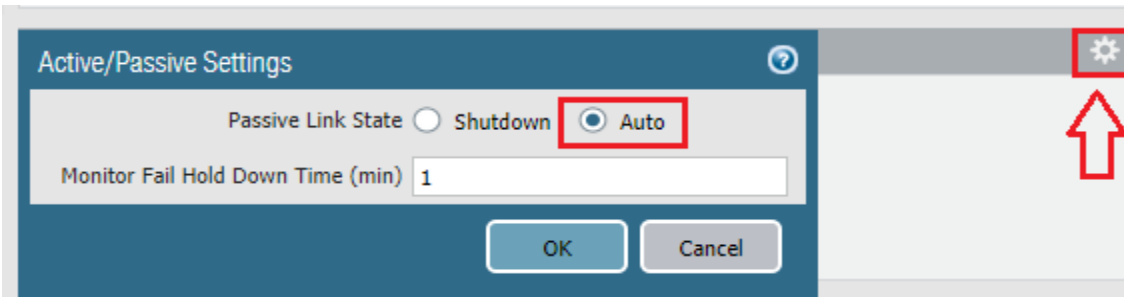
Peer HA1 IP Address: 1.1.1.1

Backup Peer HA1 IP Address: 2.2.2.1

OK Cancel

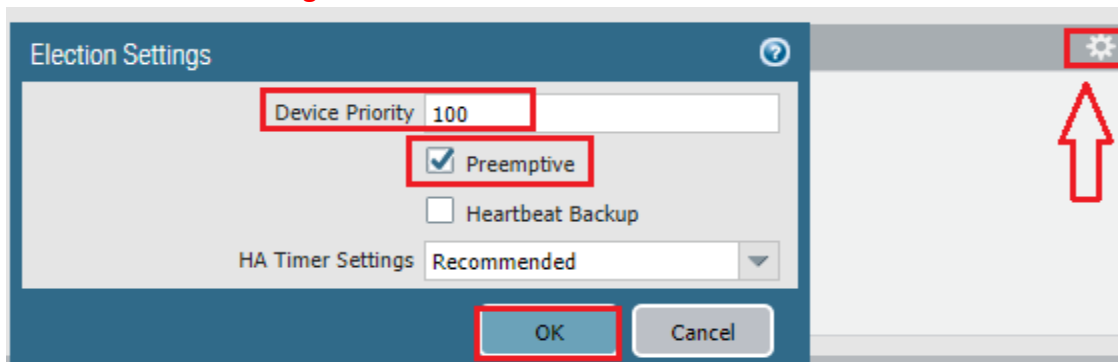
PA2-Active-Passive Settings:

In **Device>>High Availability>>General**, edit the **Active Passive Settings**.



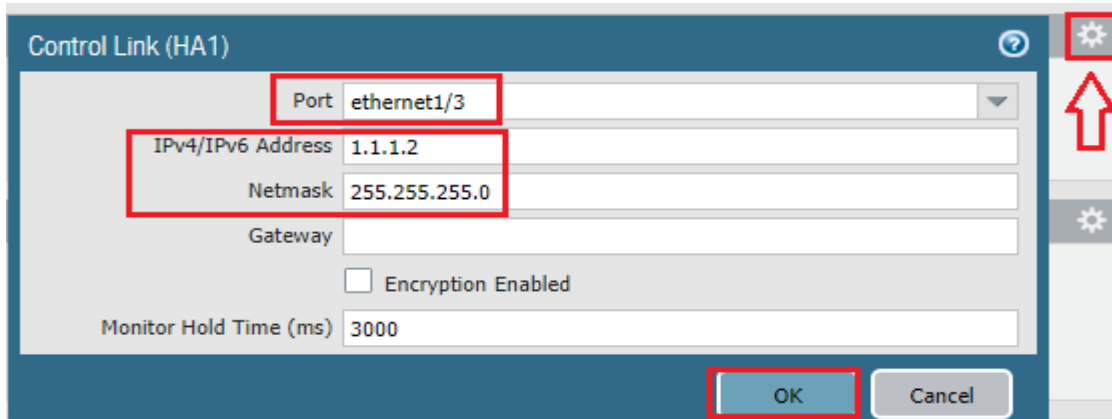
PA2-Priority and Preemption:

Add device priority to prefer PA1 as Active unit. And also, preemption will be enabled to make sure whenever PA1 firewall is up and running, it handles the traffic. The firewall with the lower value will be Active and other firewall is Passive firewall. In **Device>>High Availability>>General**, edit the **Election Settings**.



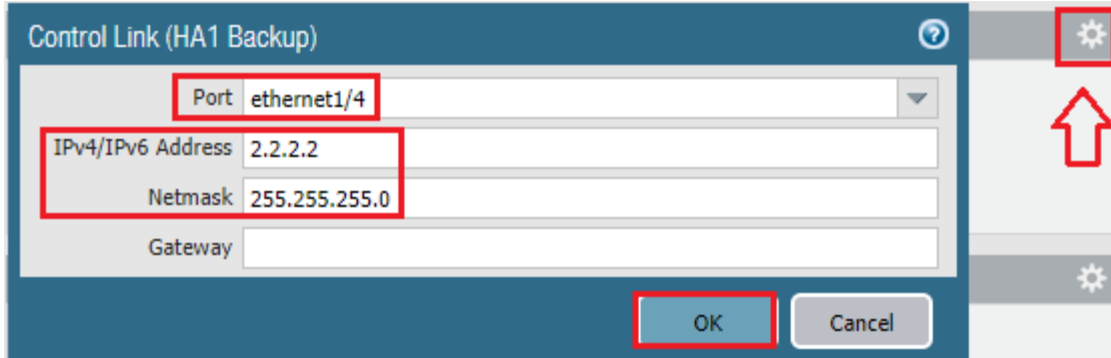
PA1-Control Plane Configuration:

Go **Device >> High Availability**. Now, by clicking on top right gear icon in **Control Link (HA1)** section, we will declare ethernet1/3 as our control plane link (HA1) as we decided earlier. We will be using 1.1.1.0/24 for HA1 link. Below are the configuration of Active and Passive nodes. It's Point-to-Point, so we do not need any gateway here.



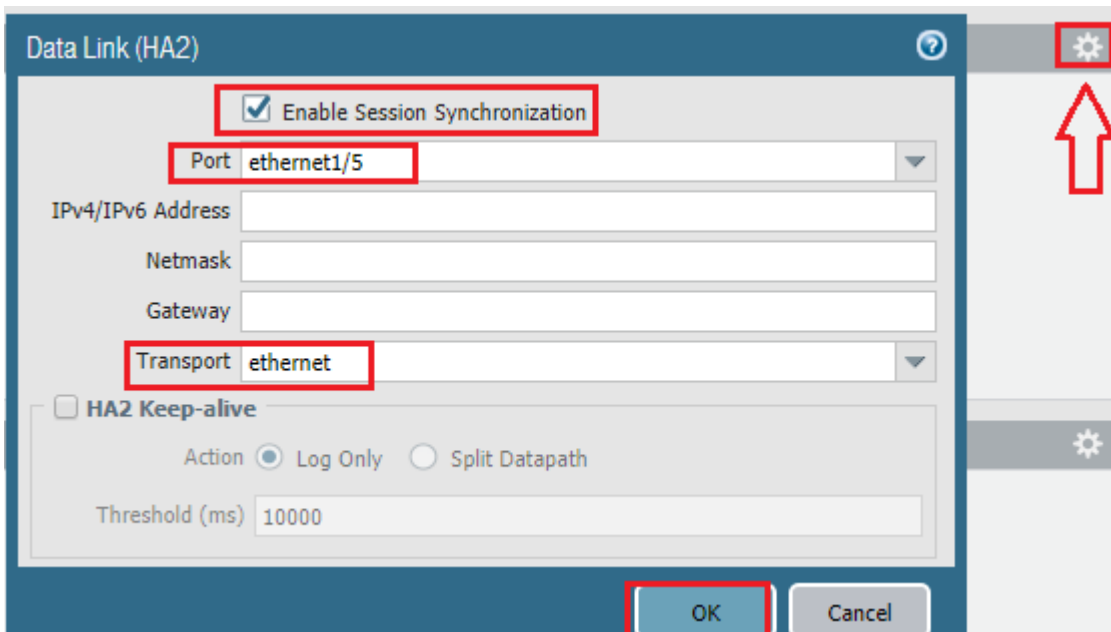
PA2-Control Plane Backup Configuration:

Go **Device >> High Availability**. Now, clicking on top right gear icon in **Control Link (HA1 Backup)** section, we will declare ethernet1/4 as our control plane link (HA1 Backup) as we decided earlier. We will be using 2.2.2.0/24 for HA1 backup link.



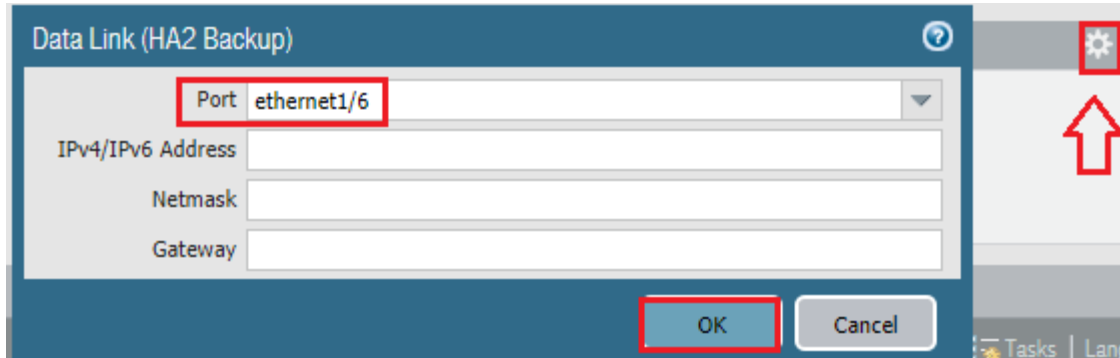
PA1-Data Link Configuration:

On the same page **Device >> High Availability**, need to click on top right gear icon in **Data Link (HA2)** section. In our case, ethernet1/5 is our HA2 link. It's directly connected, so transport mode is **ethernet**. Do not need to specify any IP address.



PA2-Data Link Backup Configuration:

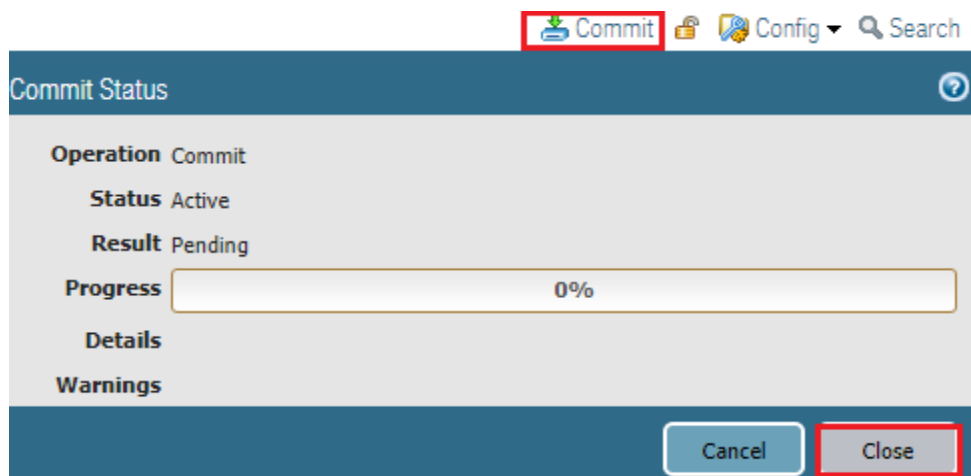
On the same page **Device >> High Availability**, need to click on top right gear icon in **Data Link (HA2 Backup)** section. In our case, ethernet1/6 is our HA2 backup link. It's directly connected, so transport mode is **ethernet**. Do not need to specify any IP address.



The image shows a configuration window titled "Data Link (HA2 Backup)". It has a "Port" dropdown menu set to "ethernet1/6". Below this are input fields for "IPv4/IPv6 Address", "Netmask", and "Gateway". At the bottom right, there are "OK" and "Cancel" buttons. A red box highlights the "Port" dropdown, and another red box highlights the "OK" button. A red arrow points upwards from the "OK" button area towards the top right corner of the window, where a gear icon is located.

PA2-Save Changes:

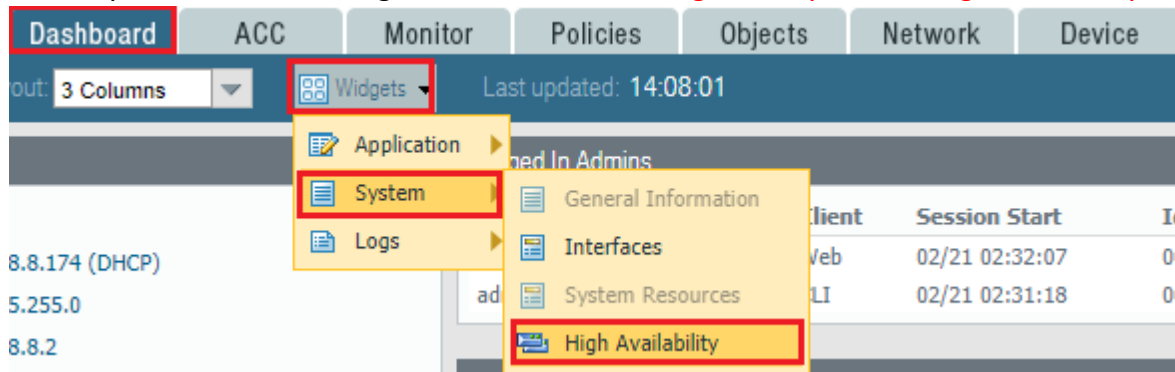
Click on top right corner Commit link to save the changes.



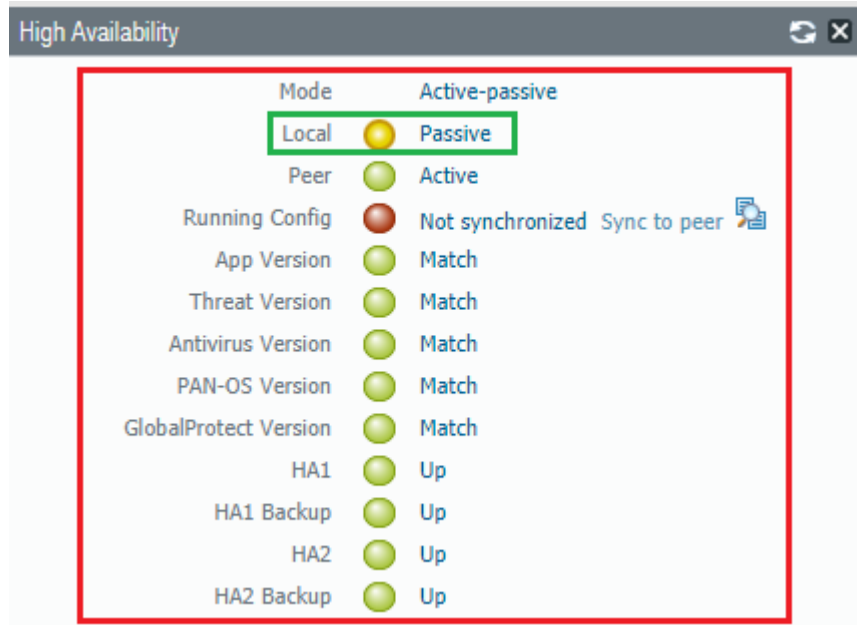
The image shows a "Commit Status" dialog box. At the top, there is a "Commit" button with a green checkmark icon, highlighted with a red box. Below this, the dialog shows the following information: "Operation: Commit", "Status: Active", "Result: Pending", and "Progress: 0%". There are sections for "Details" and "Warnings". At the bottom right, there are "Cancel" and "Close" buttons, with the "Close" button highlighted by a red box.

Verification:

To verify the HA status. Just go to **Dashboard >> Widgets >> System >> High Availability**.



You can see our Active-Passive HA is already formed. However, configuration doesn't sync yet. We can follow below to sync configuration from Active to Passive unit. We can just click on **Sync to peer**. It will automatically sync configuration from Active unit to Passive unit.



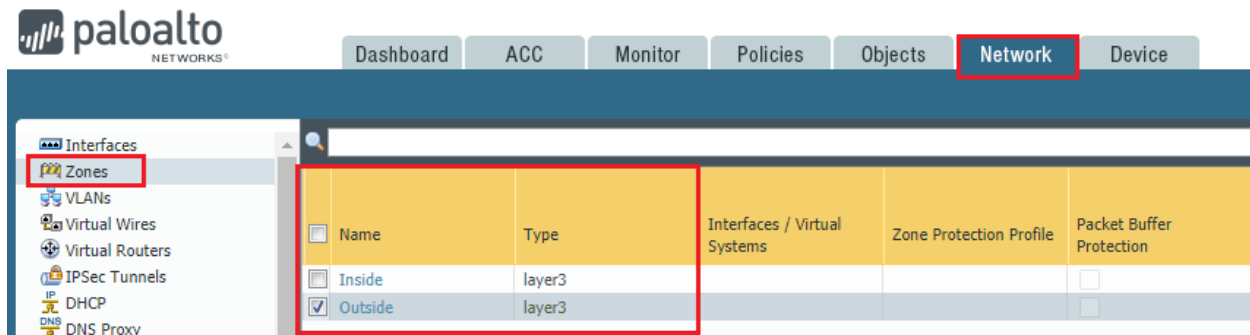
Run this command will do the same job.

admin@PA-ACTIVE(active)> request high-availability sync-to-remote running-config

Only in Active Firewall:

Create Zones:

Configure two zones names **Inside** and **Outside**. Go to **Network> Zone>Add**, Give the name **Inside**, select Type to be **Layer3** and click **OK**. Create the same way other Zone **Outside**.



Configure Security Policy:

Now, create a Security Policy to allow access from Inside to Outside zone.

Policies>Security>Add, Give the name to your Security Policy (**Inside-to-Outside**), Add Source Zone (**Inside**), Add Destination Zone (**Outside**), Allow access, in our case allowing all traffic.



Configure Interfaces:

Go to **Network>Interfaces** Click on **ethernet1/1** interface change Interface Type: **Layer3**, set Virtual Router: **default**, set Security Zone: **Outside**, Click on IPv4 tab Assign IP Address: **192.168.8.100/24** and Click **OK**. Click on **ethernet1/2** interface change Interface Type: **Layer3**, set Virtual Router: **default**, set Security Zone: **Inside**, Click on IPv4 tab Assign IP Address: **192.168.1.100/24**.

Interface	Interface Type	Management Profile	Link State	IP Address	Virtual Router	Tag
Ethernet Interface Interface Name ethernet1/1 Comment Connected to Internet Interface Type Layer3 Netflow Profile None Config IPv4 IPv6 Advanced Assign Interface To Virtual Router default Security Zone Outside OK Cancel						

Interface	Interface Type	Management Profile	Link State	IP Address	Virtual Router	Tag
Ethernet Interface Interface Name ethernet1/1 Comment Connected to Internet Interface Type Layer3 Netflow Profile None Config IPv4 IPv6 Advanced Type ☒ Static ☐ PPPoE ☐ DHCP Client IP 192.168.8.100/24 + Add - Delete Move Up Move Down IP address/netmask. Ex. 192.168.2.254/24 OK Cancel						

Ethernet Interface

Interface Name: ethernet1/2

Comment: Connected to LAN

Interface Type: Layer3

Netflow Profile: None

Config IPv4 IPv6 Advanced

Assign Interface To

Virtual Router: default

Security Zone: Inside

OK Cancel

Ethernet Interface

Interface Name: ethernet1/2

Comment: Connected to LAN

Interface Type: Layer3

Netflow Profile: None

Config IPv4 IPv6 Advanced

Type: ☒ Static ☐ PPPoE ☐ DHCP Client

IP

192.168.1.100/24

+ Add - Delete Move Up Move Down

IP address/netmask. Ex. 192.168.2.254/24

OK Cancel

Configure Routing:

Each interface must be given virtual router. **Network>Virtual Router>default** we will add default routing. **Static Routes>IPv4>Add** we will go by choosing interface> **ethernet1/1**(as Outside), put **192.168.8.2** as the next hop due to our topology.

Virtual Router - d Virtual Router - Static Route - IPv4

Router Settings

Static Routes

Redistribution Pr

RIP

OSPF

OSPFv3

BGP

Multicast

Name Default-Route

Destination 0.0.0.0/0

Interface ethernet1/1

Next Hop IP Address

192.168.8.2

Admin Distance 10 - 240

Metric 10

Route Table Unicast

BFD Profile Disable BFD

☐ Path Monitoring

Failure Condition ☒ Any ☐ All Preemptive Hold Time (min) 2

☐	Name	Enable	Source IP	Destination IP	Ping Interval(sec)	Ping Count
+ Add - Delete						

OK Cancel

Configure NAT/PAT:

Let's configure NAT using **Dynamic IP and Port** means translate all local LAN to only one IP address. I will NAT my Inside LAN 192.168.1.0/24 to 192.168.8.100 IP address of WAN.

Policies > NAT > Add Let's name it **Inside-To-Outside**.

NAT Policy Rule

General Original Packet Translated Packet

Source Address Translation

Translation Type Dynamic IP And Port

Address Type Interface Address

Interface ethernet1/1

IP Address 192.168.8.100/24

Destination Address Translation

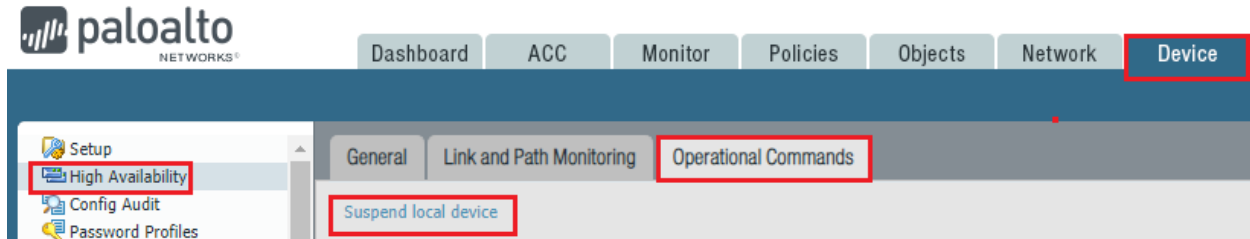
Translation Type None

OK Cancel

		Original Packet							Translated Packet
Name	Tags	Source Zone	Destination Zone	Destination Interface	Source Address	Destination Address	Service	Source Translation	
1	Inside-to-Outside	none	Inside	Outside	any	any	any	dynamic-ip-and-port ethernet1/1 192.168.8.100/24	

Verify which unit is currently active and which one is currently passive by using CLI command.
> show high-availability state

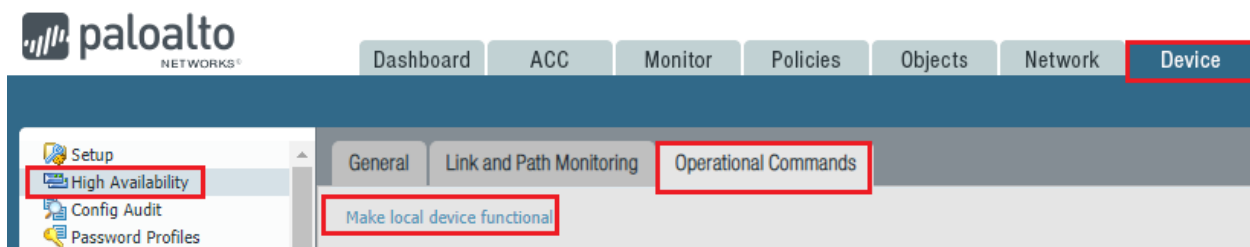
From WebGUI > Device > High Availability > Operational Commands - click Suspend local device



Verify that the firewall is now in a suspended state before a reboot and the passive member assume the active position.

High Availability			High Availability		
Mode		Active-passive	Mode		Active-passive
Local		Suspended (User requested)	Local		Active
Peer		Active	Peer		Suspended (User requested)
Running Config		Synchronized	Running Config		Synchronized
App Version		Match	App Version		Match
Threat Version		Match	Threat Version		Match
Antivirus Version		Match	Antivirus Version		Match
PAN-OS Version		Match	PAN-OS Version		Match
GlobalProtect Version		Match	GlobalProtect Version		Match
HA1		Up	HA1		Up
HA1 Backup		Up	HA1 Backup		Up
HA2		Up	HA2		Up
HA2 Backup		Up	HA2 Backup		Up

Inside of the WebGUI > Device > High Availability > Operational Commands - click on Make local device functional



Chapter 4: Barracuda Web Application Firewall (WAF)

4.1 Overview of WAFs

4.1.1 Purpose and Importance of WAFs in Protecting Web Applications

A **Web Application Firewall (WAF)** is a security tool designed to protect web applications by filtering and monitoring HTTP traffic between a web application and the Internet. WAFs specifically target the application layer (Layer 7) of the OSI model, which is where most web-based attacks occur. They act as a barrier between the client (user) and the application server, inspecting all incoming web traffic to prevent malicious attacks before they can reach the web application.

Key Purposes of a WAF:

1. **Protection Against OWASP Top Ten Threats:** The **OWASP Top Ten** is a widely accepted list of the most critical web application security risks. WAFs are specifically designed to address these risks, including threats like SQL injection, Cross-Site Scripting (XSS), Cross-Site Request Forgery (CSRF), and more. By using predefined rules and threat intelligence, WAFs can proactively block such attacks.
2. **Traffic Filtering:** WAFs inspect incoming HTTP requests and filter out malicious requests before they reach the web application. This includes filtering out harmful payloads, malicious scripts, and requests that attempt to exploit vulnerabilities.
3. **Application Layer Defense:** Traditional network firewalls focus on securing network layers (Layer 3 and 4), but a WAF operates at Layer 7, providing granular protection against application-specific vulnerabilities. It can detect and block specific attack patterns that traditional firewalls might miss.
4. **Rate Limiting and Traffic Management:** WAFs can be configured to limit the rate at which requests are made to an application. This is useful for preventing **denial-of-service (DoS)** or **distributed denial-of-service (DDoS)** attacks, where malicious actors attempt to flood the application with excessive requests to overwhelm its resources.
5. **Compliance and Security Standards:** Many industries have strict regulations regarding the protection of sensitive data. For example, the **Payment Card Industry Data Security Standard (PCI DSS)** mandates that businesses must protect cardholder data. A WAF can

help fulfill these compliance requirements by mitigating web application vulnerabilities and protecting sensitive customer data.

6. **Zero-Day Protection:** WAFs use heuristic analysis and threat intelligence to provide protection against zero-day vulnerabilities—attacks exploiting unpatched security flaws that are unknown or have not been publicly disclosed.

4.1.2 Deployment Models: Cloud-Based vs. On-Premises WAFs

WAFs can be deployed in different models based on an organization's infrastructure needs and scale. The two main deployment models are **cloud-based WAFs** and **on-premises WAFs**.

- **Cloud-Based WAFs:**
 - **Deployment:** Cloud-based WAFs are hosted by a third-party service provider and typically function as a service that customers can subscribe to. These services are typically integrated with **Content Delivery Networks (CDNs)**, which help distribute the web traffic across multiple locations, optimizing speed and availability.
 - **Advantages:**
 - **Scalability:** Cloud WAFs automatically scale to handle traffic spikes, which is particularly useful for applications with fluctuating or unpredictable traffic.
 - **Reduced Maintenance:** With a cloud-based solution, businesses don't need to manage the infrastructure, hardware, or software updates, as this is handled by the service provider.
 - **Global Reach:** Cloud-based WAFs provide protection globally, as the CDN-like architecture can mitigate risks at multiple points around the world.
 - **Disadvantages:**
 - **Less Control:** The organization has less direct control over the configuration and customization of the WAF, as it's managed by a third party.
 - **Potential Latency:** Depending on how the service routes traffic, there may be added latency compared to on-premises solutions.
- **On-Premises WAFs:**
 - **Deployment:** On-premises WAFs are installed physically within the organization's data centers or deployed as virtual appliances. The organization has full control over the deployment, configuration, and management of the WAF.
 - **Advantages:**
 - **Full Control:** Organizations have complete control over the WAF's configuration, security policies, and integration with other security tools and systems.
 - **Customization:** Since the WAF is self-hosted, businesses can tailor it to meet specific security requirements, policies, and use cases.

- **Low Latency:** On-premises WAFs typically offer lower latency as traffic doesn't need to traverse long distances to reach the cloud before being inspected.
- **Disadvantages:**
 - **Scalability Limitations:** On-premises WAFs may require additional hardware to handle increased traffic, which can be costly and require complex scaling strategies.
 - **Higher Maintenance Overhead:** The organization must manage updates, security patches, and operational maintenance, which may require dedicated IT resources.

4.1.3 Comparison of WAF Features Across Different Vendors

Different WAF vendors offer varying features, and it's important to compare these to understand the capabilities each solution brings to the table. Here's a comparison of key features across some prominent WAF vendors, including **Barracuda**, **AWS**, **Imperva**, and **F5**:

Feature	Barracuda WAF	AWS WAF	Imperva WAF	F5 WAF
OWASP Top Ten Protection	Yes	Yes	Yes	Yes
DDoS Protection	Yes	Yes	Yes	Yes
Bot Protection	Yes	No	Yes	Yes
SSL/TLS Offloading	Yes	Yes	Yes	Yes
Advanced Threat Intelligence	Yes (via Cloud)	Yes	Yes	Yes
API Security	Yes	Yes	Yes	Yes
Logging & Reporting	Advanced	Basic	Advanced	Advanced
Scalability	Cloud/Hybrid/On-Prem	Cloud-only	Cloud/On-Prem	Cloud/On-Prem
Ease of Deployment	Easy	Easy	Moderate	Complex
Custom Rule Creation	Yes	Yes	Yes	Yes

4.2 Barracuda WAF

4.2.1 Deep Dive into Barracuda WAF Capabilities and Architecture

The **Barracuda Web Application Firewall (WAF)** is a robust solution that secures web applications, APIs, and web servers against the most common and dangerous application-layer threats. Barracuda's WAF is known for its ease of deployment, effectiveness, and advanced protection capabilities.

Key Capabilities:

1. **Advanced Threat Protection:** Barracuda WAF provides **advanced protection** against common web application attacks, such as **SQL Injection**, **Cross-Site Scripting (XSS)**, **Cross-Site Request Forgery (CSRF)**, **Remote File Inclusion (RFI)**, and more. It uses signature-based detection and heuristic analysis to identify known and emerging threats.
2. **Automated Security Updates:** The Barracuda WAF continuously receives real-time threat intelligence updates from Barracuda's cloud-based security services. This ensures that the WAF is always equipped with the latest security rules and protections, minimizing the risk from newly discovered vulnerabilities.
3. **Bot Protection:** Barracuda WAF includes built-in bot detection mechanisms to defend against **credential stuffing**, **scraping**, **account takeover** attempts, and other automated attacks. By using behavioral analysis, it can detect and block malicious bots while allowing legitimate users to access the application.
4. **SSL/TLS Offloading:** Barracuda WAF supports **SSL offloading**, which reduces the burden on the backend servers by handling SSL/TLS decryption. This also enables deep packet inspection of encrypted traffic, ensuring that attacks embedded in HTTPS traffic are detected and blocked.
5. **Web Application Acceleration:** Features like **caching**, **compression**, and **content delivery network (CDN) integration** help improve the performance of web applications. These features reduce latency, optimize resource usage, and improve user experience by ensuring faster load times.
6. **API Security:** Barracuda WAF includes specific mechanisms to protect **APIs** from attacks, such as **API abuse**, **data theft**, and **API scraping**. It can be configured to enforce security policies for API calls, including input validation and threat detection.

Barracuda WAF Architecture:

Barracuda WAF can be deployed in several ways:

- **Cloud Deployment:** A cloud-based solution where the WAF is hosted on the Barracuda Cloud. This allows businesses to protect web applications without managing the underlying hardware.
- **On-Premises Deployment:** The WAF can be deployed as a physical or virtual appliance within the organization's infrastructure, providing full control over the security configuration.
- **Hybrid Deployment:** A hybrid model where the WAF is deployed both on-premises and in the cloud, offering a flexible solution for businesses with mixed infrastructure.

4.2.2 Practical: Enabling WAF Features on Barracuda WAF

- **Step 1:** Access the Barracuda WAF Management

Interface via the web browser.

- **Step 2:** Navigate to the **Security Policies** section and configure the desired protections.
- **Step 3:** Enable **Bot Protection**, **SSL Offloading**, and configure rules to block OWASP Top Ten vulnerabilities.
- **Step 4:** Test the deployment by sending requests that match known attack patterns (e.g., SQL injection).

4.2.3 Understanding Application-Layer Security Controls (OWASP Top Ten)

The **OWASP Top Ten** list identifies the ten most critical security risks to web applications. The Barracuda WAF provides out-of-the-box protection against these risks by enforcing security rules designed to prevent attacks such as:

A1: Injection

Injection attacks occur when an attacker sends malicious data (often in the form of SQL, NoSQL, OS commands, or LDAP queries) to a web application, which is then executed by the server. The most common form is **SQL injection**, where the attacker inserts malicious SQL statements into an input field (e.g., login forms). This can lead to data manipulation, unauthorized access, or even complete system compromise.

- **Prevention:** Use parameterized queries, input validation, and prepared statements.

A2: Broken Authentication

This vulnerability occurs when an attacker can bypass authentication mechanisms (e.g., through brute-force attacks or credential stuffing). It typically involves weak password policies, insecure session management, and poor implementation of multi-factor authentication (MFA).

- **Prevention:** Implement strong password policies, session management best practices, and multi-factor authentication.

A3: Sensitive Data Exposure

Sensitive data exposure happens when applications fail to protect data such as credit card numbers, passwords, or personal information during storage or transmission. Without proper encryption, attackers can access and misuse this sensitive data.

- **Prevention:** Encrypt sensitive data both in transit (using TLS/SSL) and at rest (using AES encryption), and use secure key management.

A4: XML External Entities (XXE)

XXE attacks exploit vulnerabilities in XML parsers by including external entities within XML documents. This can lead to unauthorized access to files, server-side request forgery (SSRF), and denial-of-service attacks.

- **Prevention:** Disable XML external entity processing, use safer XML parsers, and validate input thoroughly.

A5: Broken Access Control

Broken access control occurs when an attacker can gain unauthorized access to resources or functionality, often by manipulating URL parameters or accessing other users' data. This happens when proper checks are not in place to enforce user permissions based on roles.

- **Prevention:** Implement role-based access control (RBAC), ensure proper authorization checks on the server side, and validate access to resources.

A6: Security Misconfiguration

Security misconfiguration refers to improper or incomplete configuration of security settings, including default credentials, unnecessary features, or unpatched systems. This can lead to an application being exposed to attackers.

- **Prevention:** Follow the principle of least privilege, disable unused features, and apply security patches regularly.

A7: Cross-Site Scripting (XSS)

XSS attacks occur when an attacker injects malicious scripts into web pages viewed by users, often exploiting insecure web applications that fail to sanitize input properly. These scripts can steal cookies, session tokens, or perform actions on behalf of the user.

- **Prevention:** Use proper input validation and output encoding, and implement Content Security Policy (CSP) headers.

A8: Insecure Deserialization

Insecure deserialization involves manipulating serialized objects in order to execute malicious code or tamper with the integrity of data. This attack can lead to remote code execution or data corruption.

- **Prevention:** Avoid deserializing untrusted data, implement integrity checks, and use safe serialization formats.

A9: Using Components with Known Vulnerabilities

Using outdated libraries or components with known security vulnerabilities can open an application to attacks. Attackers exploit these flaws to compromise systems.

- **Prevention:** Regularly update all software components, libraries, and frameworks to the latest, secure versions, and monitor for known vulnerabilities.

A10: Insufficient Logging & Monitoring

This vulnerability occurs when an application fails to log and monitor activities appropriately, making it difficult to detect suspicious activities or breaches in real-time. This can delay response to attacks and increase damage.

- **Prevention:** Implement comprehensive logging and monitoring, set up automated alerts for suspicious activity, and ensure logs are securely stored and analyzed.

4.3 WAF Policies and Rules

4.3.1 Creating and Managing WAF Policies for Web Applications

WAF policies are essential for defining how a web application will behave in response to different types of traffic. Security policies can be applied to **allow**, **block**, or **rate-limit** specific types of traffic based on rules.

- **Steps to Create a WAF Policy:**
 - **Rule Definition:** Define the security rules that will filter traffic based on criteria such as IP address, HTTP method, and patterns in the request.
 - **Security Levels:** Set rules to specify what types of traffic are allowed, blocked, or logged for further analysis. This includes blocking dangerous input like SQL commands, scripts, or suspicious URLs.
 - **Bot Mitigation:** Apply **Bot Protection** rules to limit suspicious behavior from known bots.

4.4 Monitoring and Logging for WAF

4.4.1 Understanding WAF Logs, Metrics, and Alerting Mechanisms

WAF logs and metrics provide valuable insight into the behavior of web traffic and the effectiveness of the WAF in blocking malicious activity. Logs include:

- **Request Details:** Information such as the source IP address, timestamp, URL requested, and HTTP method.
- **Rule Violations:** Specific rules that were triggered, including details of the attack detected (e.g., SQL Injection, Cross-Site Scripting).
- **Response Actions:** Whether the request was allowed, blocked, or logged for further analysis.

4.4.2 Practical: Analyzing WAF Logs for Anomalies and Security Events

1. **Step 1:** Navigate to the **Logs** section in the Barracuda WAF console.
2. **Step 2:** Filter logs to review security events, focusing on blocked requests and failed login attempts.
3. **Step 3:** Identify potential security incidents, such as abnormal traffic spikes or requests containing malicious payloads.

4.4.3 Setting Up Custom Alerts Based on WAF Events

Alerts can be configured in the Barracuda WAF to notify administrators about potential security events. For instance:

- **Threshold Alerts:** Set up alerts to notify when a certain threshold of blocked requests is exceeded.
- **Event-Based Alerts:** Configure the WAF to alert on specific types of attacks (e.g., SQL injection attempts, bot traffic, etc.).

The Barracuda Web Application Firewall (WAF) is a comprehensive solution for protecting web applications from a variety of threats. By offering robust protections against OWASP Top Ten risks, bot attacks, DDoS, and other web-based threats, Barracuda WAF is a key component in securing web-facing applications. By implementing best practices in policy creation, log analysis, and alert configuration, organizations can ensure their applications remain protected against emerging threats while maintaining compliance with industry regulations.